

VIRENDRA SWARUP INSTITUTE OF COMPUTER STUDIES

VSICS Kanpur Master Knowledge Base

Structured Production RAG Corpus with Extracted PDF Tables & Syllabi

Target Website:	https://www.vsicskanpur.org/
Extraction Date:	September 05, 2026
Total Compiled Items:	34 Documents & Pages
Extracted Tables:	212 Structured Tables
Format:	100% Vector Text PDF with Semantic RAG Chunks

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DOCUMENT CHUNK 1: Faculty & Leadership Members [HTML Page] | URL: <https://www.vsicspanpur.org/board-of-managment.html>

Faculty & Leadership Members

Name	Designation	Qualification / Role	Experience
Mrs. Kumkum Swarup	Mrs. Kumkum Swarup	Ph.D. / Master's Degree	15+ Years
Mr. Gauravendra Swarup	Mr. Gauravendra Swarup	Ph.D. / Master's Degree	15+ Years

DOCUMENT CHUNK 2: VSICS [HTML Page] | URL: <https://www.vsicspanpur.org/infrastructure.html>

Lecture Hall

30 well designed Lecture Halls with a standard sitting capacity of 70 students at a time. The lecture hall is equipped with LCD and other modern gadgets. Few of the lecture halls are air conditioned..

Library

The heart of any Institute is its library. Our library is well stocked with a variety of books that has more than 2000 titles in over 11,000 multiple copies. It is regularly updated with the latest titles from India as well as overseas.

Seminar Hall

3 Seminar halls having seating capacity of 150 is utilized for Seminars, Conferences, Guest Lectures, Group Discussion, Debates and in-house Competitions. It is equipped with the latest audio visual gadget.

Computer Lab

8 Computer Labs, well equipped with more than 400 Pentium IV computers & other computing resources to cater to the academic needs of the Institute. All computers are networked through Novel, NT & Linux along with Internet server. The state-of-the-art centre provides hands on experience to the user & always keeps pace with the advancement of technology.

Sports & Fitness

Only a healthy body can develop and maintain a healthy mind.Hence the Institute Promotes fitness and health as a regular activity. The Institute boasts of a compound field for Football, Cricket, Hockey, Boxing & Athletics in addition to a Badminton and Handball courts. The Institute provides facilities for the conduction of the above mentioned games. The Institute also has an indoor games room with Chess, Carrom and Table Tennis facilities.

DOCUMENT CHUNK 3: Admissionform (PDF) [PDF Document] | URL: <https://www.vsicspanpur.org/docs/admissionform.pdf>

Admissionform (Official Document)

Source Document URL: <https://www.vsicspanpur.org/docs/admissionform.pdf>

Total Document Pages: 2

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Admission No. _____
Hindi, English, Maths
Science,

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&
INFORMATION

DOCUMENT CHUNK 4: Brochure (PDF) [PDF Document] | URL: <https://www.vsicskanpur.org/docs/brochure.pdf>

Brochure (Official Document)

Source Document URL: <https://www.vsicskanpur.org/docs/brochure.pdf>
Total Document Pages: 56

DOCUMENT CHUNK 5: VSICS [HTML Page] | URL: <https://www.vsicskanpur.org/mca.html>

- Overview

MCA Course is approved by AICTE & affiliated to A.K.T.U. Lucknow.

About MCA

The Master of Computer Applications (MCA) is a two-year professional postgraduate program designed for students who wish to gain advanced knowledge and expertise in computer applications, software development, and modern programming technologies. The program offers a balanced blend of theoretical concepts and practical learning, enabling students to develop strong analytical, technical, and problem-solving skills. It provides in-depth exposure to programming languages, database management, software engineering, web technologies, cloud computing, and emerging IT trends. An MCA degree equips students with the knowledge and hands-on experience required to design, develop, and manage innovative software applications efficiently. The program prepares graduates for rewarding careers in software development, system analysis, application design, IT consulting, and other technology-driven industries. The course is approved by AICTE, affiliated to AKTU, Lucknow. Where we excel? 28+ Years of Academic Excellence Placement and Career Development Support Industry Interaction and Internship Platforms Holistic Individual Development Programmes Internet-Enabled Class Rooms Well Equipped Computer Labs For Practical Learning Use Of ICT Tools For Enhanced Learning Exposure Community-Based Learning

Where we excel?

- 28+ Years of Academic Excellence
- Placement and Career Development Support
- Industry Interaction and Internship Platforms
- Holistic Individual Development Programmes
- Internet-Enabled Class Rooms
- Well Equipped Computer Labs For Practical Learning
- Use Of ICT Tools For Enhanced Learning Exposure
- Community-Based Learning
- Duration 2 Years
- Students 60

DOCUMENT CHUNK 6: VSICS [HTML Page] | URL: <https://www.vsicskanpur.org/mba.html>

- Overview

MBA Course is approved by AICTE & affiliated to A.K.T.U. Lucknow.

About MBA

The Master of Business Administration (MBA) program is designed to equip students with the knowledge, skills, and leadership qualities required to meet the evolving demands of business, industry, and commerce. The program enables students to perform effectively in diverse organizational environments and confidently address real-world business challenges. The curriculum provides comprehensive management education combined with practical exposure to contemporary business practices. In addition to academic excellence, the program emphasizes personality development, communication skills, analytical thinking, decision-making abilities, and professional ethics, preparing students to become competent, responsible, and industry-ready professionals capable of leading organizations in a dynamic business environment. The course is approved by AICTE, affiliated to AKTU, Lucknow.

Where we excel?

- 28+ Years of Academic Excellence
- Placement and Career Development Support
- Industry Interaction and Internship Platforms
- Holistic Individual Development Programmes
- Internet-Enabled Class Rooms
- Well Equipped Computer Labs For Practical Learning
- Use Of ICT Tools For Enhanced Learning Exposure
- Community-Based Learning
- Human Resources
- Marketing
- Finance
- Information Technology
- International Business
- Operations
- 28+ Years of Academic Excellence

- Placement and Career Development Support
- Industry Interaction and Internship Platforms
- Holistic Individual Development Programmes
- Internet-Enabled Class Rooms
- Well Equipped Computer Labs For Practical Learning
- Use Of ICT Tools For Enhanced Learning Exposure
- Community-Based Learning

Specializations Offered

- Human Resources
- Marketing
- Finance
- Information Technology
- International Business
- Operations
- Duration 2 Years
- Students 60

DOCUMENT CHUNK 7: VSICS [HTML Page] | URL: <https://www.vsicskanpur.org/placement.php>

Dr. Virendra Swarup Institute of Computer Studies has an impressive record of best placements with the Top Companies of the country for the past 27 years. VSICS is drawing synergistic advantages to enrich its placement potential with this rich legacy. Once students get oriented on the evolved industry requirements, they are trained rigorously through day long workshops. These workshops are handled by corporate trainers under our campus to corporate programme and the student of specific and specialized skills such as Resume Writing and Personality Development.

5**Students****75****Avg. Placement****35****Strong Alumni****Krishn Mohan****Riya Mishra****Deepanshi Soni****DAINIK JAGRAN****TCS****CAPGEMINI****WIPRO TECHNOLOGIES 2017****JUSTDIAL****WIPRO TECHNOLOGIES 2020****POLICY BAZAAR****ALIGHT SOLUTIONS****SUPREME INDUSTRIES****WIPROTECHNOLOGIES 2019****BRITISH TELECOM****uCERTIFY**

DOCUMENT CHUNK 8: Progressreport (PDF) [PDF Document] | URL:
<https://www.vsicskanpur.org/placement/ProgressReport.pdf>

Progressreport (Official Document)

****Source Document URL:**** <https://www.vsicskanpur.org/placement/ProgressReport.pdf>

****Total Document Pages:**** 15

Student Placement Record Table

Student Name	Course	Institute	Recruiter / Company
ADITI GUPTA	BBA	VSICS	JARO EDUCATION
KAUSHIKI PANDEY	BBA	VSICS	JARO EDUCATION

Student Name	Course	Institute	Recruiter / Company
GAURI PANDEY	BBA	VSICS	JARO EDUCATION
ANMOL CHAUDHARY	BBA	VSICS	4 FOX BUSINESS PVT. LTD
ANYA MISHRA	BBA	VSICS	4 FOX BUSINESS PVT. LTD
KRISHNA DUBEY	BBA	VSICS	4 FOX BUSINESS PVT. LTD
PRETYAKSHA BHATIA	BBA	VSICS	4 FOX BUSINESS PVT. LTD
VANSH MATLANI	BBA	VSICS	4 FOX BUSINESS PVT. LTD
ANSH TRIVEDI	BBA	VSICS	ACCENTURE
GAURI PANDEY	BBA	VSICS	ACCENTURE
JASLEEN KAUR	BBA	VSICS	ACCENTURE
JATIN DWIVEDI	BBA	VSICS	ACCENTURE
KAVYA GUPTA	BBA	VSICS	ACCENTURE
KHUSHI GUPTA	BBA	VSICS	ACCENTURE
MASOOM JAISWAL	BBA	VSICS	ACCENTURE
MAYANK TEKWANI	BBA	VSICS	ACCENTURE
RIDHIMA RAWAT	BBA	VSICS	ACCENTURE
SAKSHAM GUPTA	BBA	VSICS	ACCENTURE
SAUMYA SINGH SINGH	BBA	VSICS	ACCENTURE
SHUBHANGI PAL PAL	BBA	VSICS	ACCENTURE
ANSHITA GUPTA	BBA	VSICS	ARNAV INFOSOFTECH PVT. LTD.
SAUMYA BAJPAI	BBA	VSICS	ARNAV INFOSOFTECH PVT. LTD.
VANSH KAPOOR	BBA	VSICS	ARNAV INFOSOFTECH PVT. LTD.
AARADHY MISHRA	BBA	VSICS	ATHARV ONSTECH
ADITYA TIWARI	BBA	VSICS	ATHARV ONSTECH
ANSHITA GUPTA	BBA	VSICS	ATHARV ONSTECH
ANUJ TIWARI	BBA	VSICS	ATHARV ONSTECH
ANUJ YADAV	BBA	VSICS	ATHARV ONSTECH
ANUSHANSHA GAUR	BBA	VSICS	ATHARV ONSTECH
ARADHYA OMAR	BBA	VSICS	ATHARV ONSTECH
DISHA KESHWANI	BBA	VSICS	ATHARV ONSTECH
SHASHANK AHUJA	BBA	VSICS	ATHARV ONSTECH
SHASHANK DUBEY	BBA	VSICS	ATHARV ONSTECH
SHIKHA SINGH	BBA	VSICS	ATHARV ONSTECH
YOGENDRA PRATAP	BBA	VSICS	ATHARV ONSTECH
ABHISHEK TRIPATHI	BBA	VSICS	AXIS BANK
ADITI CHAURASIYA	BBA	VSICS	AXIS BANK
ANAND SACHAN	BBA	VSICS	AXIS BANK
ARYAN GUPTA	BBA	VSICS	AXIS BANK
ARYAN GUPTA	BBA	VSICS	AXIS BANK
HIMANSHU GUPTA	BBA	VSICS	AXIS BANK
KHUSHI SAHU	BBA	VSICS	AXIS BANK
MITALI KUMARI SHAKYA	BBA	VSICS	AXIS BANK

Student Name	Course	Institute	Recruiter / Company
RAUNAK GUPTA	BBA	VSICS	AXIS BANK
SHIVA SINGH	BBA	VSICS	AXIS BANK
SNEHA GAUTAM	BBA	VSICS	AXIS BANK
SNEHA GAUTAM	BBA	VSICS	AXIS BANK
UNNATI SRIVASTAVA	BBA	VSICS	AXIS BANK
UNNATI SRIVASTAVA	BBA	VSICS	AXIS BANK
UTKARSH MISHRA	BBA	VSICS	AXIS BANK
VAISHNAVI KESHARWANI	BBA	VSICS	AXIS BANK
VARUN BHARTI	BBA	VSICS	AXIS BANK
VISHAL SINGH	BBA	VSICS	AXIS BANK
VIVEK NIGAM	BBA	VSICS	AXIS BANK
ADARSH GAUR	BBA	VSICS	BRITISH TELECOM
ADITI KAUSHAL	BBA	VSICS	BRITISH TELECOM
AISHWARYA GUPTA	BBA	VSICS	BRITISH TELECOM
ANMOL CHAUDHAY	BBA	VSICS	BRITISH TELECOM
APOORV GUPTA	BBA	VSICS	BRITISH TELECOM
ARYAN GUPTA	BBA	VSICS	BRITISH TELECOM
GAURI PANDEY	BBA	VSICS	BRITISH TELECOM
GURLEEN KAUR JULKA	BBA	VSICS	BRITISH TELECOM
KARTIKEY AWASTHI	BBA	VSICS	BRITISH TELECOM
KATYAYANI SINHA	BBA	VSICS	BRITISH TELECOM
KHUSHI GUPTA	BBA	VSICS	BRITISH TELECOM
MANIK TRIPATHI	BBA	VSICS	BRITISH TELECOM
PARV BHARGAVA	BBA	VSICS	BRITISH TELECOM
PRABHAT SINGH	BBA	VSICS	BRITISH TELECOM
PRATHNA CHATURVEDI	BBA	VSICS	BRITISH TELECOM
PUNEET YADAV	BBA	VSICS	BRITISH TELECOM
RITIKA PANDEY	BBA	VSICS	BRITISH TELECOM
SAUMYA SINGH	BBA	VSICS	BRITISH TELECOM
SHIKHAR SRIVASTAVA	BBA	VSICS	BRITISH TELECOM
SHRADDHA SINGH	BBA	VSICS	BRITISH TELECOM
SHREYA NIGAM	BBA	VSICS	BRITISH TELECOM
VAIBHAV SHUKLA	BBA	VSICS	BRITISH TELECOM
VAIBHAV SINGH YADAV	BBA	VSICS	BRITISH TELECOM
VAISHNAVI GUPTA	BBA	VSICS	BRITISH TELECOM
VARUN MADNANI	BBA	VSICS	BRITISH TELECOM
ADARSHITA SINGH	BBA	VSICS	CAPGEMINI
ANUSHKA MISHRA	BBA	VSICS	CAPGEMINI
PRETYAKSHA BHATIA	BBA	VSICS	CAPGEMINI
SHREYA NIGAM	BBA	VSICS	CAPGEMINI
VANISHA KHANDUJA	BBA	VSICS	CAPGEMINI

Student Name	Course	Institute	Recruiter / Company
ADITYA TIWARI	BBA	VSICS	CMR TECHNOLOGIES
AVIRAL TIWARI	BBA	VSICS	CMR TECHNOLOGIES
SHASHANK AHUJA	BBA	VSICS	CMR TECHNOLOGIES
ANSHITA GUPTA	BBA	VSICS	FINCANA SOLUTIONS
PRETYAKSHA BHATIA	BBA	VSICS	FINCANA SOLUTIONS
SHASHANK AHUJA	BBA	VSICS	FINCANA SOLUTIONS
ANYA MISHRA	BBA	VSICS	FINCANA SOLUTIONS
ANUJ TIWARI	BBA	VSICS	FUTURZHR
PRIYANKA VERMA	BBA	VSICS	INFOSEEK SOFTWARE
GAURI PANDEY	BBA	VSICS	INTELLIPAT
AMAN SINGH CHANDEL	BBA	VSICS	JUSTDIAL
SURYANSH TRIPATHI	BBA	VSICS	KANPUR PLASTIPACK
AVIRAL TIWARI	BBA	VSICS	LOHIA CORP.
GAURI PANDEY	BBA	VSICS	MIKKY PUBLICATION
GAUTAMI KHAIRAJANI	BBA	VSICS	MIKKY PUBLICATION
ISHMEET KAUR	BBA	VSICS	MIKKY PUBLICATION

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SNO.		NAME OF STUDENT			COURSE		
1		ADITI GUPTA			BBA		
2		KAUSHIKI PANDEY			BBA		
3		GAURI PANDEY			BBA		
4		ANMOL CHAUDHARY			BBA		
5		ANYA MISHRA			BBA		
6		KRISHNA DUBEY			BBA		
7		PRETYAKSHA BHATIA			BBA		
8		VANSH MATLANI			BBA		
9		ANSH TRIVEDI			BBA		
10		GAURI PANDEY			BBA		
11		JASLEEN KAUR			BBA		

	PLACEMENT PROGRESS REPORT 2022-2025 PASSOUTS						
12		JATIN DWIVEDI			BBA		
13		KAVYA GUPTA			BBA		
14		KHUSHI GUPTA			BBA		
15		MASOOM JAISWAL			BBA		
16		MAYANK TEKWANI			BBA		
17		RIDHIMA RAWAT			BBA		
18		SAKSHAM GUPTA			BBA		
19		SAUMYA SINGH SINGH			BBA		
20		SHUBHANGI PAL PAL			BBA		
21		ANSHITA GUPTA			BBA		
22		SAUMYA BAJPAI			BBA		
23		VANSH KAPOOR			BBA		
24		AARADHY MISHRA			BBA		
25		ADITYA TIWARI			BBA		
26		ANSHITA GUPTA			BBA		
27		ANUJ TIWARI			BBA		
28		ANUJ YADAV			BBA		
29		ANUSHANSH A GAUR			BBA		
30		ARADHYA OMAR			BBA		
31		DISHA KESHWANI			BBA		
32		SHASHANK AHUJA			BBA		
33		SHASHANK DUBEY			BBA		
34		SHIKHA SINGH			BBA		
35		YOGENDRA PRATAP			BBA		

	PLACEMENT PROGRESS REPORT 2022-2025 PASSOUTS						
36			ABHISHEK TRIPATHI			BBA	
37			ADITI CHAURASIYA			BBA	
38			ANAND SACHAN			BBA	
39		ARYAN GUPTA			BBA		

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PLACEMENT PROGRESS REPORT 2022-2025 PASSOUTS

SNO.

NAME OF STUDENT

COURSE COLLEGE

COMPANY

1

ADITI GUPTA

BBA

DAMS

JARO EDUCATION

2

KAUSHIKI PANDEY

BBA

DAMS

JARO EDUCATION

3

GAURI PANDEY

BBA

VSICS

JARO EDUCATION

4

ANMOL CHAUDHARY

BBA

VSICS

4 FOX BUSINESS PVT. LTD

5

ANYA MISHRA

BBA

VSICS

4 FOX BUSINESS PVT. LTD

6

KRISHNA DUBEY

BBA

DAMS

4 FOX BUSINESS PVT. LTD

7

PRETYAKSHA BHATIA

BBA

VSICS

4 FOX BUSINESS PVT. LTD

8

VANSH MATLANI

BBA

VSICS

4 FOX BUSINESS PVT. LTD

9

ANSH TRIVEDI

BBA

VSICS

ACCENTURE

10

GAURI PANDEY

BBA

VSICS

ACCENTURE

11

JASLEEN KAUR

BBA

VSICS

ACCENTURE

12

JATIN DWIVEDI

BBA

VSICS

ACCENTURE

13

KAVYA GUPTA

BBA

VSICS

ACCENTURE

14

KHUSHI GUPTA

BBA

VSICS

ACCENTURE

15

MASOOM JAISWAL

BBA

VSICS

ACCENTURE

16

MAYANK TEKWANI

BBA

VSICS

ACCENTURE

17

RIDHIMA RAWAT

BBA

VSICS

ACCENTURE

18

SAKSHAM GUPTA

BBA

VSICS

ACCENTURE

19

SAUMYA SINGH SINGH

BBA

VSICS

ACCENTURE

20

SHUBHANGI PAL PAL

BBA

VSICS

ACCENTURE

21

ANSHITA GUPTA

BBA

VSICS

ARNAV INFOSOF PVT. LTD.

22

SAUMYA BAJPAI

BBA

DAMS

ARNAV INFOSOF PVT. LTD.

23

VANSH KAPOOR

BBA

DAMS

ARNAV INFOSOF PVT. LTD.

24

AARADHY MISHRA

BBA

VSICS

ATHARV ONSTECH

25

ADITYA TIWARI

BBA

VSICS

ATHARV ONSTECH

26

ANSHITA GUPTA

BBA

VSICS

ATHARV ONSTECH

27

ANUJ TIWARI

BBA

VSICS

ATHARV ONSTECH

28

ANUJ YADAV

BBA

VSICS

ATHARV ONSTECH

29

ANUSHANSHA GAUR

BBA

VSICS

ATHARV ONSTECH

30

ARADHYA OMA

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40		ARYAN GUPTA			BBA		
41		HIMANSHU GUPTA			BBA		
42		KHUSHI SAHU			BBA		
43		MITALI KUMARI SHAKYA			BBA		
44		RAUNAK GUPTA			BBA		
45		SHIVA SINGH			BBA		
46	SNEHA GAUTAM			BBA			DAMS
47		SNEHA GAUTAM			BBA		
48	UNNATI SRIVASTAVA			BBA			DAMS
49		UNNATI SRIVASTAVA			BBA		
50		UTKARSH MISHRA			BBA		
51		VAISHNAVI KESHARWANI			BBA		
52		VARUN BHARTI			BBA		
53		VISHAL SINGH			BBA		

40		ARYAN GUPTA			BBA		
54		VIVEK NIGAM			BBA		
55	ADARSH GAUR			BBA			DAMS
56	ADITI KAUSHAL			BBA			VSICS
57	AISHWARYA GUPTA			BBA			VSICS
58	ANMOL CHAUDHAY			BBA			VSICS
59	APOORV GUPTA			BBA			VSICS
60	ARYAN GUPTA			BBA			DAMS
61	GAURI PANDEY			BBA			VSICS
62	GURLEEN KAUR JULKA			BBA			VSICS
63	KARTIKEY AWASTHI			BBA			VSICS
64	KATYAYANI SINHA			BBA			VSICS
65	KHUSHI GUPTA			BBA			VSICS
66	MANIK TRIPATHI			BBA			VSICS
67	PARV BHARGAVA			BBA			VSICS
68	PRABHAT SINGH			BBA			VSICS
69	PRATHNA CHATURVEDI			BBA			VSICS
70	PUNEET YADAV			BBA			VSICS
71	RITIKA PANDEY			BBA			VSICS
72	SAUMYA SINGH			BBA			VSICS
73	SHIKHAR SRIVASTAVA			BBA			DAMS
74	SHRADDHA SINGH			BBA			DAMS
75	SHREYA NIGAM			BBA			VSICS
76	VAIBHAV SHUKLA			BBA			DAMS
77	VAIBHAV SINGH YADAV			BBA			DAMS

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78	VAISHNAVI GUPTA			BBA			VSICS
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80	ADARSHITA SINGH			BBA			VSICS
81	ANUSHKA MISHRA			BBA			VSICS

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ARYAN GUPTA

BBA

DAMS

AXIS BANK

41

HIMANSHU GUPTA

BBA

DAMS

AXIS BANK

42

KHUSHI SAHU

BBA

VSICS

AXIS BANK

43

MITALI KUMARI SHAKYA

BBA

VSICS

AXIS BANK

44

RAUNAK GUPTA

BBA

VSICS

AXIS BANK

45

SHIVA SINGH

BBA

VSICS

AXIS BANK

46

SNEHA GAUTAM

BBA

DAMS

AXIS BANK

47

SNEHA GAUTAM

BBA

DAMS

AXIS BANK

48

UNNATI SRIVASTAVA

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DAMS

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UNNATI SRIVASTAVA

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AXIS BANK

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UTKARSH MISHRA

BBA

VSICS

AXIS BANK

51

VAISHNAVI KESHARWANI

BBA

VSICS

AXIS BANK

52

VARUN BHARTI

BBA

VSICS

AXIS BANK

53

VISHAL SINGH

BBA

VSICS

AXIS BANK

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VIVEK NIGAM

BBA

VSICS

AXIS BANK

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ADARSH GAUR

BBA

DAMS

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AISHWARYA GUPTA

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ANMOL CHAUDHAY

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BRITISH TELECOM

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APOORV GUPTA

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82	PRETYAKSHA BHATIA	BBA	VSICS	CAPGEMINI
83	SHREYA NIGAM	BBA	VSICS	CAPGEMINI
84	VANISHA KHANDUJA	BBA	VSICS	CAPGEMINI
85	ADITYA TIWARI	BBA	VSICS	CMR TECHNOLOGIES
86	AVIRAL TIWARI	BBA	VSICS	CMR TECHNOLOGIES
87	SHASHANK AHUJA	BBA	VSICS	CMR TECHNOLOGIES
88	ANSHITA GUPTA	BBA	VSICS	FINCANA SOLUTIONS PVT.LTD.
89	PRETYAKSHA BHATIA	BBA	VSICS	FINCANA SOLUTIONS PVT.LTD.
90	SHASHANK AHUJA	BBA	VSICS	FINCANA SOLUTIONS PVT.LTD.

82	PRETYAKSHA BHATIA	BBA	VSICS	CAPGEMINI
91	ANYA MISHRA	BBA	VSICS	FINCANA SOLUTIONS PVT.LTD.
92	ANUJ TIWARI	BBA	VSICS	FUTURZHR
93	PRIYANKA VERMA	BBA	DAMS	INFOSEEK SOFTWARE SYSTEM
94	GAURI PANDEY	BBA	VSICS	INTELLIPAT
95	AMAN SINGH CHANDEL	BBA	DAMS	JUSTDIAL
96	SURYANSH TRIPATHI	BBA	VSICS	KANPUR PLASTIPACK
97	AVIRAL TIWARI	BBA	VSICS	LOHIA CORP.
98	GAURI PANDEY	BBA	VSICS	MIKKY PUBLICATION
99	GAUTAMI KHAIRAJANI	BBA	VSICS	MIKKY PUBLICATION
100	ISHMEET KAUR	BBA	DAMS	MIKKY PUBLICATION
101	NIKITA ISSRANI	BBA	VSICS	MIKKY PUBLICATION
102	SHREYA NIGAM	BBA	VSICS	MIKKY PUBLICATION
103	VAISHNAVI GUPTA	BBA	VSICS	MIKKY PUBLICATION
104	YASHASVI MISHRA	BBA	DAMS	MIKKY PUBLICATION
105	YASHI SINGH CHAUHAN	BBA	VSICS	MIKKY PUBLICATION
106	AKANKSHA KATIYAR	BBA	DAMS	MOTILAL OSWAL
107	ANUJ YADAV	BBA	DAMS	MOTILAL OSWAL
108	ARYAN	BBA	DAMS	MOTILAL OSWAL
109	HARSHITA JAISWAL	BBA	DAMS	MOTILAL OSWAL
110	KRISHNA DUBEY	BBA	DAMS	MOTILAL OSWAL
111	SAHIL GUPTA	BBA	DAMS	MOTILAL OSWAL
112	ADITYA GUPTA	BBA	DAMS	MOTILAL OSWAL
113	AFTAB ALAM	BBA	VSICS	MOTILAL OSWAL
114	AKANKSHA CHAUHAN	BBA	DAMS	MOTILAL OSWAL
115	ANVESHA SHARMA	BBA	DAMS	MOTILAL OSWAL
116	ASHISH VISHWAKARMA	BBA	DAMS	MOTILAL OSWAL
117	BHAVYA DIXIT	BBA	DAMS	MOTILAL OSWAL
118	DIVYAM KHEMANI	BBA	DAMS	MOTILAL OSWAL
119	PALAK AHUJA	BBA	VSICS	MOTILAL OSWAL

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PRETYAKSHA BHATIA

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SHREYA NIGAM

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CAPGEMINI

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VSICS

CAPGEMINI

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ADITYA TIWARI

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CMR TECHNOLOGIES

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AVIRAL TIWARI

BBA

VSICS

CMR TECHNOLOGIES

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SHASHANK AHUJA

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CMR TECHNOLOGIES

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ANSHITA GUPTA

BBA

VSICS

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PRETYAKSHA BHATIA

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VSICS

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PVT.LTD.

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SHASHANK AHUJA

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VSICS

FINCANA SOLUTIONS

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ANYA MISHRA

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AVIRAL TIWARI

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LOHIA CORP.

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GAURI PANDEY

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MIKKY PUBLICATION

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ISHMEET KAUR

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NIKITA ISSRANI

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MIKKY PUBLICATION

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SHREYA NIGAM

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VAISHNAVI GUPTA

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MOTILAL OSWAL

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HARSHITA JAISWAL

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120	PRANSHU SHUKLA	BBA	DAMS	MOTILAL OSWAL
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122	SHALINI YADAV	BBA	DAMS	MOTILAL OSWAL
123	VAISHNAVI AWASTHI	BBA	VSICS	MOTILAL OSWAL
124	VANSHIKA	BBA	DAMS	MOTILAL OSWAL
125	ANMOL CHAUDHARY	BBA	VSICS	MOVIDU TECHNOLOGIES
126	ASHUTOSH MATHUR	BBA	VSICS	MOVIDU TECHNOLOGIES
127	GAURI PANDEY	BBA	VSICS	MOVIDU TECHNOLOGIES
128	GAUTAMI KHAIRAJANI	BBA	VSICS	MOVIDU TECHNOLOGIES
129	MASOOM JAISWAL	BBA	VSICS	MOVIDU TECHNOLOGIES
130	NIKITA ISSRANI	BBA	VSICS	MOVIDU TECHNOLOGIES
131	PRIYANKA YADAV	BBA	VSICS	MOVIDU TECHNOLOGIES
132	SHREYA NIGAM	BBA	VSICS	MOVIDU TECHNOLOGIES
133	VANSH GUPTA	BBA	VSICS	MOVIDU TECHNOLOGIES
134	GAURI PANDEY	BBA	VSICS	NIVA BUPA
135	RADHIKA GUPTA	BBA	VSICS	NIVA BUPA
136	RAJVANSH DUBEY	BBA	DAMS	NIVA BUPA
137	SUDHANSHU PORWAL	BBA	VSICS	NIVA BUPA
138	SURYANSH DIXIT	BBA	VSICS	NIVA BUPA
139	VAIBHAV SHUKLA	BBA	VSICS	NIVA BUPA
140	ADITI	BBA	VSICS	PLANET SPARK
141	ADITYA SAINI	BBA	VSICS	PLANET SPARK
142	AKSHAT PANDEY	BBA	VSICS	PLANET SPARK
143	ANYA MISHRA	BBA	VSICS	PLANET SPARK
144	DEVANSHI DWIVEDI	BBA	VSICS	PLANET SPARK
145	GURLEEN KAUR JULKA	BBA	VSICS	PLANET SPARK
146	ISHA SINGH TOMAR	BBA	VSICS	PLANET SPARK
147	JASLEEN KAUR	BBA	VSICS	PLANET SPARK
148	KRATIKA SINGH	BBA	VSICS	PLANET SPARK
149	PRETYAKSHA BHATIA	BBA	VSICS	PLANET SPARK
150	ARPITA MEHROTRA	BBA	DAMS	PROFINITY
151	ASHISH GUPTA	BBA	DAMS	PROFINITY
152	KARTIKEY MISHRA	BBA	VSICS	PROFINITY
153	PRABHU NANDWANI	BBA	VSICS	PROFINITY

120	PRANSHU SHUKLA	BBA	DAMS	MOTILAL OSWAL
154	ABHAY TIWARI	BBA	DAMS	RSPL
155	ADARSH GAUR	BBA	DAMS	RSPL
156	UNNATI SRIVASTAVA	BBA	DAMS	RUPEEK
157	ABHINAV MISHRA	BBA	DAMS	SUNNY TOYOTA
158	AKRATI DWIVEDI	BBA	DAMS	SUNNY TOYOTA
159	ARPITA TRIPATHI	BBA	VSICS	SUNNY TOYOTA
160	GURJOT KAUR	BBA	VSICS	SUNNY TOYOTA
161	KARAN VEER SINGH	BBA	VSICS	SUNNY TOYOTA

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RUDRA MEHROTRA

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SHALINI YADAV

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DAMS

MOTILAL OSWAL

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VAISHNAVI AWASTHI

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VANSHIKA

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GAUTAMI KHAIRAJANI

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MASOOM JAISWAL

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MOVIDU TECHNOLOGIES

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NIKITA ISSRANI

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MOVIDU TECHNOLOGIES

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PRIYANKA YADAV

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VSICS

MOVIDU TECHNOLOGIES

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SHREYA NIGAM

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MOVIDU TECHNOLOGIES

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VANSH GUPTA

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MOVIDU TECHNOLOGIES

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GAURI PANDEY

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RADHIKA GUPTA

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RAJVANSH DUBEY

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SUDHANSHU PORWAL

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NIVA BUPA

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SURYANSH DIXIT

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VSICS

NIVA BUPA

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VAIBHAV SHUKLA

BBA

VSICS

NIVA BUPA

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ADITI

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VSICS

PLANET SPARK

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ADITYA SAINI

BBA

VSICS

PLANET SPARK

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AKSHAT PANDEY

BBA

VSICS

PLANET SPARK

143

ANYA MISHRA

BBA

VSICS

PLANET SPARK

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DEVANSHI DWIVEDI

BBA

VSICS

PLANET SPARK

145

GURLEEN KAUR JULKA

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VSICS

PLANET SPARK

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ISHA SINGH TOMAR

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PLANET SPARK

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JASLEEN KAUR

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KRATIKA SINGH

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163	SHIVAM PAL	BBA	DAMS	SUNNY TOYOTA
164	SHREYA NIGAM	BBA	VSICS	SUNNY TOYOTA
165	TANBIR SINGH	BBA	VSICS	SUNNY TOYOTA
166	ANSH TIWARI	BBA	DAMS	SUPERHOUSE
167	PUNEET YADAV	BBA	VSICS	SUPERHOUSE
168	ABHISHEK SINGH	BBA	DAMS	TALKING CROOKS
169	ANSH SINGH TOMAR	BBA	DAMS	TALKING CROOKS
170	RIYA CHAK	BBA	VSICS	TALKING CROOKS
171	PRABHAT SINGH	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
172	ADARSH GAUR	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
173	AKASH SACHAN	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
174	AKSHAT DIXIT	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
175	ANKITA SHUKLA	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
176	ANSH TIWARI	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS

162	KIRTI TIWARI	BBA	VSICS	SUNNY TOYOTA
177	ANSHIKA TIWARI	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
178	AYUSH YADAV	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
179	DARSHIKA BAJPAI	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
180	GAUTAMI KHAIRAJANI	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
181	GURLEEN KAUR JULKA	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
182	HARSHIT SINGH	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
183	JASLEEN KAUR	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
184	JATIN DWIVEDI	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
185	KARTIKEY AWASTHI	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
186	KUMKUM BAGHEL	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
187	MANIK TRIPATHI	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
188	MAYANK SHARMA	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
189	PRABHAT SINGH	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
190	PRAGATI YADAV	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
191	PRAGYA MISHRA	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
192	RAJVANSH DUBEY	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
193	ROHIT YADAV	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
194	SAUMYA BAJPAI	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
195	SAUMYA SAINI	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
196	SAUMYA SINGH	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
197	SHRADDHA SINGH	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
198	SHRESTH AGARWAL	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
199	SHREYA NIGAM	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
200	SHREYASHI TRIPATHI	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
201	SIDDHARTH PAL	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS

162	KIRTI TIWARI	BBA	VSICS	SUNNY TOYOTA
202	SUDHANSHU PORWAL	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
203	SUHANI SHUKLA	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS

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SHIVAM PAL

BBA

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SUNNY TOYOTA

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SHREYA NIGAM

BBA

VSICS

SUNNY TOYOTA

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TANBIR SINGH

BBA

VSICS

SUNNY TOYOTA

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ANSH TIWARI

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KUMKUM BAGHEL

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MANIK TRIPATHI

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204	SUMIT KRISHNA GUPTA	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
205	SURYANSH DIXIT	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
206	TANYA SRIVASTAVA	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
207	VAIBHAV SHUKLA	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS
208	VANSH MATLANI	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
209	VARUN MADNANI	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
210	VARUN VERMA	BBA	DAMS	TATA CONSULTANCY SERVICES-BPS

204	SUMIT KRISHNA GUPTA	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
211	YASHI SINGH CHAUHAN	BBA	VSICS	TATA CONSULTANCY SERVICES-BPS
212	AMAN SINGH CHANDEL	BBA	DAMS	UIFS TECHNOLOGIES
213	ANYA MISHRA	BBA	VSICS	UIFS TECHNOLOGIES
214	GAURI PANDEY	BBA	VSICS	UIFS TECHNOLOGIES
215	GURLEEN KAUR JULKA	BBA	VSICS	UIFS TECHNOLOGIES
216	SHASHANK AHUJA	BBA	VSICS	UIFS TECHNOLOGIES
217	AADITRI MISHRA	BCA	VSICS	ACCENTURE
218	AASTHA NIGAM	BCA	VSICS	ACCENTURE
219	ABHAY TRIVEDI	BCA	VSICS	ACCENTURE
220	ABHISHEK AWASTHI	BCA	VSICS	ACCENTURE
221	ABHISHEK KUMAR UPADHAYAY	BCA	VSICS	ACCENTURE
222	ABHISHEK PANDEY	BCA	VSICS	ACCENTURE
223	ADARSH DIXIT	BCA	VSICS	ACCENTURE
224	ADITYA KUMAR DWIVEDI	BCA	VSICS	ACCENTURE
225	ADITYA RAJ SRIVASTAVA	BCA	VSICS	ACCENTURE
226	ADITYA VERMA	BCA	VSICS	ACCENTURE
227	AJAY VISHWAKARMA	BCA	VSICS	ACCENTURE
228	AKSHAY PRATAP SINGH	BCA	VSICS	ACCENTURE
229	AKSHAY SHUKLA	BCA	VSICS	ACCENTURE
230	AKSHAY TIWARI	BCA	VSICS	ACCENTURE
231	AMAN RAJ GUPTA	BCA	VSICS	ACCENTURE
232	ANANT BAJPAI	BCA	VSICS	ACCENTURE
233	ANIVESH GUPTA	BCA	VSICS	ACCENTURE
234	ANSH OMAR	BCA	VSICS	ACCENTURE
235	ANSHIKA MISHRA	BCA	VSICS	ACCENTURE
236	ANSHIKA PANDEY	BCA	VSICS	ACCENTURE
237	ANSHUMAN GUPTA	BCA	VSICS	ACCENTURE
238	ANUSHKA SAINI	BCA	VSICS	ACCENTURE
239	APOORV MISHRA MISHRA	BCA	VSICS	ACCENTURE
240	APURVA DHANGAR	BCA	VSICS	ACCENTURE
241	ARCHI VERMA	BCA	VSICS	ACCENTURE
242	ARMAN SIDDIQUI SIDDIQUI	BCA	VSICS	ACCENTURE
243	ARPITA ANAND	BCA	VSICS	ACCENTURE
244	ARYAN GUPTA	BCA	VSICS	ACCENTURE
245	ASHUTOSH PATHAK	BCA	VSICS	ACCENTURE

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SUMIT KRISHNA GUPTA

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SURYANSH DIXIT

BBA

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TANYA SRIVASTAVA

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TATA CONSULTANCY SERVICES-BPS

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YASHI SINGH CHAUHAN

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AMAN SINGH CHANDEL

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GAURI PANDEY

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GURLEEN KAUR JULKA

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SHASHANK AHUJA

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AASTHA NIGAM

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ABHAY TRIVEDI

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ACCENTURE

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ABHISHEK AWASTHI

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ABHISHEK KUMAR UPADHAYAY

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ADITYA RAJ SRIVASTAVA

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AKSHAY TIWARI

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247	AYUSH DWIVEDI	BCA	VSICS	ACCENTURE
248	AYUSH SONI	BCA	VSICS	ACCENTURE

246	ASTHA SINGH	BCA	VSICS	ACCENTURE
249	BALENDU MISHRA	BCA	VSICS	ACCENTURE
250	BHUMI OMAR	BCA	VSICS	ACCENTURE
251	BHUVNESH SINGH	BCA	VSICS	ACCENTURE
252	DEVANSH AGNIHOTRI	BCA	VSICS	ACCENTURE
253	DIYA GUPTA	BCA	VSICS	ACCENTURE
254	GAURVIKA GANGWANI	BCA	VSICS	ACCENTURE
255	HARISH DWIVEDI	BCA	VSICS	ACCENTURE
256	HARSHIT SETIYA	BCA	VSICS	ACCENTURE
257	HARSHITA KANDPAL	BCA	VSICS	ACCENTURE
258	IQRA KHAN	BCA	VSICS	ACCENTURE
259	ISHA AHUJA	BCA	VSICS	ACCENTURE
260	ISHITA VERMA VERMA	BCA	VSICS	ACCENTURE
261	KARTIK VEDI	BCA	VSICS	ACCENTURE
262	KASHISH AGRAWAL	BCA	VSICS	ACCENTURE
263	KRISHANGI SRIVASTAVA	BCA	VSICS	ACCENTURE
264	LAKSHYA BAJPAI	BCA	VSICS	ACCENTURE
265	MADHVEE SHUKLA	BCA	VSICS	ACCENTURE
266	MAHAK SINGH SINGH	BCA	VSICS	ACCENTURE
267	MANAS DWIVEDI DWIVEDI	BCA	VSICS	ACCENTURE
268	MANISH SINHA	BCA	VSICS	ACCENTURE
269	MANYA SHUKLA SHUKLA	BCA	VSICS	ACCENTURE
270	NAINISH Z ZIYA	BCA	VSICS	ACCENTURE
271	NANDINI CHHOTE LAL	BCA	VSICS	ACCENTURE
272	NAVYA JAISWAL	BCA	VSICS	ACCENTURE
273	NIDHI TIWARI	BCA	VSICS	ACCENTURE
274	NIRVIKAR SHUKLA	BCA	VSICS	ACCENTURE
275	PALAK VERMA	BCA	VSICS	ACCENTURE
276	PANKHURI SHARMA	BCA	VSICS	ACCENTURE
277	PRATEEK SRIVASTAVA	BCA	VSICS	ACCENTURE
278	PURNIMA TRIPATHI	BCA	VSICS	ACCENTURE
279	RAVI KANT TRIPATHI	BCA	VSICS	ACCENTURE
280	RIDHIMAA SAXENA	BCA	VSICS	ACCENTURE
281	RISHIKA SACHAN	BCA	VSICS	ACCENTURE
282	RISHIKESH SINGH	BCA	VSICS	ACCENTURE
283	RIYA MISHRA	BCA	VSICS	ACCENTURE
284	RIYA MISHRA MISHRA	BCA	VSICS	ACCENTURE
285	SANSKRITI GUPTA	BCA	VSICS	ACCENTURE
286	SARAJ TRIVEDI	BCA	VSICS	ACCENTURE
287	SARIKA YADAV	BCA	VSICS	ACCENTURE

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ASTHA SINGH

BCA

VSICS

ACCENTURE

247

AYUSH DWIVEDI

BCA

VSICS

ACCENTURE

248

AYUSH SONI

BCA

VSICS

ACCENTURE

249

BALENDU MISHRA

BCA

VSICS

ACCENTURE

250

BHUMI OMAR

BCA

VSICS

ACCENTURE

251

BHUVNESH SINGH

BCA

VSICS

ACCENTURE

252

DEVANSH AGNIHOTRI

BCA

VSICS

ACCENTURE

253

DIYA GUPTA

BCA

VSICS

ACCENTURE

254

GAURVIKA GANGWANI

BCA

VSICS

ACCENTURE

255

HARISH DWIVEDI

BCA

VSICS

ACCENTURE

256

HARSHIT SETIYA

BCA

VSICS

ACCENTURE

257

HARSHITA KANDPAL

BCA

VSICS

ACCENTURE

258

IQRA KHAN

BCA

VSICS

ACCENTURE

259

ISHA AHUJA

BCA

VSICS

ACCENTURE

260

ISHITA VERMA VERMA

BCA

VSICS

ACCENTURE

261

KARTIK VEDI

BCA

VSICS

ACCENTURE

262

KASHISH AGRAWAL

BCA

VSICS

ACCENTURE

263

KRISHANGI SRIVASTAVA

BCA

VSICS

ACCENTURE

264

LAKSHYA BAJPAI

BCA

VSICS

ACCENTURE

265

MADHVEE SHUKLA

BCA

VSICS

ACCENTURE

266

MAHAK SINGH SINGH

BCA

VSICS

ACCENTURE

267

MANAS DWIVEDI DWIVEDI

BCA

VSICS

ACCENTURE

268

MANISH SINHA

BCA

VSICS

ACCENTURE

269

MANYA SHUKLA SHUKLA

BCA

VSICS

ACCENTURE

270

NAINISH Z ZIYA

BCA

VSICS

ACCENTURE

271

NANDINI CHHOTE LAL

BCA

VSICS

ACCENTURE

272

NAVYA JAISWAL

BCA

VSICS

ACCENTURE

273

NIDHI TIWARI

BCA

VSICS

ACCENTURE

274

NIRVIKAR SHUKLA

BCA

VSICS

ACCENTURE

275

PALAK VERMA

BCA

VSICS

ACCENTURE

276

PANKHURI SHARMA

BCA

VSICS

ACCENTURE

277

PRATEEK SRIVASTAVA

BCA

VSICS

ACCENTURE

278

PURNIMA TRIPATHI

BCA

VSICS

ACCENTURE

279

RAVI KANT TRIPATHI

BCA

VSICS

A

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288	SHAMBHAVI DWIVEDI	BCA	VSICS	ACCENTURE
289	SHANU AWASTHI AWASTHI	BCA	VSICS	ACCENTURE
290	SHIVAM SINGH RAWAT	BCA	VSICS	ACCENTURE
291	SHIVANGI SRIVASTAVA SRIVASTAVA	BCA	VSICS	ACCENTURE
292	SHIVENDRA BAJPAI	BCA	VSICS	ACCENTURE
293	SHIWANI YADAV YADAV	BCA	VSICS	ACCENTURE
294	SHLOK AGARWAL	BCA	VSICS	ACCENTURE
295	SHRUTI GUPTA	BCA	VSICS	ACCENTURE
296	SHUBHANGI SRIVASTAVA	BCA	VSICS	ACCENTURE
297	SHUBHANKAR PATEL	BCA	VSICS	ACCENTURE
298	SOUMYA MOHAN	BCA	VSICS	ACCENTURE

288	SHAMBHAVI DWIVEDI	BCA	VSICS	ACCENTURE
299	SURAJ DWIVEDI	BCA	VSICS	ACCENTURE
300	SUSHMIT SHUKLA	BCA	VSICS	ACCENTURE
301	TARANA KHAN	BCA	VSICS	ACCENTURE
302	TISHA BALANI	BCA	VSICS	ACCENTURE
303	TUSHAR SHARMA	BCA	VSICS	ACCENTURE
304	UDDESHYA SHUKLA	BCA	VSICS	ACCENTURE
305	UNNATI KESARWANI	BCA	VSICS	ACCENTURE
306	UNNATI SINGH SINGH	BCA	VSICS	ACCENTURE
307	VAIBHAVI SHARMA SHARMA	BCA	VSICS	ACCENTURE
308	VAISHALI CHAUHAN	BCA	VSICS	ACCENTURE
309	VAISHNAVI SAVITA SAVITA	BCA	VSICS	ACCENTURE
310	VARCHASW TRIPATHI	BCA	VSICS	ACCENTURE
311	VIDHI AWASTHI	BCA	VSICS	ACCENTURE
312	YASH RAJ SINGH	BCA	VSICS	ACCENTURE
313	YASHASWI DWIVEDI DWIVEDI	BCA	VSICS	ACCENTURE
314	YASHIKA SINGH	BCA	VSICS	ACCENTURE
315	ABHISHEK KUMAR	BCA	VSICS	ATHARV ONSTECH
316	ABHISHEK SHUKLA	BCA	VSICS	ATHARV ONSTECH
317	ADITYA AWASTHI	BCA	VSICS	ATHARV ONSTECH
318	AKHIL RAI	BCA	VSICS	ATHARV ONSTECH
319	ANSH GUPTA	BCA	VSICS	ATHARV ONSTECH
320	ARJUN GUPTA	BCA	VSICS	ATHARV ONSTECH
321	AYUSH BHADAURIYA	BCA	VSICS	ATHARV ONSTECH
322	BHOOMI GUPTA	BCA	VSICS	ATHARV ONSTECH
323	DIVYANSHI BHATT	BCA	VSICS	ATHARV ONSTECH
324	GAURAV SINGH	BCA	VSICS	ATHARV ONSTECH
325	GYANENDRA SHARMA	BCA	VSICS	ATHARV ONSTECH
326	HIMANSHU GUPTA	BCA	VSICS	ATHARV ONSTECH
327	KAMALPRATAP SINGH CHAUHAN	BCA	VSICS	ATHARV ONSTECH
328	KRATIKA BAJPAI	BCA	VSICS	ATHARV ONSTECH

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SHAMBHAVI DWIVEDI

BCA

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SHANU AWASTHI AWASTHI

BCA

VSICS

ACCENTURE

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SHIVAM SINGH RAWAT

BCA

VSICS

ACCENTURE

291

SHIVANGI SRIVASTAVA

SRIVASTAVA

BCA

VSICS

ACCENTURE

292

SHIVENDRA BAJPAI

BCA

VSICS

ACCENTURE

293

SHIWANI YADAV YADAV

BCA

VSICS

ACCENTURE

294

SHLOK AGARWAL

BCA

VSICS

ACCENTURE

295

SHRUTI GUPTA

BCA

VSICS

ACCENTURE

296

SHUBHANGI SRIVASTAVA

BCA

VSICS

ACCENTURE

297

SHUBHANKAR PATEL

BCA

VSICS

ACCENTURE

298

SOUMYA MOHAN

BCA

VSICS

ACCENTURE

299

SURAJ DWIVEDI

BCA

VSICS

ACCENTURE

300

SUSHMIT SHUKLA

BCA

VSICS

ACCENTURE

301

TARANA KHAN

BCA

VSICS

ACCENTURE

302

TISHA BALANI

BCA

VSICS

ACCENTURE

303

TUSHAR SHARMA

BCA

VSICS

ACCENTURE

304

UDDESHYA SHUKLA

BCA

VSICS

ACCENTURE

305

UNNATI KESARWANI

BCA

VSICS

ACCENTURE

306

UNNATI SINGH SINGH

BCA

VSICS

ACCENTURE

307

VAIBHAVI SHARMA SHARMA

BCA

VSICS

ACCENTURE

308

VAISHALI CHAUHAN

BCA

VSICS

ACCENTURE

309

VAISHNAVI SAVITA SAVITA

BCA

VSICS

ACCENTURE

310

VARCHASW TRIPATHI

BCA

VSICS

ACCENTURE

311

VIDHI AWASTHI

BCA

VSICS

ACCENTURE

312

YASH RAJ SINGH

BCA

VSICS

ACCENTURE

313

YASHASWI DWIVEDI DWIVEDI

BCA

VSICS

ACCENTURE

314

YASHIKA SINGH

BCA

VSICS

ACCENTURE

315

ABHISHEK KUMAR

BCA

VSICS

ATHARV ONSTECH

316

ABHISHEK SHUKLA

BCA

VSICS

ATHARV ONSTECH

317

ADITYA AWASTHI

BCA

VSICS

ATHARV ONSTECH

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AKHIL RAI

BCA

VSICS

ATHARV ONSTECH

319

ANSH GUPTA

BCA

VSICS

ATHARV ONSTECH

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329	NAINSI PATEL	BCA	VSICS	ATHARV ONSTECH
330	PRIYANSHU YADAV	BCA	VSICS	ATHARV ONSTECH
331	RAJ GUPTA	BCA	VSICS	ATHARV ONSTECH
332	RAVI VERMA	BCA	VSICS	ATHARV ONSTECH
333	RISHITA	BCA	VSICS	ATHARV ONSTECH
334	RITIK SAXENA	BCA	VSICS	ATHARV ONSTECH
335	RITIK TIWARI	BCA	VSICS	ATHARV ONSTECH
336	SHREYA GHOSH	BCA	VSICS	ATHARV ONSTECH
337	SHREYA SAPRA	BCA	VSICS	ATHARV ONSTECH
338	SHUBHAM SHARMA	BCA	VSICS	ATHARV ONSTECH
339	SIDDHARTH RAO	BCA	VSICS	ATHARV ONSTECH
340	UDAY PRATAP SINGH	BCA	VSICS	ATHARV ONSTECH
341	YASH KISHOR CHAUDHARY	BCA	VSICS	ATHARV ONSTECH
342	AAQILA JAMAL	BCA	VSICS	CAPGEMINI
343	AASTHA NIGAM	BCA	VSICS	CAPGEMINI
344	ADITI DUBEY	BCA	VSICS	CAPGEMINI
345	ANCHAL VERMA	BCA	VSICS	CAPGEMINI
346	ANSHIKA AWASTHI	BCA	VSICS	CAPGEMINI
347	ANSHIKA YADAV	BCA	VSICS	CAPGEMINI
348	ARPITA ANAND	BCA	VSICS	CAPGEMINI
349	KALYANI SHUKLA	BCA	VSICS	CAPGEMINI
350	NAVYA JAISWAL	BCA	VSICS	CAPGEMINI
351	PANKHURI SHARMA	BCA	VSICS	CAPGEMINI
352	PURNIMA TRIPATHI	BCA	VSICS	CAPGEMINI
353	SANSKRITI DWIVEDI	BCA	DAMS	CAPGEMINI
354	TISHA BALANI	BCA	VSICS	CAPGEMINI
355	VAIBHAVI SHARMA	BCA	VSICS	CAPGEMINI
356	VAISHNAVI	BCA	VSICS	CAPGEMINI
357	YASHASWI DWIVEDI	BCA	VSICS	CAPGEMINI

329	NAINSI PATEL	BCA	VSICS	ATHARV ONSTECH
358	ABHAAS RANJAN	BCA	VSICS	JUSTDIAL
359	ABHAY DWIVEDI	BCA	VSICS	JUSTDIAL
360	ABHAY TRIVEDI	BCA	VSICS	JUSTDIAL
361	ABHISHEK TRIPATHI	BCA	VSICS	JUSTDIAL
362	AKHIL SACHAN	BCA	VSICS	JUSTDIAL
363	AKSHAT GUPTA	BCA	VSICS	JUSTDIAL
364	AMAAN QAMAR	BCA	DAMS	JUSTDIAL
365	ANURAG SINGH	BCA	VSICS	JUSTDIAL
366	ASHISH KUMAR SHUKLA	BCA	VSICS	JUSTDIAL
367	DHRUV RAJ MISHRA	BCA	VSICS	JUSTDIAL
368	HARISH DWIVEDI	BCA	VSICS	JUSTDIAL
369	NAMAN TIWARI	BCA	VSICS	JUSTDIAL
370	NAYAN MISHRA	BCA	VSICS	JUSTDIAL

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NAINSI PATEL

BCA

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ATHARV ONSTECH

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PRIYANSHU YADAV

BCA

VSICS

ATHARV ONSTECH

331

RAJ GUPTA

BCA

VSICS

ATHARV ONSTECH

332

RAVI VERMA

BCA

VSICS

ATHARV ONSTECH

333

RISHITA

BCA

VSICS

ATHARV ONSTECH

334

RITIK SAXENA

BCA

VSICS

ATHARV ONSTECH

335

RITIK TIWARI

BCA

VSICS

ATHARV ONSTECH

336

SHREYA GHOSH

BCA

VSICS

ATHARV ONSTECH

337

SHREYA SAPRA

BCA

VSICS

ATHARV ONSTECH

338

SHUBHAM SHARMA

BCA

VSICS

ATHARV ONSTECH

339

SIDDHARTH RAO

BCA

VSICS

ATHARV ONSTECH

340

UDAY PRATAP SINGH

BCA

VSICS

ATHARV ONSTECH

341

YASH KISHOR CHAUDHARY

BCA

VSICS

ATHARV ONSTECH

342

AAQILA JAMAL

BCA

VSICS

CAPGEMINI

343

AASTHA NIGAM

BCA

VSICS

CAPGEMINI

344

ADITI DUBEY

BCA

VSICS

CAPGEMINI

345

ANCHAL VERMA

BCA

VSICS

CAPGEMINI

346

ANSHIKA AWASTHI

BCA

VSICS

CAPGEMINI

347

ANSHIKA YADAV

BCA

VSICS

CAPGEMINI

348

ARPITA ANAND

BCA

VSICS

CAPGEMINI

349

KALYANI SHUKLA

BCA

VSICS

CAPGEMINI

350

NAVYA JAISWAL

BCA

VSICS

CAPGEMINI

351

PANKHURI SHARMA

BCA

VSICS

CAPGEMINI

352

PURNIMA TRIPATHI

BCA

VSICS

CAPGEMINI

353

SANSKRITI DWIVEDI

BCA

DAMS

CAPGEMINI

354

TISHA BALANI

BCA

VSICS

CAPGEMINI

355

VAIBHAVI SHARMA

BCA

VSICS

CAPGEMINI

356

VAISHNAVI

BCA

VSICS

CAPGEMINI

357

YASHASWI DWIVEDI

BCA

VSICS

CAPGEMINI

358

ABHAAS RANJAN

BCA

VSICS

JUSTDIAL

359

ABHAY DWIVEDI

BCA

VSICS

JUSTDIAL

360

ABHAY TRIVEDI

BCA

VSICS

JUSTDIAL

361

ABHISHEK TRIPATHI

BCA

VSICS

JUSTDIAL

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AKHIL SAC

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371	NIKHIL AWASTHI	BCA	VSICS	JUSTDIAL
372	NITEESH TIWARI	BCA	VSICS	JUSTDIAL
373	NITESH KUMAR GUPTA	BCA	DAMS	JUSTDIAL
374	PRASHANT YADAV	BCA	VSICS	JUSTDIAL
375	PRATEEK SHARMA	BCA	VSICS	JUSTDIAL
376	PRATHAM AWASTHI	BCA	VSICS	JUSTDIAL
377	PRATHAM DWIVEDI	BCA	VSICS	JUSTDIAL
378	RAHUL SONI	BCA	DAMS	JUSTDIAL
379	RATNESH MITTAL	BCA	VSICS	JUSTDIAL
380	RAUNAK GUPTA	BCA	VSICS	JUSTDIAL
381	SPARSH DIXIT	BCA	DAMS	JUSTDIAL
382	YASH PANDEY	BCA	VSICS	JUSTDIAL
383	ANSHIKA SHARMA	BCA	VSICS	PLANET SPARK
384	ANSHIKA SHUKLA	BCA	VSICS	PLANET SPARK
385	ARJUN GUPTA	BCA	VSICS	PLANET SPARK
386	SHREYA SRIVASTAVA	BCA	DAMS	PLANET SPARK
387	TEENA DIXIT	BCA	VSICS	PLANET SPARK
388	AASTHA NIGAM	BCA	VSICS	TATA CONSULTANCY SERVICES
389	ABHAY PRATAP SINGH	BCA	VSICS	TATA CONSULTANCY SERVICES
390	ABHISHEK AWASTHI	BCA	VSICS	TATA CONSULTANCY SERVICES
391	ABHISHEK KUMAR UPADHAYAY	BCA	VSICS	TATA CONSULTANCY SERVICES
392	ADITI PANDEY	BCA	VSICS	TATA CONSULTANCY SERVICES
393	ADITI TIWARI	BCA	VSICS	TATA CONSULTANCY SERVICES
394	ADITYA GAUR	BCA	DAMS	TATA CONSULTANCY SERVICES
395	AJAY VISHWAKARMA	BCA	VSICS	TATA CONSULTANCY SERVICES
396	AKHIL SACHAN	BCA	VSICS	TATA CONSULTANCY SERVICES
397	AKSHAT GUPTA	BCA	VSICS	TATA CONSULTANCY SERVICES
398	AKSHAY SHUKLA	BCA	VSICS	TATA CONSULTANCY SERVICES
399	AMARDEEP YADAV	BCA	VSICS	TATA CONSULTANCY SERVICES
400	ANANYA SHUKLA	BCA	VSICS	TATA CONSULTANCY SERVICES
401	ANANYA YADAV	BCA	VSICS	TATA CONSULTANCY SERVICES
402	ANIKET SINGH CHANDEL	BCA	VSICS	TATA CONSULTANCY SERVICES

371	NIKHIL AWASTHI	BCA	VSICS	JUSTDIAL
403	ANSH OMAR	BCA	VSICS	TATA CONSULTANCY SERVICES
404	ANSHIKA AWASTHI	BCA	VSICS	TATA CONSULTANCY SERVICES
405	ANSHIKA SINGH	BCA	VSICS	TATA CONSULTANCY SERVICES
406	ANURAG SINGH	BCA	VSICS	TATA CONSULTANCY SERVICES
407	ANUSHKA SAINI	BCA	VSICS	TATA CONSULTANCY SERVICES
408	ARJUN GUPTA	BCA	VSICS	TATA CONSULTANCY SERVICES
409	ARPIT SRIVASTAVA	BCA	VSICS	TATA CONSULTANCY SERVICES
410	ARYAN CHAURASIA	BCA	VSICS	TATA CONSULTANCY SERVICES
411	ASHISH JAISWAL	BCA	VSICS	TATA CONSULTANCY SERVICES
412	ASHUTOSH PATHAK	BCA	VSICS	TATA CONSULTANCY SERVICES

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NITEESH TIWARI

BCA

VSICS

JUSTDIAL

373

NITESH KUMAR GUPTA

BCA

DAMS

JUSTDIAL

374

PRASHANT YADAV

BCA

VSICS

JUSTDIAL

375

PRATEEK SHARMA

BCA

VSICS

JUSTDIAL

376

PRATHAM AWASTHI

BCA

VSICS

JUSTDIAL

377

PRATHAM DWIVEDI

BCA

VSICS

JUSTDIAL

378

RAHUL SONI

BCA

DAMS

JUSTDIAL

379

RATNESH MITTAL

BCA

VSICS

JUSTDIAL

380

RAUNAK GUPTA

BCA

VSICS

JUSTDIAL

381

SPARSH DIXIT

BCA

DAMS

JUSTDIAL

382

YASH PANDEY

BCA

VSICS

JUSTDIAL

383

ANSHIKA SHARMA

BCA

VSICS

PLANET SPARK

384

ANSHIKA SHUKLA

BCA

VSICS

PLANET SPARK

385

ARJUN GUPTA

BCA

VSICS

PLANET SPARK

386

SHREYA SRIVASTAVA

BCA

DAMS

PLANET SPARK

387

TEENA DIXIT

BCA

VSICS

PLANET SPARK

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AASTHA NIGAM

BCA

VSICS

TATA CONSULTANCY SERVICES

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ABHAY PRATAP SINGH

BCA

VSICS

TATA CONSULTANCY SERVICES

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ABHISHEK AWASTHI

BCA

VSICS

TATA CONSULTANCY SERVICES

391

ABHISHEK KUMAR UPADHAYAY

BCA

VSICS

TATA CONSULTANCY SERVICES

392

ADITI PANDEY

BCA

VSICS

TATA CONSULTANCY SERVICES

393

ADITI TIWARI

BCA

VSICS

TATA CONSULTANCY SERVICES

394

ADITYA GAUR

BCA

DAMS

TATA CONSULTANCY SERVICES

395

AJAY VISHWAKARMA

BCA

VSICS

TATA CONSULTANCY SERVICES

396

AKHIL SACHAN

BCA

VSICS

TATA CONSULTANCY SERVICES

397

AKSHAT GUPTA

BCA

VSICS

TATA CONSULTANCY SERVICES

398

AKSHAY SHUKLA

BCA

VSICS

TATA CONSULTANCY SERVICES

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AMARDEEP YADAV

BCA

VSICS

TATA CONSULTANCY SERVICES

400

ANANYA SHUKLA

BCA

VSICS

TATA

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413	AVANI SHARMA	BCA	VSICS	TATA CONSULTANCY SERVICES
414	AWADHESH KUMAR	BCA	DAMS	TATA CONSULTANCY SERVICES
415	BALENDU MISHRA	BCA	VSICS	TATA CONSULTANCY SERVICES
416	DEVANSHI VERMA	BCA	VSICS	TATA CONSULTANCY SERVICES
417	DHRUV VISHWAKARMA	BCA	VSICS	TATA CONSULTANCY SERVICES
418	DIVYA TRIPATHI	BCA	VSICS	TATA CONSULTANCY SERVICES
419	EKTA SINGH CHAUHAN	BCA	VSICS	TATA CONSULTANCY SERVICES
420	GARIMA SHARMA	BCA	VSICS	TATA CONSULTANCY SERVICES
421	GAURAV SINGH	BCA	VSICS	TATA CONSULTANCY SERVICES
422	HARSH TIWARI	BCA	VSICS	TATA CONSULTANCY SERVICES
423	HARSHIT SETIYA	BCA	VSICS	TATA CONSULTANCY SERVICES

413	AVANI SHARMA	BCA	VSICS	TATA CONSULTANCY SERVICES
424	HRIDYANSH SHUKLA	BCA	VSICS	TATA CONSULTANCY SERVICES
425	ISHITA VERMA	BCA	VSICS	TATA CONSULTANCY SERVICES
426	JEEVANSI GAUTAM	BCA	VSICS	TATA CONSULTANCY SERVICES
427	KAUSHIKI YADAV	BCA	VSICS	TATA CONSULTANCY SERVICES
428	KHUSHI TRIPATHI	BCA	VSICS	TATA CONSULTANCY SERVICES
429	MANISH AWASTHI	BCA	VSICS	TATA CONSULTANCY SERVICES
430	MUKHTAR HUSSAIN	BCA	VSICS	TATA CONSULTANCY SERVICES
431	MUSKAN GIRI	BCA	DAMS	TATA CONSULTANCY SERVICES
432	NAVAL SHARMA	BCA	VSICS	TATA CONSULTANCY SERVICES
433	NAVYA JAISWAL	BCA	VSICS	TATA CONSULTANCY SERVICES
434	NIKHIL AWASTHI	BCA	VSICS	TATA CONSULTANCY SERVICES
435	NISHTHA AGARWAL	BCA	VSICS	TATA CONSULTANCY SERVICES
436	PRASHANT YADAV	BCA	VSICS	TATA CONSULTANCY SERVICES
437	PRASOON MITTAL	BCA	VSICS	TATA CONSULTANCY SERVICES
438	PRATEEK KUMAR GUPTA	BCA	DAMS	TATA CONSULTANCY SERVICES
439	PRATHAM DWIVEDI	BCA	VSICS	TATA CONSULTANCY SERVICES
440	PRATHAM TIWARI	BCA	VSICS	TATA CONSULTANCY SERVICES
441	PRIYA RATHORE	BCA	VSICS	TATA CONSULTANCY SERVICES
442	RAGINI SONI	BCA	VSICS	TATA CONSULTANCY SERVICES
443	RAHUL SONI	BCA	DAMS	TATA CONSULTANCY SERVICES
444	RAVI KANT TRIPATHI	BCA	VSICS	TATA CONSULTANCY SERVICES
445	RISHU SHARMA	BCA	VSICS	TATA CONSULTANCY SERVICES
446	RITIK SAINI	BCA	VSICS	TATA CONSULTANCY SERVICES
447	ROBIN KUMAR	BCA	VSICS	TATA CONSULTANCY SERVICES

413	AVANI SHARMA	BCA	VSICS	TATA CONSULTANCY SERVICES
448	SAMEER SINGH	BCA	VSICS	TATA CONSULTANCY SERVICES
449	SARIKA YADAV	BCA	VSICS	TATA CONSULTANCY SERVICES
450	SAUMYA KAUSHAL	BCA	VSICS	TATA CONSULTANCY SERVICES
451	SAURABH SINGH PATEL	BCA	DAMS	TATA CONSULTANCY SERVICES
452	SHASHANK YADAV	BCA	VSICS	TATA CONSULTANCY SERVICES
453	SHIVAM SINGH	BCA	DAMS	TATA CONSULTANCY SERVICES
454	SHREYA GUPTA	BCA	VSICS	TATA CONSULTANCY SERVICES

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413

AVANI SHARMA

BCA

VSICS

TATA CONSULTANCY SERVICES

414

AWADHESH KUMAR

BCA

DAMS

TATA CONSULTANCY SERVICES

415

BALENDU MISHRA

BCA

VSICS

TATA CONSULTANCY SERVICES

416

DEVANSHI VERMA

BCA

VSICS

TATA CONSULTANCY SERVICES

417

DHRUV VISHWAKARMA

BCA

VSICS

TATA CONSULTANCY SERVICES

418

DIVYA TRIPATHI

BCA

VSICS

TATA CONSULTANCY SERVICES

419

EKTA SINGH CHAUHAN

BCA

VSICS

TATA CONSULTANCY SERVICES

420

GARIMA SHARMA

BCA

VSICS

TATA CONSULTANCY SERVICES

421

GAURAV SINGH

BCA

VSICS

TATA CONSULTANCY SERVICES

422

HARSH TIWARI

BCA

VSICS

TATA CONSULTANCY SERVICES

423

HARSHIT SETIYA

BCA

VSICS

TATA CONSULTANCY SERVICES

424

HRIDYANSH SHUKLA

BCA

VSICS

TATA CONSULTANCY SERVICES

425

ISHITA VERMA

BCA

VSICS

TATA CONSULTANCY SERVICES

426

JEEVANSHI GAUTAM

BCA

VSICS

TATA CONSULTANCY SERVICES

427

KAUSHIKI YADAV

BCA

VSICS

TATA CONSULTANCY SERVICES

428

KHUSHI TRIPATHI

BCA

VSICS

TATA CONSULTANCY SERVICES

429

MANISH AWASTHI

BCA

VSICS

TATA CONSULTANCY SERVICES

430

MUKHTAR HUSSAIN

BCA

VSICS

TATA CONSULTANCY SERVICES

431

MUSKAN GIRI

BCA

DAMS

TATA CONSULTANCY SERVICES

432

NAVAL SHARMA

BCA

VSICS

TATA CONSULTANCY SERVICES

433

NAVYA JAISWAL

BCA

VSICS

TATA CONSULTANCY SERVICES

434

NIKHIL AWASTHI

BCA

VSICS

TATA CONSULTANCY SERVICES

435

NISHTHA AGARWAL

BCA

VSICS

TATA CONSULTANCY SERVICES

436

PRASHANT YADAV

BCA

VSICS

TATA CONSULTANCY SERVICES

437

PRASOON MITTAL

BCA

VSICS

TATA CONSULTANCY SERVICES

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Page 12 Data Table

455	SHRUTI GUPTA	BCA	VSICS	TATA CONSULTANCY SERVICES
456	SHUBHI DWIVEDI	BCA	VSICS	TATA CONSULTANCY SERVICES
457	SIMRAN GUPTA	BCA	VSICS	TATA CONSULTANCY SERVICES
458	SNEHA SHARMA	BCA	VSICS	TATA CONSULTANCY SERVICES
459	SOMIL TIWARI	BCA	VSICS	TATA CONSULTANCY SERVICES

455	SHRUTI GUPTA	BCA	VSICS	TATA CONSULTANCY SERVICES
460	TARANA KHAN	BCA	VSICS	TATA CONSULTANCY SERVICES
461	TEENA DIXIT	BCA	VSICS	TATA CONSULTANCY SERVICES
462	UNNATI GUPTA	BCA	VSICS	TATA CONSULTANCY SERVICES
463	UNNATI KESARWANI	BCA	VSICS	TATA CONSULTANCY SERVICES
464	VAISHALI CHAUHAN	BCA	VSICS	TATA CONSULTANCY SERVICES
465	VAISHNAVI KESHARWANI	BCA	VSICS	TATA CONSULTANCY SERVICES
466	VANDITA BHARTI	BCA	VSICS	TATA CONSULTANCY SERVICES
467	VISHAL SINGH	BCA	VSICS	TATA CONSULTANCY SERVICES
468	YASH PANDEY	BCA	VSICS	TATA CONSULTANCY SERVICES
469	AASTHA NIGAM	BCA	VSICS	WIPRO
470	ABHAY TRIVEDI	BCA	VSICS	WIPRO
471	ADEEBA HADIYA	BCA	VSICS	WIPRO
472	ANANYA YADAV	BCA	VSICS	WIPRO
473	ANCHAL VERMA	BCA	VSICS	WIPRO
474	ARJUN GUPTA	BCA	VSICS	WIPRO
475	ARJUN SRIVASTAVA	BCA	VSICS	WIPRO
476	ARMAN SIDDIQUI	BCA	VSICS	WIPRO
477	ARPIT DWIVEDI	BCA	VSICS	WIPRO
478	AYUSH SINGH	BCA	VSICS	WIPRO
479	BALENDU MISHRA	BCA	VSICS	WIPRO
480	DEEPANSHI SHUKLA	BCA	VSICS	WIPRO
481	DEVANSH AGNIHOTRI	BCA	VSICS	WIPRO
482	DHRUV RAJ MISHRA	BCA	VSICS	WIPRO
483	GAURAV SINGH CHAUHAN	BCA	VSICS	WIPRO
484	JAHNAVI SINGH	BCA	DAMS	WIPRO
485	JAHNVI SHUKLA	BCA	DAMS	WIPRO
486	KANISHKA SINGH	BCA	VSICS	WIPRO
487	KARTIKAY SETHI	BCA	VSICS	WIPRO
488	MANISH AWASTHI	BCA	VSICS	WIPRO
489	MANISH SINHA	BCA	VSICS	WIPRO
490	MUSKAN GIRI	BCA	DAMS	WIPRO
491	NIKITA MISHRA	BCA	VSICS	WIPRO
492	NIRVIKAR SHUKLA	BCA	VSICS	WIPRO
493	NISHTHA AGARWAL	BCA	VSICS	WIPRO

455	SHRUTI GUPTA	BCA	VSICS	TATA CONSULTANCY SERVICES
494	PARNIKA SHUKLA	BCA	VSICS	WIPRO
495	PRATEEK SHARMA	BCA	VSICS	WIPRO
496	PRATHAM DWIVEDI	BCA	VSICS	WIPRO

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455

SHRUTI GUPTA

BCA

VSICS

TATA CONSULTANCY SERVICES

456

SHUBHI DWIVEDI

BCA

VSICS

TATA CONSULTANCY SERVICES

457

SIMRAN GUPTA

BCA

VSICS

TATA CONSULTANCY SERVICES

458

SNEHA SHARMA

BCA

VSICS

TATA CONSULTANCY SERVICES

459

SOMIL TIWARI

BCA

VSICS

TATA CONSULTANCY SERVICES

460

TARANA KHAN

BCA

VSICS

TATA CONSULTANCY SERVICES

461

TEENA DIXIT

BCA

VSICS

TATA CONSULTANCY SERVICES

462

UNNATI GUPTA

BCA

VSICS

TATA CONSULTANCY SERVICES

463

UNNATI KESARWANI

BCA

VSICS

TATA CONSULTANCY SERVICES

464

VAISHALI CHAUHAN

BCA

VSICS

TATA CONSULTANCY SERVICES

465

VAISHNAVI KESHARWANI

BCA

VSICS

TATA CONSULTANCY SERVICES

466

VANDITA BHARTI

BCA

VSICS

TATA CONSULTANCY SERVICES

467

VISHAL SINGH

BCA

VSICS

TATA CONSULTANCY SERVICES

468

YASH PANDEY

BCA

VSICS

TATA CONSULTANCY SERVICES

469

AASTHA NIGAM

BCA

VSICS

WIPRO

470

ABHAY TRIVEDI

BCA

VSICS

WIPRO

471

ADEEBA HADIYA

BCA

VSICS

WIPRO

472

ANANYA YADAV

BCA

VSICS

WIPRO

473

ANCHAL VERMA

BCA

VSICS

WIPRO

474

ARJUN GUPTA

BCA

VSICS

WIPRO

475

ARJUN SRIVASTAVA

BCA

VSICS

WIPRO

476

ARMAN SIDDIQUI

BCA

VSICS

WIPRO

477

ARPIT DWIVEDI

BCA

VSICS

WIPRO

478

AYUSH SINGH

BCA

VSICS

WIPRO

479

BALENDU MISHRA

BCA

VSICS

WIPRO

480

DEEPANSHI SHUKLA

BCA

VSICS

WIPRO

481

DEVANSH AGNIHOTRI

BCA

VSICS

WIPRO

482

DHRUV RAJ MISHRA

BCA

VSICS

WIPRO

483

GAURAV SINGH CHAUHAN

BCA

VSICS

WIPRO

484

JAHNAVI SINGH

BCA

DAMS

WIPRO

485

JAHNVI SHUKLA

BCA

DAMS

WI

Page 13 Data Table

497	PRIYA RATHORE	BCA	VSICS	WIPRO
498	RAUNAK GUPTA	BCA	VSICS	WIPRO
499	RISHABH GUPTA	BCA	VSICS	WIPRO
500	RISHIKESH SINGH	BCA	VSICS	WIPRO
501	RISHU SHARMA	BCA	VSICS	WIPRO
502	SAMEER SINGH	BCA	VSICS	WIPRO
503	SAMEER VERMA	BCA	VSICS	WIPRO
504	SARANYA SHARMA	BCA	VSICS	WIPRO
505	SARIKA YADAV	BCA	VSICS	WIPRO
506	SAUMYA KAUSHAL	BCA	VSICS	WIPRO
507	SHIVANGI SRIVASTAVA	BCA	VSICS	WIPRO
508	SHIVANSH GUPTA	BCA	VSICS	WIPRO
509	SHIVENDRA BAJPAI	BCA	VSICS	WIPRO
510	SHRESTHA JAISWAL	BCA	DAMS	WIPRO
511	SHUBHAM SHARMA	BCA	VSICS	WIPRO
512	SHUBHANGI SRIVASTAVA	BCA	VSICS	WIPRO
513	SHUBHANKAR PATEL	BCA	VSICS	WIPRO
514	SHUBHI DWIVEDI	BCA	VSICS	WIPRO
515	SONALI TRIVEDI	BCA	VSICS	WIPRO
516	STUTI SAXENA	BCA	VSICS	WIPRO
517	SURAJ DWIVEDI	BCA	VSICS	WIPRO
518	TANISH JHA	BCA	VSICS	WIPRO
519	TISHA BALANI	BCA	VSICS	WIPRO
520	TUSHAR SHARMA	BCA	VSICS	WIPRO

497	PRIYA RATHORE	BCA	VSICS	WIPRO
521	UTKARSH MISHRA	BCA	VSICS	WIPRO
522	VAISHNAVI OM PRAKASH	BCA	VSIPS	WIPRO
523	VAISHNAVI TIWARI	BCA	VSIPS	WIPRO
524	VIKAS GUPTA	BCA	VSICS	WIPRO
525	VINAYAK SHARMA	BCA	VSICS	WIPRO
526	VISHAL SINGH	BCA	VSICS	WIPRO
527	GAURI SRIVASTAVA	MBA	DAMS	JARO EDUCATION
528	DIVYANSHI MISHRA	MBA	DAMS	CLASSMATE CONSULTANT
529	RITIKA PANDEY	MBA	DAMS	CLASSMATE CONSULTANT
530	ALOK SHUKLA	MBA	VSICS	HDFC LIFE
531	ISHA PANDEY	MBA	DAMS	HDFC LIFE
532	ANJU TRIVEDI	MBA	VSICS	HUMAN POTENTIAL CONSULTANT
533	DEEPALI NISHAD	MBA	VSICS	HUMAN POTENTIAL CONSULTANT
534	ISHA PANDEY	MBA	DAMS	HUMAN POTENTIAL CONSULTANT
535	SHREYASH TRIPATHI	MBA	VSICS	HUMAN POTENTIAL CONSULTANT
536	AVANTIKA PAL	MBA	VSICS	MIKKY PUBLICATION

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497

PRIYA RATHORE

BCA

VSICS

WIPRO

498

RAUNAK GUPTA

BCA

VSICS

WIPRO

499

RISHABH GUPTA

BCA

VSICS

WIPRO

500

RISHIKESH SINGH

BCA

VSICS

WIPRO

501

RISHU SHARMA

BCA

VSICS

WIPRO

502

SAMEER SINGH

BCA

VSICS

WIPRO

503

SAMEER VERMA

BCA

VSICS

WIPRO

504

SARANYA SHARMA

BCA

VSICS

WIPRO

505

SARIKA YADAV

BCA

VSICS

WIPRO

506

SAUMYA KAUSHAL

BCA

VSICS

WIPRO

507

SHIVANGI SRIVASTAVA

BCA

VSICS

WIPRO

508

SHIVANSH GUPTA

BCA

VSICS

WIPRO

509

SHIVENDRA BAJPAI

BCA

VSICS

WIPRO

510

SHRESTHA JAISWAL

BCA

DAMS

WIPRO

511

SHUBHAM SHARMA

BCA

VSICS

WIPRO

512

SHUBHANGI SRIVASTAVA

BCA

VSICS

WIPRO

513

SHUBHANKAR PATEL

BCA

VSICS

WIPRO

514

SHUBHI DWIVEDI

BCA

VSICS

WIPRO

515

SONALI TRIVEDI

BCA

VSICS

WIPRO

516

STUTI SAXENA

BCA

VSICS

WIPRO

517

SURAJ DWIVEDI

BCA

VSICS

WIPRO

518

TANISH JHA

BCA

VSICS

WIPRO

519

TISHA BALANI

BCA

VSICS

WIPRO

520

TUSHAR SHARMA

BCA

VSICS

WIPRO

521

UTKARSH MISHRA

BCA

VSICS

WIPRO

522

VAISHNAVI OM PRAKASH

BCA

VSIPS

WIPRO

523

VAISHNAVI TIWARI

BCA

VSIPS

WIPRO

524

VIKAS GUPTA

BCA

VSICS

WIPRO

525

VINAYAK SHARMA

BCA

VSICS

WIPRO

526

VISHAL SINGH

BCA

VSICS

WIPRO

527

GAURI SRIVASTAVA

MBA

DAMS

JARO EDUCATION

528

DIVYANSHI MISHRA

MBA

DAMS

CLASSMATE CONSULTANT

529

RITIKA PANDEY

MBA

DAMS

CLASSMATE CONSULTANT

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ALOK SHUKLA

MBA

VSICS

HDFC LIFE

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ISHA PANDEY

MBA

DAMS

HDFC LIFE

532

ANJU TRIVEDI

MBA

VSICS

HUMAN POTENTIAL

CONSULTANT

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537	DIVYANSHI KUSHWAHA	MBA	VSICS	MIKKY PUBLICATION
538	KRATI MISHRA	MBA	VSIPS	MIKKY PUBLICATION
539	ALISHA RAIS	MBA	VSIPS	MOTILAL OSWAL
540	AMAN KUMAR	MBA	DAMS	MOTILAL OSWAL
541	AYUSH DWIVEDI	MBA	VSICS	MOTILAL OSWAL
542	DIVYA KAPOOR	MBA	VSIPS	MOTILAL OSWAL
543	SHRADDHA SHARMA	MBA	VSICS	MOTILAL OSWAL
544	TANYA GUPTA	MBA	VSIPS	MOTILAL OSWAL
545	ABHISHEK KUMAR	MBA	VSIPS	MOVIDU TECHNOLOGIES
546	ANJU TRIVEDI	MBA	VSICS	MOVIDU TECHNOLOGIES
547	AYUSHI SRIVASTAVA	MBA	VSICS	MOVIDU TECHNOLOGIES
548	SANDESH MISHRA	MBA	VSIPS	MOVIDU TECHNOLOGIES
549	SHALINI SINGH	MBA	VSICS	MOVIDU TECHNOLOGIES
550	SHIVAM SINGH	MBA	VSICS	MOVIDU TECHNOLOGIES
551	SHIVANSH AWASTHI	MBA	VSICS	MOVIDU TECHNOLOGIES
552	SURBHI KUSHWAHA	MBA	VSICS	MOVIDU TECHNOLOGIES

537	DIVYANSHI KUSHWAHA	MBA	VSICS	MIKKY PUBLICATION
553	VAIBHAV TIWARI	MBA	VSIPS	MOVIDU TECHNOLOGIES
554	VAISHNAVI GUPTA	MBA	VSICS	MOVIDU TECHNOLOGIES
555	VISHAL DWIVEDI	MBA	VSICS	MOVIDU TECHNOLOGIES
556	SHIVANSH AWASTHI	MBA	VSICS	NIVA BUPA
557	AANCHAL SHUKLA	MBA	VSIPS	PROFINITY
558	ANIRUDDHA PANDEY	MBA	VSICS	PROFINITY
559	SHUBHANGI SINGH	MBA	VSIPS	PROFINITY
560	ASHISH KUMAR SINGH	MBA	VSIPS	RSPL
561	KARAN KUMAR	MBA	DAMS	RSPL
562	DIPALI NISHAD	MBA	VSICS	SRVA EDUCATION
563	VANSHIKA	MBA	VSICS	SRVA EDUCATION
564	HARSHIT DWIVEDI	MBA	VSICS	SUPERHOUSE
565	SHIVANSH AWASTHI	MBA	VSICS	SUPERHOUSE
566	ABHISHEK KUMAR	MBA	VSIPS	UIFS TECHNOLOGIES
567	ANSHIKA KUMARI SINHA	MBA	VSICS	UIFS TECHNOLOGIES
568	ANSHIKA SHUKLA	MBA	VSICS	UIFS TECHNOLOGIES
569	HARSHIT DWIVEDI	MBA	VSICS	WFM EXPERTS
570	SADDAM HUSSAIN	MCA	VSICS	AMEYAA NXT TECHNOLOGY PVT. LTD.
571	TARUN MISHRA	MCA	VSICS	AMEYAA NXT TECHNOLOGY PVT. LTD.
572	YASH NIGAM	MCA	VSICS	AMEYAA NXT TECHNOLOGY PVT. LTD.
573	ANAND KUMAR	MCA	VSICS	DAINIK JAGRAN
574	ROHIT MISHRA	MCA	VSICS	DAINIK JAGRAN
575	SHANTANU DEV	MCA	VSICS	DAINIK JAGRAN
576	SONAL GUPTA	MCA	VSICS	DAINIK JAGRAN

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DIVYANSHI KUSHWAHA

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KRATI MISHRA

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ALISHA RAIS
MBA
VSIPS
MOTILAL OSWAL
540
AMAN KUMAR
MBA
DAMS
MOTILAL OSWAL
541
AYUSH DWIVEDI
MBA
VSICS
MOTILAL OSWAL
542
DIVYA KAPOOR
MBA
VSIPS
MOTILAL OSWAL
543
SHRADDHA SHARMA
MBA
VSICS
MOTILAL OSWAL
544
TANYA GUPTA
MBA
VSIPS
MOTILAL OSWAL
545
ABHISHEK KUMAR
MBA
VSIPS

MOVIDU TECHNOLOGIES

546

ANJU TRIVEDI

MBA

VSICS

MOVIDU TECHNOLOGIES

547

AYUSHI SRIVASTAVA

MBA

VSICS

MOVIDU TECHNOLOGIES

548

SANDESH MISHRA

MBA

VSIPS

MOVIDU TECHNOLOGIES

549

SHALINI SINGH

MBA

VSICS

MOVIDU TECHNOLOGIES

550

SHIVAM SINGH

MBA

VSICS

MOVIDU TECHNOLOGIES

551

SHIVANSH AWASTHI

MBA

VSICS

MOVIDU TECHNOLOGIES

552

SURBHI KUSHWAHA

MBA

VSICS

MOVIDU TECHNOLOGIES

553

VAIBHAV TIWARI

MBA

VSIPS

MOVIDU TECHNOLOGIES

554

VAISHNAVI GUPTA

MBA

VSICS

MOVIDU TECHNOLOGIES

555

VISHAL DWIVEDI

MBA

VSICS

MOVIDU TECHNOLOGIES

556

SHIVANSH AWASTHI

MBA

VSICS

NIVA BUPA

557

AANCHAL SHUKLA

MBA

VSIPS

PROFINITY

558

ANIRUDDHA PANDEY

MBA

VSICS

PROFINITY

559

SHUBHANGI SINGH

MBA

VSIPS

PROFINITY

560

ASHISH KUMAR SINGH

MBA

VSIPS
RSPL
561
KARAN KUMAR
MBA
DAMS
RSPL
562
DIPALI NISHAD
MBA
VSICS
SRVA EDUCATION
563
VANSHIKA
MBA
VSICS
SRVA EDUCATION
564
HARSHIT DWIVEDI
MBA
VSICS
SUPERHOUSE
565
SHIVANSH AWASTHI
MBA
VSICS
SUPERHOUSE
566
ABHISHEK KUMAR
MBA
VSIPS
UIFS TECHNOLOGIES
567
ANSHIKA KUMAR

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578	YASH BANSAL	MCA	VSICS	DAINIK JAGRAN
579	AMAN SHUKLA	MCA	VSICS	TECHNOSYS PVT. LTD
580	ANSHIKA SINGH	MCA	VSICS	TECHNOSYS PVT. LTD
581	ESHA SAINI	MCA	VSICS	TECHNOSYS PVT. LTD
582	ROHIT MISHRA	MCA	VSICS	WFMEXPERTS
583	SHANTANU DEV	MCA	VSICS	WFMEXPERTS
584	SONAL GUPTA	MCA	VSICS	WFMEXPERTS
585	STUTI MISHRA	MCA	VSICS	WFMEXPERTS
586	YASH BANSAL	MCA	VSICS	WFMEXPERTS

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577

STUTI MISHRA

MCA

VSICS

DAINIK JAGRAN

578

YASH BANSAL

MCA

VSICS

DAINIK JAGRAN

579

AMAN SHUKLA

MCA

VSICS

TECHNOSYS PVT. LTD

580

ANSHIKA SINGH

MCA

VSICS

TECHNOSYS PVT. LTD

581

ESHA SAINI

MCA

VSICS

TECHNOSYS PVT. LTD

582

ROHIT MISHRA

MCA
VSICS
WFMEXPERTS
583
SHANTANU DEV
MCA
VSICS
WFMEXPERTS
584
SONAL GUPTA
MCA
VSICS
WFMEXPERTS
585
STUTI MISHRA
MCA
VSICS
WFMEXPERTS
586
YASH BANSAL
MCA
VSICS
WFMEXPERTS

DOCUMENT CHUNK 9: Plhl (PDF) [PDF Document] | URL:
<https://www.vsicskanpur.org/assets/placement/plhl.pdf>

Plhl (Official Document)

****Source Document URL:**** <https://www.vsicskanpur.org/assets/placement/plhl.pdf>

****Total Document Pages:**** 23

DOCUMENT CHUNK 10: Plhl23 (PDF) [PDF Document] | URL:
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Plhl23 (Official Document)

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****Total Document Pages:**** 14

DOCUMENT CHUNK 11: Samaj Kalyan Scholarship - Important Notice [HTML Page] | URL: <https://www.vsicsskanpur.org/important notice>

Kindly read the norms and guidelines of the U.P. government for the same.
The Current Guidelines are attached herewith :Admin28/03/2025
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Dvc (Official Document)

Source Document URL: <https://www.vsicsskanpur.org/docs/dvc.pdf>
Total Document Pages: 1

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Md (Official Document)

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Total Document Pages: 21

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Appendix8	
	MandatoryDisclosure

Page 2 Content

Locationmap
of theInstitution
Updated on

28/10/2025

AICTE Permanent ID

1-5543171

Date & Period of last approval

10April2025

2025-26

Accreditation Status

Not Accredited

Name of the Institution

Dr Virendra Swarup Institute of Computer Studies

Address of the Institution

810,W-1Block,Juhi,Saket Nagar,Kanpur-208014

City & Pin Code

Kanpur-208014

State / UT

Uttar Pradesh

Phone number with STD code 0512-2603080-81,0512-2603077

Office hours at the Institution

9:00 AM -5:00 PM

Academic hours at the Institution

9:30 AM -4:30 PM

Email

arpitawasthi17@gmail.com

Website

www.vsicskanpur.org

Nearest Railway Station (dist in Km)

Kanpur Cental-06 Km

Nearest Airport (dist in Km)

ChakeriAirport-15km

Type of Institution

Private-Self Financed

Category (1) of the Institution

Non Minority

Category (2) of the Institution

Co-Ed

Name of the organization running

Dr Virendra Swarup Educational Foundation

the Institution

Type of the organization

Society

Address of the organization

15/96, CivilLines, Kanpur-208001

Registered with

SocietyReg. Act 1860

Registration date

9/8/1989

Website of the organization

Name of the affiliating University

Dr. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, UTTER PRADESH

Address

VistarYojna, NayaKhera, Jankipuram, Lucknow, UttarPradesh226031

Website

www.aktu.ac.in

Latest affiliation period

2024-27

Name of Director

Dr. Pragya Awasthi

Exact Designation

Direct

Page 3 Content

Frequency of meetings & date of Once in a quarter, last meeting heldon 11 JULY,2025

last meeting

Governing Body Members

1 Dr. Rajnish Kumar

2 Dr. Ashok Kumar Saxena

3 Mr. Gauravendra Swarup

4 Mr. Ranjit Singh

5 Dr. Ramesh Verma

6 Mrs. Sharmila Nandi

7 Sr. Vinod Bahal

8 Dr. Prem Narain Nigam

9 Dr. Sushil Kumar

10 Dr. Manjula Srivastava

11 Nominee of Affiliating University

12 Nominee of State Government

13 An Industrialist/Technologist/Educationist for the region Nominated by the state government/UT

14 Dr. Pragya Awasthi

15 Mr. Kailash Kandpal

16 Mr. Ajay Nath Dubey

Academic Advisory Body

1 Kumkum Swarup(Member)

2 Gauravendra Swarup(Secretary)

3 Dr Vinay Pathak(V.C. CSJMU, Kanpur

4 Dr G.P .Srivastava(Prof C.S.J.M,University,Kanpur)

5 Dr. Sanjay Srivastava - V.C. Nehru Gram Bharti Uni. Allahabad(Chairman)

6 Dr Arpit Awasthi(Executive Director of the Institute)

7 Dr. S.K. Singh Professor Amity university

Frequency of meetings & date of Once in a quarter, last meeting heldon 11 JULY,2025

last meeting

Organizational Chart

Chairman

SecretaryJoint

Secretary

Director

Teaching Staff

Administration

Training &Placement

Dean Academics

Admin Officer

Training&PlacementOfficer

Faculty

Lab Assistant

Counselor

Librarian

Accounts Assistant

House Keeping Staff

9

8

7

Page 4 Content

Student feedback mechanism on

Regular Feedback from Students for improvement of Institute is taken

Institutional Governance/faculty

and Constructive Suggestions are considered for implementation

performance

Grievance redressal mechanism

Different Committees have been formed by Board of Management.

For faculty, staff and students

Anti-Ragging Measures:-

1.

Anti-Ragging Committee and Squad constituted as per AICTE norms.

2.

Declaration of Zero Tolerance Policy towards Ragging.

Grievance Redressal and Internal Committees

1.

Grievance Redressal Committee

2.

Internal Complaint Committee (ICC)

3.

SC/ST Committee

4.

Academic and Disciplinary Committees

Programs Offered:-

Course Duration Intake Actual Admissions Year of Starting

Level

Approved by AICTE/UGC

Professional activities

MCA 2 Years [60] 75 (EWS+FW) [1998]

Postgraduate

Yes

Seminars,Workshops,Industrial training conducted

regularly

MBA 2 Years [60] 61 [2013]

Postgraduate

Yes

Seminars,Workshops,Industrial training conducted

regularly

BCA 3 Years [600] 660 [1998]

Undergraduate Yes

Seminars,Workshops,Industrial training conducted
regula

Page 5 Content

15Name of Teaching Staff*

Dr. PRAGYA AWASTHI

Designation

DIRECTOR

Date of Joining the Institution

27-03-2016

Qualifications with Class/Grade

UG-B.Sc-

PG-MBA,M.Com

PhD- YES

Total Experience in Years

Teaching -23

Industry-02

Research-12

Papers Published

National-09

International -04

Papers Presented in Conferences

National -01

International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs-27 Projects at Masters level -41

Books Published / IPRs/ Patents

2

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional IIT Kanpur, HBTI Kanpur,

MNNITAllahabadInstitutions

16Name of Teaching Staff*

Dr. ARPIT AWASTHI

Designation

PRINCIPAL

Date of Joining the Institution

05-07-2006

Qualifications with Class/Grade

UG-B.Sc-

PG-M.COM, PGDBM, MMM

PhD- YES

Total Experience in Years

Teaching -25

Industry-02

Research-7

Papers Published

National-05

International -02

Papers Presented in Conferences

National -01

International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs-27 Projects at Masters level -20

Books Published / IPRs/ Patents

2

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional CSJ

Page 6 Content

hoto

Photo

Name of Teaching Staff*

RAHUL AGNIHOTRI

Designation

ASSISTANT PROFESSOR

Department

COMPUTER APPLICATION

Date of Joining the Institution

1/8/2006

Qualifications with Class/Grade

UG-B.Sc-II Di PG-MCA-I Div, MBA ,M.Tech-I Div UGC NET

PhD

Total Experience in Years

Teaching -18

Industry-

Research-3

Papers Published

National-1

International

Papers Presented in Conferences

National-1International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional IITKanpur, HBTI Kanpur, MNNITAllahabad

Institutions

Name of Teaching Staff*

NITIN MISHRA

Designation

ASSISTANT PROFESSOR

Department

COMPUTER APPLICATION

Date of Joining the Institution

14/8/2006

Qualifications with Class/Grade

UG-B.Sc-72%PG-MCA-76%, MBA, M-TECH

PhD

Total Experience in Years

Teaching -20

Industry-2

Research

Papers Published

National

International

Papers Presented in Conferences

National01International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

JECJabalpur,PIMGwalior,HBTKanpur

In

Page 7 Content

Name of Teaching Staff*

KAILASH KANDPAL

Designation

ASSISTANT PROFESSOR

Department

COMPUTER APPLICATION

Date of Joining the Institution

1/8/2006

Qualifications with Class/Grade

UG-B.C.A-I DIV PG-MCA-I DIV, M.PHIL-IIDIV, MBA, UGCNET PhD

Total Experience in Years

Teaching -18

Industry-

Research

Papers Published

National

International

Papers Presented in Conferences

National

International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

HBTIKanpur

Institutions

Name of Teaching Staff*

IQBAL MASOOD AHMAD

Designation

ASSISTANT PROFESSOR

Department

COMPUTER APPLICATION

Date of Joining the Institution

24/07/2011

Qualifications with Class/Grade

UG-B.Sc-I DIV

PG-MCA-I DIV

PhD

Total Experience in Years

Teaching -24

Industry-

Research

Papers Published

National

International

Papers Presented in Conferences

National

International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional HBTIKanpur, CSJM

KanpurInstitutions

Name of Teaching Staff*

JOYDEEP DAS

Page 8 Content

HOTO

23 Name of Teaching Staff

Designation

SUMEGHA BHATIYA

ASSISTANT PROFESSOR

Department

BUSINESS MANAGEMENT

Date of Joining the Institution

1/7/2008

Qualifications with Class/Grade

UG-B.COM- I DIVPG-M.B.A, UGCNET

PhD-

Total Experience in Years

Teaching- 16

Industry -1

Research-2

Papers Published

National

International

Papers Presented in Conferences

National-3

International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

HBTI Kanpur,CSJM Kanpur

Institutions

Name of Teaching Staff*

RAMESH KR SRIVASTAVA

Designation

ASSISTANT PROFESSOR

Department

BUSINESS MANAGEMENT

Date of Joining the Institution

7/8/2006

Qualifications with Class/Grade

UG-B.Sc-II DIV

PG-MCA-I DIV

PhD

Total Experience in Years

Teaching -15

Industry-

Research

Papers Published

National

International

Papers Presented in Conferences

National

International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

HBTKanpur,MPECKanpur

Page 9 Content

Name of Teaching Staff*

ASHISH PANDEY

Designation

Assistant Professor

Department

BUSINESS MANAGEMENT

Date of Joining the Institution

2/7/2010

Qualifications with Class/Grade

UG-B.Sc.

PG-M.B.A, M.Sc.

PhD-

Total Experience in Years

Teaching-19

Industry -02

Research-01

Papers Published

National

International

Papers Presented in Conferences

National 04

International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

HBTIKanpur,CSJM Kanpur

Institutions

25Name of Teaching Staff*

ANCHAL KHATRI

Designation

Department

Date of Joining the Institution

Qualifications with Class/Grade

ASSISTANT

PROFESSORBUSINES

MANAGEMENT 15/07/2019

UG-B.C.A

PG-PGDM

PhD-

TotalExperienceinYears Papers

Published

Papers Presented in Conferences

PhD Guide? Give field &University

Teaching-10

National

National

Field

Industry -00

Research

International

International

University

PhDs / Projects Guided

PhDs

Projects at Masters level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

InteractionwithProfessional

Institutions

NIL

N.A

26

Page 10 Content

Name of Teaching Staff*

AJAY NATH DUBEY

Designation

Assistant Professor

Department

BUSINESS MANAGEMENT

Date of Joining the Institution

13/09/2012

Qualifications with Class/Grade

UG-B.SC- I DIV

PG-M.B.A

PhD-PURSUINGPH.D

Total Experience in Years

Teaching-17

Industry -02

Research-02

Papers Published

National

International

Papers Presented in Conferences

National 04

International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

HBTKanpur, CSJMKanpur

Institutions

Name of Teaching Staff*

POOJA MISHRA

Designation

Assistant Professor

Department

BUSINESS MANAGEMENT

Date of Joining the Institution

11/08/2021

Qualifications with Class/Grade

PG-M.B.A

PhD-

Total Experience in Years

Teaching-7

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National

International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

N.A

Institutions

Name of Teaching Staff*

KM KIRTI

Designatio

Page 11 Content

Name of Teaching Staff*

MOHD. FAHEEM

Designation

Assistant Professor

Department

BUSINESS MANAGEMENT

Date of Joining the Institution

01/08/2014

Qualifications with Class/Grade

PG-M.B.A, M.A

PhD

Total Experience in Years

Teaching-12

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

HBTKanpur,CSJMKanpur

Institutions

Name of Teaching Staff*

VINEET BAJPAI

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

13/10/2008

Qualifications with Class/Grade

PG-M.C.A, M.SC (MATH)

PhD

Total Experience in Years

Teaching-16

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

CSJMKanpur

Institutions

Page 12 Content

Name of Teaching Staff*

RAM AWTAR

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

08/01/2008

Qualifications with Class/Grade

PG-M.C.A,

PhD

Total Experience in Years

Teaching-20

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

HBTIKanpur,CSJMKanpur

Institutions

Name of Teaching Staff*

SANJAY KUMAR TIWARI

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

08/01/2011

Qualifications with Class/Grade

PG-M.TECH, M.C.A

PhD Total Experience in

Years

Teaching-17

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with

Page 13 Content

Name of Teaching Staff*

PAWASI SAXENA

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

08/01/2006

Qualifications with Class/Grade

UG-B.TECH PG-M.TECH

PhD

Total Experience in Years

Teaching-18

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

HBTIKanpur,CSJMKanpur

Institutions

Name of Teaching Staff*

SATYA PRATAP SINGH

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

20/11/2011

Qualifications with Class/Grade

PG- M.SC (MATH), M.SC (CS), PGDCA, PGDBMS

PhD

Total Experience in Years

Teaching-20

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

Page 14 Content

Name of Teaching Staff*

AVNEESH YADAV

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

21/08/2023

Qualifications with Class/Grade

PG-M.C.A

PhD Total Experience in

Years

Teaching-6

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

CSJMKanpur

Institutions

Name of Teaching Staff*

ANURAG CHANDRA MISHRA

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

01/10/2014

Qualifications with Class/Grade

PG-M.C.A

PhD

Total Experience in Years

Teaching-13

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Pr

Page 15 Content

Name of Teaching Staff*

VIVEK NIGAM

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

07/01/2008

Qualifications with Class/Grade

PG-M.C.A

PhD

Total Experience in Years

Teaching-17

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional CSJM Kanpur

Institutions

Name of Teaching Staff*

ASHISH KUMAR

Designation

Assistant Professor

Department

BUSINESS MANAGEMENT

Date of Joining the Institution

05/07/2019

Qualifications with Class/Grade

UG-B.SC PG-M.B.A, M.COM PhD

Total Experience in Years

Teaching-5

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Page 16 Content

Name of Teaching Staff*

SHAILENDRA DIXIT

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

07/01/2008

Qualifications with Class/Grade

PG-M.C.A, M.SC, UGCNET

PhD

Total Experience in Years

Teaching-16

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

CSJM Kanpur

Institutions

Name of Teaching Staff*

AKARSHI TIWARI

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

19/10/2023

Qualifications with Class/Grade

UG- B.C.A PG-M.C.A PhD

Total Experience in Years

Teaching-2

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books

Page 17 Content

Name of Teaching Staff*

HARSHITA YADAV

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

07/08/2024

Qualifications with Class/Grade

UG-BCA PG-M.C.A PhD

Total Experience in Years

Teaching-1

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

N.A

Institutions

Name of Teaching Staff*

ARCHITA DUBEY

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

15/07/2024

Qualifications with Class/Grade

UG-B.C.A PG-M.C.A PhD

Total Experience in Years

Teaching-1

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Publi

Page 18 Content

Name of Teaching Staff*

SHRUTI SHUKLA

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

15/01/2013

Qualifications with Class/Grade

PG-M.B.A

PhD

Total Experience in Years

Teaching-14

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

N.A

Institutions

Name of Teaching Staff*

Dr. ATUL KUMAR SINGH

Designation

Assistant Professor

Department

BUSINESS MANAGEMENT

Date of Joining the Institution

15/07/2024

Qualifications with Class/Grade

UG-B.COM PG-M.COM PhD –YES ((Commerce)

Total Experience in Years

Teaching-13

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Pat

Page 19 Content

Name of Teaching Staff*

ANUJ SINGH

Designation

Assistant Professor

Department

BUSINESS MANAGEMENT

Date of Joining the Institution

15/07/2023

Qualifications with Class/Grade

UG-B.APG-MBA PhD

Total Experience in Years

Teaching-2

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

N.A

Institutions

Name of Teaching Staff*

ALOK SHUKLA

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

02/01/2024

Qualifications with Class/Grade

PG-M.C.A PhD

Total Experience in Years

Teaching-1

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books P

Page 20 Content

Name of Teaching Staff*

SHALINI MISHRA

Designation

Assistant Professor

Department

COMPUTER APPLICATION

Date of Joining the Institution

04/08/2014

Qualifications with Class/Grade

PG-M.C.A PhD

Total Experience in Years

Teaching-11

Industry

Research

Papers Published

National

International

Papers Presented in Conferences

National International

PhD Guide? Give field &University

Field

University

PhDs / Projects Guided

PhDs

Projects at Masters Level

Books Published / IPRs/ Patents

NIL

Professional Memberships

NIL

Consultancy Activities

NIL

Awards

NIL

Grants fetched

NIL

Interaction with Professional

N.A

Institutions

Laboratory details

YES

Computer Centre facilities

05 Computer Labs with 238 terminals

Library facilities

02 Libraries

Auditorium / Seminar Halls / Amphi

03 (Conference+ Seminar)

Cafeteria

Canteen Available

Indoor Sports facilities

Yes-Table Tennis, Carom ,Chess, Badminton etc.

Outdoor Sports facilities

Yes-Cricket Society has got its own ground of 8 acres

57

Page 21 Content

Gymnasium facilities

No

Facilities for disabled

If required will be made available

Any other facilities

No

Boys Hostel

NO

Girls Hostel

NO

Medical & other Facilities at Hostel

No

Counseling / Mentoring

By Pshychatrist Dr Alok Bajpai

Career Counseling

Through various Seminars by Industry Professionals

Medical facilities

Sr Medical Consultant Appointed

Student Insurance

ByAKTU

Students Activity Body

BY Different Committees

Cultural activities

Unnayan - The cultural Fest is held every year

Sports activities

Cricket,TableTennis,Chess, Badminton, Carromconducted

Literary activities

Conducted throughout the year- Debates, Quizzes, seminars etc.

Magazine / Newsletter

Cynosure - News letter Published Annually

Technical activities / TechFest

AVISKAR-MNIT,INCORD-HBTI

Industrial Visits / Tours

conducted

Alumni activities

Conducted Annually Students are taken for Industrial trips

Name of the Information Officer for RTI Mr Viresh Saxena

Designation Senior Admin Officer

Phone number with STD code 0512-2603080-81, 0512-2603077

Email viresh_saxena@rediffmail.com

58

61

59

60

DOCUMENT CHUNK 14: Faculty & Leadership Members [HTML Page] | URL:
https://www.vsicskanpur.org/long_journey/index4.html

Faculty & Leadership Members

Name	Designation	Qualification / Role	Experience
Prof.(Dr.) Arpit Awasthi	Executive Director	Ph.D. / Master's Degree	15+ Years
Dr.Rekh Nath Singh	Head Research & FDP	Ph.D. / Master's Degree	15+ Years
Mr.Viresh Saxena	Administrative Head	Ph.D. / Master's Degree	15+ Years
Mr.Sunil Saxena	Administrative Head	Ph.D. / Master's Degree	15+ Years
Mr.Chandrajeet Singh	Maintainence Head	Ph.D. / Master's Degree	15+ Years

DOCUMENT CHUNK 15: Mca Fee (PDF) [PDF Document] | URL:
<https://www.vsicskanpur.org/docs/MCA-FEE.pdf>

Mca Fee (Official Document)

Source Document URL: <https://www.vsicskanpur.org/docs/MCA-FEE.pdf>

Total Document Pages: 1

Page 1 Data Table

HEADS	I SEM	II SEM	III SEM	IV SEM
Time to Pay	(on Admission)	Jan-2026	July-2026	Jan-2027
Tuition Fees	Rs. 27500/-	Rs. 27500/-	Rs. 27500/-	Rs. 27500/-

Page 1 Content

Dr. Virendra Swarup Institute of Computer Studies

MCA fees* (2026 – 2028)

HEADS

I SEM

II SEM

III SEM

IV SEM

Time to Pay

(on Admission)

Jan-2026
July-2026
Jan-2027
Tuition Fees
Rs. 27500/-
Rs. 27500/-
Rs. 27500/
Rs. 27500/-
Security
Rs. 3000/-

*Other fees as applicable

DOCUMENT CHUNK 16: Mba Fee (PDF) [PDF Document] | URL: <https://www.vsicskanpur.org/docs/MBA-Fee.pdf>

Mba Fee (Official Document)

Source Document URL: <https://www.vsicskanpur.org/docs/MBA-Fee.pdf>

Total Document Pages: 1

Page 1 Data Table

HEADS	I SEM	II SEM	III SEM	IV SEM
Time to Pay	(on Admission)	Jan-2027	July-2027	Jan-2028
Tuition Fees	Rs. 29500/-	29500	29500	29500
Total	Rs. 32500/-	Rs. 29500/-	Rs. 29500/-	Rs. 29500/-

Page 1 Content

Dr. Virendra Swarup Institute of Computer Studies
MBA fees* (2026 – 2028)
HEADS
I SEM
II SEM
III SEM
IV SEM
Time to Pay
(on Admission)
Jan-2027
July-2027

Jan-2028

Tuition Fees

Rs. 29500/-

29500

29500

29500

Security

Rs. 3000/-

Total

Rs. 32500/-

Rs. 29500/-

Rs. 29500/-

Rs. 29500/-

*Other fees as applicable

DOCUMENT CHUNK 17: Circularsecurity (PDF) [PDF Document] | URL:
<https://www.vsicskanpur.org/docs/circularsecurity.pdf>

Circularsecurity (Official Document)

Source Document URL: <https://www.vsicskanpur.org/docs/circularsecurity.pdf>

Total Document Pages: 1

DOCUMENT CHUNK 18: Antiragging (PDF) [PDF Document] | URL:
<https://www.vsicskanpur.org/docs/antiragging.pdf>

Antiragging (Official Document)

Source Document URL: <https://www.vsicskanpur.org/docs/antiragging.pdf>

Total Document Pages: 1

DOCUMENT CHUNK 19: VSICS [HTML Page] | URL: <https://www.vsicskanpur.org/gsc.html>

Our Schools

- Dr. VSEC, H-Block , Kidwai Nagar, Kanpur
- Dr. VSEC, N-Block ,Bara Devi, Kanpur
- Dr. VSEC, Awadhपुरi , Kanpur
- Dr. VSEC, Sharda Nagar, Kanpur
- Dr. VSEC, Panki, Kanpur

- Dr. VSEC, Shyam Nagar(J), Kanpur
- Dr. VSEC, Shyam Nagar(S), Kanpur
- Dr. VSEC, Chandpur, Farrukhabad
- Dr. VSEC, Unnao
- Dr. VSEC, Jajmau, Kanpur

Our Colleges

- DAV College
- DBS College
- DG College
- Mahila Mahavidyalaya
- DB College Raebareli

DOCUMENT CHUNK 20: Excellence (PDF) [PDF Document] | URL: <https://www.vsicskanpur.org/docs/excellence.pdf>

Excellence (Official Document)

Source Document URL: <https://www.vsicskanpur.org/docs/excellence.pdf>

Total Document Pages: 2

DOCUMENT CHUNK 21: List of Academic Toppers [HTML Page] | URL: <https://www.vsicskanpur.org/toppers.html>

List of Academic Toppers

Course	Batch	Top Rankers / Rank Holder Students
MBA	2023-25	SANDLI SRIVASTAVA, DEEPALI NISHAD, NANDINI VERMA
BBA	2022-25	NIKITA ISSRANI, PRIYANKA YADAV, AVIRAL TIWAI & KASHISH ATHWAI
MCA	2023-25	ANSHIKI KHARE, BOBBY KASHYAP, ABHINAY PANDEY
BCA	2022-25	SUSHMIT SHUKLA, GAURVIKA GANGWANI, DEVANSHI VERMA

DOCUMENT CHUNK 22: VSICS [HTML Page] | URL: <https://www.vsicskanpur.org/bba.html>

- Overview

Bachelor of Business Administration (BBA) Course is approved by AICTE ,affiliated to CSJMU, Kanpur

About BBA

VSEF offers a three-year full-time Bachelor of Business Administration (BBA) program. The program is a comprehensive professional course designed to provide students with a strong foundation in managerial concepts and business practices. It aims to develop managerial competence, leadership qualities, and entrepreneurial skills, enabling students to meet the evolving demands of the corporate world. The BBA program is structured to prepare students for successful careers in business and management by combining theoretical knowledge with practical exposure. This government-approved professional degree program also serves as an excellent platform for pursuing higher studies in management or securing rewarding career opportunities through placements. The course is approved by AICTE ,affiliated to CSJMU, Kanpur

Where we excel?

- 28+ Years of Academic Excellence
- Placement and Carrer Development Support
- Industry Interaction and Internship Platforms
- Holistic Individual Development Programmes
- Internet-Enabled Class Rooms
- Well Equipped Computer Labs For Practical Learning
- Use Of ICT Tools For Enhanced Learning Exposure
- Community-Based Learning
- Duration3 Years
- Students240

DOCUMENT CHUNK 23: VSICS [HTML Page] | URL: <https://www.vsicsskanpur.org/bca.html>

- Overview

Bachelor of Computer Application (BCA) Course is approved by AICTE, affiliated to CSJMU, Kanpur

About BCA

VSEF offers a three-year full-time Bachelor of Computer Applications (BCA) program. The course is designed to provide students with a strong foundation in computer applications, programming, software development, and emerging technologies. It emphasizes practical learning and industry-oriented training to meet the evolving requirements of the IT and software sectors. The BCA program introduces students to the practical aspects of software development while equipping them with the technical skills and knowledge required to build successful careers in the technology industry. Graduates of the program have excellent opportunities in software development, web development, database management, system administration, and other IT-related fields. This government-approved professional degree program also provides an ideal platform for pursuing higher studies in computer science, information technology, and related disciplines, as well as securing rewarding employment opportunities through placements. The course is approved by AICTE ,affiliated to CSJMU, Kanpur

Where we excel?

- 28+ Years of Academic Excellence
- Placement and Carrer Development Support
- Industry Interaction and Internship Platforms
- Holistic Individual Development Programmes
- Internet-Enabled Class Rooms

- Well Equipped Computer Labs For Practical Learning
- Use Of ICT Tools For Enhanced Learning Exposure
- Community-Based Learning
- Duration 3 Years
- Students 600

DOCUMENT CHUNK 24: VSICS [HTML Page] | URL: <https://www.vsicskanpur.org/events.html>

Srijan- An Innovation to Creativity

Ever since the advent of human beings, they have been ornamented with an attribute called creativity. To adorn it more, DR. VIRENDRA SWARUP INSTITUTE OF COMPUTER STUDIES created SRIJAN - THE ANNUAL CULTURAL FESTIVAL. It's a compendium of courage, valour, talent, dance, drama, singing, innovation to new ideas, are been portrayed by our students. The extravagant show, been nurtured by sheer talent and enthusiasm live up to the expectations of the spectators. Organized in the winters, where the stage is all set to have a larger than life experience. Students, staff, faculties & the management whole heartedly give their best to make it a mega success.

Winter Carnival - Fun Fiesta

Among the various avenues & moments of learning, the students of Dr. VSICS organized Winter Carnival- Fun Fiesta. Fun Fiesta Envision, Enter prize & Enthral, one day event to provide our students with opportunity to give space to their ideas & explore their professional aptitude of creating & managing with exuberance & frolic. The events was a practical exposure to in- class studies marking the day by plethora of activities, amusement stalls, lucky draw & many other besides.

Energy - The Athletic Meet

VSEF organizes two-day athletic meet - Energy providing students with the opportunity to relax & rejuvenate exploring an untouched realm in their lives - sports & athletics. The two day event was marked by plethora of track & field events participated by eight teams of VSEF with immense vigour & fervour.

DOCUMENT CHUNK 25: VSICS [HTML Page] | URL: <https://www.vsicskanpur.org/abhivyakti.php>

Abhivyakti - An Expression of Care

Deepawali celebration with UP Police

The true essence of Celebrating festivals lies in sharing n caring. Having well understood students of VSICS Mgmt. Dept. made an effort to bring smiles on the faces of elderly members of society by making a visit to Vradhaasharam n those sparkling joys of the little ones knew no bounds when the students spend fulfilled moments with them on their visit to Orphanage

DOCUMENT CHUNK 26: VSICS [HTML Page] | URL: <https://www.vsicskanpur.org/expert.html>

It is great to see the quality success achieved by Dr.Virendra Swarup Institute of Computer Studies in this quantitative world. I visited, I viewed, I was overwhelmed&it was wonderful to see the highly motivated shining faces of young professionals of tomorrow.

Very inspiring arena and learning environment. I wish them all the best.

Excellent leadership and obligation of promoters will undoubtedly take VSICS to greater heights.

What a nice Institute! All the best for total growth in all directions for Technical and Management Development.

My best wishes for the success of this institute, it has a wonderful infrastructure and I sincerely wish to see VSICS in time to gain glory unlimitedly.

The pass –outs of the Institute actually reflected the commitment and dedication of those shaping them. The amount of confidence inculcated makes us feel good to have such management institute in the city.

We are proud to observe the dynamism & sincerity in the young management graduates of the Institute. Besides the conceptual knowledge, the qualities of team spirit & diligence are worth reckoning.

I had a wonderful experience from the institute and I pray to God, that VSICS flourishes more in the time to come.

Hope to see this institute add more gems in its journey to come.

DOCUMENT CHUNK 27: Mcasyllabusnew (PDF) [PDF Document] | URL: <https://www.vsicskanpur.org/docs/mcasyllabusnew.pdf>

Mcasyllabusnew (Official Document)

Source Document URL: <https://www.vsicskanpur.org/docs/mcasyllabusnew.pdf>

Total Document Pages: 85

Page 1 Content

MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Page 1

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY

UTTAR PRADESH, LUCKNOW

EVALUATION SCHEME & SYLLABUS

First Year

FOR

MASTER OF COMPUTER APPLICATIONS

(MCA)

(Two Year Course)

Based On

NEP2020

(Effective from the Session: 2024-25)

Page 2 Data Table

S.No .	Subject Code	Subject Name	Periods			Sessional	
			L	T	P	CT	TA
1.	BMC101	Fundamental of Computers & Emerging Technologies	3	0	0	20	10
2.	BMC102	Problem Solving using C	3	1	0	20	10

S.No .	Subject Code	Subject Name	Periods			Sessional	
3.	BMC103	Principles of Management & Communication	3	0	0	20	10
4.	BMC104	Discrete Mathematics	3	0	0	20	10
5.	BMC105	Computer Organization & Architecture	3	1	0	20	10
6.	BMC151	Problem Solving using C Lab	0	0	4	30	20
7.	BMC152	Computer Organization & Architecture Lab	0	0	3	30	20
8.	BMC153	Professional Communication Lab	0	0	2	30	20
9.	BMC106	Cyber Security*	2	0	0	20	10
		Total					

Page 2 Data Table

S.No .	Subject Code	Subject Name	Periods			Sessional	
			L	T	P	CT	TA
1.	BMC201	Web Technology	3	0	0	20	10
2.	BMC202	Object Oriented Programming	3	0	0	20	10
3.	BMC203	Operating Systems	3	0	0	20	10
4.	BMC204	Database Management Systems	3	0	0	20	10
5.	BMC205	Data Structures & Analysis of Algorithms	4	0	0	20	10
6.	BMC251	Web Technology Lab	0	0	3	30	20
7.	BMC252	Object Oriented Programming Lab	0	0	3	30	20
8.	BMC253	DBMS Lab	0	0	2	30	20
9.	BMC254	Data Structures & Analysis of Algorithms Lab	0	0	3	30	20

S.No .	Subject Code	Subject Name	Periods			Sessional	
10.	BVA251	Sports and Yoga**	0	0	3	-	100
		Total					

Page 2 Content

MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Page 2

MCA (MASTER OF COMPUTER APPLICATIONS)

MCA FIRST YEAR, 2024-25

SEMESTER-I

S.No

.

Subject

Code

Subject Name

Periods

Sessional

ESE

Total

Credit

L

T

P

CT

TA

Total

1.

BMC101

Fundamental of Computers &

Emerging Technologies

3

0

0

20

10

30

70

100

3

2.

BMC102

Problem Solving using C

3

1

0

20

10

30

70

100

4

3.

BMC103

Principles of Management &

Communication

3

0

0

20

10

30

70

100

3

4.

BMC104

Discrete Mathematics

3

0

0

20

10

30

70

100

3

5.

BMC105

Computer Organization &

Architecture

3

1

0

20

10

30

70

100

4

6.

BMC151

Problem Solving using C

Lab

0

0

4

30

20

50

50

100

2

7.

BMC152

Computer Organization &

Architecture Lab

0

0

3

30

20

50

50

100

2

8.

BMC153

Professional Communication

Lab

0

0

2

30

20

50

50

100

2

9.

BMC106

Cyber Security*

2

0

0

20

10

30

70

100

0

Total

330

570

900

23

CT: Class Test TA: Teacher Assessment

L/T/P: Lecture/ Tutorial/ Practical

- Qualifying Non-credit Course

SEMESTER-II

S.No

.

Subject

Code

Subject Name

Periods

Sessional

ESE

Total

Credit

L

T

P

CT

TA

Total

1.

BMC201

Web Technology

3

0

0

20

10

30

70

100

3

2.

BMC202

Object Oriented Programming

3

0

0

20

10

30

70

100

3

3.

BMC203

Operating Systems

3

0

0

20

10

30

70

100

3

4.

BMC204

Database Management

Systems

3

0

0

20

10

30

70

100

Page 3 Content

MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Page 3

Syllabus

MCA 1st Year

Ist Semester

Page 4 Data Table

BMC101 : FUNDAMENTAL OF COMPUTERS & EMERGING TECHNOLOGIES			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to			
CO 1	Demonstrate the knowledge of the basic structure, components, features and generations of computers.		K , K 1 2
CO 2	Describe the concept of computer languages, language translators and construct algorithms to solve problems using programming concepts.		K K 2, 3
CO 3	Compare and contrast features, functioning & types of operating system and computer networks.		K 4
CO 4	Demonstrate architecture, functioning & services of the Internet and basics of multimedia.		K 2
CO 5	Illustrate the emerging trends and technologies in the field of Information Technology.		K , K 1 2
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Introduction to Computer: Definition, Computer Hardware & Computer Software Components: Hardware – Introduction, Input devices, Output devices, Central Processing Unit, Memory- Primary and Secondary. Software - Introduction, Types – System and Application. Computer Languages: Introduction, Concept of Compiler, Interpreter & Assembler Problem solving concept: Algorithms – Introduction, Definition, Characteristics, Limitations, Conditions in pseudo-code, Loops in pseudo code.		08
II	Operating system: Definition, Functions, Types, Classification, Elements of command based and GUI based operating system. Computer Network: Overview, Types (LAN, WAN and MAN), Data communication, topologies.		08
III	Internet: Overview, Architecture, Functioning, Basic services like WWW, FTP, Telnet, Gopher etc., Search engines, E-mail, Web Browsers. Internet of Things (IoT): Definition, Sensors, their types and features, Smart Cities, Industrial Internet of Things.		08

BMC101 : FUNDAMENTAL OF COMPUTERS & EMERGING TECHNOLOGIES			
IV	Block chain: Introduction, overview, features, limitations and application areas fundamentals of Block Chain. Crypto currencies: Introduction, Applications and use cases Cloud Computing: nature and benefits, AWS, Google, Microsoft & IBM Services		08
V	Emerging Technologies: Introduction, overview, features, limitations and application areas of Augmented Reality, Virtual Reality, Grid computing, Green computing, Big data analytics, Quantum Computing and Brain Computer Interface		08
Suggested Readings: 1. Rajaraman V., "Fundamentals of Computers", Prentice-Hall of India. 2. Norton P., "Introduction to Computers", McGraw Hill Education. 3. Goel A., "Computer Fundamentals", Pearson. 4. Balagurusamy E., "Fundamentals of Computers", McGraw Hill. 5. Thareja R., "Fundamentals of Computers", Oxford University Press. 6. Bindra J., "The Tech Whisperer- on Digital Transformation and the Technologies that Enable it", Penguin Books.			

Page 4 Content

MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Page 4

MCA (MASTER OF COMPUTER APPLICATIONS)

FIRST YEAR SYLLABUS

SEMESTER-I

BMC101 : FUNDAMENTAL OF COMPUTERS & EMERGING TECHNOLOGIES

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to

CO 1

Demonstrate the knowledge of the basic structure, components, features and generations of computers.

K1, K2

CO 2

Describe the concept of computer languages, language translators and construct algorithms to solve problems using programming concepts.

K2, K3

CO 3

Compare and contrast features, functioning & types of operating system and computer networks.

K4

CO 4

Demonstrate architecture, functioning & services of the Internet and basics of multimedia.

K2

CO 5

Illustrate the emerging trends and technologies in the field of Information Technology.

K1, K2

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Introduction to Computer: Definition, Computer Hardware & Computer Software

Components: Hardware – Introduction, Input devices, Output devices, Central Processing Unit, Memory- Primary and Secondary. Software - Introduction, Types – System and Application.

Computer Languages: Introduction, Concept of Compiler, Interpreter & Assembler

Problem solving concept: Algorithms – Introduction, Definition, Characteristics, Limitations, Conditions in pseudo-code, Loops in pseudo code.

08

II

Operating system: Definition, Functions, Typ

Page 5 Data Table

BMC102 : PROBLEM SOLVING USING C			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to			
CO 1	Describe the functional components and fundamental concepts of a digital computer system including number systems.		K , K 1 2
CO 2	Construct flowchart and write algorithms for solving basic problems.		K , K 2 3
CO 3	Write 'C' programs that incorporate use of variables, operators and expressions along with data types.		K , K 2 3
CO 4	Write simple programs using the basic elements like control statements, functions, arrays and strings.		K , K 2 3
CO 5	Write advanced programs using the concepts of pointers, structures, unions and enumerated data types.		K , K 2 3
CO 6	Apply pre-processor directives and basic file handling and graphics operations in advanced programming.		K , K 2 3
DETAILED SYLLABUS			3-1-0
Unit	Topic		Proposed Lecture
I	Basics of programming: Approaches to problem solving, Use of high- level programming language for systematic development of programs, Concept of algorithm and flowchart, Concept and role of structured programming. Basics of C: History of C, Salient features of C, Structure of C Program, Compiling C Program, Link and Run C Program, Character set, Tokens, Keywords, Identifiers, Constants, Variables, Instructions, Data types, Standard Input/Output, Operators and expressions.		08

BMC102 : PROBLEM SOLVING USING C			
II	Conditional Program Execution: if, if-else, and nested if-else statements, Switch statements, Restrictions on switch values, Use of break and default with switch, Comparison of switch and if-else. Loops and Iteration: for, while and do-while loops, Multiple loop variables, Nested loops, Assignment operators, break and continue statement. Functions: Introduction, Types, Declaration of a Function, Function calls, Defining functions, Function Prototypes, Passing arguments to a function Return value		08
III	Arrays: Array notation and representation, Declaring one-dimensional array, Initializing arrays, Accessing array elements, Manipulating array elements, Arrays of unknown or varying size, Two-dimensional arrays, Multidimensional arrays. Pointers: Introduction, Characteristics, * and & operators, Pointer type declaration and assignment, Pointer arithmetic, Call by reference, Passing pointers to functions, array of pointers, Pointers to functions, Pointer to pointer, Array of pointers. Strings: Int		08

Page 5 Content

MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Page 5

BMC102 : PROBLEM SOLVING USING C

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to

CO 1

Describe the functional components and fundamental concepts of a digital computer system including number systems.

K1, K2

CO 2

Construct flowchart and write algorithms for solving basic problems.

K2, K3

CO 3

Write 'C' programs that incorporate use of variables, operators and

expressions along with data types.

K2, K3

CO 4

Write simple programs using the basic elements like control statements, functions, arrays and strings.

K2, K3

CO 5

Write advanced programs using the concepts of pointers, structures, unions and enumerated data types.

K2, K3

CO 6

Apply pre-processor directives and basic file handling and graphics operations in advanced programming.

K2, K3

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Basics of programming: Approaches to problem solving, Use of high-level programming language for systematic development of programs, Concept of algorithm and flowchart, Concept and role of structured programming.

Basics of C: History of C, Salient features of C, Structure of C Program, Compiling C Program, Link and Run C Program, Character set, Tokens, Keywords, Identifiers, Constants, Variables, Instructions, Data types, Standard Input/Output, Operators and expressions.

08

II

Conditional Program Execution: if,

Page 6 Data Table

IV	Structure: Introduction, Initializing, defining and declaring structure, Accessing members, Operations on individual members, Operations on structures, Structure within structure, Array of structure, Pointers to structure. Union: Introduction, Declaring union, Usage of unions, Operations on union. Enumerated data types Storage classes: Introduction, Types- automatic, register, static and external.	08
V	Dynamic Memory Allocation: Introduction, Library functions – malloc, calloc, realloc and free. File Handling: Basics, File types, File operations, File pointer, File opening modes, File handling functions, File handling through command line argument, Record I/O in files. Graphics: Introduction, Constant, Data types and global variables used in graphics, Library functions used in drawing, Drawing and filling images, GUI interaction within the program.	08
Suggested Readings: 1. Kanetkar Y., "Let Us C", BPB Publications. 2. Hanly J. R. and Koffman E. B., "Problem Solving and Program Design in C", Pearson Education. 3. Schildt H., "C- The Complete Reference", Tata McGraw-Hill. 4. Goyal K. K. and Pandey H.M., "Trouble Free C", University Science Press 5. Gottfried B., "Schaum's Outlines- Programming in C", Tata McGraw-Hill Publications. 6. Kochan S.G., "Programming in C", Addison-Wesley. 7. Dey P. and Ghosh M., "Computer Fundamentals and Programming		

Page 6 Content

MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Page 6

IV

Structure: Introduction, Initializing, defining and declaring structure, Accessing members, Operations on individual members, Operations on structures, Structure within structure, Array of structure, Pointers to structure.

Union: Introduction, Declaring union, Usage of unions, Operations on union. Enumerated data types

Storage classes: Introduction, Types- automatic, register, static and external.

08

V

Dynamic Memory Allocation: Introduction, Library functions – malloc, calloc, realloc and free.

File Handling: Basics, File types, File operations, File pointer, File

opening modes, File handling functions, File handling through command line argument, Record I/O in files.

Graphics: Introduction, Constant, Data types and global variables used in graphics, Library functions used in drawing, Drawing and filling images, GUI interaction within the program.

08

Suggested Readings:

1.

Kanetkar Y., "Let Us C", BPB Publications.

2.

Hanly J. R. and Koffman E. B., "Problem Solving and Program Design in C", Pearson Education.

3.

Schildt H., "C- The Complete Reference", Tata McGraw-Hill.

4.

Goyal K. K. and Pandey H.M., "Trouble Free C", University Science Press

5.

Gottfried B., "Schaum's Outlines- Programming in C", Tata McGraw-Hill Publications.

6.

Kochan S.G., "Programming in C", Addison-Wesley.

7.

Dey P. and Ghosh M., "Computer Fundamentals and Programming in C", Oxfo

Page 7 Data Table

BMC103 : PRINCIPLES OF MANAGEMENT & COMMUNICATION			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to			
CO 1	Describe primary features, processes and principles of management.		K , K 1 2
CO 2	Explain functions of management in terms of planning, decision making and organizing.		K , K 3 4
CO 3	Illustrate key factors of leadership skill in directing and controlling business resources and processes.		K , K 5 6
CO 4	Exhibit adequate verbal and non-verbal communication skills		K , K 1 3
CO 5	Demonstrate effective discussion, presentation and writing skills.		K , K 3 5
DETAILED SYLLABUS			3-0-0

BMC103 : PRINCIPLES OF MANAGEMENT & COMMUNICATION			
Unit	Topic		Proposed Lecture
I	Management: Need, Scope, Meaning and Definition. The process of Management, Development of Management thought F.W. Taylor and Henry Fayol, Harthorne Studies, Qualities of an Efficient Management.		08
II	Planning & Organizing: Need, Scope and Importance of Planning, Steps in planning, Decision making model. Organizing need and Importance, Organizational Design, Organizational structure, centralization and Decentralization, Delegation.		08
III	Directing & Controlling: Motivation—Meaning, Importance, need. Theories of Motivation, Leadership—meaning, need and importance, leadership style, Qualities of effective leader, principles of directing, Basic control process, Different control Techniques.		08
IV	Introduction to Communication: What is Communication, Levels of communication, Barriers to communication, Process of Communication, Non-verbal Communication, The flow of Communication: Downward, Upward, Lateral or Horizontal (Peer group) Communication, Technology Enabled communication, Impact of Technology, Selection of appropriate communication Technology, Importance of Technical communication.		08
V	Business letters: Sales & Credit letters; Claim and Adjustment Letters; Job application and Resumes. Reports: Types; Structure, Style & Writing of Reports. Technical Proposal: Parts; Types; Writing of Proposal; Significance. Nuances of Delivery; Body Language; Dimensions of Speech: Syllable; Accent; Pitch; Rhythm; Intonation; Paralinguistic features of voice; Communication skills, Presentation strategies, Group Discussion; Interview skills; Workshop; Conference; Seminars.		08

BMC103 : PRINCIPLES OF MANAGEMENT & COMMUNICATION			
Suggested Readings: 1. Tripathi P.C. and Reddy P.N., "Principles of Management", McGraw Hill Education. 2. Gupta C. B., "Management Principles and Practice", Sultan Chand & Sons. 3. Chhabra T.N., "Business Communication", Sun India Publication. 4. Arora V.N. and Chandra L., "Improve Your Writing", Oxford University Press. 5. Rani M. and Verma S., "Technical Communication: A Practical Approach", Acme Learning. 6. Raman M. and Sharma S., "Technical Communication- Principles and Practices", Oxford			

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Page 7

BMC103 : PRINCIPLES OF MANAGEMENT & COMMUNICATION

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to

CO 1

Describe primary features, processes and principles of management.

K1, K2

CO 2

Explain functions of management in terms of planning, decision making and organizing.

K3, K4

CO 3

Illustrate key factors of leadership skill in directing and controlling business resources and processes.

K5, K6

CO 4

Exhibit adequate verbal and non-verbal communication skills

K1, K3

CO 5

Demonstrate effective discussion, presentation and writing skills.

K3, K5

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Management: Need, Scope, Meaning and Definition. The process of Management, Development of Management thought F.W. Taylor and Henry Fayol, Harthorne Studies, Qualities of an Efficient Management.

08

II

Planning & Organizing: Need, Scope and Importance of Planning, Steps in planning, Decision making model. Organizing need and Importance,

Organizational

Design,

Organizational

structure,

centralization and Decentralization, Delegation.

08

III

Directing & Controlling: Motivation—Meaning, Importance, need.

Theories of Motivation, Leadership—meaning, need and importance, leadership style, Qualities of effective leader, principles of directing,

Basic control process, Different control Techniques.

08

IV

Introduction to Co

Page 8 Data Table

BMC104 : DISCRETE MATHEMATICS			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to			

BMC104 : DISCRETE MATHEMATICS			
CO 1	Use mathematical and logical notation to define and formally reason about basic discrete structures such as Sets, Relations and Functions		K , K 1 2
CO 2	Apply mathematical arguments using logical connectives and quantifiers to check the validity of an argument through truth tables and propositional and predicate logic		K2, K3
CO 3	Identify and prove properties of Algebraic Structures like Groups, Rings and Fields		K , K 3 4
CO 4	Formulate and solve recurrences and recursive functions		K , K 3 4
CO 5	Apply the concept of combinatorics to solve basic problems in discrete mathematics		K , K 1 3
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Set Theory: Introduction, Size of sets and Cardinals, Venn diagrams, Combination of sets, Multisets, Ordered pairs and Set Identities. Relation: Definition, Operations on relations, Composite relations, Properties of relations, Equality of relations, Partial order relation. Functions: Definition, Classification of functions, Operations on functions, Recursively defined functions.		08
II	Posets, Hasse Diagram and Lattices: Introduction, Partial ordered sets, Combination of Partial ordered sets, Hasse diagram, Introduction of lattices, Properties of lattices – Bounded, Complemented, Modular and Complete lattice. Boolean Algebra: Introduction, Axioms and Theorems of Boolean algebra, Boolean functions. Simplification of Boolean functions, Karnaugh maps, Logic gates.		08
III	Propositional: Propositions, Truth tables, Tautology, Contradiction, Algebra of Propositions, Theory of Inference and Natural Detection. Predicate Logic: Theory of Predicates, First order predicate, Predicate formulas, Quantifiers, Inference theory of predicate logic.		08

BMC104 : DISCRETE MATHEMATICS			
IV	Algebraic Structures: Introduction to algebraic Structures and properties. Types of algebraic structures: Semi group, Monoid, Group, Abelian group and Properties of group. Subgroup, Cyclic group, Cosets, Permutation groups, Homomorphism and Isomorphism of groups. Rings and Fields: Definition and elementary properties of Rings and Fields.		08
V	Natural Numbers: Introduction, Pano's axioms, Mathematical Induction, Strong Induction and Induction with Nonzero Base cases. Recurrence Relation & Generating functions: Introduction and properties of Generating Functions. Simple Recurrence relation with constant coefficients and Linear recurrence relation without constant coefficients. Methods of solving recurrences. Combinatorics: Introduction, Counting techniques and Pigeonhole principle, Polya's Counting theorem.		08

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Page 8

BMC104 : DISCRETE MATHEMATICS

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to

CO 1

Use mathematical and logical notation to define and formally reason about basic discrete structures such as Sets, Relations and Functions

K1, K2

CO 2

Apply mathematical arguments using logical connectives and quantifiers to check the validity of an argument through truth tables and propositional and predicate logic

K2, K3

CO 3

Identify and prove properties of Algebraic Structures like Groups, Rings and Fields

K3, K4

CO 4

Formulate and solve recurrences and recursive functions

K3, K4

CO 5

Apply the concept of combinatorics to solve basic problems in discrete mathematics

K1, K3

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Set Theory: Introduction, Size of sets and Cardinals, Venn diagrams, Combination of sets, Multisets, Ordered pairs and Set Identities.

Relation: Definition, Operations on relations, Composite relations, Properties of relations, Equality of relations, Partial order relation.

Functions: Definition, Classification of functions, Operations on functions, Recursively defined functions.

08

II

Posets, Hasse Diagram and Lattices: Introduction, Partial ordered sets, Combination of Partial ordered sets, Hasse diagram, Introduction of lattices, Properties of lattices – Bounded, Complemented, Modular and Complete I

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

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Suggested Readings:

1. Rosen K. H., "Discrete Mathematics and Its Applications", McGraw Hill.
2. Kolman B., Busby R.C and Ross S.C., "Discrete Mathematics Structures", Prentice Hall.
3. Girmaldi R.P, "Discrete and Combinatorial Mathematics", Addison Wesley.
4. Singh Y.N., "Discrete Mathematical Structures", Wiley- India.
5. Sarkar S., "A Textbook of Discrete Mathematics", S. Chand & Company PVT. LTD.
6. Krishnamurthy V., "Combinatorics Theory & Application", East-West Press Pvt. Ltd.

7. Liptschutz S. and Lipson M.C., "Discrete Mathematics", McGraw Hill.

8. Trembely J.P. and Manohar R., "Discrete Mathematical Structure with application to Computer Science", McGraw Hill.

Page 10 Data Table

BMC105 : COMPUTER ORGANIZATION & ARCHITECTURE			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to			
CO 1	Describe functional units of digital system and explain how arithmetic and logical operations are performed by computers		K , K 2 3
CO 2	Describe the operations of control unit and write sequence of instructions for carrying out simple operation using various addressing modes.		K , K 2 4
CO 3	Design various types of memory and its organization.		K 3
CO 4	Describe the various modes in which IO devices communicate with CPU and memory.		K , K 2 3
CO 5	List the criteria for classification of parallel computer and describe various architectural schemes.		K , K 1 2
DETAILED SYLLABUS			
Unit	Topic		Proposed Lecture
I	Introduction: Functional units of digital system and their interconnections, buses, bus architecture, types of buses and bus arbitration. Register, bus and memory transfer. Processor organization: general registers organization, stack organization and addressing modes.		08
II	Arithmetic and logic unit: Look ahead carries adders. Multiplication: Signed operand multiplication, Booths algorithm and array multiplier. Division and logic operations. Floating point arithmetic operation, Arithmetic & logic unit design. IEEE Standard for Floating Point Numbers.		08

BMC105 : COMPUTER ORGANIZATION & ARCHITECTURE			
III	Control Unit: Instruction types, formats, instruction cycles and sub cycles (fetch and execute etc.), micro-operations, execution of a complete instruction. Program Control, Reduced Instruction Set Computer, Pipelining. Hardwire and micro programmed control: micro-program sequencing, concept of horizontal and vertical microprogramming.		08
IV	Memory: Basic concept and hierarchy, semiconductor RAM memories, 2D & 2 1/2D memory organization. ROM memories. Cache memories: concept and design issues & performance, address mapping and replacement Auxiliary memories: magnetic disk, magnetic tape and optical disks Virtual memory: concept implementation.		08
V	Input / Output: Peripheral devices, I/O interface, I/O ports, Interrupts: interrupt hardware, types of interrupts and exceptions. Modes of Data Transfer: Programmed I/O, interrupt initiated I/O and Direct Memory Access., I/O channels and processors. Serial Communication: Synchronous & asynchronous communication, standard communication interfaces.		08

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Page 10

BMC105 : COMPUTER ORGANIZATION & ARCHITECTURE

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to

CO 1

Describe functional units of digital system and explain how arithmetic and logical operations are performed by computers

K2, K3

CO 2

Describe the operations of control unit and write sequence of instructions for carrying out simple operation using various addressing modes.

K2, K4

CO 3

Design various types of memory and its organization.

K3

CO 4

Describe the various modes in which IO devices communicate with CPU and memory.

K2, K3

CO 5

List the criteria for classification of parallel computer and describe various architectural schemes.

K1, K2

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Introduction:

Functional

units

of

digital

system

and

their

interconnections, buses, bus architecture, types of buses and bus arbitration. Register, bus and memory transfer.

Processor

organization:

general

registers organization,

stack

organization and addressing modes.

08

II

Arithmetic and logic unit: Look ahead carries adders. Multiplication:
Signed operand multiplication, Booths algorithm and array multiplier.
Division and logic operations. Floating point arithmetic operation,
Arithmetic & logic unit design. IEEE Standard for Floating Point
Numbers.

08

III

Control Unit: Instruction types, formats, instru

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Suggested Readings:

1. Mano M. M., "Computer System Architecture", PHI.
2. Hamacher C., Vranesic Z. and Zaky S., "Computer Organization", McGraw-Hill.
3. Hayes J. P., "Computer Architecture and Organization", Tata McGraw Hill.
4. Stallings W., "Computer Organization and Architecture-Designing for Performance", Pearson Education.
5. Parahami B., "Computer Architecture", Oxford University Press.
6. Patterson D. A. and Hennessy J. L., "Computer Architecture-A Quantitative Approach", Elsevier Pub.
7. Tannenbaum A.S., "Structured Computer Organization", PHI.

Page 12 Data Table

BMC151 : PROBLEM SOLVING USING C LAB		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO1	Write, compile, debug and execute programs in a C programming environment.	K 3
CO2	Write programs that incorporate use of variables, operators and expressions along with data types.	K 3
CO3	Write programs for solving problems involving use of decision control structures and loops.	K 3
CO4	Write programs that involve the use of arrays, structures and user defined functions.	K 3
CO5	Write programs using graphics and file handling operations.	K 3

BMC151 : PROBLEM SOLVING USING C LAB		
1. Program to implement conditional statements in C language. 2. Program to implement switch-case statement in C language 3. Program to implement looping constructs in C language. 4. Program to perform basic input-output operations in C language. 5. Program to implement user defined functions in C language. 6. Program to implement recursive functions in C language. 7. Program to implement one-dimensional arrays in C language. 8. Program to implement two-dimensional arrays in C language. 9. Progr		

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Page 12

BMC151 : PROBLEM SOLVING USING C LAB

Course Outcome (CO)

Bloom's

Knowledge

Level (KL)

At the end of course, the student will be able to

CO1

Write, compile, debug and execute programs in a C programming environment.

K3

CO2

Write programs that incorporate use of variables, operators and expressions along with data types.

K3

CO3

Write programs for solving problems involving use of decision control structures and loops.

K3

CO4

Write programs that involve the use of arrays, structures and user defined functions.

K3

CO5

Write programs using graphics and file handling operations.

K3

1. Program to implement conditional statements in C language.
2. Program to implement switch-case statement in C language
3. Program to implement looping constructs in C language.
4. Program to perform basic input-output operations in C language.
5. Program to implement user defined functions in C language.
6. Program to implement recursive functions in C language.
7. Program to implement one-dimensional arrays in C language.
8. Program to implement two-dimensional arrays in C language.
9. Program to perform various operations on two-dimensional arrays in C language.
10. Program to implement multi-dimensional arrays in C language.
11. Program to implement string manipulation functions in C language.
12. Program to implement structure in C language.
13. Program to implement union in C

Page 13 Data Table

BMC152 : COMPUTER ORGANIZATION & ARCHITECTURE LAB		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO1	Design and verify combinational circuits (adder, code converter, decoder, multiplexer) using basic gates.	K 6
CO2	Design and verify various flip-flops.	K 3
CO3	Design I/O system and ALU.	K 3
CO4	Demonstrate combinational circuit using simulator	K 2
1. Implement HALF ADDER, FULL ADDER using basic logic gates. 2. Implement Binary -to -Gray, Gray -to -Binary code conversions. 3. Implement 3-8 line DECODER. Implementing 4x1 and 8x1 MULTIPLEXERS. 4. Verify the excitation tables of various FLIP-FLOPS. 5. Design of an 8-bit Input/ Output system with four 8-bit Internal Registers. 6. Design of an 8-bit ARITHMETIC LOGIC UNIT. 7. Design the data path of a computer from its register transfer language description. 8. Design the control unit of a compu		

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BMC152 : COMPUTER ORGANIZATION & ARCHITECTURE LAB

Course Outcome (CO)

Bloom's

Knowledge

Level (KL)

At the end of course, the student will be able to

CO1

Design and verify combinational circuits (adder, code converter, decoder, multiplexer) using basic gates.

K6

CO2

Design and verify various flip-flops.

K3

CO3

Design I/O system and ALU.

K3

CO4

Demonstrate combinational circuit using simulator

K2

1. Implement HALF ADDER, FULL ADDER using basic logic gates.
2. Implement Binary -to -Gray, Gray -to -Binary code conversions.
3. Implement 3-8 line DECODER. Implementing 4x1 and 8x1 MULTIPLEXERS.
4. Verify the excitation tables of various FLIP-FLOPS.
5. Design of an 8-bit Input/ Output system with four 8-bit Internal Registers.
6. Design of an 8-bit ARITHMETIC LOGIC UNIT.
7. Design the data path of a computer from its register transfer language description.
8. Design the control unit of a computer using either hardwiring or microprogramming based on its register transfer language description.
9. Implement a simple instruction set computer with a control unit and a data path.

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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BMC153 : PROFESSIONAL COMMUNICATION LAB		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO1	Develop the ability to work as a team member as an integral activity in the workplace.	K 3
CO2	Increase confidence in their ability to read, comprehend, organize, and retain written information. Improve reading fluency.	K 4

BMC153 : PROFESSIONAL COMMUNICATION LAB		
CO3	Write coherent speech outlines that demonstrate their ability to use organizational formats with a specific purpose; Deliver effective speeches that are consistent with and appropriate for the audience and purpose.	K ,K 5 6
CO4	Develop proper listening skills; articulate and enunciate words and sentences clearly and efficiently.	K 3
CO5	Show confidence and clarity in public speaking projects; be schooled in preparation and research skills for oral presentations.	K 5
1. Group Discussion: participating in group discussions- understanding group dynamics. 2. GD strategies-activities to improve GD skills. Practical based on Accurate and Current Grammatical Patterns. 3. Interview Etiquette-dress code, body language attending job interview – Telephone/Skype interview one to one interview &Panel interview. 4. Communication Skills for Seminars/Conferences/Workshops with emphasis on Paralinguistic/ Kinesics, practicing word stress, rhythm in sentences, weak forms, in		

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Page 14

BMC153 : PROFESSIONAL COMMUNICATION LAB

Course Outcome (CO)

Bloom's

Knowledge

Level (KL)

At the end of course, the student will be able to

CO1

Develop the ability to work as a team member as an integral activity in the workplace.

K3

CO2

Increase confidence in their ability to read, comprehend, organize, and retain written information. Improve reading fluency.

K4

CO3

Write coherent speech outlines that demonstrate their ability to use organizational formats with a specific purpose; Deliver effective

speeches that are consistent with and appropriate for the audience and purpose.

K5,K6

CO4

Develop proper listening skills; articulate and enunciate words and sentences clearly and efficiently.

K3

CO5

Show confidence and clarity in public speaking projects; be schooled in preparation and research skills for oral presentations.

K5

1. Group Discussion: participating in group discussions- understanding group dynamics.

2. GD strategies-activities to improve GD skills. Practical based on Accurate and Current Grammatical Patterns.

3. Interview Etiquette-dress code, body language attending job interview – Telephone/Skype interview one to one interview & Panel interview.

4. Communication Skills for Seminars/Conferences/Workshops with emphasis on Paralinguistic/ Kinesics, practicing word stress, rhythm in sentences, weak forms, intonation.

5. Oral Presentation Skills for Tec

Page 15 Data Table

BMC106 : CYBER SECURITY			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to			
CO 1	Identify and analyze nature & inherent difficulties in the security of the Information System.		K 3
CO 2	Analyze various threats and attacks, corresponding counter measures and various vulnerability assessment and security techniques in an organization.		K 3
CO 3	Applications of cyber based policies and use of IPR and patent law for software-based design. Define E-commerce types and threats to E-commerce.		K , K 1 2
CO 4	Explain concepts and theories of networking and apply them to various situations, classifying networks, analyzing performance.		K 2
DETAILED SYLLABUS			2-0-0

BMC106 : CYBER SECURITY			
Unit	Topic		Proposed Lecture
I	Introduction- Introduction to Information Systems, Types of Information Systems, Development of Information Systems, Introduction to Information Security and CIA triad, Need for Information Security, Threats to Information Systems, Information Assurance and Security Risk Analysis, Cyber Security.		08
II	Application Security- (Database, E-mail and Internet), Data Security Considerations (Backups, Archival Storage and Disposal of Data), Security Technology (Firewall, VPNs, Intrusion Detection System), Access Control. Security Threats -Viruses, Worms, Trojan Horse, Bombs, Trapdoors, Spoofs, E-mail Viruses, Macro Viruses, Malicious Software, Network and Denial of Services Attack.		08
III	Introduction to E-Commerce- Threats to E-Commerce, Electronic Payment System, e-Cash, Credit/Debit Cards. Digital Signature, Cryptography Developing Secure Information Systems, Application Development Security, Information Security Governance & Risk Management, Security Architecture & Design Security Issues in Hardware, Data Storage & Downloadable Devices, Physical Security of IT Assets - Access Control, CCTV, Backup Security Measures.		08
IV	Security Policies- Why policies should be developed, Policy Review Process, Publication and Notification Requirement of policies, Types of policies – WWW policies, Email Security policies, Corporate Policies, Sample Security Policies. Case Study – Corporate Security		08
V	Information Security Standards-ISO, IT Act, Copyright Act, IPR. Cyber Crimes, Cyber Laws in India; IT Act 2000 Provisions, Intellectual Property Law, Copy Right Law, Semiconductor Law and Patent Law, Software Piracy and Software License.		08

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Page 15

BMC106 : CYBER SECURITY

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to

CO 1

Identify and analyze nature & inherent difficulties in the security of the Information System.

K3

CO 2

Analyze various threats and attacks, corresponding counter measures and various vulnerability assessment and security techniques in an organization.

K3

CO 3

Applications of cyber based policies and use of IPR and patent law for software-based design. Define E-commerce types and threats to E-commerce.

K1, K2

CO 4

Explain concepts and theories of networking and apply them to various situations, classifying networks, analyzing performance.

K2

DETAILED SYLLABUS

2-0-0

Unit

Topic

Proposed

Lecture

I

Introduction- Introduction to Information Systems, Types of Information Systems, Development of

Information

Systems,

Introduction to Information Security and CIA triad, Need for Information Security, Threats to Information Systems, Information Assurance and Security Risk Analysis, Cyber Security.

08

II

Application Security- (Database, E-mail and Internet), Data Security Considerations (Backups, Archival Storage and Disposal of Data), Security Technology (Firewall, VPNs, Intrusion Detection System), Access Control. Security Threats -Viruses, Worms, Trojan Horse, Bombs, Trapdoors, Spoofs, E-mail Viruses, Macro Viruses, Malicious Softwa

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Suggested Readings:

1. Pfleeger C. P. and Pfleeger S. L., "Analyzing Computer Security", Pearson Education India.
2. Pachghare V. K., "Cryptography and information Security", PHI Learning Private Limited.
3. Goyal K. K. and Garg A. "Cyber Security", University Science Press.
4. Whitman M. E. and Mattord H. J, "Principle of Information Security" Cengage.
5. Chapple M. and Seidl D., "Cyberwarfare: Information operations in a connected world" Jones & Bartlett Learning.
6. Goyal K. K., Garg A. and Singhal S., "Cyber Security & Data Privacy", HP Hamilton.
7. Schou C.D. and Shoemaker D. P., "Information Assurance for the Enterprise", Tata McGraw Hill.

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

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Syllabus

MCA 1st Year

IInd Semester

Page 18 Data Table

BMC201 : WEB TECHNOLOGY			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to:			

BMC201 : WEB TECHNOLOGY			
CO 1	Apply the knowledge of HTML and CSS to develop web application and analyze the insights of internet programming to implement complete application over the web.		K3, K6
CO 2	Understand, analyze and apply the role of JavaScript in the workings of the web and web applications.		K2, K3
CO 3	Understand, analyze and build dynamic web applications using servlet and JSP.		K , K 2 3
CO 4	Develop Spring-based Java applications using Java configuration, XML configuration, annotation-based configuration, beans and their scopes, and properties.		K2, K4, K6
CO 5	Develop web application using Spring Boot and RESTful Web Services		K , K 3 6
DETAILED SYLLABUS			3-1-0
Unit	Topic		Proposed Lecture
I	Web Page Designing: Introduction and Web Development Strategies, History of Web and Internet, Protocols Governing Web, HTML-Introduction, HTML Tags, HTML-Grouping Using Div & Span, HTML-Lists, HTML-Images, HTML- Hyperlink, HTML- Table, HTML-Iframe, HTML-Form, Introduction of CSS, CSS Syntax, External Style Sheet using < link >, Multiple Style Sheets, Value Lengths and Percentages, CSS-Selectors, CSS-Box Model, Floats, Clear, Introduction to Bootstrap.		08
II	Scripting: Introduction to JavaScript, Creating Variables in JavaScript, Creating Functions in JavaScript, UI Events, Returning Data from Functions, Working with Conditions, looping in JavaScript, Block Scope Variables, Working with Objects, Creating Object using Object Literals, Manipulating DOM Elements with JavaScript		08

BMC201 : WEB TECHNOLOGY			
III	Web Application development using JSP & Servlets: Servlet Overview and Architecture, Interface Servlet and the Servlet Life Cycle, Handling HTTP get Requests, Handling HTTP post Requests, Redirecting Requests to Other Resources, Session Tracking, Cookies, Session Tracking with Http Session. Java Server Pages (JSP): Introduction, Java Server Pages Overview, A First Java Server Page Example, Implicit Objects, Scripting, Standard Actions, Directives, Custom Tag Libraries.		08

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MCA (MASTER OF COMPUTER APPLICATIONS)

FIRST YEAR SYLLABUS

SEMESTER-II

BMC201 : WEB TECHNOLOGY

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to:

CO 1

Apply the knowledge of HTML and CSS to develop web application and analyze the insights of internet programming to implement complete application over the web.

K3, K6

CO 2

Understand, analyze and apply the role of JavaScript in the workings of the web and web applications.

K2, K3

CO 3

Understand, analyze and build dynamic web applications using servlet and JSP.

K2, K3

CO 4

Develop Spring-based Java applications using Java configuration, XML configuration, annotation-based configuration, beans and their

scopes, and properties.

K2, K4, K6

CO 5

Develop web application using Spring Boot and RESTful Web

Services

K3, K6

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Web Page Designing: Introduction and Web Development

Strategies, History of Web and Internet, Protocols Governing Web,

HTML-Introduction, HTML Tags, HTML-Grouping Using Div &

Span, HTML-Lists, HTML-Images, HTML- Hyperlink, HTML-

Table, HTML-Iframe, HTML-Form, Introduction of CSS, CSS

Syntax, External Style Sheet using < link >, Multiple Style Sheets,

Value Lengths and Percentages, CSS-Selectors, CSS-Box Model,

Floats, Clear, Introduction to Bootstrap.

08

II

Scripting: Introduction

Page 19 Data Table

IV	Spring: Spring Core Basics-Spring Dependency Injection concepts, Introduction to Design patterns, Factory Design Pattern, Strategy Design pattern, Spring Inversion of Control, AOP, Bean Scopes- Singleton, Prototype, Request, Session, Application, WebSocket, Auto wiring, Annotations, Life Cycle Call backs, Bean Configuration styles	08
V	Spring Boot: Spring Boot- Spring Boot Configuration, Spring Boot Annotations, Spring Boot Actuator, Spring Boot Build Systems, Spring Boot Code Structure, Spring Boot Runners, Logger, BUILDING RESTFUL WEB SERVICES, Rest Controller, Request Mapping, Request Body, Path Variable, Request Parameter, GET, POST, PUT, DELETE APIs, Build Web Applications	08

IV	Spring: Spring Core Basics-Spring Dependency Injection concepts, Introduction to Design patterns, Factory Design Pattern, Strategy Design pattern, Spring Inversion of Control, AOP, Bean Scopes- Singleton, Prototype, Request, Session, Application, WebSocket, Auto wiring, Annotations, Life Cycle Call backs, Bean Configuration styles	08
Suggested Readings: 1. Burdman J., "Collaborative Web Development – Strategies and Best practices for Web Teams", Addison-Wesley. 2. Xavier C, "Web Technology & Design", New Age international Publishers. 3. Bayross I., "Web Technologies", BPB Publications. 4. Schieldth H., "The Complete Reference – HTML & CSS", McGraw Hill. 5. Bergsten H., "Java Server Pages", SPD O' Reilly. 6. Walls C., "Spring Boot in Action", Manning Publications. 7. Bakliwal S., "Hands-on Application Development using Spring		

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IV

Spring: Spring Core Basics-Spring Dependency Injection concepts, Introduction to Design patterns, Factory Design Pattern, Strategy Design pattern, Spring Inversion of Control, AOP, Bean Scopes- Singleton, Prototype, Request, Session, Application, WebSocket, Auto wiring, Annotations, Life Cycle Call backs, Bean Configuration styles

08

V

Spring Boot: Spring Boot- Spring Boot Configuration, Spring Boot Annotations, Spring Boot Actuator, Spring Boot Build Systems, Spring Boot Code Structure, Spring Boot Runners, Logger, BUILDING RESTFUL WEB SERVICES, Rest Controller, Request Mapping, Request Body, Path Variable, Request Parameter, GET, POST, PUT, DELETE APIs, Build Web Applications

08

Suggested Readings:

1.

Burdman J., "Collaborative Web Development – Strategies and Best practices for Web Teams", Addison-Wesley.

2.

Xavier C., "Web Technology & Design", New Age international Publishers.

3.

Bayross I., "Web Technologies", BPB Publications.

4.

Schieldth H., "The Complete Reference – HTML & CSS", McGraw Hill.

5.

Bergsten H., "Java Server Pages", SPD O' Reilly.

6.

Walls C., "Spring Boot in Action", Manning Publications.

7.

Bakliwal S., "Hands-on Application Development using Spring Boot", BPB Publications.

Page 20 Data Table

BMC202 : OBJECT ORIENTED PROGRAMMING			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to			
CO 1	List the significance and key features of object-oriented programming and modeling using UML		K 4
CO 2	Construct basic structural, behavioral and architectural models using object oriented software engineering approach.		K 6
CO 3	Integrate object-oriented modeling techniques for analysis and design of a system.		K4, K5
CO 4	Use the basic features of data abstraction and encapsulation in C++ programs.		K 4
CO 5	Use the advanced features such as Inheritance, polymorphism and virtual function in C++ programs.		K , K 3 4
DETAILED SYLLABUS			3-1-0
Unit	Topic		Proposed Lecture

BMC202 : OBJECT ORIENTED PROGRAMMING			
I	Introduction: Object Oriented Programming: objects, classes, Abstraction, Encapsulation, Inheritance, Polymorphism, OOP in Java, Characteristics of Java, The Java Environment, Java Source File Structure, and Compilation. Fundamental Programming Structures in Java: Defining classes in Java, constructors, methods, access specifies, static members, Comments, Data Types, Variables, Operators, Control Flow, Arrays.		08
II	Inheritance, Interfaces, and Packages: Inheritance: Super classes, sub classes, Protected members, constructors in sub classes, Object class, abstract classes and methods. Interfaces: defining an interface, implementing interface, differences between classes and interfaces and extending interfaces, Object cloning, inner classes. Packages: Defining Package, CLASSPATH Setting for Packages, Making JAR Files for Library Packages, Import and Static Import Naming Convention For Packages, Networking ja		08
III	Exception Handling, I/O: Exceptions: exception hierarchy, throwing and catching exceptions, built-in exceptions, creating own exceptions, Stack Trace Elements. Input / Output Basics: Byte streams and Character streams, Reading and Writing, Console Reading and Writing Files.		08
IV	Multithreading and Generic Programming: Differences between multi-threading and multitasking, thread life cycle, creating threads, synchronizing threads, Inter-thread communication, daemon threads, thread groups. Generic Programming: Generic classes, generic methods, Bounded Types: Restrictions and Limitations.		08

BMC202 : OBJECT ORIENTED PROGRAMMING			
V	Event Driven Programming: Graphics programming: Frame, Components, working with 2D shapes, Using colors, fonts, and images. Basics of event handling: event handlers, adapter classes, actions, mouse events, AWT event hierarchy. Introduction to Swing: layout management, Swing Components: Text Fields, Text Areas, Buttons, Check Boxes, Radio Buttons, Lists, choices, Scrollbars, Windows Menus and Dialog Boxes.		08

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Page 20

BMC202 : OBJECT ORIENTED PROGRAMMING

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to

CO 1

List the significance and key features of object-oriented programming and modeling using UML

K4

CO 2

Construct basic structural, behavioral and architectural models using object oriented software engineering approach.

K6

CO 3

Integrate object-oriented modeling techniques for analysis and design of a system.

K4, K5

CO 4

Use the basic features of data abstraction and encapsulation in C++ programs.

K4

CO 5

Use the advanced features such as Inheritance, polymorphism and virtual function in C++ programs.

K3, K4

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Introduction: Object Oriented Programming: objects, classes, Abstraction, Encapsulation, Inheritance, Polymorphism, OOP in Java, Characteristics of Java, The Java Environment, Java Source File Structure, and Compilation. Fundamental Programming Structures in Java: Defining classes in Java, constructors, methods, access specifiers, static members, Comments, Data Types, Variables, Operators, Control Flow, Arrays.

08

II

Inheritance, Interfaces, and Packages: Inheritance: Super classes, sub classes, Protected members, constructors in sub classes, Object class, abstract classes and methods. Interfaces: defining an interface, implementing interface, difference

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Page 21

Suggested Readings:

1.

Schildt H., "Java - The complete Reference", McGraw Hill Education.

2.

Horstmann C. S. and Cornell G., "Core Java Volume I Fundamentals", Prentice Hall.

3.

Holzner S., "Java - Black Book", Dreamtech.

4.

Balagurusamy E., "Programming in Java", Tata McGraw Hill.

5.

Mughal K., "A Programmer's Guide to Java SE 8 Oracle Certified Associate (OCA)", Addison-Wesley.

Page 22 Data Table

BMC203 : OPERATING SYSTEMS			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to			
CO 1	Explain main components, services, types and structure of Operating Systems.		K 2
CO 2	Apply the various algorithms and techniques to handle the various concurrency control issues.		K 3
CO 3	Compare and apply various CPU scheduling algorithms for process execution.		K 2
CO 4	Identify occurrence of deadlock and describe ways to handle it.		K 3
CO 5	Explain and apply various memory, I/O and disk management techniques.		K 5
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Introduction: Operating System Structure- Layered structure, System Components, Operating system functions, Classification of Operating systems- Batch, Interactive, Time-sharing, Real-Time System, Multiprocessor Systems, Multiuser Systems, Multi process Systems, Multithreaded Systems, Operating System services, Reentrant Kernels, Monolithic and Microkernel Systems.		08
II	Concurrent Processes: Process Concept, Principle of Concurrency, Producer / Consumer Problem, Mutual Exclusion, Critical Section Problem, Dekker's solution, Peterson's solution, Semaphores, Test and Set operation, Classical Problem in Concurrency- Dining Philosopher Problem, Sleeping Barber Problem, Inter Process Communication models and Schemes, Process generation.		08

BMC203 : OPERATING SYSTEMS			
III	CPU Scheduling: Scheduling Concepts, Performance Criteria, Process States, Process Transition Diagram, Schedulers, Process Control Block (PCB), Process address space, Process identification information, Threads and their management, Scheduling Algorithms, Multiprocessor Scheduling. Deadlock: System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock.		08
IV	Memory Management: Basic bare machine, Resident monitor, Multiprogramming with fixed partitions, Multiprogramming with variable partitions, Protection schemes, Paging, Segmentation, Paged segmentation, Virtual memory concepts, Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing, Cache memory organization, Locality of reference.		08
V	I/O Management and Disk Scheduling: I/O devices, and I/O subsystems, I/O buffering, Disk storage and disk scheduling, RAID. File System: File concept, File organization and access mechanism, File directories, and File sharing, File system implementation issues, File system protection and security.		08

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

BMC203 : OPERATING SYSTEMS

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to

CO 1

Explain main components, services, types and structure of Operating Systems.

K2

CO 2

Apply the various algorithms and techniques to handle the various concurrency control issues.

K3

CO 3

Compare and apply various CPU scheduling algorithms for process execution.

K2

CO 4

Identify occurrence of deadlock and describe ways to handle it.

K3

CO 5

Explain and apply various memory, I/O and disk management techniques.

K5

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Introduction: Operating System Structure- Layered structure, System Components, Operating system functions, Classification of Operating systems- Batch, Interactive, Time-sharing, Real-Time System, Multiprocessor Systems, Multiuser Systems, Multi process Systems, Multithreaded Systems, Operating System services, Reentrant Kernels, Monolithic and Microkernel Systems.

08

II

Concurrent Processes: Process Concept, Principle of Concurrency, Producer / Consumer Problem, Mutual Exclusion, Critical Section Problem, Dekker's solution, Peterson's solution, Semaphores, Test and Set operation, Classical Problem in Concurrency- Dining Philosopher Problem, Sleeping Barber Problem, Inter Process Communication models and Schemes, Process generation.

08

III

CPU Scheduling: Scheduling C

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Suggested Readings:

1. Silberschatz A., Galvin P. B. and Gagne G., "Operating Systems Concepts", Wiley Pub.
2. Halder S. and Arvind A. A "Operating Systems", Pearson Education.
3. Dietel H. M, " An Introduction to Operating System", Pearson Education.
4. Stallings W., "Operating Systems: Internals and Design Principles", Pearson Education.
5. Harris J.A., "Operating Systems (Schaum's Outlines)", McGraw Hill Education.

Page 24 Data Table

BMC204 : DATABASE MANAGEMENT SYSTEMS			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to			
CO 1	Describe the features of a database system and its application and compare various types of data models.		K 2
CO 2	Construct an ER Model for a given problem and transform it into a relation database schema.		K , K 5 6
CO 3	Formulate solution to a query problem using SQL Commands, relational algebra, tuple calculus and domain calculus.		K , K 5 6
CO 4	Explain the need of normalization and normalize a given relation to the desired normal form.		K , K 2 3
CO 5	Explain different approaches of transaction processing and concurrency control.		K 2
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Introduction: Overview, Database System vs File System, Database System Concept and Architecture, Data Model Schema and Instances, Data Independence and Database Language and Interfaces, Data Definitions Language, DML, Overall Database Structure. Data Modeling Using the Entity Relationship Model: ER Model Concepts, Notation for ER Diagram, Mapping Constraints, Keys, Concepts of Super Key, Candidate Key, Primary Key, Generalization, Aggregation, Reduction of an ER Diagrams to Tables, Extended ER		08

BMC204 : DATABASE MANAGEMENT SYSTEMS			
II	Relational data Model and Language: Relational Data Model Concepts, Integrity Constraints, Entity Integrity, Referential Integrity, Keys Constraints, Domain Constraints, Relational Algebra, Relational Calculus, Tuple and Domain Calculus. Introduction to SQL: Characteristics of SQL, Advantage of SQL. SQL Data Type and Literals. Types of SQL Commands. SQL Operators and their Procedure. Tables, Views and Indexes. Queries and Sub Queries. Aggregate Functions. Insert, Update and Delete Operations, Jo		08
III	Data Base Design & Normalization: Functional dependencies, normal forms, first, second, third normal forms, BCNF, inclusion dependence, loss less join decompositions, normalization using FD, MVD, and JDs, alternative approaches to database design		08
IV	Transaction Processing Concept: Transaction System, Testing of Serializability, Serializability of Schedules, Conflict & View Serializable Schedule, Recoverability, Recovery from Transaction Failures, Log Based Recovery, Checkpoints, Deadlock Handling. Distributed Database: Distributed Data Storage, Concurrency Control, Directory System		08
V	Concurrency Control Techniques: Concurrency Control, Locking Techniques for Concurrency Control, Time Stamping Protocols for Concurrency Control, Validation Based Protocol, Multiple Granularity, Multi Version Schemes, Recovery with Concurrent		08

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

BMC204 : DATABASE MANAGEMENT SYSTEMS

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to

CO 1

Describe the features of a database system and its application and compare various types of data models.

K2

CO 2

Construct an ER Model for a given problem and transform it into a relation database schema.

K5, K6

CO 3

Formulate solution to a query problem using SQL Commands, relational algebra, tuple calculus and domain calculus.

K5, K6

CO 4

Explain the need of normalization and normalize a given relation to the desired normal form.

K2, K3

CO 5

Explain different approaches of transaction processing and concurrency control.

K2

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Introduction: Overview, Database System vs File System, Database System Concept and Architecture, Data Model Schema and Instances, Data Independence and Database Language and Interfaces, Data Definitions Language, DML, Overall Database Structure. Data Modeling Using the Entity Relationship Model: ER Model Concepts, Notation for ER Diagram, Mapping Constraints, Keys, Concepts of Super Key, Candidate Key, Primary Key, Generalization, Aggregation, Reduction of an ER Diagrams to Tables, Extended ER Model, Relationship of Higher Degree.

08

II

Relational data Model and Language: Relational Data Model

Concepts, Integrity Constraints, Entity Integrity,

Page 25 Data Table

	Transaction, Case Study of Oracle.	
Suggested Readings: 1. Silbertschatz A., Korth H. and Sudarshan S., "Database Concepts", McGraw Hill. 2. Date C. J., "An Introduction to Database Systems", Addison Wesley. 3. Elmasri R. and Navathe S., "Fundamentals of Database Systems", Pearson Education. 4. O'Neil P., "Databases", Elsevier Publications. 5. Ramakrishnan R. Gehrke J., "Database Management Systems", McGraw Hill. 6. Leon A. and Leon M., "Database Management Systems", Vikas Publishing House. 7. Desai B.C., "An Introduction to Data		

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Transaction, Case Study of Oracle.

Suggested Readings:

1. Silbertschatz A., Korth H. and Sudarshan S., "Database Concepts", McGraw Hill.
2. Date C. J., "An Introduction to Database Systems", Addison Wesley.
3. Elmasri R. and Navathe S., "Fundamentals of Database Systems", Pearson Education.
4. O'Neil P., "Databases", Elsevier Publications.
5. Ramakrishnan R. Gehrke J., "Database Management Systems", McGraw Hill.
6. Leon A. and Leon M., "Database Management Systems", Vikas Publishing House.
7. Desai B.C., "An Introduction to Database Systems", Galgotia Publications.
8. Majumdar A. K. and Bhattacharya P., "Database Management System", Tata McGraw Hill.

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BMC205 : DATA STRUCTURES & ANALYSIS OF ALGORITHMS		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO 1	Explain the concept of data structure, abstract data types, algorithms, analysis of algorithms and basic data organization schemes such as arrays and linked lists.	K 2
CO 2	Describe the applications of stacks and queues and implement various operations on them using arrays and linked lists.	K 3
CO 3	Describe the properties of graphs and trees and implement various operations such as searching and traversal on them.	K 3

BMC205 : DATA STRUCTURES & ANALYSIS OF ALGORITHMS		
CO 4	Compare incremental and divide-and-conquer approaches of designing algorithms for problems such as sorting and searching.	K 4
CO 5	Apply and analyze various design approaches such as Divide-and- Conquer, greedy and dynamic for problem solving.	K 4
DETAILED SYLLABUS		4-0-0
Unit	Topic	Proposed Lecture
I	Introduction to data structure: Data, Entity, Information, Difference between Data and Information, Data type , Build in data type, Abstract data type, Definition of data structures, Types of Data Structures: Linear and Non-Linear Data Structure, Introduction to Algorithms: Definition of Algorithms, Difference between algorithm and programs, properties of algorithm, Algorithm Design Techniques, Performance Analysis of Algorithms, Complexity of various code structures, Order of Growth, Asymptotic	08
II	Stacks: Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack in C, Application of stack: Prefix and Postfix Expressions, Evaluation of postfix expression, Iteration and Recursion- Principles of recursion, Tail recursion, Removal of recursion Problem solving using iteration and recursion with examples such as binary search, Fibonacci numbers, and Hanoi towers. Queues: Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array a	08

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BMC205 : DATA STRUCTURES & ANALYSIS OF ALGORITHMS

Course Outcome (CO)

Bloom's

Knowledge

Level (KL)

At the end of course, the student will be able to

CO 1

Explain the concept of data structure, abstract data types, algorithms, analysis of algorithms and basic data organization schemes such as arrays and linked lists.

K2

CO 2

Describe the applications of stacks and queues and implement

various operations on them using arrays and linked lists.

K3

CO 3

Describe the properties of graphs and trees and implement various operations such as searching and traversal on them.

K3

CO 4

Compare incremental and divide-and-conquer approaches of designing algorithms for problems such as sorting and searching.

K4

CO 5

Apply and analyze various design approaches such as Divide-and-Conquer, greedy and dynamic for problem solving.

K4

DETAILED SYLLABUS

4-0-0

Unit

Topic

Proposed

Lecture

I

Introduction to data structure: Data, Entity, Information,
Difference between Data and Information, Data type , Build in data
type, Abstract data type, Definition of data structures, Types of Data
Structures: Linear and Non-Linear Data Structure, Introduction to
Algorithms: Definition of Algorithms, Difference between algorithm
and programs, properties of algorithm, Algorithm Design Techniques,
Performance Analysis of Algorithms, Complexity of various code
structures, Order of Growth, Asymptotic No

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	Collision resolution Techniques used in Hashing.	
III	Sorting: Insertion Sort, Selection Sort, Bubble Sort, Heap Sort, Comparison of Sorting Algorithms, Sorting in Linear Time: Counting Sort and Bucket Sort. Graphs: Terminology used with Graph, Data Structure for Graph Representations: Adjacency Matrices, Adjacency List, Adjacency. Graph Traversal: Depth First Search and Breadth First Search, Connected Component.	08
IV	Trees: Basic terminology used with Tree, Binary Trees, Binary Tree Representation: Array Representation and Pointer (Linked List) Representation, Binary Search Tree, Complete Binary Tree, A Extended Binary Trees, Tree Traversal algorithms: Inorder, Preorder and Postorder, Constructing Binary Tree from given Tree Traversal, Operation of Insertion, Deletion, Searching & Modification of data in Binary Search Tree. Threaded Binary trees, Huffman coding using Binary Tree, AVL Tree and B Tree.	08
V	Divide and Conquer with Examples Such as Merge Sort, Quick Sort, Matrix Multiplication: Strassen's Algorithm Dynamic Programming: Dijkstra Algorithm, Bellman Ford Algorithm, All- pair Shortest Path: Warshal Algorithm, Longest Common Sub-sequence Greedy Programming: Prims and Kruskal algorithm.	08
Suggested Readings: 1. Cormen T. H., Leiserson C. E., Rivest R. L. and Stein C., "Introduction to Algorithms", PHI. 2. Horowitz E., Sahni S. and Rajasekharan S., "Fundamentals of Computer Algorithms", Universities Press. 3. Dave P. H. and Dave H. B., "Design and Analysis of Algorithms", Pearson Education. 4. Lipschuts S., "Theory and Problems of Data Structures Schaum's Series)", Tata McGraw-Hill. 5. Goyal K. K., Sharma S. and Gupta A., "Data Structures and Analysis of Algorithms", HP Hamilton. 6		

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

Collision resolution Techniques used in Hashing.

III

Sorting: Insertion Sort, Selection Sort, Bubble Sort, Heap Sort,

Comparison of Sorting Algorithms, Sorting in Linear Time: Counting Sort and Bucket Sort.

Graphs: Terminology used with Graph, Data Structure for Graph

Representations: Adjacency Matrices, Adjacency List, Adjacency.

Graph Traversal: Depth First Search and Breadth First Search,

Connected Component.

08

IV

Trees: Basic terminology used with Tree, Binary Trees, Binary Tree Representation: Array Representation and Pointer (Linked List) Representation, Binary Search Tree, Complete Binary Tree, A Extended Binary Trees, Tree Traversal algorithms: Inorder, Preorder and Postorder, Constructing Binary Tree from given Tree Traversal, Operation of Insertion, Deletion, Searching & Modification of data in Binary Search Tree. Threaded Binary trees, Huffman coding using Binary Tree, AVL Tree and B Tree.

08

V

Divide and Conquer with Examples Such as Merge Sort, Quick Sort, Matrix Multiplication: Strassen's Algorithm Dynamic Programming: Dijkstra Algorithm, Bellman Ford Algorithm, All- pair Shortest Path: Warshal Algorithm, Longest Common Sub-sequence Greedy Programming: Prims and Kruskal algorithm.

08

Suggested Readings:

1.

Cormen T. H., Leiserson C. E., Rivest R. L. and Stein C., "Introduction to Algorithms", PHI.

2.

Horowitz E., Sahni S. and Rajasekharan S., "Fundament

Page 28 Data Table

BMC251 : WEB TECHNOLOGY LAB		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO1	Design web pages using HTML, DHTML and Cascading Styles sheets.	K 2
CO2	Develop a dynamic web pages using JavaScript.	K 3
CO3	Develop an interactive web applications using JSP.	K 3
CO4	Create web applications using Spring Boot.	K 2

BMC251 : WEB TECHNOLOGY LAB		
1. Create a simple webpage using HTML. 2. Create a HTML page, which has properly aligned paragraphs with image along with it. 3. Write a program to display list of items in different styles. 4. Use frames to Include Images and Videos. 5. Add a Cascading Style sheet for designing the web page. 6. Design a dynamic web page with validation using JavaScript. 7. Write a program using JavaScript to demonstrate the concept of built-in array methods. 8. Write a program using JavaScript to demonstrate th		

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

BMC251 : WEB TECHNOLOGY LAB

Course Outcome (CO)

Bloom's

Knowledge

Level (KL)

At the end of course, the student will be able to

CO1

Design web pages using HTML, DHTML and Cascading Styles sheets.

K2

CO2

Develop a dynamic web pages using JavaScript.

K3

CO3

Develop an interactive web applications using JSP.

K3

CO4

Create web applications using Spring Boot.

K2

1. Create a simple webpage using HTML.
2. Create a HTML page, which has properly aligned paragraphs with image along with it.
3. Write a program to display list of items in different styles.
4. Use frames to Include Images and Videos.
5. Add a Cascading Style sheet for designing the web page.
6. Design a dynamic web page with validation using JavaScript.
7. Write a program using JavaScript to demonstrate the concept of built-in array methods.
8. Write a program using JavaScript to demonstrate the concept of nested functions.

9. Write programs using JavaScript for Web Page to display browsers information.
10. Write a program using JavaScript to merge property of two objects.
11. Write a program using JavaScript to include a JS file into another JS file.
12. Develop a Servlet to validate user name and password stored in database. Display authorized user is she/he is authorized else display unauthorized user.
13. Write JSP & Servlet program to store student details sent from registration form in to database tab

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BMC252 : OBJECT ORIENTED PROGRAMMING LAB		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO1	Use the Concept of Data Abstraction and Encapsulation in C++ programs.	K 3
CO2	Design and Develop C++ program using the concept such as polymorphism, virtual function, exception handling and template.	K 3
CO3	Apply object- o r i e n t e d techniques to analyze, design and develop a complete solution for a given problem.	K 3
1. Use Java compiler and eclipse platform to write and execute java program. 2. Write programs to implement conditional statements and looping constructs. 3. Write programs to implement basic input / output operations. 4. Write programs using inheritance and polymorphism. 5. Write programs to implement error-handling techniques using exception handling and multithreading. 6. Write programs to demonstrate the use of java packages. 7. Write programs to demonstrate the concept of file handling and		

Page 29 Data Table

BMC253 : DATABASE MANAGEMENT SYSTEMS LAB		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO1	Use the Concept of Data Abstraction and Encapsulation in C++ programs.	K 6
CO2	Write SQL commands to query a database.	K 3
CO3	Write PL/SQL programs for implementing stored procedures, stored functions, cursors, trigger and packages.	K 6

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BMC252 : OBJECT ORIENTED PROGRAMMING LAB

Course Outcome (CO)

Bloom's

Knowledge

Level (KL)

At the end of course, the student will be able to

CO1

Use the Concept of Data Abstraction and Encapsulation in C++ programs.

K3

CO2

Design and Develop C++ program using the concept such as polymorphism, virtual function, exception handling and template.

K3

CO3

Apply object- oriented techniques to analyze, design and develop a complete solution for a given problem.

K3

1. Use Java compiler and eclipse platform to write and execute java program.
2. Write programs to implement conditional statements and looping constructs.
3. Write programs to implement basic input / output operations.
4. Write programs using inheritance and polymorphism.
5. Write programs to implement error-handling techniques using exception handling and multithreading.
6. Write programs to demonstrate the use of java packages.
7. Write programs to demonstrate the concept of file handling and establishment of database connection.
8. Write program to develop a calculator application in Java.
9. Write program to develop a Client Server Application.
10. Write program to develop GUI applications using Swing components.

Note: The instructor may add/delete/modify experiments, wherever he/she feels in a justified manner.

BMC253 : DATABASE MANAGEMENT SYSTEMS LAB

Course Outcome (CO)

Bloom's

Knowledge

Le

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1. Install oracle/ MYSQL.
2. Create Entity-Relationship Diagram using case tools.
3. Write SQL statements Using ORACLE /MYSQL:
 - a) Write basic SQL SELECT statements.
 - b) Restrict and sort data.
 - c) Display data from multiple tables.
 - d) Aggregate data using group function.
 - e) Manipulate data.
 - f) Create and manage tables.
4. Normalization.
5. Create cursor.
6. Create procedure and functions.
7. Create packages and triggers.
8. Design and implement payroll processing system.
9. Design and implement Library Information System.
10. Design and implement Student Information System.
11. Automatic Backup of Files and Recovery of Files

Note: The instructor may add/delete/modify experiments, wherever he/she feels in a justified manner.

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BMC254 : DATA STRUCTURES & ANALYSIS OF ALGORITHMS LAB		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO1	Write and execute programs to implement various searching and sorting algorithms.	K 3
CO2	Write and execute programs to implement various operations on two-dimensional arrays.	K 3
CO3	Implement various operations of Stacks and Queues using both arrays and linked lists data structures.	K 3
CO4	Implement graph algorithm to solve the problem of minimum spanning tree	K 3

BMC254 : DATA STRUCTURES & ANALYSIS OF ALGORITHMS LAB		
Program in C or C++ for following: 1. To implement addition and multiplication of two 2D arrays. 2. To transpose a 2D array. 3. To implement stack using array. 4. To implement queue using array. 5. To implement circular queue using array. 6. To implement stack using linked list. 7. To implement queue using linked list. 8. To implement BFS using linked list. 9. To implement DFS using linked list. 10. To implement Linear Search. 11. To implement Binary Search. 12. To implement Bubble Sorting. 13.		

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

BMC254 : DATA STRUCTURES & ANALYSIS OF ALGORITHMS LAB

Course Outcome (CO)

Bloom's

Knowledge

Level (KL)

At the end of course, the student will be able to

CO1

Write and execute programs to implement various searching and sorting algorithms.

K3

CO2

Write and execute programs to implement various operations on two-dimensional arrays.

K3

CO3

Implement various operations of Stacks and Queues using both arrays and linked lists data structures.

K3

CO4

Implement graph algorithm to solve the problem of minimum spanning tree

K3

Program in C or C++ for following:

1. To implement addition and multiplication of two 2D arrays.
2. To transpose a 2D array.
3. To implement stack using array.

4. To implement queue using array.
5. To implement circular queue using array.
6. To implement stack using linked list.
7. To implement queue using linked list.
8. To implement BFS using linked list.
9. To implement DFS using linked list.
10. To implement Linear Search.
11. To implement Binary Search.
12. To implement Bubble Sorting.
13. To implement Selection Sorting.
14. To implement Insertion Sorting.
15. To implement Merge Sorting.
16. To implement Heap Sorting.
17. To implement Matrix Multiplication by Strassen's algorithm.
18. Find Minimum Spanning Tree using Kruskal's Algorithm.

Note: The instructor may add/delete/modify experiments, wherever he/she feels in a justified

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

BVA251 : SPORTS AND YOGA

Objective of the Course:

- To maintain mental and physical wellness upright and develop ability in the students to cope up with the stress arising in the life.
- To create space in the curriculum to nurture the potential of the students in sports/games/yoga etc.
- To introduce a practice oriented introductory course on the subject.

More involved / advanced course may come up in subsequent years of study

Syllabus/ Guidelines

Part A: Sports/Games

Some form of Athletics would be compulsory for all students, unless restricted due to medical / physical reasons. In addition to this, student has to opt for at least one game out of the remaining mentioned below.

A fair theoretical knowledge and a reasonable amount of field / site practice of the chosen games will be essential.

1. Athletics

Compulsory

2. Volleyball
3. Basketball
4. Handball
5. Football
6. Badminton
7. Kabaddi
8. Kho-kho
9. Table tennis
10. Cricket

Part B: Yoga

a. Introduction of Yoga

Introduction of Yoga, Origin of Yoga, Aims and Objective of Yoga, Patanjali

Yoga darshan, Hath yoga, Gheranda Samhita, Karm yoga, Gyan yoga.

b. Asanas, Pranayam and Meditation Practices

Meaning of Asanas, Objective of Asanas, rules and regulations of Asanas and

Pranayams, Types of Yogasana

Yogic postures: Standing Posture, Sitting posture, Supine posture, Prone posture, balancing Postures,

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MASTER OF COMPUTER APPLICATIONS (Two Year Course) MCA Ist Year 2024-25

c. Science of Yoga

Physiological effects of Asanas- Paranayama and meditation, stress management and yoga, Mental health and yoga practice, Health and Personality Development.

General Guidelines

1. Institutes must assign minimum of three periods in the Time Table for the activities of Sports/Yoga.
2. Institutes must provide field/facility and offer a minimum of five choices of the Games/Sports.
3. Institutes are required to provide sports instructor / yoga teacher to mentor the students.
4. Student must be made familiar with the terminologies, rules/regulations, dimension/ marking of the play field/area and general knowledge of national/ international level facts/figures related to the chosen game.

Assessment:

The Institute must assign coordinator / subject teacher for the subject, for every batch/group of the students, who would be responsible for coordinating the required activities and keep watch on the level of student's participation in the chosen game.

Coordinator/mentor would be responsible for the award of the sessional marks based upon following components.

I.

Level of understanding and general awareness

(20 %)

II.

Involvements in the Practice Sessions

(50 %)

III.

Regularity, Sincerity and Discipline

(20 %)

IV.

Participation in University level / District level / State level / National Level events

(10 %)

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Page 35 Data Table

S. No.	Subject Code	Subject Name	Periods			Sessional	
			L	T	P	CT	TA
1.	KCA301	Artificial Intelligence	3	0	0	30	20
2.	KCA302	Software Engineering	4	0	0	30	20
3.	KCA303	Computer Network	3	1	0	30	20
4.		Elective – 1	3	0	0	30	20
5.		Elective – 2	3	1	0	30	20
6.	KCA351	Artificial Intelligence Lab	0	0	3	30	20
7.	KCA352	Software Engineering Lab	0	0	3	30	20
8.	KCA353	Mini Project**	0	0	4	30	20
		Total					

Page 35 Data Table

S. No.	Subject Code	Subject Name	Periods			Sessional	
			L	T	P	CT	TA
1.		Elective – 3	3	0	0	30	20
2.		Elective – 4	3	0	0	30	20
3.		Elective – 5	3	0	0	30	20
4.	KCA451	Project	-	-	-	-	200
		Total					

Page 35 Data Table

Elective-1	KCA011	Cryptography & Network Security
	KCA012	Data Warehousing & Data Mining
	KCA013	Software Project Management
	KCA014	Cloud Computing
	KCA015	Compiler Design

Page 35 Data Table

Elective-2	KCA021	Web Technology
	KCA022	Big Data
	KCA023	Simulation & Modeling
	KCA024	Software Testing & Quality Assurance
	KCA025	Digital Image Processing

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MASTER OF COMPUTER APPLICATION (MCA)

Curriculum & Evaluation Scheme MCA(III & IV semester)

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MCA SECOND YEAR, 2021-22

SEMESTER-III

S. No.

Subject

Code

Subject Name

Periods

Sessional

ESE

Total

Credit

L

T

P

CT

TA

Total

1.

KCA301

Artificial Intelligence

3

0

0

30

20

50

100

150

3

2.

KCA302

Software Engineering

4

0

0

30

20

50

100

150

4

3.

KCA303

Computer Network

3

1

0

30

20

50

100

150

4

4.

Elective – 1

3

0

0

30

20

50

100

150

3

5.

Elective – 2

3

1

0

30

20

50

100

150

3

6.

KCA351

Artificial Intelligence Lab

0

0

3

30

20

50

50

100

2

7.

KCA352

Software Engineering Lab

0

0

3

30

20

50

50

100

2

8.

KCA353

Mini Project**

0

0

4

30

20

50

50

100

2

Total

1050

23

CT: Class Test TA: Teacher Assessment

L/T/P: Lecture/ Tutorial/ Practical

SEMESTER-IV

S. No.

Subject

Code

Subject Name

Periods

Sessional

ESE

Total

Credit

L

T

P

CT

TA

Total

1.

Elective – 3

3

0

0

30

20

50

100

150

3

2.

Elective – 4

3

0

0

30

20

50

100

150

3

3.

Elective – 5

3

0

0

30

20

50

100

150

3

4.

KCA451

Project

-

-

-

-

200

200

500

700

14

Total

1050

23

CT: Class Test TA: Teacher Assessment

L/T/P: Lecture/ Tutorial/ Practical

** The Mini Project (6 weeks) conducted during summer break after II semester and will be assessed during III semester. The Course will be carried out at the Institute under the guidance of a F

Page 36 Data Table

Elective-3	KCA031	Privacy & Security in Online Social Media
	KCA032	Soft Computing
	KCA033	Pattern Recognition
	KCA034	Data Analytics
	KCA035	Software Quality Engineering

Page 36 Data Table

Elective-4	KCA041	Blockchain Architecture
	KCA042	Neural Network
	KCA043	Internet of Things
	KCA044	Modern Application Development
	KCA045	Distributed Database Systems

Page 36 Data Table

Elective-5	KCA051	Mobile Computing
	KCA052	Computer Graphics and Animation
	KCA053	Natural Language Processing
	KCA054	Machine Learning
	KCA055	Quantum Computing

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MASTER OF COMPUTER APPLICATION (MCA)

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Elective-3

KCA031

Privacy & Security in Online Social Media

KCA032

Soft Computing

KCA033

Pattern Recognition

KCA034

Data Analytics

KCA035

Software Quality Engineering

Elective-4

KCA041

Blockchain Architecture

KCA042

Neural Network

KCA043

Internet of Things

KCA044

Modern Application Development

KCA045

Distributed Database Systems

Elective-5

KCA051

Mobile Computing

KCA052

Computer Graphics and Animation

KCA053

Natural Language Processing

KCA054

Machine Learning

KCA055

Quantum Computing

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MASTER OF COMPUTER APPLICATION (MCA)

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SECOND YEAR SYLLABUS

SEMESTER-III

Page 38 Data Table

KCA301: Artificial Intelligence			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Define the meaning of intelligence and study various intelligent agents.		K 1
CO 2	Understand, analyze and apply AI searching algorithms in different problem domains.		K , K , K 2 3 4
CO 3	Study and analyze various models for knowledge representation.		K , K 1 3
CO 4	Understand the basic concepts of machine learning to analyze and implement widely used learning methods and algorithms.		K , K , K 2 4 6
CO 5	Understand the concept of pattern recognition and evaluate various classification and clustering techniques		K , K 2 5
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Artificial Intelligence: Introduction to artificial intelligence, Historical development and foundation areas of artificial intelligence, Tasks and application areas of artificial intelligence. Introduction, types and structure of intelligent agents, Computer Vision, Natural language processing.		08
II	Searching Techniques: Introduction, Problem solving by searching, Searching for solutions, Uniformed searching techniques, Informed searching techniques, Local search algorithms, Adversarial search methods, Search techniques used in games, Alpha-Beta pruning.		08
III	Knowledge Representation and Reasoning: Propositional logic, Predicate logic, First order logic, Inference in first order logic, Clause form conversion, Resolution. Chaining- concept, forward chaining and backward chaining, Utility theory and Probabilistic reasoning, Hidden Markov model, Bayesian networks.		08

KCA301: Artificial Intelligence			
IV	Machine Learning: Introduction, types and application areas, Decision trees, Statistical learning methods, Learning with complete data - concept and Naïve Bayes models, Learning with hidden data- concept and EM algorithm, Reinforcement learning.		08
V	Pattern Recognition: Introduction and design principles, Statistical pattern recognition, Parameter estimation methods - Principle component analysis and Linear discrimination analysis, Classification techniques - Nearest neighbor rule and Bayes classifier, K-means clustering, Support vector machine.		08
Suggested Readings: 1. Russell S. and Norvig P., "Artificial Intelligence – A Modern Approach", Pearson Education. 2. Rich E. and Knight K., "Artificial Intelligence", McGraw Hill Publications. 3. Charnik E. and McDermott D., "Introduction to Artificial Intelligence", Pearson Education. 4. Patterson D. W., "Artificial Intelligence and Expert Systems", Prentice Hall of India Publications. 5. Khemani D., "A First Course in Artificial Intelligence", McGraw Hill. 6. Winston P. H., "Artificial Intell			

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KCA301: Artificial Intelligence

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Define the meaning of intelligence and study various intelligent agents.

K1

CO 2

Understand, analyze and apply AI searching algorithms in different problem domains.

K2 , K3, K4

CO 3

Study and analyze various models for knowledge representation.

K1, K3

CO 4

Understand the basic concepts of machine learning to analyze and implement widely used learning methods and algorithms.

K2 , K4, K6

CO 5

Understand the concept of pattern recognition and evaluate various classification and clustering techniques

K2 , K5

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Artificial Intelligence: Introduction to artificial intelligence, Historical development and foundation areas of artificial intelligence, Tasks and application areas of artificial intelligence. Introduction, types and structure of intelligent agents, Computer Vision, Natural language processing.

08

II

Searching Techniques: Introduction, Problem solving by searching, Searching for solutions, Uniformed searching techniques, Informed searching techniques, Local search algorithms, Adversarial search methods, Search techniques used in games, Alpha-Beta pruning.

08

III

Knowledge Representation and Reasoning: Propositional logic, Predic

Page 39 Data Table

KCA302: Software Engineering			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			

KCA302: Software Engineering			
CO 1	Explain various software characteristics and analyze different software Development Models.		K , K 1 2
CO 2	Demonstrate the contents of a SRS and apply basic software quality assurance practices to ensure that design, development meet or exceed applicable standards.		K , K 1 2
CO 3	Compare and contrast various methods for software design.		K , K 2 3
CO 4	Formulate testing strategy for software systems, employ techniques such as unit testing, Test driven development and functional testing.		K 3
CO 5	Manage software development process independently as well as in teams and make use of various software management tools for development, maintenance and analysis.		K 5
DETAILED SYLLABUS			3-1-0
Unit	Topic		Proposed Lecture
I	Introduction: Introduction to Software Engineering, Software Components, Software Characteristics, Software Crisis, Software Engineering Processes, Similarity and Differences from Conventional Engineering Processes, Software Quality Attributes. Software Development Life Cycle (SDLC) Models: Water Fall Model, Prototype Model, Spiral Model, Evolutionary Development Models, Iterative Enhancement Models.		08
II	Software Requirement Specifications (SRS): Requirement Engineering Process: Elicitation, Analysis, Documentation, Review and Management of User Needs, Feasibility Study, Information Modelling, Data Flow Diagrams, Entity Relationship Diagrams, Decision Tables, SRS Document, IEEE Standards for SRS. Software Quality Assurance (SQA): Verification and Validation, SQA Plans, Software Quality Frameworks, ISO 9000 Models, SEI-CMM Model.		08

KCA302: Software Engineering			
III	Software Design: Basic Concept of Software Design, Architectural Design, Low Level Design: Modularization, Design Structure Charts, Pseudo Codes, Flow Charts, Coupling and Cohesion Measures, Design Strategies: Function Oriented Design, Object Oriented Design, Top- Down and Bottom-Up Design. Software Measurement and Metrics: Various Size Oriented Measures: Halstead's Software Science, Function Point (FP) Based Measures, Cyclomatic Complexity Measures: Control Flow Graphs.		08
IV	Software Testing: Testing Objectives, Unit Testing, Integration Testing, Acceptance Testing, Regression Testing, Testing for Functionality and Testing for Performance, Top Down and Bottom- Up Testing Strategies: Test Drivers and Test Stubs, Structural Testing (White Box Testing), Functional Testing (Black Box Testing), Test Data Suit Preparation, Alpha and Beta Testing of Products. Static Testing Strategies: Formal Technical Reviews (Peer Reviews), Walk Through,		08

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MASTER OF COMPUTER APPLICATION (MCA)

Curriculum & Evaluation Scheme MCA(III & IV semester)

Page 6

KCA302: Software Engineering

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Explain various software characteristics and analyze different software Development Models.

K1, K2

CO 2

Demonstrate the contents of a SRS and apply basic software quality assurance practices to ensure that design, development meet or exceed applicable standards.

K1, K2

CO 3

Compare and contrast various methods for software design.

K2, K3

CO 4

Formulate testing strategy for software systems, employ techniques such as unit testing, Test driven development and functional testing.

K3

CO 5

Manage software development process independently as well as in teams and make use of various software management tools for development, maintenance and analysis.

K5

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Introduction:

Introduction

to

Software

Engineering,

Software

Components, Software Characteristics, Software Crisis, Software Engineering Processes, Similarity and Differences from Conventional Engineering

Processes,

Software

Quality

Attributes.

Software

Development Life Cycle (SDLC) Models: Water Fall Model, Prototype Model, Spiral Model, Evolutionary Development Models, Iterative Enhancement Models.

08

II

Software

Requirement

Specifications

(SRS):

Requirement

En

Page 40 Data Table

	Code Inspection, Compliance with Design and Coding Standards.	
V	Software Maintenance and Software Project Management: Software as an Evolutionary Entity, Need for Maintenance, Categories of Maintenance: Preventive, Corrective and Perfective Maintenance, Cost of Maintenance, Software Re-Engineering, Reverse Engineering. Software Configuration Management Activities, Change Control Process, Software Version Control, An Overview of CASE Tools. Estimation of Various Parameters such as Cost, Efforts, Schedule/Duration, Constructive Cost Models (COCOMO), Resource A	08
Suggested Readings: 1. R S Pressman, "Software Engineering: A Practitioners Approach", McGraw Hill. 2. Pankaj Jalote, "Software Engineering", Wiley 3. Rajib Mall, "Fundamentals of Software Engineering", PHI Publication. 4. K K Aggarwal and Yogesh Singh, "Software Engineering", New Age International Publishers. 5. Ghezzi, M. Jarayeri, D. Manodrioli, "Fundamentals of Software Engineering", PHI Publication. 6. Ian Sommerville, "Software Engineering", Addison Wesley. 7. Kassem Saleh, "Software Engin		

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Code Inspection, Compliance with Design and Coding Standards.

V

Software Maintenance and Software Project Management:

Software as an Evolutionary Entity, Need for Maintenance, Categories of Maintenance: Preventive, Corrective and Perfective Maintenance, Cost of Maintenance, Software Re-Engineering, Reverse Engineering.

Software Configuration Management Activities, Change Control Process, Software Version Control, An Overview of CASE

Tools. Estimation of Various Parameters such as Cost, Efforts, Schedule/Duration, Constructive Cost Models (COCOMO), Resource Allocation Models, Software Risk Analysis and Management.

08

Suggested Readings:

1. R S Pressman, "Software Engineering: A Practitioners Approach", McGraw Hill.
2. Pankaj Jalote, "Software Engineering", Wiley
3. Rajib Mall, "Fundamentals of Software Engineering", PHI Publication.
4. K K Aggarwal and Yogesh Singh, "Software Engineering", New Age International Publishers.
5. Ghezzi, M. Jarayeri, D. Manodrioli, "Fundamentals of Software Engineering", PHI Publication.
6. Ian Sommerville, "Software Engineering", Addison Wesley.
7. Kassem Saleh, "Software Engineering", Cengage Learning
8. Pfleeger, "Software Engineering", Macmillan Publication

Page 41 Data Table

KCA303: Computer Networks			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Describe communication models TCP/IP, ISO-OSI model, network topologies along with communicating devices and connecting media.		K2
CO 2	Apply knowledge of error detection, correction and learn concepts of flow control along with error control.		K3
CO 3	Classify various IP addressing techniques, subnetting along with network routing protocols and algorithms.		K4
CO 4	Understand various transport layer protocols and their design considerations along with congestion control to maintain Quality of Service.		K2

KCA303: Computer Networks			
CO 5	Understand applications-layer protocols and elementary standards of cryptography and network security.		K2
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Data Communications: Introduction: Data communication Components and characteristics, Data representation and Data flow. Networks: LAN, WAN, MAN, Topologies. Protocols and Standards: ISO-OSI model and TCP-IP Model. Network Connecting Devices: HUB, Bridge, Switch, Router and Gateways. Transmission Media: Guided and unguided Media Classification and Arrangement: Wired LANs and Wireless LANs		08
II	Data Link Layer: Error Detection and Error Correction: Types of errors, LRC, VRC, Checksum, CRC, and Hamming Code. Flow Control and Error Control: Stop and Wait Protocol, Sliding Window, Go-back-N-ARQ Protocol and Selective-Repeat ARQ Protocol. Channel Allocation Protocols: Random Access, Controlled and Channelization techniques such as ALOHA, CSMA, CSMA/CD, CDMA/CA, TDMA, FDMA, Token Passing, etc.		08
III	Network Layer: Switching Techniques: Circuit Switching, Packet Switching, and Message Switching. Logical addressing: IPv4 and IPv6 Address schemes, Classes and subnetting Network Layer Protocols: ARP, RARP, BOOTP and DHCP Routing Techniques: Interdomain and Intradomain routing with examples.		08
IV	Transport Layer: Introduction to Transport Layer: Process-to-Process Delivery:		08

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KCA303: Computer Networks

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Describe communication models TCP/IP, ISO-OSI model, network topologies along with communicating devices and connecting media.

K2

CO 2

Apply knowledge of error detection, correction and learn concepts of flow control along with error control.

K3

CO 3

Classify various IP addressing techniques, subnetting along with network routing protocols and algorithms.

K4

CO 4

Understand various transport layer protocols and their design considerations along with congestion control to maintain Quality of Service.

K2

CO 5

Understand applications-layer protocols and elementary standards of cryptography and network security.

K2

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Data

Communications:

Introduction:

Data

communication

Components and characteristics, Data representation and Data flow.

Networks: LAN, WAN, MAN, Topologies.

Protocols and Standards: ISO-OSI model and TCP-IP Model.

Network Connecting Devices: HUB, Bridge, Switch, Router and Gateways.

Transmission Media: Guided and unguided Media

Classification and Arrangement: Wired LANs and Wireless LANs

08

II

Data Link Layer:

Error Detection and Error Correction: Types of errors, LRC, VRC,

Checksum, CRC, and Hamming Code.

Flow Control and

Page 42 Data Table

	Reliable and unreliable Connection, Port and Socket Addressing Transport Layer Protocols with packet formats: User Datagram Protocol (UDP), Transmission Control Protocol (TCP), Stream Control Transmission Protocol (SCTP). Congestion Control: Techniques for handling the Congestion Control. Quality of Service (QoS): Flow Characteristics and techniques to improve QoS.	
V	Application Layer: Basic Concept of Application Layer: Domain Name System, World Wide Web, Hyper Text Transfer Protocol, Electronic mail, File Transfer Protocol, Remote login. Introduction to Cryptography: Definition, Goal, Applications, Attacks, Encryption, decryption, public-key and private key cryptography.	08
Suggested Readings: 1. Behrouz Forouzan, "Data Communication and Networking", McGraw Hill 2. Andrew Tanenbaum "Computer Networks", Prentice Hall. 3. William Stallings, "Data and Computer Communication", Pearson. 4. Kurose and Ross, "Computer Networking- A Top-Down Approach", Pearson. 5. Peterson and Davie, "Computer Networks: A Systems Approach", Morgan Kaufmann 6. W. A. Shay, "Understanding Communications and Networks", Cengage Learning. 7. D. Comer, "Computer Networks and Internets", Pearson.		

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Reliable and unreliable Connection, Port and Socket Addressing

Transport Layer Protocols with packet formats: User Datagram

Protocol (UDP), Transmission Control Protocol (TCP), Stream Control Transmission Protocol (SCTP).

Congestion Control: Techniques for handling the Congestion Control.

Quality of Service (QoS): Flow Characteristics and techniques to improve QoS.

V

Application Layer:

Basic Concept of Application Layer: Domain Name System, World

Wide Web, Hyper Text Transfer Protocol, Electronic mail, File Transfer Protocol, Remote login.

Introduction to Cryptography: Definition, Goal, Applications,

Attacks,

Encryption,

decryption,

public-key

and

private

key

cryptography.

08

Suggested Readings:

1. Behrouz Forouzan, "Data Communication and Networking", McGraw Hill
2. Andrew Tanenbaum "Computer Networks", Prentice Hall.
3. William Stallings, "Data and Computer Communication", Pearson.
4. Kurose and Ross, "Computer Networking- A Top-Down Approach", Pearson.
5. Peterson and Davie, "Computer Networks: A Systems Approach", Morgan Kaufmann
6. W. A. Shay, "Understanding Communications and Networks", Cengage Learning.
7. D. Comer, "Computer Networks and Internets", Pearson.
8. Behrouz Forouzan, "TCP/IP Protocol Suite", McGraw Hill.

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ELECTIVE-1

Page 44 Data Table

KCA011: Cryptography & Network Security			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Understand various security attacks and their protection mechanism.		K 2
CO 2	Apply and analyze various encryption algorithms.		K , K 3 4
CO 3	Understand functions and algorithms to authenticate messages and study and apply different digital signature techniques.		K , K , K 1 2 3
CO 4	Analyze different types of key distributions.		K 4
CO 5	Study and appraise different IP and system security mechanism.		K , K 1 5
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Introduction to security attacks, Services and mechanism, Classical encryption techniques substitution ciphers and transposition ciphers, Cryptanalysis, Steganography, Stream and block ciphers. Modern Block Ciphers: Block ciphers principles, Shannon's theory of confusion and diffusion, Feistel structure, Data encryption standard(DES), Strength of DES, Idea of differential cryptanalysis, Block cipher modes of operations, Triple DES		08
II	Introduction to group, field, finite field of the form $GF(p)$, Modular arithmetic, Prime and relative prime numbers, Extended Euclidean Algorithm, Advanced Encryption Standard (AES). Fermat's and Euler's theorem, Primality testing, Chinese Remainder theorem, Discrete Logarithmic Problem, Principals of public key crypto systems, RSA algorithm, Security of RSA		08

KCA011: Cryptography & Network Security			
III	Message Authentication Codes: Authentication requirements, Authentication functions, Message authentication code, Hash functions, Birthday attacks, Security of hash functions, Secure hash algorithm (SHA). Digital Signatures: Digital Signatures, Elgamal Digital Signature Techniques, Digital signature standards (DSS), Proof of digital signature algorithm.		08
IV	Key Management and distribution: Symmetric key distribution, Diffie-Hellman Key Exchange, Public key distribution, X.509 Certificates, Public key Infrastructure. Authentication Applications: Kerberos Electronic mail security: pretty good privacy (PGP), S/MIME.		08
V	IP Security: Architecture, Authentication header, Encapsulating security payloads, Combining security associations, Key management. Introduction to Secure Socket Layer, Secure electronic transaction (SET). System Security: Introductory idea of Intrusion, Intrusion detection, Viruses and related threats, firewalls.		08
Suggested Readings: 1. Stallings W., "Cryptography and Network Security: Principals and Practice", Pearson Education. 2. Frouzan B. A., "Cryptography and Network Security", McGraw Hill. 3. Kahate A., "Cryptography and Network Security", Tata McGraw Hill.			

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KCA011: Cryptography & Network Security

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Understand various security attacks and their protection mechanism.

K2

CO 2

Apply and analyze various encryption algorithms.

K3, K4

CO 3

Understand functions and algorithms to authenticate messages and study and apply different digital signature techniques.

K1, K2, K3

CO 4

Analyze different types of key distributions.

K4

CO 5

Study and appraise different IP and system security mechanism.

K1, K5

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Introduction to security attacks, Services and mechanism, Classical encryption techniques substitution ciphers and transposition ciphers, Cryptanalysis, Steganography, Stream and block ciphers.

Modern Block Ciphers: Block ciphers principles, Shannon's theory of confusion and diffusion, Feistel structure, Data encryption standard(DES), Strength of DES, Idea of differential cryptanalysis, Block cipher modes of operations, Triple DES

08

II

Introduction to group, field, finite field of the form $GF(p)$, Modular arithmetic, Prime and relative prime numbers, Extended Euclidean Algorithm, Advanced Encryption Standard (AES).

Fermat's and Euler's theorem, Primality testing, Chinese Remainder theorem, Discrete Logarithmic Problem, Principals of

Page 45 Data Table

KCA012: Data Warehousing & Data Mining			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO1	Demonstrate knowledge of Data Warehouse and its components.		K , K 1 2
CO2	Discuss the process of Warehouse Planning and Implementation.		K , K 1 2
CO3	Discuss and implement various supervised and Non supervised learning algorithms on data.		K 6
CO4	Explain the various process of Data Mining and decide best according to type of data.		K , K 2 5
CO5	Explain process of knowledge discovery in database (KDD). Design Data Mining model.		K , K 2 5
	DETAILED SYLLABUS		4-0-0
Unit	Topic		Proposed Lecture
I	Data Warehousing: Overview, Definition, Data Warehousing Components, Building a Data Warehouse, Warehouse Database, Mapping the Data Warehouse to a Multiprocessor Architecture, Difference between Database System and Data Warehouse, Multi Dimensional Data Model, Data Cubes, Stars, Snow Flakes, Fact Constellations, Concept.		08
II	Data Warehouse Process and Technology: Warehousing Strategy, Warehouse /management and Support Processes, Warehouse Planning and Implementation, Hardware and Operating Systems for Data Warehousing, Client/Server Computing Model & Data Warehousing. Parallel Processors & Cluster Systems, Distributed DBMS implementations, Warehousing Software, Warehouse Schema Design		08

KCA012: Data Warehousing & Data Mining			
III	Data Mining: Overview, Motivation, Definition & Functionalities, Data Processing, Form of Data Pre-processing, Data Cleaning: Missing Values, Noisy Data, (Binning, Clustering, Regression, Computer and Human inspection), Inconsistent Data, Data Integration and Transformation. Data Reduction:-Data Cube Aggregation, Dimensionality reduction, Data Compression, Numerosity Reduction, Discretization and Concept hierarchy generation, Decision Tree		08
IV	Classification: Definition, Data Generalization, Analytical Characterization, Analysis of attribute relevance, Mining Class comparisons, Statistical measures in large Databases, Statistical-Based Algorithms, Distance-Based Algorithms, Decision Tree-Based Algorithms. Clustering: Introduction, Similarity and Distance Measures, Hierarchical and Partitional Algorithms. Hierarchical Clustering- CURE and Chameleon. Density Based Methods DBSCAN, OPTICS. Grid Based Methods- STING, CLIQUE. Model Based Me		08
V	Data Visualization and Overall Perspective: Aggregation, Historical information, Query Facility, OLAP function and Tools. OLAP Servers, ROLAP, MOLAP, HOLAP, Data Mining interface, Security, Backup and		

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KCA012: Data Warehousing & Data Mining

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO1

Demonstrate knowledge of Data Warehouse and its components.

K1, K2

CO2

Discuss the process of Warehouse Planning and Implementation.

K1, K2

CO3

Discuss and implement various supervised and Non supervised learning algorithms on data.

K6

CO4

Explain the various process of Data Mining and decide best according to type of data.

K2, K5

CO5

Explain process of knowledge discovery in database (KDD). Design Data Mining model.

K2, K5

DETAILED SYLLABUS

4-0-0

Unit

Topic

Proposed

Lecture

I

Data

Warehousing:

Overview,

Definition,

Data

Warehousing

Components, Building a Data Warehouse, Warehouse Database, Mapping the Data Warehouse to a Multiprocessor Architecture, Difference between Database System and Data Warehouse, Multi Dimensional Data Model, Data Cubes, Stars, Snow Flakes, Fact Constellations, Concept.

08

II

Data Warehouse Process and Technology: Warehousing Strategy,

Warehouse /management and Support Processes, Warehouse Planning and

Implementation, Hardware and Operating Systems for Data Warehousing,
 Client/Server Computing Model & Data Warehousing. Parallel Processors
 & Cluster Systems, Distributed DBMS implementations, Warehousing
 Software, Warehouse Schema Design
 08

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	Recovery, Tuning Data Warehouse, Testing Data Warehouse. Warehousing applications and Recent Trends: Types of Warehousing Applications, Web Mining, Spatial Mining and Temporal Mining.	08
Suggested Readings: 1. Alex Berson, Stephen J. Smith "Data Warehousing, Data-Mining & OLAP", TMH. 2. Mark Humphries, Michael W. Hawkins, Michelle C. Dy, "Data Warehousing: Architecture and Implementation", Pearson. 3. I.Singh, "Data Mining and Warehousing", Khanna Publishing House. 4. Margaret H. Dunham, S. Sridhar, "Data Mining: Introductory and Advanced Topics" Pearson Education 5. Arun K. Pujari, "Data Mining Techniques" Universities Press. 5. Pieter Adriaans, Dolf Zantinge, "Data-Mining", Pear		

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Recovery,

Tuning

Data

Warehouse,

Testing

Data

Warehouse.

Warehousing applications and Recent Trends: Types of Warehousing

Applications, Web Mining, Spatial Mining and Temporal Mining.

08

Suggested Readings:

1. Alex Berson, Stephen J. Smith "Data Warehousing, Data-Mining & OLAP", TMH.
2. Mark Humphries, Michael W. Hawkins, Michelle C. Dy, "Data Warehousing: Architecture and Implementation", Pearson.
3. I.Singh, "Data Mining and Warehousing", Khanna Publishing House.

4. Margaret H. Dunham, S. Sridhar, "Data Mining: Introductory and Advanced Topics" Pearson Education
5. Arun K. Pujari, "Data Mining Techniques" Universities Press.
5. Pieter Adriaans, Dolf Zantinge, "Data-Mining", Pearson Education

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KCA013: Software Project Management			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Identify project planning objectives, along with various cost/effort estimation models.		K 3
CO 2	Organize & schedule project activities to compute critical path for risk analysis		K 3
CO 3	Monitor and control project activities.		K , K 4 5
CO 4	Formulate testing objectives and test plan to ensure good software quality under SEI- CMM		K 6
CO 5	Configure changes and manage risks using project management tools.		K , K 2 4
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Project Evaluation and Project Planning: Importance of Software Project Management – Activities – Methodologies – Categorization of Software Projects – Setting objectives – Management Principles – Management Control – Project portfolio Management – Cost-benefit evaluation technology – Risk evaluation – Strategic program Management – Stepwise Project Planning.		08
II	Project Life Cycle and Effort Estimation: Software process and Process Models – Choice of Process models – Rapid Application development – Agile methods – Dynamic System Development Method – Extreme Programming – Managing interactive processes – Basics of Software estimation – Effort and Cost estimation techniques – COSMIC Full function points – COCOMO II – a Parametric Productivity Model.		08

KCA013: Software Project Management			
III	Activity Planning and Risk Management: Objectives of Activity planning – Project schedules – Activities – Sequencing and scheduling – Network Planning models – Formulating Network Model – Forward Pass & Backward Pass techniques – Critical path (CRM) method – Risk identification – Assessment – Risk Planning – Risk Management – PERT technique – Monte Carlo simulation – Resource Allocation – Creation of Critical paths – Cost schedules.		08
IV	Project Management and Control: Framework for Management and control – Collection of data – Visualizing progress – Cost monitoring – Earned Value Analysis – Prioritizing Monitoring – Project tracking – Change control Software Configuration Management – Managing contracts – Contract Management.		08
V	Staffing in Software Projects: Managing people – Organizational behavior – Best methods of staff selection – Motivation – The Oldham – Hackman job characteristic model – Stress – Health and Safety – Ethical and Professional concerns – Working in teams – Decision making – Organizational structures – Dispersed and Virtual teams – Communications genres – Communication plans – Leadership.		08
Suggested Readings: 1. Bob Hughes, Mike Cotterell and Rajib Mall: "Software Project Management" – Fifth Edition, McGraw Hill, New Delhi, 2012. 2. Robert K. Wysocki ■ "Effective Software Project Management" – Wiley Publication, 2011. 3. Walker Royce: ■ "Software Project Management" - Addison-Wesley, 1998. 4. Gopalaswamy Ramesh, ■ "Managing Global Software Projects" – McGraw Hill Education (India), Fourteenth Reprint 2013. 5. Koontz Harold & Weihrich Heinz, "Essentials of Management", McGraw Hill			

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KCA013: Software Project Management

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Identify project planning objectives, along with various cost/effort estimation models.

K3

CO 2

Organize & schedule project activities to compute critical path for risk analysis

K3

CO 3

Monitor and control project activities.

K4, K5

CO 4

Formulate testing objectives and test plan to ensure good software quality under SEI-

CMM

K6

CO 5

Configure changes and manage risks using project management tools.

K2, K4

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Project Evaluation and Project Planning: Importance of Software Project

Management – Activities – Methodologies – Categorization of Software Projects –

Setting objectives – Management Principles – Management Control – Project

portfolio Management – Cost-benefit evaluation technology – Risk evaluation –

Strategic program Management – Stepwise Project Planning.

08

II

Project Life Cycle and Effort Estimation: Software process and Process Models –
 Choice of Process models – Rapid Application development – Agile methods –
 Dynamic System Development Method – Extreme Programming– Managing
 interactive processes – Basics of Software estimation – Effort and Cost
 estimation techniques – COSMIC Full function points – COCOMO II – a Parametric

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KCA014: Cloud Computing			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Understand the concepts of Cloud Computing, key technologies, strengths and limitations of cloud computing.		K K 1 , 2
CO 2	Develop the ability to understand and use the architecture to compute and storage cloud, service and models.		K K 1, 3
CO 3	Understand the application in cloud computing.		K K 4, 5
CO 4	Learn the key and enabling technologies that help in the development of cloud.		K K 3, 4
CO 5	Explain the core issues of cloud computing such as resource management and security.		K , K 2 6
DETAILED SYLLABUS			3-1-0
Unit	Topic		Proposed Lecture
I	Introduction: Cloud Computing – Definition of Cloud – Evolution of Cloud Computing – Underlying Principles of Parallel and Distributed, History of Cloud Computing - Cloud Architecture - Types of Clouds - Business models around Clouds – Major Players in Cloud Computing- issues in Clouds - Eucalyptus - Nimbus - Open Nebula, CloudSim.		08
II	Cloud Services: Types of Cloud services: Software as a Service- Platform as a Service –Infrastructure as a Service - Database as a Service - Monitoring as a Service –Communication as services. Service providers- Google, Amazon, Microsoft Azure, IBM, Sales force.		08

KCA014: Cloud Computing			
III	Collaborating Using Cloud Services: Email Communication over the Cloud - CRM Management – Project Management-Event Management - Task Management – Calendar - Schedules - Word Processing – Presentation – Spreadsheet - Databases – Desktop - Social Networks and Groupware.		08
IV	Virtualization for Cloud: Need for Virtualization – Pros and cons of Virtualization – Types of Virtualization –System VM, Process VM, Virtual Machine monitor – Virtual machine properties - Interpretation and binary translation, HLL VM - supervisors – Xen, KVM, VMware, Virtual Box, Hyper-V.		08
V	Security, Standards and Applications: Security in Clouds: Cloud security challenges – Software as a Service Security, Common Standards: The Open Cloud Consortium – The Distributed management Task Force – Standards for application Developers – Standards for Messaging – Standards for Security, End user access to cloud computing, Mobile Internet devices and the cloud. Hadoop – MapReduce – Virtual Box – Google App Engine – Programming Environment for Google App Engine		08

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KCA014: Cloud Computing

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Understand the concepts of Cloud Computing, key technologies, strengths and limitations of cloud computing.

K1 , K2

CO 2

Develop the ability to understand and use the architecture to compute

and storage cloud, service and models.

K1, K3

CO 3

Understand the application in cloud computing.

K4, K5

CO 4

Learn the key and enabling technologies that help in the development of cloud.

K3, K4

CO 5

Explain the core issues of cloud computing such as resource management and security.

K2, K6

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Introduction: Cloud Computing – Definition of Cloud – Evolution of Cloud Computing – Underlying Principles of Parallel and Distributed, History of Cloud Computing - Cloud Architecture - Types of Clouds - Business models around Clouds – Major Players in Cloud Computing- issues in Clouds - Eucalyptus - Nimbus - Open Nebula, CloudSim.

08

II

Cloud Services: Types of Cloud services: Software as a Service- Platform as a Service –Infrastructure as a Service - Database as a Service - Monitoring as a Service –Communication as services. Service providers- Google, Amazon, Microsoft Azure, IBM, Sales force.

08

III

Collaborating Using Cloud Services: Email Communication over t

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Suggested Readings:

1. David E.Y. Sarna, "Implementing and Developing Cloud Application", CRC press 2011.
2. Lee Badger, Tim Grance, Robert Patt-Corner, Jeff Voas, NIST, Draft cloud computing synopsis and recommendation, May 2011.
3. Anthony T Velte, Toby J Velte, Robert Elsenpeter, "Cloud Computing : A Practical Approach", Tata McGraw-Hill 2010.
4. Haley Beard, "Best Practices for Managing and Measuring Processes for On-demand Computing, Applications and Data Centers in the Cloud with SLAs", Emereo Pty Limited, July 2008.
5. G. J. Popek, R.P. Goldberg, "Formal requirements for virtualizable third generation Architectures, Communications of the ACM", No.7 Vol.17, July 1974

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KCA015 : Compiler Design			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course , the student will be able to:			
CO 1	Acquire knowledge of different phases and passes of the compiler and also able to use the compiler tools like LEX, YACC, etc. Students will also be able to design different types of compiler tools to meet the requirements of the realistic constraints of compilers.		K , K 3 6
CO 2	Understand the parser and its types i.e. Top-Down and Bottom-up parsers and construction of LL, SLR, CLR, and LALR parsing table.		K , K 2 6
CO 3	Implement the compiler using syntax-directed translation method and get knowledge about the synthesized and inherited attributes.		K , K 4 5
CO 4	Acquire knowledge about run time data structure like symbol table organization and different techniques used in that.		K , K 2 3
CO 5	Understand the target machine's run time environment, its instruction set for code generation and techniques used for code optimization.		K , K 2 4
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture

KCA015 : Compiler Design			
I	Introduction to Compiler: Phases and passes, Bootstrapping, Finite state machines and regular expressions and their applications to lexical analysis, Optimization of DFA-Based Pattern Matchers implementation of lexical analyzers, lexical-analyzer generator, LEX compiler, Formal grammars and their application to syntax analysis, BNF notation, ambiguity, YACC. The syntactic specification of programming languages: Context free grammars, derivation and parse trees, capabilities of CFG.		08
II	Basic Parsing Techniques: Parsers, Shift reduce parsing, operator precedence parsing, top down parsing, predictive parsers Automatic Construction of efficient Parsers: LR parsers, the canonical Collection of LR(0) items, constructing SLR parsing tables, constructing Canonical LR parsing tables, Constructing LALR parsing tables, using ambiguous grammars, an automatic parser generator, implementation of LR parsing tables.		08
III	Syntax-directed Translation: Syntax-directed Translation schemes, Implementation of Syntax-directed Translators, Intermediate code, postfix notation, Parse trees & syntax trees, three address code, quadruple & triples, translation of assignment statements, Boolean expressions, statements that alter the flow of control, postfix translation, translation with a top down parser. More about translation: Array references in arithmetic expressions, procedures call, declarations and case statements.		08
IV	Symbol Tables: Data structure for symbols tables, representing scope information. Run- Time Administration: Implementation of simple stack allocation scheme, storage allocation in block structured language. Error Detection & Recovery: Lexical Phase errors, syntactic phase errors semantic errors.		08

KCA015 : Compiler Design			
V	Code Generation: Design Issues, the Target Language. Addresses in the Target Code, Basic Blocks and Flow Graphs, Optimization of Basic Blocks, Code Generator. Code optimization: Machine-Independent Optimizations, Loop optimization, DAG representation of basic blocks, value numbers and algebraic laws, Global Data-Flow analysis.		08

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KCA015 : Compiler Design

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course , the student will be able to:

CO 1

Acquire knowledge of different phases and passes of the compiler and also able to use the compiler tools like LEX, YACC, etc. Students will also be able to design different types of compiler tools to meet the requirements of the realistic constraints of compilers.

K3, K6

CO 2

Understand the parser and its types i.e. Top-Down and Bottom-up parsers and construction of LL, SLR, CLR, and LALR parsing table.

K2, K6

CO 3

Implement the compiler using syntax-directed translation method and get knowledge about the synthesized and inherited attributes.

K4, K5

CO 4

Acquire knowledge about run time data structure like symbol table organization and different techniques used in that.

K2, K3

CO 5

Understand the target machine's run time environment, its instruction set for code generation and techniques used for code optimization.

K2, K4

DETAILED SYLLABUS

3-0-0

Unit

Topic

Propose

d

Lecture

I

Introduction to Compiler: Phases and passes, Bootstrapping, Finite state machines and regular expressions and their applications to lexical analysis, Optimization of DFA-Based Pattern Matchers implementation of lexical analyzers, lexical-analyzer generator, LEX compiler, Formal grammars and their application to syntax analysis,

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Text books:

1. K. Muneeswaran, Compiler Design, First Edition, Oxford University Press.
2. J.P. Bennet, "Introduction to Compiler Techniques", Second Edition, Tata McGraw-Hill, 2003.
3. Henk Alblas and Albert Nymeyer, "Practice and Principles of Compiler Building with C", PHI, 2001.
4. Aho, Sethi & Ullman, "Compilers: Principles, Techniques and Tools", Pearson Education
5. V Raghvan, "Principles of Compiler Design", TMH
6. Kenneth Loudon, "Compiler Construction", Cengage Learning.
7. Charles Fischer and Ricard LeBlanc, "Crafting a Compiler with C", Pearson Education

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ELECTIVE-2

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KCA021: Web Technology							
	Course Outcome (CO)						

KCA021: Web Technology							
	At the end of course, the student will be able to:						
CO 1							Apply the knowledge of HTML and CSS to develop web application and
							analyze the insights of internet programming to implement complete
							application over the web.
CO 2							Understand, analyze and apply the role of JavaScript in the workings of the
							web and web applications.
	CO 3						Understand, analyze and build dynamic web applications using servlet and JSP.
CO 4							Develop Spring-based Java applications using Java configuration, XML
							configuration, annotation-based configuration, beans and their scopes, and
							properties.
	CO 5						Develop web application using Spring Boot and RESTful Web Services
	DETAILED SYLLABUS						

KCA021: Web Technology							
	Unit			Topic			
I				Web Page Designing: Introduction and Web Development Strategies, History of			
				Web and Internet, Protocols Governing Web, HTML-Introduction, HTML Tags,			
				HTML-Grouping Using Div & Span, HTML-Lists, HTML-Images, HTML-			
				Hyperlink, HTML-Table, HTML-Iframe, HTML-Form, Introduction of CSS, CSS			
				Syntax, External Style Sheet using < link >, Multiple Style Sheets, Value Lengths			
				and Percentages, CSS-Selectors, CSS-Box Model, Floats, Clear, Introduction to			
				Bootstrap.			
II				Scripting: Introduction to JavaScript, Creating Variables in JavaScript, Creating			
				Functions in JavaScript, UI Events, Returning Data from Functions, Working with			

KCA021: Web Technology							
				Conditions, looping in JavaScript, Block Scope Variables, Working with Objects,			
				Creating Object using Object Literals, Manipulating DOM Elements with			
				JavaScript			
III				Web Application development using JSP & Servlets: Servlet Overview and			
				Architecture, Interface Servlet and the Servlet Life Cycle, Handling HTTP get			
				Requests, Handling HTTP post Requests, Redirecting Requests to Other			
				Resources, Session Tracking, Cookies, Session Tracking with Http Session. Java			
				Server Pages (JSP): Introduction, Java Server Pages Overview, A First Java Server			
				Page Example, Implicit Objects, Scripting, Standard Actions, Directives, Custom			

KCA021: Web Technology							
				Tag Libraries.			
IV				Spring: Spring Core Basics-Spring Dependency Injection concepts, Introduction			
				to Design patterns, Factory Design Pattern, Strategy Design pattern, Spring			
				Inversion of Control, AOP, Bean Scopes-Singleton, Prototype, Request, Session,			
				Application, WebSocket, Auto wiring, Annotations, Life Cycle Call backs, Bean			
				Configuration styles			
V				Spring Boot: Spring Boot-Spring Boot Configuration, Spring Boot Annotations,			
				Spring Boot Actuator, Spring Boot Build Systems, Spring Boot Code Structure,			
				Spring Boot Runners, Logger, BUILDING RESTFUL WEB SERVICES, Rest			
				Controller, Request Mapping, Request Body, Path Variable, Request Parameter,			

KCA021: Web Technology							
				GET, POST, PUT, DELETE APIs, Build Web Applications			

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KCA021: Web Technology

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to:

CO 1

Apply the knowledge of HTML and CSS to develop web application and analyze the insights of internet programming to implement complete application over the web.

K3, K6

CO 2

Understand, analyze and apply the role of JavaScript in the workings of the web and web applications.

K2, K3

CO 3

Understand, analyze and build dynamic web applications using servlet and JSP.

K2, K3

CO 4

Develop Spring-based Java applications using Java configuration, XML configuration, annotation-based configuration, beans and their scopes, and properties.

K2, K4, K6

CO 5

Develop web application using Spring Boot and RESTful Web Services

K3, K6

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Web Page Designing: Introduction and Web Development Strategies, History of Web and Internet, Protocols Governing Web, HTML-Introduction, HTML Tags, HTML-Grouping Using Div & Span, HTML-Lists, HTML-Images, HTML-Hyperlink, HTML-Table, HTML-Iframe, HTML-Form, Introduction of CSS, CSS Syntax, External Style Sheet using < link >, Multiple Style Sheets, Value Lengths and Percentages, CSS-Selectors, CSS-Box Model, Floats, Clear, Introduction to Bootstrap.

08

II

Scripting: Introduction to JavaScript, Creating Variables in JavaScript, Creating Functio

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	Text books:	
	1. Burdman, Jessica, "Collaborative Web Development" Addison Wesley	
	2. Xavier, C, "Web Technology and Design" , New Age International	
	3. Ivan Bayross," HTML, DHTML, Java Script, Perl & CGI", BPB Publication	
	4. Bhave, "Programming with Java", Pearson Education	
	6. Hans Bergsten, "Java Server Pages", SPD O'Reilly	
	7. Naughton, Schildt, "The Complete Reference JAVA2", TMH	
	8. Craig Walls, "Spring Boot in Action"	

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Text books:

1. Burdman, Jessica, "Collaborative Web Development" Addison Wesley
2. Xavier, C, "Web Technology and Design" , New Age International
3. Ivan Bayross," HTML, DHTML, Java Script, Perl & CGI", BPB Publication
4. Bhave, "Programming with Java", Pearson Education
6. Hans Bergsten, "Java Server Pages", SPD O'Reilly

7. Naughton, Schildt, “The Complete Reference JAVA2”, TMH

8. Craig Walls, “Spring Boot in Action”

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KCA022: Big Data			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO1	Demonstrate knowledge of Big Data Analytics concepts and its applications in business.		K , K 1 2
CO2	Demonstrate functions and components of Map Reduce Framework and HDFS.		K , K 1 2
CO3	Develop queries in NoSQL environment.		K 6
CO4	Explain process of developing Map Reduce based distributed processing applications.		K , K 2 5
CO5	Explain process of developing applications using HBASE, Hive, Pig etc.		K ,K 2 5
	DETAILED SYLLABUS		4-0-0
Unit	Topic		Proposed Lecture
I	Introduction to Big Data: Types of digital data, history of Big Data innovation, introduction to Big Data platform, drivers for Big Data, Big Data architecture and characteristics, 5 Vs of Big Data, Big Data technology components, Big Data importance and applications, Big Data features – security, compliance, auditing and protection, Big Data privacy and ethics, Big Data Analytics, Challenges of conventional systems, intelligent data analysis, nature of data, analytic processes and tools, analys		08

KCA022: Big Data			
II	Hadoop: History of Hadoop, Apache Hadoop, the Hadoop Distributed File System, components of Hadoop, data format, analyzing data with Hadoop, scaling out, Hadoop streaming, Hadoop pipes, Hadoop Echo System. Map-Reduce: Map-Reduce framework and basics, how Map Reduce works, developing a Map Reduce application, unit tests with MR unit, test data and local tests, anatomy of a Map Reduce job run, failures, job scheduling, shuffle and sort, task execution, Map Reduce types, input formats, output format		08
III	HDFS (Hadoop Distributed File System): Design of HDFS, HDFS concepts, benefits and challenges, file sizes, block sizes and block abstraction in HDFS, data replication, how does HDFS store, read, and write files, Java interfaces to HDFS, command line interface, Hadoop file system interfaces, data flow, data ingest with Flume and Scoop, Hadoop archives, Hadoop I/O: Compression, serialization, Avro and file-based data structures. Hadoop Environment: Setting up a Hadoop cluster, cluster specification		08
IV	Hadoop Eco System and YARN: Hadoop ecosystem components, schedulers, fair and capacity, Hadoop 2.0 New Features – Name Node high availability, HDFS federation, MRv2, YARN, Running MRv1 in YARN. NoSQL Databases: Introduction to NoSQL MongoDB: Introduction, data types, creating, updating and deleting documents, querying, introduction to indexing, capped collections Spark: Installing spark, spark applications, jobs, stages and tasks, Resilient Distributed Databases, anatomy of a Spark job run, Spark		08
V	Hadoop Eco System Frameworks: Applications on Big Data using Pig, Hive and HBase Pig : Introduction to PIG, Execution Modes of Pig, Comparison of Pig with Databases, Grunt, Pig Latin, User Defined Functions, Data Processing operators, Hive - Apache Hive architecture and installation, Hive shell, Hive services, Hive		08

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KCA022: Big Data

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO1

Demonstrate knowledge of Big Data Analytics concepts and its applications in business.

K1, K2

CO2

Demonstrate functions and components of Map Reduce Framework and HDFS.

K1, K2

CO3

Develop queries in NoSQL environment.

K6

CO4

Explain process of developing Map Reduce based distributed processing applications.

K2, K5

CO5

Explain process of developing applications using HBASE, Hive, Pig etc.

K2,K5

DETAILED SYLLABUS

4-0-0

Unit

Topic

Proposed

Lecture

I

Introduction to Big Data: Types of digital data, history of Big Data innovation, introduction to Big Data platform, drivers for Big Data, Big Data architecture and characteristics, 5 Vs of Big Data, Big Data technology components, Big Data importance and applications, Big Data features – security, compliance, auditing and

protection, Big Data privacy and ethics, Big Data Analytics, Challenges of conventional systems, intelligent data analysis, nature of data, analytic processes and tools, analysis vs reporting, modern data analytic tools.

08

II

Hadoop: History of Hadoop, Apache Hadoop, the Hadoop Distributed File System, components of Hadoop, data format, analyzing data with Hadoop, scaling out, Hadoop streaming, Hadoop pipes, Hadoop Echo System.

Map-Reduce:

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	metastore, comparison with traditional databases, HiveQL, tables, querying data and user defined functions, sorting and aggregating, Map Reduce scripts, joins & subqueries. HBase – Hbase concepts, clients, example, Hbase vs RDBMS, advanced usage, schema design, advance indexing, Zookeeper – how it helps in monitoring a cluster, how to build applications with Zookeeper. IBM Big Data strategy, introduction to Infosphere, BigInsights and Big Sheets, introduction to Big SQL.	
Suggested Readings: 1. Michael Minelli, Michelle Chambers, and Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley. 2. Big-Data Black Book, DT Editorial Services, Wiley. 3. Dirk deRoos, Chris Eaton, George Lapis, Paul Zikopoulos, Tom Deutsch, "Understanding Big Data Analytics for Enterprise Class Hadoop and Streaming Data", McGrawHill. 4. Thomas Erl, Wajid Khattak, Paul Buhler, "Big Data Fundamentals: Concepts, Drivers and Te		

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metastore, comparison with traditional databases, HiveQL, tables, querying data and user defined functions, sorting and aggregating, Map Reduce scripts, joins & subqueries.

HBase – Hbase concepts, clients, example, Hbase vs RDBMS, advanced usage, schema design, advance indexing, Zookeeper – how it helps in monitoring a cluster, how to build applications with Zookeeper. IBM Big Data strategy, introduction to Infosphere, BigInsights and Big Sheets, introduction to Big SQL.

Suggested Readings:

1. Michael Minelli, Michelle Chambers, and Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley.
2. Big-Data Black Book, DT Editorial Services, Wiley.
3. Dirk deRoos, Chris Eaton, George Lapis, Paul Zikopoulos, Tom Deutsch, "Understanding Big Data Analytics for Enterprise Class Hadoop and Streaming Data", McGrawHill.
4. Thomas Erl, Wajid Khattak, Paul Buhler, "Big Data Fundamentals: Concepts, Drivers and Techniques", Prentice Hall.
5. Bart Baesens "Analytics in a Big Data World: The Essential Guide to Data Science and its Applications (WILEY Big Data Series)", John Wiley & Sons
6. Arshdeep Bahga, Vijay Madisetti, "Big Data Science & Analytics: A Hands On Approach ", VPT
7. Anand Rajaraman and Jeffrey David Ullman, "Mining of Massive Datasets", CUP
8. Tom White, "Hadoop: The Definitive Guide", O'

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KCA023 : Simulation and Modelling			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course , the student will be able to understand			
CO 1	Study the concept of system, its components and types.		K 1
CO 2	Understand and analyze nature and techniques of major simulation models.		K , K 2 4
CO 3	Study and analyze the idea of continuous and discrete system simulation.		K , K 1 4
CO 4	Understand the notion of system dynamics and system dynamics diagrams.		K 2
CO 5	Finding critical path computation and understanding PERT networks		K , K 1 4
DETAILED SYLLABUS			3-1-0
Unit	Topic		Proposed Lecture
I	System definition and components, stochastic activities, continuous and discrete systems, System modeling, Types of models, static and dynamic physical models, static and dynamic mathematical models, full corporate model, types of system study.		08

KCA023 : Simulation and Modelling			
II	System simulation, Need of simulation, Basic nature of simulation, techniques of simulation, comparison of simulation and analytical methods, types of system Simulation, real time simulation, hybrid simulation, simulation of pursuit problem, single-server queuing system and an inventory problem, Monte-Carlo simulation, Distributed Lag model, Cobweb model.		08
III	Simulation of continuous Systems, analog vs digital simulation, simulation of water reservoir system, simulation of a servo system, simulation of an auto-pilot. Discrete system simulation, fixed time step vs. event-to-event model, generation of random numbers, test of randomness, Monte-Carlo computation vs. stochastic simulation.		08
IV	System dynamics, exponential growth models, exponential decay models, logistic curves, system dynamics diagrams, world model.		08
V	Simulation of PERT networks, critical path computation, uncertainties in activity duration, resource allocation and consideration, Simulation languages, object oriented simulation		08
Suggested Readings: 1. Geoffrey Gordon, "System Simulation", PHI 2. Narsingh Deo, "System Simulation with digital computer", PHI. 3. Averill M. Law and W. David Kelton, "Simulation Modelling and Analysis", TMH.			

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KCA023 : Simulation and Modelling

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course , the student will be able to understand

CO 1

Study the concept of system, its components and types.

K1

CO 2

Understand and analyze nature and techniques of major simulation models.

K2, K4

CO 3

Study and analyze the idea of continuous and discrete system simulation.

K1, K4

CO 4

Understand the notion of system dynamics and system dynamics diagrams.

K2

CO 5

Finding critical path computation and understanding PERT networks

K1, K4

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

System definition and components, stochastic activities, continuous and discrete systems, System modeling, Types of models, static and dynamic physical models, static and dynamic mathematical models, full corporate model, types of system study.

08

II

System simulation, Need of simulation, Basic nature of simulation, techniques of simulation, comparison of simulation and analytical methods, types of system Simulation, real time simulation, hybrid simulation, simulation of pursuit problem, single-server queuing system and an inventory problem, Monte-Carlo simulation, Distributed Lag model, Cobweb model.

08

III

Simulation of continuous Systems, analog vs digital simulation,
simulation of water reservoir system, simulation of a servo system,
simula

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KCA024: Software Testing &						Quality Assurance	
Course Outcome (CO)				Bloom's Knowledge Level (KL)			
At the end of course, the student will be able to understand							
CO 1	Test the software by applying testing techniques to deliver a product free from bugs.						
CO 2	Investigate the scenario and select the proper testing technique.						
CO 3	Explore the test automation concepts and tools and estimation of cost, schedule based on standard metrics.						
CO 4	Understand how to detect, classify, prevent and remove defects.						

KCA024: Software Testing &						Quality Assurance	
CO 5	Choose appropriate quality assurance models and develop quality. Ability to conduct formal inspections, record and evaluate results of inspections.						
DETAILED SYLLABUS							
Unit	Topic						
I	Software Testing Basics: Testing as an engineering activity, Role of process in software quality, Testing as a process, Basic definitions, Software testing principles, The tester's role in a software development organization, Origins of defects, Defect classes, The defect repository and test design, Defect examples, Developer / Tester support for developing a defect repository.						

KCA024: Software Testing &						Quality Assurance	
II	Testing Techniques and Levels of Testing: Using White Box Approach to Test design– Static Testing Vs. Structural Testing, Code Functional Testing, Coverage and Control Flow Graphs, Using Black Box Approaches to Test Case Design, Random Testing, Requirements based testing, Decision tables, State-based testing, Cause-effect graphing, Error guessing, Compatibility testing, Levels of Testing -Unit Testing, Integration Testing, Defect Bash Elimination. System Testing - Usability and Accessibility Tes						

KCA024: Software Testing &						Quality Assurance	
III	Software Test Automation And Quality Metrics: Software Test Automation, Skills needed for Automation, Scope of Automation, Design and Architecture for Automation, Requirements for a Test Tool, Challenges in Automation Tracking the Bug, Debugging. Testing Software System Security - Six-Sigma, TQM - Complexity Metrics and Models, Quality Management Metrics, Availability Metrics, Defect Removal Effectiveness, FMEA, Quality Function Deployment, Taguchi Quality Loss Function, Cost of Quality.						

KCA024: Software Testing &						Quality Assurance	
IV	Fundamentals of Software Quality Assurance: SQA basics, Components of the Software Quality Assurance System, software quality in business context, planning for software quality assurance, product quality and process quality, software process models, 7 QC Tools and Modern Tools.						
V		Software Assurance Models:	Models for Quality Assurance, ISO-9000 series,				
		CMM, CMMI, Test Maturity Models, SPICE, Malcolm Baldrige Model- P-					
		CMM.					
		Software Quality Assurance Trends:			Software Process- PSP and TSP, OO		
		Methodology, Clean room software engineering, Defect Injection and					
		prevention, Internal Auditing and Assessments, Inspections & Walkthroughs,					

KCA024: Software Testing &						Quality Assurance	
		Case Tools and their affect on Software Quality.					
Suggested Readings: 1. Srinivasan Desikan, Gopalaswamy Ramesh, "Software Testing: Principles and Practices", Pearson. 2. Daniel Galin, "Software Quality Assurance: From Theory to Implementat ion", Pearson							

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KCA024: Software Testing & Quality Assurance

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Test the software by applying testing techniques to deliver a product free from bugs.

K3

CO 2

Investigate the scenario and select the proper testing technique.

K1, K4

CO 3

Explore the test automation concepts and tools and estimation of cost, schedule based on standard metrics.

K2, K4

CO 4

Understand how to detect, classify, prevent and remove defects.

K1, K2

CO 5

Choose appropriate quality assurance models and develop quality. Ability to conduct formal inspections, record and evaluate results of inspections.

K3, K4

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Software Testing Basics: Testing as an engineering activity, Role of process in software quality, Testing as a process, Basic definitions, Software testing principles, The tester's role in a software development organization, Origins of defects, Defect classes, The defect repository and test design, Defect examples, Developer / Tester support for developing a defect repository.

08

II

Testing Techniques and Levels of Testing: Using White Box Approach to Test design– Static Testing Vs. Structural Testing, Code Functional Testing, Coverage and Control Flow Graphs, Using Black Box Approaches to Test

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Addison Wesley.

3. Aditya P. Mathur, "Foundations of Software Testing", Pearson.
4. Paul Ammann, Jeff Offutt, "Introduction to Software Testing", Cambridge University Press.
5. Paul C. Jorgensen, "Software Testing: A Craftsman's Approach", Auerbach Publications.
6. William Perry, "Effective Methods of Software Testing", Wiley Publishing, Third Edition.
7. Renu Rajani, Pradeep Oak, "Software Testing – Effective Methods, Tools and Techniques", Tata McGraw Hill.
8. Stephen Kan, "Metrics and Models in Software Quality", Addison – Wesley, Second Edition.
9. S. A. Kelkar, "Software quality and Testing", PHI Learning Pvt, Ltd.
10. Watts S Humphrey, "Managing the Software Process", Pearson Education Inc.

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KCA025: Digital Image Processing			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Explain the basic concepts of two-dimensional signal acquisition, sampling, quantization and color model.		K , K 1 2
CO 2	Apply image processing techniques for image enhancement in both the spatial and frequency domains.		K , K 2 3
CO 3	Apply and compare image restoration techniques in both spatial and frequency domain.		K , K 2 3
CO 4	Compare edge based and region based segmentation algorithms for ROI extraction.		K , K 3 4
CO 5	Explain compression techniques and descriptors for image processing.		K , K 2 3
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Digital Image Fundamentals: Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels – Color image fundamentals – RGB, HSI models, Two-dimensional mathematical preliminaries, 2D transforms – DFT, DCT.		08
II	Image Enhancement: Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering–Smoothing and Sharpening Spatial Filtering, Frequency Domain: Introduction to Fourier Transform– Smoothing and Sharpening frequency domain filters – Ideal, Butterworth and Gaussian filters, Homomorphic filtering, Color image enhancement.		08
III	Image Restoration: Image Restoration – degradation model, Properties, Noise models – Mean Filters – Order Statistics –Adaptive filters – Band reject Filters – Band pass Filters – Notch Filters – Optimum Notch Filtering – Inverse Filtering – Wiener filtering		08

KCA025: Digital Image Processing			
IV	Image Segmentation: Edge detection, Edge linking via Hough transform – Thresholding – Region based segmentation – Region growing – Region splitting and merging – Morphological processing- erosion and dilation, Segmentation by morphological watersheds – basic concepts – Dam construction – Watershed segmentation algorithm.		08
V	Image Compression and Recognition: Need for data compression, Huffman, Run Length Encoding, Shift codes, Arithmetic coding, JPEG standard, MPEG. Boundary representation, Boundary description, Fourier Descriptor, Regional Descriptors – Topological feature, Texture – Patterns and Pattern classes – Recognition based on matching.		08
Suggested Readings: 1. Rafael C. Gonzalez, Richard E. Woods, "Digital Image Processing", Pearson, Third Edition, 2010. 2. Anil K. Jain, "Fundamentals of Digital Image Processing", Pearson, 2002. 3. Kenneth R. Castleman, "Digital Image Processing" Pearson, 2006. 4. D, E. Dudgeon and R M. Mersereau, "Multidimensional Digital Signal Processing", Prentice Hall Professional Technical Reference, 1990. 5. William K. Pratt, "Digital Image Processing" John Wiley, New York, 2002. 6. Milan Sonka et al, "Im			

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KCA025: Digital Image Processing

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Explain the basic concepts of two-dimensional signal acquisition, sampling, quantization and color model.

K1, K2

CO 2

Apply image processing techniques for image enhancement in both the spatial and frequency domains.

K2, K3

CO 3

Apply and compare image restoration techniques in both spatial and frequency domain.

K2, K3

CO 4

Compare edge based and region based segmentation algorithms for ROI extraction.

K3, K4

CO 5

Explain compression techniques and descriptors for image processing.

K2, K3

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Digital Image Fundamentals: Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels – Color image fundamentals – RGB, HSI models, Two-dimensional mathematical preliminaries, 2D transforms – DFT, DCT.

08

II

Image Enhancement: Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering–Smoothing and Sharpening Spatial Filtering, Frequency Domain: Introduction to Fourier Transform– Smoothing and Sharpening frequency domain filters – Ideal, Butterworth and Gaussian filters, Homomorphic f

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KCA351: Artificial Intelligence Lab			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Study and understand AI tools such as Python / MATLAB.		K ,K 1 2
CO 2	Apply AI tools to analyze and solve common AI problems.		K , K 3 4
CO 3	Implement and compare various AI searching algorithms.		K 6
CO 4	Implement various machine learning algorithms.		K 6
CO 5	Implement various classification and clustering techniques.		K 6
DETAILED SYLLABUS			
1. Installation and working on various AI tools such as Python / MATLAB. 2. Programs to solve basic AI problems. 3. Implementation of different AI searching techniques. 4. Implementation of different game playing techniques. 5. Implementation of various knowledge representation techniques. 6. Program to demonstrate the working of Bayesian network. 7. Implementation of pattern recognition problems such as handwritten character/ digit recognition, speech recognition, etc. 8. Implementation of diff			
Note: TheInstructormayadd/delete /modify/tuneexperiments,whereverhe/shefeelsinajustifiedmanner.			

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KCA351: Artificial Intelligence Lab

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Study and understand AI tools such as Python / MATLAB.

K1,K2

CO 2

Apply AI tools to analyze and solve common AI problems.

K3, K4

CO 3

Implement and compare various AI searching algorithms.

K6

CO 4

Implement various machine learning algorithms.

K6

CO 5

Implement various classification and clustering techniques.

K6

DETAILED SYLLABUS

1. Installation and working on various AI tools such as Python / MATLAB.
2. Programs to solve basic AI problems.
3. Implementation of different AI searching techniques.
4. Implementation of different game playing techniques.
5. Implementation of various knowledge representation techniques.
6. Program to demonstrate the working of Bayesian network.
7. Implementation of pattern recognition problems such as handwritten character/ digit recognition, speech recognition, etc.
8. Implementation of different classification techniques.
9. Implementation of various clustering techniques.
10. Natural language processing tool development.

Note:

The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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KCA352: Software Engineering Lab			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Identify ambiguities, inconsistencies and incompleteness from a requirements specification and state functional and non-functional requirement.		K , K 2 4
CO 2	Identify different actors and use cases from a given problem statement and draw use case diagram to associate use cases with different types of relationship.		K , K 3 5

KCA352: Software Engineering Lab			
CO 3	Draw a class diagram after identifying classes and association among them.		K , K 4 5
CO 4	Graphically represent various UML diagrams and associations among them and identify the logical sequence of activities undergoing in a system, and represent them pictorially.		K , K 4 5
CO 5	Able to use modern engineering tools for specification, design, implementation and testing.		K , K 3 4
DETAILED SYLLABUS			
For any given case/ problem statement do the following; 1. Prepare a SRS document in line with the IEEE recommended standards. 2. Draw the use case diagram and specify the role of each of the actors. 3. Prepare state the precondition, post condition and function of each use case. 4. Draw the activity diagram. 5. Identify the classes. Classify them as weak and strong classes and draw the class diagram. 6. Draw the sequence diagram for any two scenarios. 7. Draw the collaboration diagram. 8. Draw			
Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner. Draw the deployment diagram			

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KCA352: Software Engineering Lab

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Identify ambiguities, inconsistencies and incompleteness from a requirements specification and state functional and non-functional requirement.

K2, K4

CO 2

Identify different actors and use cases from a given problem statement and draw use case diagram to associate use cases with different types of relationship.

K3, K5

CO 3

Draw a class diagram after identifying classes and association among them.

K4, K5

CO 4

Graphically represent various UML diagrams and associations among them and identify the logical sequence of activities undergoing in a system, and represent them pictorially.

K4, K5

CO 5

Able to use modern engineering tools for specification, design, implementation and testing.

K3, K4

DETAILED SYLLABUS

For any given case/ problem statement do the following;

1. Prepare a SRS document in line with the IEEE recommended standards.
2. Draw the use case diagram and specify the role of each of the actors.
3. Prepare state the precondition, post condition and function of each use case.
4. Draw the activity diagram.
5. Identify the classes. Classify them as weak and strong classes and draw the class diagram.
6. Draw the sequence diagram for any two scenarios.
7. Draw the

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SECOND YEAR SYLLABUS

SEMESTER-IV

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ELECTIVE-3

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KCA031: Privacy and Security in Online Social Media							
	Course Outcome (CO)						
	At the end of course, the student will be able to:						
	CO 1						Understand working of online social networks
	CO 2					Describe privacy policies of online social media	
CO 3							Analyse count ermeasures to control information sharing in Online social
							networks.
	CO 4					Apply knowledge of identity management in Online social networks	
	CO 5						Compare various privacy issues associated with popular social media.
	DETAILED SYLLABUS						
	Unit			Topic			
I				Introduction to Online Social Networks: Introduction to Social Networks, From			

KCA031: Privacy and Security in Online Social Media							
				offline to Online Communities, Online Social Networks, Evolution of Online			
				Social Networks, Analysis and Properties, Security Issues in Online Social			
				Networks, Trust Management in Online Social Networks, Controlled Information			
				Sharing in Online Social Networks, Identity Management in Online Social			
				Networks, data collection from social networks, challenges, opportunities, and			
				pitfalls in online social networks, APIs; Collecting data from Online Social Media.			
II				Trust Management in Online Social Networks: Trust and Policies, Trust and			

KCA031: Privacy and Security in Online Social Media							
				Reputation Systems, Trust in Online Social, Trust Properties, Trust Components,			
				Social Trust and Social Capital, Trust Evaluation Models, Trust, credibility, and			
				reputations in social systems; Online social media and Policing, Information			
				privacy disclosure, revelation, and its effects in OSM and online social networks;			
				Phishing in OSM & Identifying fraudulent entities in online social networks			
III				Controlled Information Sharing in Online Social Networks: Access Control			
				Models, Access Control in Online Social Networks, Rela tionship-Based Access			
				Control, Privacy Settings in Commercial Online Social Networks, Existing Access			
				Control Approaches			

KCA031: Privacy and Security in Online Social Media							
IV				Identity Management in Online Social Networks: Identity Management, Digital			
				Identity, Identity Management Models: From Identity 1.0 to Identity 2.0, Identity			
				Management in Online Social Networks, Identity as Self -Presentation, Identity			
				thefts, Open Security Issues in Online Social Networks			
V				Case Study: Privacy and security issues associated with various social media such			
				as Facebook, Instagram, Twitter, LinkedIn etc.			
	Textbooks:						
	1. Security and Privacy-Preser ving in Social Networks, Editors: Chbeir, Richard, Al Bouna,						
	Bechara (Eds.), Springer, 2013.						

KCA031: Privacy and Security in Online Social Media							
	2. Security and Trust in Online Social Networks, Barbara Carminati, Elena Ferrari, Marco Viviani,						
	Morgan & Claypool publications.						
	3. Security and Privacy in Social Networks, Editors: Altshuler, Y., Elovici, Y., Cremers, A.B.,						
	Aharony, N., Pentland, A. (Eds.), Springer, 2013						
	4. Security and privacy preserving in social networks, Elie Raad & Richard Chbeir, Richard						
	Chbeir& Bechara Al Bouna, 2013						
	5. Social Media Security: Leveraging Social Networking While Mitigating Risk, Michael Cross,						
	2013						

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KCA031: Privacy and Security in Online Social Media

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to:

CO 1

Understand working of online social networks

K2

CO 2

Describe privacy policies of online social media

K2

CO 3

Analyse countermeasures to control information sharing in Online social networks.

K3

CO 4

Apply knowledge of identity management in Online social networks

K3

CO 5

Compare various privacy issues associated with popular social media.

K3

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Introduction to Online Social Networks: Introduction to Social Networks, From offline to Online Communities, Online Social Networks, Evolution of Online Social Networks, Analysis and Properties, Security Issues in Online Social Networks, Trust Management in Online Social Networks, Controlled Information Sharing in Online Social Networks, Identity Management in Online Social Networks, data collection from social networks, challenges, opportunities, and pitfalls in online social networks, APIs; Collecting data from Online Social Media.

08

II

Trust Management in Online Social Networks: Trust and Policies, Trust and Reputation Systems, Trust in Online Social, Trust Properties, Trust Components,

Social Trust and Social Capital, Trust Evaluation Models, Trust, credibility, and reputations in so

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KCA032: Soft Computing				
Course Outcome (CO)		Bloom's Knowledge Level (KL)		
At the end of course, the student will be able to understand				
CO 1	Recognize the need of soft computing and study basic concepts and techniques of soft computing.		K , K 1 2	
CO 2	Understand the basic concepts of artificial neural network to analyze widely used neural networks.		K , K 2 4	
CO 3	Apply fuzzy logic to handle uncertainty in various real-world problems.		K 3	
CO 4	Study various paradigms of evolutionary computing and evaluate genetic algorithm in solving optimization problems.		K , K 1 5	
CO 5	Apply hybrid techniques in applications of soft computing.		K 3	
DETAILED SYLLABUS			3-0-0	
Unit	Topic		Proposed Lecture	
I	Introduction to Soft Computing: Introduction, Comparison with hard computing, Concept of learning and adaptation, Constituents of soft computing, Applications of soft computing. Artificial Neural Networks: Basic concepts of neural networks, Human brain, Biological neural network, History of artificial neural networks, Basic building blocks of an artificial neuron, Neural network architectures, Activation functions, Characteristics and limitation of neural networks.		08	

KCA032: Soft Computing				
II	Artificial Neural Networks: Learning methods - Supervised, Unsupervised, Reinforcement, Hebbian, Gradient descent, Competitive, Stochastic. Major classes of neural networks: Perceptron networks, Multilayer perceptron model, Back-propagation network, Radial basis function network, Recurrent neural network, Hopfield networks, Kohonen self-organizing feature maps.		08	
III	Fuzzy Logic: Introduction to Fuzzy Logic, Comparison with crisp logic, Properties of classical sets, Operations on classical sets, Properties of fuzzy sets, Operations on fuzzy sets, Classical relations, Fuzzy relations, Features and types of fuzzy membership functions, Fuzzy arithmetic, Fuzzy measures. Fuzzy Systems: Crisp logic, Predicate logic, Fuzzy logic, Fuzzy propositions, Inference rules, Fuzzy inference systems- Fuzzification, Inference, Defuzzification, Types of inference engines.		08	
V	Evolutionary Computing: Introduction, Evolutionary algorithm, Biological evolutionary process, Paradigms of evolutionary computing – Genetic algorithm and Genetic programming, Evolutionary strategies, Evolutionary programming. Genetic Algorithm: Introduction, Traditional optimization and search techniques, Comparison with traditional algorithms, Operations- Encoding, Selection, Crossover and Mutation, Classification of Genetic algorithm.		08	

KCA032: Soft Computing				
V	Hybrid Soft Computing Techniques: Introduction, Classification of hybrid systems, Neuro-fuzzy hybrid systems, Neuro-genetic hybrid systems, Fuzzy- genetic hybrid systems. Other Soft Computing Techniques: Tabu Search, Ant colony based		08	

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KCA032: Soft Computing

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Recognize the need of soft computing and study basic concepts and techniques of soft computing.

K1, K2

CO 2

Understand the basic concepts of artificial neural network to analyze widely used neural networks.

K2, K4

CO 3

Apply fuzzy logic to handle uncertainty in various real-world problems.

K3

CO 4

Study various paradigms of evolutionary computing and evaluate genetic algorithm in solving optimization problems.

K1, K5

CO 5

Apply hybrid techniques in applications of soft computing.

K3

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Introduction to Soft Computing: Introduction, Comparison with hard computing, Concept of learning and adaptation, Constituents of soft computing, Applications of soft computing.

Artificial Neural Networks: Basic concepts of neural networks, Human brain, Biological neural network, History of artificial neural networks, Basic building blocks of an artificial neuron, Neural network architectures, Activation functions, Characteristics and limitation of neural networks.

08

II

Artificial Neural Networks: Learning methods - Supervised, Unsupervised, Reinforcement, Hebbian, Gradient descent, Competitive, Stochastic.

Major classes of neural networks: Perceptron ne

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	optimization, Swarm Intelligence.	
Suggested Readings: 1. Sivanandam S.N. and Deepa S.N., "Principles of Soft Computing", Wiley-India. 2. Rajasekaran S. and Vijayalakshmi Pai G.A., "Neural Networks, Fuzzy Logic and Genetic Algorithms- Synthesis and Applications", PHI Learning. 3. Chakraverty S., Sahoo D.M. and Mahato N. R., "Concepts of Soft Computing- Fuzzy and ANN with Programming", Springer. 4. Kaushik S. and Tiwari S., "Soft Computing – Fundamentals, Techniques and Applications", McGrawHill Education. 5. Jang J.-S.R., Sun C.-		

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optimization, Swarm Intelligence.

Suggested Readings:

1. Sivanandam S.N. and Deepa S.N., "Principles of Soft Computing", Wiley-India.
2. Rajasekaran S. and Vijayalakshmi Pai G.A., "Neural Networks, Fuzzy Logic and Genetic Algorithms- Synthesis and Applications", PHI Learning.

3. Chakraverty S., Sahoo D.M. and Mahato N. R., “Concepts of Soft Computing- Fuzzy and ANN with Programming”, Springer.
4. Kaushik S. and Tiwari S., “Soft Computing – Fundamentals, Techniques and Applications’, McGrawHill Education.
5. Jang J.-S.R., Sun C.-T. and Mizutani E., “Neuro-Fuzzy and Soft Computing”, Prentice-Hall of India.
6. Karray F. O. and Silva C. D., “Soft Computing and Intelligent Systems Design – Theory, Tools and Applications”, Pearson Education.
7. Freeman J. A. and Skapura D. M., “Neural Networks: Algorithms, Applications and Programming Techniques”, Pearson.
8. Siman H., “Neural Netowrks”, Prentice Hall of India.

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KCA033: Pattern Recognition			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Study of basics of Pattern recognition. Understand the designing principles and Mathematical foundation used in pattern recognition.		K , K 1 2
CO 2	Analysis the Statistical Patten Recognition.		K K 3, 4
CO 3	Understanding the different Parameter estimation methods.		K , K 1 2
CO 4	Understanding the different Nonparametric Techniques.		K , K 1 2,
CO 5	Understand and Make use of unsupervised learning and Clustering in Pattern recognition.		K K K 2 3, 4
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Introduction: Basics of pattern recognition, Design principles of pattern recognition system, Learning and adaptation, Pattern recognition approaches, Mathematical foundations – Linear algebra, Probability Theory, Expectation, mean and covariance, Normal distribution, multivariate normal densities, Chi squared test.		08
II	Statistical Patten Recognition: Bayesian Decision Theory, Classifiers, Normal density and discriminant functions		08

KCA033: Pattern Recognition			
III	Parameter estimation methods: Maximum-Likelihood estimation, Bayesian Parameter estimation, Dimension reduction methods - Principal Component Analysis (PCA), Fisher Linear discriminant analysis, Expectation-maximization (EM), Hidden Markov Models (HMM), Gaussian mixture models.		08
IV	Nonparametric Techniques: Density Estimation, Parzen Windows, K- Nearest Neighbor Estimation, Nearest Neighbor Rule, Fuzzy classification.		08
V	Unsupervised Learning & Clustering: Criterion functions for clustering, Clustering Techniques: Iterative square - error partitional clustering – K means, agglomerative hierarchical clustering, Cluster validation.		08
Suggested Readings: 1. Duda R. O., Hart P. E. and Stork D. G., "Pattern Classification", John Wiley. 2. Bishop C. M., "Neural Network for Pattern Recognition", Oxford University Press. 3. Singhal R., "Pattern Recognition: Technologies & Applications", Oxford University Press. 4. Theodoridis S. and Koutroumbas K., "Pattern Recognition", Academic Press.			

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KCA033: Pattern Recognition

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Study of basics of Pattern recognition. Understand the designing principles and Mathematical foundation used in pattern recognition.

K1, K2

CO 2

Analysis the Statistical Patten Recognition.

K3, K4

CO 3

Understanding the different Parameter estimation methods.

K1, K2

CO 4

Understanding the different Nonparametric Techniques.

K1, K2,

CO 5

Understand and Make use of unsupervised learning and Clustering in Pattern recognition.

K2 K3, K4

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Introduction: Basics of pattern recognition, Design principles of pattern recognition system, Learning and adaptation, Pattern recognition approaches, Mathematical foundations – Linear algebra, Probability Theory, Expectation, mean and covariance, Normal distribution, multivariate normal densities, Chi squared test.

08

II

Statistical Pattern Recognition: Bayesian Decision Theory, Classifiers, Normal density and discriminant functions

08

III

Parameter estimation methods: Maximum-Likelihood estimation, Bayesian Parameter estimation, Dimension reduction methods - Principal Component

Analysis

(PCA),

Fisher

Linear

discriminant

analysis,

Expectation-

maximization (EM), Hidden Markov Models (HMM), Gaussian mixt

Page 69 Data Table

KCA034: Data Analytics				
Course Outcome (CO)		Bloom's Knowledge Level (KL)		
At the end of course, the student will be able to understand				
CO1	Describe the life cycle phases of Data Analytics through discovery, planning and building.		K , K 1 2	
CO2	Understand and apply Data Analysis Techniques.		K , K 2 3	
CO3	Implement various Data streams.		K 3	
CO4	Understand item sets, Clustering, frame works & Visualizations.		K 2	
CO5	Apply R tool for developing and evaluating real time applications.		K , K , K 3 5 6	
	DETAILED SYLLABUS		4-0-0	
Unit	Topic		Proposed Lecture	
I	Introduction to Data Analytics: Sources and nature of data, classification of data (structured, semi-structured, unstructured), characteristics of data, introduction to Big Data platform, need of data analytics, evolution of analytic scalability, analytic process and tools, analysis vs reporting, modern data analytic tools, applications of data analytics. Data Analytics Lifecycle: Need, key roles for successful analytic projects, various phases of data analytics lifecycle – discovery, data prepa		08	

KCA034: Data Analytics				
II	Data Analysis: Regression modeling, multivariate analysis, Bayesian modeling, inference and Bayesian networks, support vector and kernel methods, analysis of time series: linear systems analysis & nonlinear dynamics, rule induction, Neural Networks: Learning and generalisation, competitive learning, principal component analysis and neural networks, fuzzy logic: extracting fuzzy models from data, fuzzy decision trees, stochastic search methods.		08	
III	Mining Data Streams: Introduction to streams concepts, stream data model and architecture, stream computing, sampling data in a stream, filtering streams, counting distinct elements in a stream, estimating moments, counting oneness in a window, decaying window, Real-time Analytics Platform (RTAP) applications, Case studies – Real time sentiment analysis, stock market predictions.		08	
IV	Frequent Itemsets and Clustering: Mining frequent itemsets, market based modelling, Apriori algorithm, handling large data sets in main memory, limited pass algorithm, counting frequent itemsets in a stream, Clustering techniques: hierarchical, K-means, clustering high dimensional data, CLIQUE and ProCLUS, frequent pattern based clustering methods, clustering in non-euclidean space, clustering for streams and parallelism.		08	

KCA034: Data Analytics				
V	Frame Works and Visualization: MapReduce, Hadoop, Pig, Hive, HBase, MapR, Sharding, NoSQL Databases, S3, Hadoop Distributed File Systems, Visualization: visual data analysis techniques, interaction techniques, systems and applications. Introduction to R - R graphical user interfaces, data import and export, attribute and data types, descriptive statistics, exploratory data analysis, visualization before analysis, analytics for unstructured data.		08	
Suggested Readings: 1. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer. 2. Anand Rajaraman and Jeffrey David Ullman, "Mining of Massive Datasets", Cambridge University Press. 3. Bill Franks, "Taming the Big Data Tidal wave: Finding Opportunities in Huge Data Streams				

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KCA034: Data Analytics

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO1

Describe the life cycle phases of Data Analytics through discovery, planning and building.

K1, K2

CO2

Understand and apply Data Analysis Techniques.

K2, K3

CO3

Implement various Data streams.

K3

CO4

Understand item sets, Clustering, frame works & Visualizations.

K2

CO5

Apply R tool for developing and evaluating real time applications.

K3, K5, K6

DETAILED SYLLABUS

4-0-0

Unit

Topic

Proposed

Lecture

I

Introduction to Data Analytics: Sources and nature of data, classification of data (structured, semi-structured, unstructured), characteristics of data, introduction to Big Data platform, need of data analytics, evolution of analytic scalability, analytic process and tools, analysis vs reporting, modern data analytic tools, applications of data analytics.

Data Analytics Lifecycle: Need, key roles for successful analytic projects, various phases of data analytics lifecycle – discovery, data preparation, model planning, model building, communicating results, operationalization

08

II

Data Analysis: Regression modeling, multivariate analysis, Bayesian modeling, inference and Bayesian networks, support vector and kernel methods, analysis of time series: linear systems analysis & nonlinear dynamics, rule

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with Advanced Analytics", John Wiley & Sons.

4. John Garrett, "Data Analytics for IT Networks : Developing Innovative Use Cases", Pearson Education.

5. Michael Minelli, Michelle Chambers, and Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley.

6. David Dietrich, Barry Heller, Beibei Yang, "Data Science and Big Data Analytics", EMC

Education Series, John Wiley.

7. Frank J Ohlhorst, "Big Data Analytics: Turning Big Data into Big Money", Wiley and SAS Business Series.

8. Colleen Mccue, "Data Mining and Predictive Analysis: Intelligence Gathering and Crime Analysis", Elsevier.

9. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer.

10. Paul Zikopoulos, Chris Eaton, Paul Zikopoulos, "Understanding Big Data: Analytics for Enterprise Class Hadoop and Streaming Data", McGraw Hill.

11. Trevor Hastie, Robert Tibshirani, Jerome Friedman, "The Elements of Statistical Learning", Springer.

12. Mark Gardner, "Beginning R: The Statistical Programming Language", Wrox Publication.

13. Pete Warden, "Big Data Glossary", O'Reilly.

14. Glenn J. Myatt, "Making Sense of Data", John Wiley & Sons.

15. Peter Bühlmann, Petros Drineas, Michael Kane, Mark van der Laan, "Handbook of Big Data", CRC Press.

16. Jiawei Han, Micheline Kamber "Data Mining Concepts and Techniques", Second Edition,

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KCA035: Software Quality Engineering							
	Course Outcome (CO)						
	At the end of course, the student will be able to:						
CO 1							Understand basic concepts of Software Quality along with its documents and
							process
	CO 2					Apply knowledge of Software Quality in various types of software	
	CO 3						Compare the various reliability models for different scenarios

KCA035: Software Quality Engineering							
	CO 4						Illustrate the software Quality Planning and Assurance
	CO 5						Make use of various testing techniques in software implementation
	DETAILED SYLLABUS						
	Unit			Topic			
I				Software Quality: Definition, Software Quality Attributes and Specification, Cost			
				of Quality, Defects, Faults, Failures, Defect Rate and Reliability, Defect			
				Prevention, Reduction, and Containment, Overview of Different Types of Software			
				Review, Introduction to Measurement and Inspection Process, Documents and			
				Metrics.			
II				Software Quality Metrics Product Quality Metrics: Defect Density, Customer			

KCA035: Software Quality Engineering							
				Problems Metric, Customer Satisfaction Metrics, Function Points, In-Process			
				Quality Metrics: Defect Arrival Pattern, Phase-Based Defect Removal Pattern,			
				Defect Removal Effectiveness, Metrics for Software Maintenance: Backlog			
				Management Index, Fix Response Time, Fix Quality, Software Quality Indicators.			
III				Software Quality Management and Models: Modeling Process, Software			
				Reliability Models: The Rayleigh Model, Exponential Distribution and Software			
				Reliability Growth Models, Software Reliability Allocation Models, Criteria for			

KCA035: Software Quality Engineering							
				Model Evaluation, Software Quality Assessment Models: Hierarchical Model of			
				Software Quality Assessment.			
IV				Software Quality Assurance: Quality Planning and Control, Quality Improvement			
				Process, Evolution of Software Quality Assurance (SQA), Major SQA Activities,			
				Major SQA Issues, Zero Defect Software, SQA Techniques, Statistical Quality			
				Assurance, Total Quality Management, Quality Standards and Processes.			
V				Software Verification, Validation & Testing: Verification and Validation,			
				Evolutionary Nature of Verification and Validation, Impracticality of Testing all			

KCA035: Software Quality Engineering							
				Data and Paths, Proof of Correctness, Software Testing, Functional, Structural and			
				Error-Oriented Analysis & Testing, Static and Dynamic Testing Tools,			
				Characteristics of Modern Testing Tools.			
	Text books:						
	1. Jeff Tian, Software Quality Engineering (SQE), Wiley-I nterscience, 2005; ISBN 0-471-						
	71345 -7						
	2. Metrics and Models in Software Quality Engineering, Stephen H. Kan, Addison Wesley						
	(2002), ISBN: 0201729156						
	3. Norman E. Fenton and Shari Lawrence Pfleeger, "Software Metrics" Thomson, 2003						
	4. Mordechai Ben – Menachem and Garry S.Marliss, "Software Quality", Thomson Asia						
	Pte Ltd, 2003.						

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KCA035: Software Quality Engineering

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to:

CO 1

Understand basic concepts of Software Quality along with its documents and process

K2

CO 2

Apply knowledge of Software Quality in various types of software

K3

CO 3

Compare the various reliability models for different scenarios

K4

CO 4

Illustrate the software Quality Planning and Assurance

K2

CO 5

Make use of various testing techniques in software implementation

K3

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Software Quality: Definition, Software Quality Attributes and Specification, Cost of Quality, Defects, Faults, Failures, Defect Rate and Reliability, Defect Prevention, Reduction, and Containment, Overview of Different Types of Software Review, Introduction to Measurement and Inspection Process, Documents and Metrics.

08

II

Software Quality Metrics Product Quality Metrics: Defect Density, Customer Problems Metric, Customer Satisfaction Metrics, Function Points, In-Process Quality Metrics: Defect Arrival Pattern, Phase-Based Defect Removal Pattern, Defect Removal Effectiveness, Metrics for Software Maintenance: Backlog Management Index, Fix Response Time, Fix Quality, Software Quality Indicators.

08

III

Software Quality Management and Models: Modeling Process, Software Reliability

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ELECTIVE-4

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KCA041: Blockchain Architecture			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO1	Study and understand basic concepts of blockchain architecture.		K , K 1 2
CO2	Analyze various requirements for consensus protocols.		K 4
CO3	Apply and evaluate the consensus process.		K , K 3 5
CO4	Understand the concepts of Hyperledger fabric.		K 1
CO5	Analyze and evaluate various use cases in financial software and supply chain.		K , K 4 5
	DETAILED SYLLABUS		4-0-0
Unit	Topic		Proposed Lecture
I	Introduction to Blockchain: Digital Money to Distributed Ledgers, Design Primitives: Protocols, Security, Consensus, Permissions, Privacy. Blockchain Architecture and Design: Basic crypto primitives: Hash, Signature, Hashchain to Blockchain, Bitcoin Basic, Basic consensus mechanisms.		08

KCA041: Blockchain Architecture			
II	Consensus: Requirements for the consensus protocols, Proof of Work (PoW), Scalability aspects of Blockchain consensus protocols, distributed consensus, consensus in Bitcoin. Permissioned Blockchains: Design goals, Consensus protocols for Permissioned Blockchains		08
III	Hyperledger Fabric: Decomposing the consensus process, Hyperledger fabric components. Chaincode Design and Implementation Hyperledger Fabric: Beyond Chaincode: fabric SDK and Front End, Hyperledger composer tool.		08
IV	Use case 1: Blockchain in Financial Software and Systems (FSS): (i) Settlements, (ii) KYC, (iii) Capital markets, (iv) Insurance. Use case 2: Blockchain in trade/supply chain: (i) Provenance of goods, visibility, trade/supply chain finance, invoice management discounting, etc.		08
V	Use case 3: Blockchain for Government: (i) Digital identity, land records and other kinds of record keeping between government entities, (ii) public distribution system social welfare systems, Blockchain Cryptography, Privacy and Security on Blockchain		08
Suggested Readings: 1. Andreas Antonopoulos, "Mastering Bitcoin: Unlocking Digital Cryptocurrencies", O'Reilly 2. Melanie Swa, "Blockchain", O'Reilly 3. "Hyperledger Fabric", https://www.hyperledger.org/projects/fabric 4. Bob Dill, David Smits, "Zero to Blockchain - An IBM Redbooks course", https://www.redbooks.ibm.com/Redbooks.nsf/RedbookAbstracts/crse0401.html			

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KCA041: Blockchain Architecture

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO1

Study and understand basic concepts of blockchain architecture.

K1, K2

CO2

Analyze various requirements for consensus protocols.

K4

CO3

Apply and evaluate the consensus process.

K3, K5

CO4

Understand the concepts of Hyperledger fabric.

K1

CO5

Analyze and evaluate various use cases in financial software and supply chain.

K4, K5

DETAILED SYLLABUS

4-0-0

Unit

Topic

Proposed

Lecture

I

Introduction to Blockchain: Digital Money to Distributed Ledgers, Design

Primitives: Protocols, Security, Consensus, Permissions, Privacy.

Blockchain Architecture and Design: Basic crypto primitives: Hash, Signature,

Hashchain to Blockchain, Bitcoin Basic, Basic consensus mechanisms.

08

II

Consensus: Requirements for the consensus protocols, Proof of Work (PoW),

Scalability aspects of Blockchain consensus protocols, distributed consensus, consensus in Bitcoin.

Permissioned Blockchains: Design goals, Consensus protocols for Permissioned Blockchains

08

III

Hyperledger Fabric: Decomposing the consensus process, Hyperledger fabric components.

Chaincode Design and Implementation Hyperledger Fabric: Beyond Chaincode: fabric SDK and Front End, Hyperledger composer tool.

08

IV

Use case 1: Blockchain in Financi

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KCA042: Neural Networks			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Study of basic concepts of Neuro Computing, Neuroscience and ANN. Understand the different supervised and unsupervised and neural networks performance.		K , K 1 2
CO 2	Study of basic Models of neural network. Understand the Perception network. and Compare neural networks and their algorithm.		K K 2, 3
CO 3	Study and Demonstrate different types of neural network. Make use of neural networks for specified problem domain.		K K K 2 3, 4
CO 4	Understand and Identify basic design requirements of recurrent network and Self- organizing feature map.		K , K 1 2
CO 5	Able to understand the some special network. Able to understand the concept of Soft computing.		K , K K 1 2 3
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Neurocomputing and Neuroscience: The human brain, biological neurons, neural processing, biological neural network. Artificial Neural Networks: Introduction, historical notes, neuron model, knowledge representation, comparison with biological neural network, applications. Learning process: Supervised learning, unsupervised learning, error correction learning, competitive learning, adaptation learning, Statistical nature of the learning process.		08

KCA042: Neural Networks			
II	Basic Models: McCulloch-Pitts neuron model, Hebb net, activation functions, aggregation functions. Perceptron networks: Perceptron learning, single layer perceptron networks, multilayer perceptron networks. Least mean square algorithm, gradient descent rule, nonlinearly separable problems and benchmark problems in NN.		08
III	Multilayer neural network: Introduction, comparison with single layer networks. Back propagation network: Architecture, back propagation algorithm, local minima and global minima, heuristics for making back propagation algorithm performs better, applications. Radial basis function network: Architecture, training algorithm, approximation properties of RBF networks, comparison of radial basis function network and back propagation networks.		08
IV	Recurrent network: Introduction, architecture and types. Self-organizing feature map: Introduction, determining winner, Kohonen Self Organizing feature maps (SOM) architecture, SOM algorithm, properties of feature map; Learning vector quantization-architecture and algorithm. Principal component and independent component analysis.		08
V	Special networks: Cognitron, Support vector machines. Complex valued NN and complex valued BP. Soft computing: Introduction, Overview of techniques, Hybrid soft computing techniques.		08
Suggested Readings: 1. Kumar S., "Neural Networks- A Classroom Approach", McGraw Hill. 2. Haykin S., "Neural Networks – A Comprehensive Foundation", Pearson Education. 3. Yegnanarayana B. "Artificial Neural Networks", Prentice Hall of India. 4. Freeman J. A., "Neural Networks", Pearson Education. 5. James F., "Neural Networks – Algorithms, Applications and Programming Techniques", Pearson Education.			

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KCA042: Neural Networks

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Study of basic concepts of Neuro Computing, Neuroscience and ANN. Understand the different supervised and unsupervised and neural networks performance.

K1, K2

CO 2

Study of basic Models of neural network. Understand the Perception network. and Compare neural networks and their algorithm.

K2, K3

CO 3

Study and Demonstrate different types of neural network. Make use of neural networks for specified problem domain.

K2 K3, K4

CO 4

Understand and Identify basic design requirements of recurrent network and Self-organizing feature map.

K1, K2

CO 5

Able to understand the some special network. Able to understand the concept of Soft computing.

K1, K2 K3

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Neurocomputing and Neuroscience: The human brain, biological neurons, neural processing, biological neural network.

Artificial Neural Networks: Introduction, historical notes, neuron model, knowledge representation, comparison with biological neural network, applications.

Learning process: Supervised learning, unsupervised learning, error correction learning, competitive learning, adaptation learning, Statistical nature of the learning process.

08

II

Basic Models: McCulloch-Pitts neuron model,

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KCA043: Internet of Things				
Course Outcome (CO)			Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand				
CO 1		Demonstrate basic concepts, principles and challenges in IoT.		K1,K2
CO 2		Illustrate functioning of hardware devices and sensors used for IoT.		K2
CO 3		Analyze network communication aspects and protocols used in IoT.		K4
CO 4		Apply IoT for developing real life applications using Arduino programming.		K3
CP 5		To develop IoT infrastructure for popular applications		K K 2, 3
DETAILED SYLLABUS				3-1-0
Unit	Topic			Proposed Lecture
I	Internet of Things (IoT): Vision, Definition, Conceptual Framework, Architectural view, technology behind IoT, Sources of the IoT, M2M Communication, IoT Examples. Design Principles for Connected Devices: IoT/M2M systems layers and design standardization, communication technologies, data enrichment and consolidation, ease of designing and affordability			08

KCA043: Internet of Things				
II	Hardware for IoT: Sensors, Digital sensors, actuators, radio frequency identification (RFID) technology, wireless sensor networks, participatory sensing technology. Embedded Platforms for IoT: Embedded computing basics, Overview of IOT supported Hardware platforms such as Arduino, NetArduino, Raspberry pi, Beagle Bone, Intel Galileo boards and ARM cortex.			08
III	Network & Communication aspects in IoT: Wireless Medium access issues, MAC protocol survey, Survey routing protocols, Sensor deployment & Node discovery, Data aggregation & dissemination			08
IV	Programming the Arduino: Arduino Platform Boards Anatomy, Arduino IDE, coding, using emulator, using libraries, additions in arduino, programming the arduino for IoT.			08
V	Challenges in IoT Design challenges: Development Challenges, Security Challenges, Other challenges IoT Applications: Smart Metering, E-health, City Automation, Automotive Applications, home automation, smart cards, communicating data with H/W units, mobiles, tablets, Designing of smart street lights in smart city.			08

KCA043: Internet of Things				
Text books: 1. Olivier Hersent, David Boswarthick, Omar Elloumi "The Internet of Things key applications and protocols", Wiley 2. Jeeva Jose, Internet of Things, Khanna Publishing House 3. Michael Miller "The Internet of Things" by Pearson 4. Raj Kamal "INTERNET OF THINGS", McGraw-Hill, 1ST Edition, 2016 5. Arshdeep Bahga, Vijay Madisetti "Internet of Things (A hands on approach)" 1ST edition, VPI publications, 2014 6. Adrian McEwen, Hakin Cassimally "Designing the Internet of Things" Wiley India				

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KCA043: Internet of Things

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Demonstrate basic concepts, principles and challenges in IoT.

K1, K2

CO 2

Illustrate functioning of hardware devices and sensors used for IoT.

K2

CO 3

Analyze network communication aspects and protocols used in IoT.

K4

CO 4

Apply IoT for developing real life applications using Arduino programming.

K3

CP 5

To develop IoT infrastructure for popular applications

K2, K3

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Internet of Things (IoT): Vision, Definition, Conceptual Framework, Architectural view, technology behind IoT, Sources of the IoT, M2M Communication, IoT Examples. Design Principles for Connected Devices: IoT/M2M systems layers and design standardization, communication technologies, data enrichment and consolidation, ease of designing and affordability

08

II

Hardware for IoT: Sensors, Digital sensors, actuators, radio frequency identification (RFID) technology, wireless sensor networks, participatory sensing technology. Embedded Platforms for IoT: Embedded computing basics, Overview of IOT supported Hardware platforms such as Arduino, NetArduino, Raspberry pi, Beagle Bone, Intel Galileo boards and ARM cortex.

08

III

Network & Communication aspects in IoT: Wireless Medium acc

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	KCA044: Modern Application Development						
	Course Outcome (CO)						Bloom's Knowledge Level (KL)
	At the end of course , the student will be able to:						
CO 1				Understand the fundamental of Kotlin Programing for Android Application			
				Development.			

	KCA044: Modern Application Development						
	CO 2		Describe the UI Layout and architecture of Android Operating System.				
CO 3				Designing android application using Jetpack Library based on MVVM			
				Architecture.			
CO 4				Developing android application based on REST API using Volley and Retrofit			
				Library.			
	CO 5			Ability to debug the Performance and Security of Android Applications.			
	DETAILED SYLLABUS						
	Unit			Topic			
I				Kotlin Fundamental: Introduction to Kotlin, Basic Syntax, Idioms, Coding			
				Conventions, Basics, Basic Types, Packages, Control Flow, Returns and Jumps,			
				Classes and Objects, Classes and Inheritance, Properties and Fields, Interfaces,			

	KCA044: Modern Application Development						
				Visibility Modifiers, Extensions, Data Classes, Generics, Nested Classes, Enum			
				Classes, Objects, Delegation, Delegated Properties, Functions and Lambdas,			
				Functions, Lambdas, Inline Functions, Higher-Order Functions, Scope Functions,			
				Collections, Ranges, Type Checks and Casts, This expressions, Equality, Operator			
				overloading, Null Safety, Exceptions, Annotations, Reflection.			
II				Android Fundamental: Android Architecture: Introduction to Android,			
				Layouts, Views and Resources, Activities and Intents, Activity Lifecycle and			
				Saving State, Implicit or Explicit Intents.			
				User Interaction and Intuitive Navigation: Material Design, Theme, Style and			

	KCA044: Modern Application Development						
				Attributes, Input Controls, Menus, Widgets, Screen Navigation, Recycler View,			
				ListView, Adap ters, Drawables , Notifications.			
III				Storing, Sharing and Retrieving Data in Android Applications: Overview to			
				storing data, shared preferences, App settings, Store and query data in Android's			
				SQLite database, Content Providers, Content Resolver, Loading data using			
				loaders.			
				Jetpack Components : Fragments, Jetpack Navigation, Lifecycle, Lifecycle			
				Observer, Lifecycle Owner, View Model, View Model Factory, View Model			
				Provider, LiveData, Room API, Data Binding, View Binding, MVVM			
				Architecture Basics			

	KCA044: Modern Application Development						
IV				Asynchronous Data Handling, Networking and Files: Asynchronous Task,			
				Coroutines, API Handling, JSON Parsing, Volley Library, Retrofit Library, File			
				Handling, HTML and XML Parsing, Broadcast receivers, Services			

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KCA044: Modern Application Development

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course , the student will be able to:

CO 1

Understand the fundamental of Kotlin Programing for Android Application Development.

K2

CO 2

Describe the UI Layout and architecture of Android Operating System.

K3

CO 3

Designing android application using Jetpack Library based on MVVM Architecture.

K6

CO 4

Developing android application based on REST API using Volley and Retrofit Library.

K6

CO 5

Ability to debug the Performance and Security of Android Applications.

K5

DETAILED SYLLABUS

3-1-0

Unit

Topic

Proposed

Lecture

I

Kotlin Fundamental: Introduction to Kotlin, Basic Syntax, Idioms, Coding Conventions, Basics, Basic Types, Packages, Control Flow, Returns and Jumps, Classes and Objects, Classes and Inheritance, Properties and Fields, Interfaces, Visibility Modifiers, Extensions, Data Classes, Generics, Nested Classes, Enum Classes, Objects, Delegation, Delegated Properties, Functions and Lambdas, Functions, Lambdas, Inline Functions, Higher-Order Functions, Scope Functions, Collections, Ranges, Type Checks and Casts, This expressions, Equality, Operator overloading, Null Safety, Exceptions, Annotations, Reflection.

08

II

Android Fundamental: Android Architecture: Introduction to Android, Layouts, Views and Resources, Activities and Inte

Page 77 Data Table

V			Permissions, Performance and Security:		08	
			Firebase, AdMob, APK Singing, Publish App, Packaging and deployment,			
			Google Maps, GPS and Wi-Fi, Download Manager, Work Manager, Alarms,			
			Location, Map and Sensors, APK Singing, Publish App			
	Text books:					

V			Permissions, Performance and Security:		08	
	1. Meier R., "Professional Android 2 Application Development", Wiley.					
	2. Hashimi S., Komatineni S. and MacLean D., "Pro Android 2", Apress.					
	3. Murphy M., "Beginning Android 2", Apress.					
	4. Delessio C. and Darcey L., "Android Application Development", Pearson Education.					
	5. DiMarzio J.F., "Android a Programming Guide", Tata McGraw Hill.					

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V

Permissions, Performance and Security:

Firestore, AdMob, APK Signing, Publish App, Packaging and deployment, Google Maps, GPS and Wi-Fi, Download Manager, Work Manager, Alarms, Location, Map and Sensors, APK Signing, Publish App

08

Text books:

1. Meier R., "Professional Android 2 Application Development", Wiley.
2. Hashimi S., Komatineni S. and MacLean D., "Pro Android 2", Apress.
3. Murphy M., "Beginning Android 2", Apress.
4. Delessio C. and Darcey L., "Android Application Development", Pearson Education.
5. DiMarzio J.F., "Android a Programming Guide", Tata McGraw Hill.

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	KCA045: Distributed Database Systems						
	Course Outcome (CO)						
	At the end of course , the student will be able to:						
	CO 1			Understand theoretical and practical aspects of distributed database systems.			
CO 2				Study and identify various issues related to the development of distributed			
				database system			
CO 3				Understand the design aspects of object-oriented database system and related			
				development			
CO 4			Equip students with principles and knowledge of distributed reliability.	Equip students with principles and knowledge of distributed reliability.			
CO 5				Equip students with principles and knowledge of parallel and object-oriented			
				databases.			
	DETAILED SYLLABUS						
Unit	Unit		Topic	Topic			
I				Introduction: Distributed Data Processing, Distributed Database System,			

	KCA045: Distributed Database Systems						
				Promises of DDBSs, Problem areas. Distributed DBMS Architecture:			
				Architectural Models for Distributed DBMS, DDMBs Architecture. Distributed			
				Database Design: Alternative Design Strategies, Distribution Design issues,			
				Fragmentation, Allocation.			
II				Query processing and decomposition: Query processing objectives,			
				characterizatio n of query processors, layers of query processing, query			
				decomposition, localization of distributed data. Distributed query Optimization:			
				Query optimization, centralized query optimization, distributed query			
				optimization algorithms.			
III				Transaction Management: Definition, properties of transaction, types of			

	KCA045: Distributed Database Systems						
				transactions, distributed concurrency control: Serializability, concurrency control			
				mechanisms & algorithms, time - stamped & optimistic concurrency control			
				Algorithms, deadlock Management.			
IV				Distributed DBMS Reliability: Reliability concepts and measures, fault-			
				tolerance in distributed systems, failures in Distributed DBMS, local & distributed			
				reliability protocols, site failures and network partitioning. Parallel Database			
				Systems: Parallel database system architectures, parallel data placement, parallel			
				query processing, load balancing, database clusters			
V				Distributed object Database Management Systems: Fundamental object			

	KCA045: Distributed Database Systems						
				concepts and models, object distributed design, architectural issues, object			
				management, distributed object storage, object query Processing.			
				Object Oriented Data Model: Inheritance, object identity, persistent			
				programming languages, persistence of objects, comparison OODBMS and			
				ORDBMS			
	Text books:						
	M. Tamer OZSU and Patuck Valduriez: Principles of Distributed Database Systems, Pearson Edn. Asia,						
	2001. 2. Stefano Ceri and Giuseppe Pelagatti: Distributed Databases, McGraw Hill. REFERENCE						
	BOOKS: 1. Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom: "Database Systems: The						

	KCA045: Distributed Database Systems						
	Complete Book", Second Edition, Pearson International Edition						

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KCA045: Distributed Database Systems

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course , the student will be able to:

CO 1

Understand theoretical and practical aspects of distributed database systems.

K2

CO 2

Study and identify various issues related to the development of distributed database system

K3

CO 3

Understand the design aspects of object-oriented database system and related development

K4

CO 4

Equip students with principles and knowledge of distributed reliability.

K3

CO 5

Equip students with principles and knowledge of parallel and object-oriented databases.

K5

DETAILED SYLLABUS

4-0-0

Unit

Topic

Proposed

Lecture

I

Introduction: Distributed Data Processing, Distributed Database System,
Promises of DDBSSs, Problem areas. Distributed DBMS Architecture:
Architectural Models for Distributed DBMS, DDMBS Architecture. Distributed
Database Design: Alternative Design Strategies, Distribution Design issues,
Fragmentation, Allocation.

08

II

Query

processing

and

decomposition:

Query

processing

objectives,

characterization of query processors, layers of query processing, query
decomposition, localization of distributed data. Distributed query Optimization:

Query

optimization,

centralized

query

optimization,

distributed

query

optimization algorithms.

08

III

Transaction Management: Definition, properties

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ELECTIVE-5

Page 80 Data Table

KCA051: Mobile Computing			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Study and aware fundamentals of mobile computing.		K K 1, 2
CO 2	Study and analyze wireless networking protocols, applications and environment.		K K 1, 4
CO 3	Understand various data management issues in mobile computing.		K 2
CO 4	Analyze different type of security issues in mobile computing environment.		K 4
CO 5	Study, analyze, and evaluate various routing protocols used in mobile computing.		K , K K 1 4, 5
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Introduction, Issues in mobile computing, Overview of wireless telephony, Cellular concept, GSM-air interface, channel structure; Location management- HLR-VLR, hierarchical, handoffs; Channel allocation in cellular systems, CDMA, GPRS, MAC for cellular system.		08
II	Wireless Networking, Wireless LAN Overview- MAC issues, IEEE 802.11, Blue Tooth, Wireless multiple access protocols, TCP over wireless, Wireless applications, Data broadcasting, Mobile IP, WAP- architecture, protocol stack, application environment, applications.		08
III	Data management issues in mobile computing, data replication for mobile computers, adaptive clustering for mobile wireless networks, File system, Disconnected operations.		08
IV	Mobile Agents computing, Security and fault tolerance, Transaction processing in mobile computing environment.		08
V	Adhoc networks, Localization, MAC issues, Routing protocols, Global state routing (GSR), Destination sequenced distance vector routing (DSDV), Dynamic source routing (DSR), Adhoc on demand distance vector routing (AODV), Temporary ordered routing algorithm (TORA), QoS in Adhoc Networks, applications		08

KCA051: Mobile Computing			
Suggested Readings: 1. Schiller J., "Mobile Communications", Pearson 2. Upadhyaya S. and Chaudhury A., "Mobile Computing", Springer 3. Kamal R., "Mobile Computing", Oxford University Press. 4. Talukder A. K. and Ahmed H., "Mobile Computing Technology, Applications and Service Creation", McGraw Hill Education 5. Garg K., "Mobile Computing Theory and Practice", Pearson. 6. Kumar S., "Wireless and Mobile Communication", New Age International Publishers 7. Manvi S. S. and Kakkasageri M. S., "Wireless			

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KCA051: Mobile Computing

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Study and aware fundamentals of mobile computing.

K1, K2

CO 2

Study and analyze wireless networking protocols, applications and environment.

K1, K4

CO 3

Understand various data management issues in mobile computing.

K2

CO 4

Analyze different type of security issues in mobile computing environment.

K4

CO 5

Study, analyze, and evaluate various routing protocols used in mobile computing.

K1, K4, K5

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Introduction, Issues in mobile computing, Overview of wireless telephony, Cellular concept, GSM- air interface, channel structure; Location management- HLR-VLR, hierarchical, handoffs; Channel allocation in cellular systems, CDMA, GPRS, MAC for cellular system.

08

II

Wireless Networking, Wireless LAN Overview- MAC issues, IEEE 802.11, Blue Tooth, Wireless multiple access protocols, TCP over wireless, Wireless applications, Data broadcasting, Mobile IP, WAP- architecture, protocol stack, application environment, applications.

08

III

Data management issues in mobile computing, data replication for mobile computers, adaptive clustering for mobile wireless networks, File system, Disconnected operations.

08

IV

Mobile Agents computing, Securit

Page 81 Data Table

KCA052: Computer Graphics and Animation			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Understand the graphics hardware used in field of computer graphics.		K 2
CO 2	Understand the concept of graphics primitives such as lines and circle based on different algorithms.		K , K 2 4

KCA052: Computer Graphics and Animation			
CO 3	Apply the 2D graphics transformations, composite transformation and Clipping concepts.		K 4
CO 4	Apply the concepts and techniques used in 3D computer graphics, including viewing transformations, projections, curve and hidden surfaces.		K , K 2 3
CO 5	Perform the concept of multimedia and animation in real life.		K , K 2 3
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Introduction and Line Generation: Types of computer graphics, Graphic Displays- Random scan displays, Raster scan displays, Frame buffer and video controller, Points and lines, Line drawing algorithms, Circle generating algorithms, Mid-point circle generating algorithm, and parallel version of these algorithms.		08
II	Transformations: Basic transformation, Matrix representations and homogenous coordinates, Composite transformations, Reflections and shearing. Windowing and Clipping: Viewing pipeline, Viewing transformations, 2-D Clipping algorithms- Line clipping algorithms such as Cohen Sutherland line clipping algorithm, Liang Barsky algorithm, Line clipping against non rectangular clip windows; Polygon clipping – Sutherland Hodgeman polygon clipping, Weiler and Atherton polygon clipping, Curve clipping, Tex		08
III	Three Dimensional: 3-D Geometric Primitives, 3-D Object representation, 3- D Transformation, 3-D viewing, projections, 3-D Clipping. Curves and Surfaces: Quadric surfaces, Spheres, Ellipsoid, Blobby objects, Introductory concepts of Spline, Bspline and Bezier curves and surfaces.		08

KCA052: Computer Graphics and Animation			
IV	Hidden Lines and Surfaces: Back Face Detection algorithm, Depth buffer method, A- buffer method, Scan line method, basic illumination models– Ambient light, Diffuse reflection, Specular reflection and Phong model, Combined approach, Warn model, Intensity Attenuation, Color consideration, Transparency and Shadows.		08
V	Multimedia Systems: Design Fundamentals, Back ground of Art, Color theory overview, Sketching & illustration, Storyboarding, different tools for animation. Animation: Principles of Animations, Elements of animation and their use, Power of Motion, Animation Techniques, Animation File Format, Making animation for Rolling Ball, making animation for a Bouncing Ball, Animation for the web, GIF, Plugins and Players, Animation tools for World Wide Web.		08
Suggested Readings: 1. Hearn D. and Baker M. P., "Computer Graphics C Version", Pearson Education 2. Foley, Vandan, Feiner, Hughes,"Computer Graphics principle", Pearson Education. 3. Rogers, "Procedural Elements of Computer Graphics", McGraw Hill 4. Newman W. M., Sproull R. F., "Principles of Interactive computer Graphics", McGraw Hill. 5. Sinha A. N. and Udai A. D., "Computer Graphics", McGraw Hill. 6. Mukherjee, "Fundamentals of Computer graphics & Multimedia", PHI Learning Private Limited.			

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KCA052: Computer Graphics and Animation

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Understand the graphics hardware used in field of computer graphics.

K2

CO 2

Understand the concept of graphics primitives such as lines and circle based on different algorithms.

K2, K4

CO 3

Apply the 2D graphics transformations, composite transformation and Clipping concepts.

K4

CO 4

Apply the concepts and techniques used in 3D computer graphics, including viewing transformations, projections, curve and hidden surfaces.

K2, K3

CO 5

Perform the concept of multimedia and animation in real life.

K2, K3

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Introduction and Line Generation: Types of computer graphics, Graphic Displays- Random scan displays, Raster scan displays, Frame buffer and video controller, Points and lines, Line drawing algorithms, Circle generating algorithms, Mid-point circle generating algorithm, and parallel version of these algorithms.

08

II

Transformations:

Basic

transformation,

Matrix

representations

and

homogenous coordinates, Composite transformations, Reflections and

shearing.

Windowing and Clipping: Viewing pipeline, Viewing transformations, 2-D

Clipping algorithms- Line clipping algorithms such as Cohen Sutherland line clipping algor

Page 82 Data Table

KCA053: Natural Language Processing			
Course Outcome (CO)		Bloom's Knowledge Level (KL)	
At the end of course, the student will be able to understand			
CO 1	Study and understand basic concepts, background and representations of natural language.		K , K 1 2
CO 2	Analyze various real-world applications of NLP.		K 4
CO 3	Apply different parsing techniques in NLP.		K 3
CO 4	Understand grammatical concepts and apply them in NLP.		K , K 2 3
CO 5	Apply various statistical and probabilistic grammar methods to handle and evaluate ambiguity.		K , K 3 5
DETAILED SYLLABUS			3-0-0
Unit	Topic		Proposed Lecture
I	Introduction to Natural Language Understanding: The study of Language, Applications of NLP, Evaluating Language Understanding Systems, Different levels of Language Analysis, Representations and Understanding, Organization of Natural language Understanding Systems, Linguistic Background: An outline of English syntax.		08
II	Introduction to semantics and knowledge representation, some applications like machine translation, database interface.		08
III	Grammars and Parsing: Grammars and sentence Structure, Top-Down and Bottom-Up Parsers, Transition Network Grammars, Top- Down Chart Parsing. Feature Systems and Augmented Grammars: Basic Feature system for English, Morphological Analysis and the Lexicon, Parsing with Features, Augmented Transition Networks.		08

KCA053: Natural Language Processing			
IV	Grammars for Natural Language: Auxiliary Verbs and Verb Phrases, Movement Phenomenon in Language, Handling questions in Context-Free Grammars. Human preferences in Parsing, Encoding uncertainty, Deterministic Parser.		08
V	Ambiguity Resolution: Statistical Methods, Probabilistic Language Processing, Estimating Probabilities, Part-of Speech tagging, Obtaining Lexical Probabilities, Probabilistic Context-Free Grammars, Best First Parsing. Semantics and Logical Form, Word senses and Ambiguity, Encoding Ambiguity in Logical Form.		08
Suggested Readings: 1. Akshar Bharti, Vineet Chaitanya and Rajeev Sangal, "NLP: A Paninian Perspective", Prentice Hall, New Delhi. 2. James Allen, "Natural Language Understanding", Pearson Education. 3. D. Jurafsky, J. H. Martin, "Speech and Language Processing", Pearson Education. 4. L. M. Ivansca, S. C. Shapiro, "Natural Language Processing and Language Representation", AAAI Press, 2000. 5. T. Winograd, Language as a Cognitive Process, Addison-Wesley.			

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KCA053: Natural Language Processing

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course, the student will be able to understand

CO 1

Study and understand basic concepts, background and representations of natural language.

K1, K2

CO 2

Analyze various real-world applications of NLP.

K4

CO 3

Apply different parsing techniques in NLP.

K3

CO 4

Understand grammatical concepts and apply them in NLP.

K2, K3

CO 5

Apply various statistical and probabilistic grammar methods to handle and evaluate ambiguity.

K3, K5

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Introduction to Natural Language Understanding: The study of Language, Applications of NLP, Evaluating Language Understanding Systems, Different levels of Language Analysis, Representations and Understanding, Organization of Natural language Understanding Systems, Linguistic Background: An outline of English syntax.

08

II

Introduction to semantics and knowledge representation, some applications like machine translation, database interface.

08

III

Grammars and Parsing: Grammars and sentence Structure, Top-Down and Bottom-Up Parsers, Transition Network Grammars, Top- Down Chart Parsing. Feature Systems and Augmented Grammars: Basic Feature system for English, Morphological Analysis and the Lexicon, Parsing with Features, Augmented Transition Networks.

08

IV

Page 83 Data Table

KCA054: Machine Learning Techniques				
Course Outcome (CO)			Bloom's Knowledge Level (KL)	
At the end of course , the student will be able:				
CO 1		To understand the need for machine learning for various problem solving		K , K 1 2
CO 2		To understand a wide variety of learning algorithms and how to evaluate models generated from data		K , K 1 3
CO 3		To understand the latest trends in machine learning		K , K 2 3
CO 4		To design appropriate machine learning algorithms and apply the algorithms to a real-world problems		K , K 4 6
CO 5		To optimize the models learned and report on the expected accuracy that can be achieved by applying the models		K K 4, 5
DETAILED SYLLABUS				3-0-0
Unit	Topic			Proposed Lecture
I	INTRODUCTION – Learning, Types of Learning, Well defined learning problems, Designing a Learning System, History of ML, Introduction of Machine Learning Approaches – (Artificial Neural Network, Clustering, Reinforcement Learning, Decision Tree Learning, Bayesian networks, Support Vector Machine, Genetic Algorithm), Issues in Machine Learning and Data Science Vs Machine Learning;			08

KCA054: Machine Learning Techniques				
II	REGRESSION: Linear Regression and Logistic Regression BAYESIAN LEARNING - Bayes theorem, Concept learning, Bayes Optimal Classifier, Naïve Bayes classifier, Bayesian belief networks, EM algorithm. SUPPORT VECTOR MACHINE: Introduction, Types of support vector kernel – (Linear kernel, polynomial kernel, and Gaussian kernel), Hyperplane – (Decision surface), Properties of SVM, and Issues in SVM.			08
III	DECISION TREE LEARNING - Decision tree learning algorithm, Inductive bias, Inductive inference with decision trees, Entropy and information theory, Information gain, ID-3 Algorithm, Issues in Decision tree learning. INSTANCE-BASED LEARNING – k-Nearest Neighbour Learning, Locally Weighted Regression, Radial basis function networks, Case-based learning.			08
IV	ARTIFICIAL NEURAL NETWORKS – Perceptron's, Multilayer perceptron, Gradient descent and the Delta rule, Multilayer networks, Derivation of Backpropagation Algorithm, Generalization, Unsupervised Learning – SOM Algorithm and its variant; DEEP LEARNING - Introduction, concept of convolutional neural network, Types of layers – (Convolutional Layers, Activation function, pooling, fully connected), Concept of Convolution (1D and 2D) layers, Training of network, Case study of CNN for eg on Diabetic			08

KCA054: Machine Learning Techniques				
V	REINFORCEMENT LEARNING–Introduction to Reinforcement Learning , Learning Task,Example of Reinforcement Learning in Practice, Learning Models for Reinforcement – (Markov Decision process , Q Learning - Q Learning function, Q Learning Algorithm), Application of Reinforcement Learning,Introduction to Deep Q Learning.			08

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KCA054: Machine Learning Techniques

Course Outcome (CO)

Bloom's Knowledge Level

(KL)

At the end of course , the student will be able:

CO 1

To understand the need for machine learning for various problem solving

K1 , K2

CO 2

To understand a wide variety of learning algorithms and how to evaluate models generated from data

K1 , K3

CO 3

To understand the latest trends in machine learning

K2 , K3

CO 4

To design appropriate machine learning algorithms and apply the algorithms to a real-world problems

K4 , K6

CO 5

To optimize the models learned and report on the expected accuracy that can be achieved by applying the models

K4, K5

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

INTRODUCTION – Learning, Types of Learning, Well defined learning problems, Designing a Learning System, History of ML, Introduction of Machine Learning Approaches – (Artificial Neural Network, Clustering, Reinforcement Learning, Decision Tree Learning, Bayesian networks, Support Vector Machine, Genetic Algorithm), Issues in Machine Learning and Data Science Vs Machine Learning;

08

II

REGRESSION: Linear Regression and Logistic Regression

BAYESIAN LEARNING - Bayes theorem, Concept learning, Bayes Optimal Classifier, Naïve Bayes classifier, Bayesian belief networks, EM algorithm.

SUPPORT VECTOR MACHINE: Introduction, Types of support vector kernel – (Linear

Page 84 Data Table

	GENETIC ALGORITHMS: Introduction, Components, GA cycle of reproduction, Crossover, Mutation, Genetic Programming, Models of Evolution and Learning, Applications.	
Text books: 1. Tom M. Mitchell, ■Machine Learning, McGraw-Hill Education (India) Private Limited, 2013. 2. Ethem Alpaydin, ■Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press 2004. 3. Stephen Marsland, ■Machine Learning: An Algorithmic Perspective, CRC Press, 2009. 4. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag. 5. M. Gopal, "Applied Machine Learning", McGraw Hill Education		

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GENETIC

ALGORITHMS:

Introduction, Components,

GA

cycle

of

reproduction, Crossover, Mutation, Genetic Programming, Models of Evolution

and Learning, Applications.

Text books:

1. Tom M. Mitchell, ■Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
2. Ethem Alpaydin, ■Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press 2004.
3. Stephen Marsland, ■Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
4. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.
5. M. Gopal, "Applied Machine Learning", McGraw Hill Education

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KCA055: Quantum Computing				
Course Outcome (CO)		Bloom's Knowledge Level (KL)		
At the end of course , the student will be able to understand				
CO 1	Distinguish problems of different computational complexity and explain why certain			K , K 1 2
	problems are rendered tractable by quantum computation with reference to the relevant			
	concepts in quantum theory.			
CO 2	Demonstrate an understanding of a quantum computing algorithm by simulating it on a			K , K 2 3
	classical computer, and state some of the practical challenges in building a quantum			
	computer.			
CO 3	Contribute to a medium-scale application program as part of a co-operative team, making			K , K 2 3

KCA055: Quantum Computing				
	use of appropriate collaborative development tools (such as version control systems).			
CO 4	Produce code and documentation that is comprehensible to a group of different			K , K 3 4
	programmers and present the theoretical background and results of a project in written and			
	verbal form.			
CO 5	Apply knowledge, skills, and understanding in executing a defined project of research,			K K 3, 6
	development, or investigation and in identifying and implementing relevant outcomes.			
DETAILED SYLLABUS				3-0-0
Unit	Topic			Proposed Lecture
I	Fundamental Concepts: Global Perspectives, Quantum Bits, Quantum Computation, Quantum Algorithms, Quantum Information, Postulates of Quantum Mechanisms.			08
II	Quantum Computation: Quantum Circuits – Quantum algorithms, Single Orbit operations, Control Operations, Measurement, Universal Quantum Gates, Simulation of Quantum Systems, Quantum Fourier transform, Phase estimation, Applications, Quantum search algorithms – Quantum counting – Speeding up the solution of NP – complete problems – Quantum Search for an unstructured database.			08

KCA055: Quantum Computing				
III	Quantum Computers: Guiding Principles, Conditions for Quantum Computation, Harmonic Oscillator Quantum Computer, Optical Photon Quantum Computer – Optical cavity Quantum electrodynamics, Ion traps, Nuclear Magnetic resonance			08
IV	Quantum Information: Quantum noise and Quantum Operations – Classical Noise and Markov Processes, Quantum Operations, Examples of Quantum noise and Quantum Operations – Applications of Quantum operations, Limitations of the Quantum operations formalism, Distance Measures for Quantum information.			08
V	Quantum Error Correction: Introduction, Shor code, Theory of Quantum Error – Correction, Constructing Quantum Codes, Stabilizer codes, Fault – Tolerant Quantum Computation, Entropy and information – Shannon Entropy, Basic properties of Entropy, Von Neumann, Strong Sub Additivity, Data Compression, Entanglement as a physical resource .			08

KCA055: Quantum Computing				
Text books: 1. Micheal A. Nielsen. & Issac L. Chiang, "Quantum Computation and Quantum Information", Cambridge University Press, Fint South Asian edition, 2002. 2. Eleanor G. Rieffel , Wolfgang H. Polak , "Quantum Computing - A Gentle Introduction" (Scientific and Engineering Computation) Paperback – Import, 3 Oct 2014 3. Computing since Democritus by Scott Aaronson 4. Computer Science: An Introduction by N. David Mermin 5. Yanofsky's and Mannucci, Quantum Computing for Computer Scientists.				

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KCA055: Quantum Computing

Course Outcome (CO)

Bloom's Knowledge Level (KL)

At the end of course , the student will be able to understand

CO

1

Distinguish problems of different computational complexity and explain why certain problems are rendered tractable by quantum computation with reference to the relevant concepts in quantum theory.

K1 , K2

CO

2

Demonstrate an understanding of a quantum computing algorithm by simulating it on a classical computer, and state some of the practical challenges in building a quantum computer.

K2 , K3

CO

3

Contribute to a medium-scale application program as part of a co-operative team, making use of appropriate collaborative development tools (such as version control systems).

K2 , K3

CO

4

Produce code and documentation that is comprehensible to a group of different programmers and present the theoretical background and results of a project in written and verbal form.

K3 , K4

CO

5

Apply knowledge, skills, and understanding in executing a defined project of research, development, or investigation and in identifying and implementing relevant outcomes.

K3, K6

DETAILED SYLLABUS

3-0-0

Unit

Topic

Proposed

Lecture

I

Fundamental Concepts: Global Perspectives, Quantum Bits, Quantum Computation, Quantum Algorithms, Quantum Information, Postulates of Quantum Mechanisms.

08

II

Quantum Computation: Quantu

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Total Document Pages: 127

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1

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY

UTTAR PRADESH, LUCKNOW

Evaluation Scheme & Syllabus

MBA

(Common)

First Year

AS PER

AICTE MODEL CURRICULUM

& NEP 2020

(Effective from the Session: 2024-25)

Page 2 Data Table

S. No.	Codes	SUBJECT	PERIODS			INTERNAL EVALUATION SCHEME	
			L	T	P	CT	TA
1	BMB 101	MANAGEMENT CONCEPTS & ORGANISATIONAL BEHAVIOUR	4	0	0	20	10
2	BMB 102	MANAGERIAL ECONOMICS	4	0	0	20	10
3	BMB 103	FINANCIAL ACCOUNTING & ANALYSIS	3	1	0	20	10
4	BMB 104	BUSINESS STATISTICS & ANALYTICS	3	1	0	20	10
5	BMB 105	MARKETING MANAGEMENT	4	0	0	20	10
6	BMB 106	CREATIVITY, INNOVATION AND ENTREPRENEURSHIP	2	0	0	20	10
7	BMB 107	BUSINESS COMMUNICATION	3	1	0	20	10
8	BMB 151	IT SKILLS-1	0	0	3	20	10
9	BMB 152	MINI PROJECT -1	0	0	3	20	10
10	BVC 151	SPORTS & YOGA *	0	0	3	0	100

Page 2 Content

2

MBA 1st Year Course Structure in accordance with

AICTE Model Curriculum & NEP 2020 Effective w.e.f.

Academic Session 2024-25

Semester I

S.

No.

Codes

SUBJECT

PERIODS

INTERNAL EVALUATION

SCHEME

END

SEMESTER

EVALUATION

TOTAL

CREDIT

L

T

P

CT

TA

PS

TOTAL

TE

PE

1

MANAGEMENT CONCEPTS

4

0

0

20

10

0

30

70

0

100

3

BMB 101

& ORGANISATIONAL

BEHAVIOUR

2

BMB 102

MANAGERIAL ECONOMICS

4

0

0

20

10

0

30

70

0

100

3

3

BMB 103

FINANCIAL ACCOUNTING

& ANALYSIS

3

1

0

20

10

0

30

70

0

100

3

4

BMB 104

BUSINESS STATISTICS &

ANALYTICS

3

1

0

20

10

0

30

70

0

100

3

5

BMB 105

MARKETING

MANAGEMENT

4

0

0

20

10

0

30

70

0

100

3

6

BMB 106

CREATIVITY, INNOVATION

AND ENTREPRENEURSHIP

2

0

0

20

10

0

30

70

100

2

7

BMB 107

BUSINESS

COMMUNICATION

3

1

0

20

10

0

30

70

0

100

3

8

BMB 151

IT SKILLS-1

0

0

3

20 10

0

30

-

70

100

3

9

BMB 152

MINI PROJECT -1

0

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3

20

10

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30

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70

100

3

10

BVC 151

SPORTS & YOGA *

0

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3

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26

•

Qualifying but Non Credit Course

Page 3 Data Table

S. No	CODE	SUBJECT	PERIODS			INTERNAL EVALUATION SCHEME	
			L	T	P	CT	TA
1	BMB 201	BUSINESS EN VIRONMENT & LEGAL ASPECT OF BUSINESS	4	0	0	20	10
2	BMB 202	HUMAN RESOURCE M ANAGEMENT	4	0	0	20	10
3	BMB 203	BUSINESS RESEARCH METHODS	3	1	0	20	10

S. No	CODE	SUBJECT	PERIODS			INTERNAL EVALUATION SCHEME	
4	BMB 204	FINANCIAL M ANAGEMENT & CORPORATE FINANCE	4	0	0	20	10
5	BMB 205	OPERATIONS MANAGEMENT	4	0	0	20	10
6	BMB 206	QUANTITATIVE TECHNIQUES FOR MANAGERS	3	1	0	20	10
7	BMB 207	COST & MAN AGEMENT ACCOUNTING	4	0	0	20	10
8	BMB 208	MANAGEMENT INFORMATION SYSTEMS	2	0	0	20	10
9	BMB 251	IT SKILLS-2	0	0	2	20	10
10	BMB 252	MINI PROJECT -2	0	0	3	20	10

Page 3 Content

3

Semester II

S.

No

CODE

SUBJECT

PERIODS

INTERNAL

EVALUATION

SCHEME

END

SEMESTER

EVALUATION

TOTAL CREDIT

L

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TA

PS

TOTA

L

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PE

1

BMB 201

BUSINESS ENVIRONMENT

& LEGAL ASPECT OF

BUSINESS

4

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0

20

10

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30

70

0

100

3

2

BMB 202

HUMAN RESOURCE

MANAGEMENT

4

0

0

20

10

0

30

70

0

100

3

3

BMB 203

BUSINESS RESEARCH

METHODS

3

1

0

20

10

0

30

70

0

100

3

4

BMB 204

FINANCIAL MANAGEMENT

& CORPORATE FINANCE

4

0

0

20

10

0

30

70

0

100

3

5

BMB 205

OPERATIONS

MANAGEMENT

4

0

0

20

10

0

30

70

0

100

3

6

BMB 206

QUANTITATIVE

TECHNIQUES FOR

MANAGERS

3

1

0

20

10

0

30

70

0

100

3

7

BMB 207

COST & MANAGEMENT

ACCOUNTING

4

0

0

20

10

0

30

70

0

100

3

8

BMB 208

MANAGEMENT

INFORMATION SYSTEMS

2

0

0

20

10

0

30

70

0

100

2

9

BMB 251

IT SKILLS-2

0

0

2

20

10

0

30

0

70

100

1

10

BMB 252

MINI PROJECT -2

0

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3

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26

L/T/P – Lecture/Tutorial/Practical, CT/TA/PS- Class Test/Teachers Assessment/Practical Session, TE/PE-
Term End/ Practical End

Page 4 Content

4

SEMESTER I

MANAGEMENT CONCEPTS AND ORGANISATIONAL BEHAVIOUR

BMB 101

Course Credit: 3

Contact Hours: 40

Course Objectives:

1. To provide fundamentals of management concepts and theories.
2. To help the students understand the concepts of organizational behaviour
3. To apply the concepts of management and organizational behaviors in real world situations
4. Familiarizing the students with the contemporary challenges and prospects in management.
5. Developing managerial and leadership skills among students

UNIT I (8 Lectures)

Fundamentals of Management: Management practices from past to present, Different levels of management, Managerial skills and Managerial Functions, Case Studies

Planning- Objective of planning, Planning process, Types of planning, Types of plans, Management by Objective, Decision-making- types, process & techniques, Case Studies

UNIT-II (8 Lectures)

Organising & Staffing- Types of organization, Organization structure, delegation and decentralization of authority, Meaning of staffing, Recruitment, selection & placement, Training & development..

Directing & Controlling- Principle of directing, Essence of coordination, Different control techniques, Management by exception. Case Studies.

UNIT III(8 Lectures)

Fundamentals of individual behavior: Concepts of OB,OB Model, Components of OB, Personality,

types of personality, Personal effectiveness, meaning of Attitudes, Types, Components, attitude formation and attitude

Page 5 Data Table

Course Outcomes	Bloom's taxonomy
CO 1: Developing understanding of managerial practices and their perspectives.	Knowledge (K 2) Remembering (K1)
CO2: Understanding and Applying the concepts of organizational behavior	Knowledge (K 2) Applying (K 4)
CO 3: Applying the concepts of management and analyze organizational behaviors in real world situations	Applying (K 4) Analyzing (K 5)
CO 4: Comprehend and practice contemporary issues in management.	Comprehending (K 3)
CO 5: Applying managerial and leadership skills among students	Applying (K4)

Page 5 Content

5

COURSE OUTCOME

Course Outcomes

Bloom's taxonomy

CO 1: Developing understanding of managerial practices and their perspectives.

Knowledge (K 2)

Remembering (K1)

CO2: Understanding and Applying the concepts of organizational behavior

Knowledge (K 2)

Applying (K 4)

CO 3: Applying the concepts of management and analyze organizational behaviors in real world situations

Applying (K 4)

Analyzing (K 5)

CO 4: Comprehend and practice contemporary issues in management.

Comprehending (K 3)

CO 5: Applying managerial and leadership skills among students

Applying (K4)

Suggested Readings

1. Koontz Harold &Weihrich Heinz – Essentials of management (Tata McGraw Hill, 5th Edition,2008)

2. L. M. Prasad- Principles and Practices of Management, Sulatn Chand & Sons, 7th edition, 2007.
3. Stephen P. Robbins, ■Organizational Behaviour■, 12th Edition, Prentice Hall
4. Dr. Premvir Kapoor, Principles and Practices of Management, Khanna Publishing House, Delhi
5. Robbins & Coulter - Management (Prentice Hall of India, 9th Edition)
6. Principles of Management, George R. Terry & S.G. Franklin, AITBS, Delhi.
7. N M Khandelwal- Indian Ethos & Values for Management- Himalyan Publishing
8. Fred Luthans, ■Organizational Behaviour■, 12th Edition, McGraw Hill International Edition
9. Aswathappa K, ■Organizational Behaviour (Text, Cases and Games)■, Himalaya Publication
10. UdaiPareek, ■Organizational Behavior■, Oxford University Press

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6

MANAGERIAL ECONOMICS

BMB 102

Course Credit: 3

Contact Hours: 40

Course Objective:

1. To understand the importance of Managerial Economics in management and businesses
2. To apply the principles of managerial economics in achieving business objectives
3. Be equipped with the tools necessary in forecasting product demand
4. Understand and be able to apply latest pricing strategies
5. Understand and analyze the macro environment affecting the business decision making.

UNIT –I (6 Hours)

Basic Concepts and principles: Definition, Nature and Scope of Economics-Micro Economics and Macro Economics, Managerial Economics and its relevance in business decisions. Fundamental Principles of Managerial Economics - Incremental Principle, Marginal Principle, Opportunity Cost Principle, Discounting Principle, Concept of Time Perspective, Equi-Marginal Principle, Utility Analysis, Cardinal Utility and Ordinal Utility. Case Studies

UNIT –II (8Hours)

Demand and Supply Analysis: Theory of Demand, Types of Demand. Determinants of demand, Demand Function, Demand Schedule, Demand curve, Law of Demand, Exceptions to the law of Demand, Shifts in demand curve, Elasticity of Demand and its measurement. Price Elasticity, Income Elasticity, Arc Elasticity. Cross Elasticity and Advertising Elasticity. Uses of Elasticity of Demand for managerial decision making, Demand forecasting meaning, significance and methods.(numerical Exercises) Case Studies
Supply

Page 7 Data Table

Course Outcomes	Bloom's taxonomy
CO1: Students will be able to remember the concepts of micro economics and also able to understand the various micro economic principles to make effective economic decisions under conditions of risk and uncertainty.	• Knowledge (K 2) • Remembering (k1)
CO2: The students would be able to understand the law of demand & supply & their elasticities , evaluate & analyse these concepts and apply them in various changing situations in industry . Students would be able to apply various techniques to forecast demand for better utilization of resources.	• Knowledge (K 2) • Applying (K 4) • Synthesizing (K6) • Evaluating (K7)
CO3: The students would be able to understand the production concept and how the production output changes with the change in inputs and able to analyse the effect of cost to business and their relation to analyze the volatility in the business world	• Comprehending (K 3) • Applying (K 4) • Analyzing (K 5) • Evaluating (K7)
CO4: The students would be able to understand & evaluate the different market structure and their different equilibriums for industry as well as for consumers for the survival in the industry by the application of various pricing strategic	• Applying (K 4) • Analyzing (K 5) • Synthesizing (K6)
CO5: The students would be able to analyse the macroeconomic concepts & their relation to micro economic concept & how they affect the business & economy.	• Knowledge (K 2) • Comprehending (K 3)

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7

Course Outcomes:

Course Outcomes

Bloom's taxonomy

CO1: Students will be able to remember the concepts of micro economics and also able to understand the various micro economic principles to make effective economic decisions under conditions of risk and uncertainty.

•

Knowledge

(K 2)

•

Remembering (k1)

CO2: The students would be able to understand the law of demand & supply & their elasticities , evaluate & analyse these concepts and apply them in various changing situations in industry . Students would be able to apply various techniques to forecast demand for better utilization of resources.

•

Knowledge (K 2)

•

Applying (K 4)

-

Synthesizing (K6)

-

Evaluating (K7)

CO3: The students would be able to understand the production concept and how the production output changes with the change in inputs and able to analyse the effect of cost to business and their relation to analyze the volatility in the business world

-

Comprehending (K 3)

-

Applying (K 4)

-

Analyzing (K 5)

-

Evaluating (K7)

CO4: The students would be able to understand & evaluate the different market structure and their different equilibriums for industry as well as for consumers for the survival in the industry by the application of various pricing strategic

-

Applying (K 4)

-

Analyzing (K 5)

-

Synthesizing (K6)

CO5: The students would be able to analyse the macroeconomic concepts & their relation to micro economic concept &

Page 8 Content

8

FINANCIAL ACCOUNTING AND ANALYSIS

BMB 103

Course Credit: 3

Contact Hours: 40

Course Objectives:

- 1) To understand the fundamentals, basic theory and concepts of financial accounting.
- 2) To have a knowledge about various Accounting Standards used in preparation of financial statements.
- 3) To have an understanding of preparation and presentation of financial statements.
- 4) To acquire knowledge about various techniques used for analysing financial statements with its application.
- 5) To enable students acquainted with current trends and social responsibility accounting.

UNIT I (4 Hrs)

Meaning and Scope of Accounting: Evolution and Users of Accounting, Basic Accounting terminologies, Principles of Accounting, Accounting Concepts & Conventions, Accounting Equation, Deprecation Accounting. GAAP(introduction).

UNIT II (10 Hrs)

Mechanics of Accounting: Accounting Standards and IFRS: International Accounting Principles and Standards; Matching of Indian Accounting Standards with International Accounting Standards, Double entry system of Accounting, journalizing of transactions; Ledger posting and Trial Balance.

UNIT III (12 Hrs)

Presentation of Financial Statement: Preparation of final accounts (Profit & Loss Account and Balance Sheet) according to companies act 2013 (vertical format), Excel Application to make Balance sheet, Case studies and Workshops, Preparation of Cash Flow Statement and its analysis.

UNIT IV (8 Hrs)

Analysis of financi

Page 9 Data Table

S.No	Course Outcome	Bloom's Taxonomy
1 2 3 4 5	CO1.Understand and apply accounting concepts, principles and conventions for their routine monetary transaction. CO2. Understand about IFRS, Ind AS and IAS for preparation and reporting of financial statements. CO3. Create and prepare financial statements and Cash flow in accordance with Generally Accepted Accounting Principles CO4. Analyse, interpret and communicate the information contained in basic financial statements and explain the limitations of such statements. CO5. Recognising various t	Knowledge (K2)/ Comprehending (K 3) Knowledge (K2) Synthesizing (K6)/ Remembering (k1) Analysing (K 4) / Evaluating (K7)) Knowledge (K2) Applying (K 4)

Page 9 Content

9

Course Outcome:

After successful completion of this course students will be able to

S.No

Course Outcome

Bloom's Taxonomy

1
2
3
4
5

CO1.Understand and apply accounting concepts, principles and conventions for their routine monetary transaction.

CO2. Understand about IFRS, Ind AS and IAS for preparation and reporting of financial statements.

CO3. Create and prepare financial statements and Cash flow in accordance with Generally Accepted Accounting Principles

CO4. Analyse, interpret and communicate the information contained in basic financial statements and explain the limitations of such statements.

CO5. Recognising various types of accounting and utilize the technology and social responsibility in facilitating and enhancing accounting and financial reporting processes

Knowledge (K2)/

Comprehending (K 3)

Knowledge (K2)

Synthesizing (K6)/

Remembering (k1)

Analysing (K 4) / Evaluating

(K7))

Knowledge (K2)

Applying (K 4)

Suggested Readings

1. Maheshwari S.N &Maheshwari S K – A text book of Accounting for Management (Vikas, 10th Edition)
2. Essentials of Financial Accounting (based on IFRS), Bhattacharya (PHI,3rd Ed)
3. Khan and Jain - Financial Management (Tata McGraw Hill, 7th Ed.)
4. PC Tulsian- Financial Accounting (Pearson, 2016)
5. Dhamija - Financial Accounting for managers: (Prentice Hall, 2nd Edition).
6. Narayanswami - Financial Accounting: A Managerial Perspective (PHI,5th Ed)
7. DhaneshkKhatri- Financial Accounting (T

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BUSINESS STATISTICS & ANALYTICS

BMB 104

Course Credit: 3

Contact Hours: 40 hours

Course Objectives

1. Understand the different basic concept / fundamentals of business statistics.
2. Understand the importance of measures of Descriptive statistics which includes measures of central tendency, Measures of Dispersion, Time Series Analysis, Index Number, Correlation and Regression analysis and their implication on Business performance.
3. Understand the concept of Probability and its usage in various business applications.
4. Understand the Hypothesis Testing concepts and use inferential statistics- t, F, Z Test and Chi Square Test
5. Understand the practical application of Descriptive and Inferential Statistics concepts and their uses for Business Analytics.

Unit I (10 Sessions): Descriptive Statistics

Meaning, Scope, types, functions and limitations of statistics, Measures of Central tendency – Mean, Median, Mode, Quartiles, Measures of Dispersion – Range, Inter quartile range, Mean deviation, Standard deviation, Variance, Coefficient of Variation, Skewness and Kurtosis.

Unit II (8 Sessions): Time Series & Index Number

Time series analysis: Concept, Additive and Multiplicative models, Components of time series, Trend analysis: Least Square method - Linear and Non- Linear equations, Applications in business decision-making.

Index Numbers:- Meaning , Types of index numbers, uses of index numbers, Construction of Price, Quantity and Volume

Page 11 Data Table

Course Outcome	Blooms Taxonomy
CO1. Gaining Knowledge of basic concept / Fundamentals of business statistics.	• Knowledge (K 2)
CO2. To compute various measures of central tendency, Measures of Dispersion, Time Series Analysis, Index Number, Correlation and Regression analysis and their implication on Business performance.	• Remembering (K1) • Applying (K 4)
CO3. Evaluating basic concepts of probability and perform probability theoretical distributions	• Comprehending (K 3) • Applying (K 4)
CO4. To apply Hypothesis Testing concepts and able to apply inferential statistics- t, F, Z Test and Chi Square Test	• Analyzing (K 5) • Synthesizing (K6)
CO5. To perform practical application by taking managerial decision and evaluating the Concept of Business Analytics.	• Evaluating (K7) • Applying (K 4)

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analyze data-Descriptive analytics and Predictive analytics.

Course Outcome:

Course Outcome

Blooms Taxonomy

CO1. Gaining Knowledge of basic concept /

Fundamentals of business statistics.

-

Knowledge (K 2)

CO2. To compute various measures of central tendency, Measures of Dispersion, Time Series

Analysis,

Index

Number,

Correlation

and

Regression analysis and their implication on

Business performance.

-

Remembering (K1)

-

Applying (K 4)

CO3. Evaluating basic concepts of probability and perform probability theoretical distributions

-

Comprehending (K 3)

-

Applying (K 4)

CO4. To apply Hypothesis Testing concepts and able to apply inferential statistics- t, F, Z Test and Chi Square Test

-

Analyzing (K 5)

-

Synthesizing (K6)

CO5. To perform practical application by taking managerial decision and evaluating the Concept of Business Analytics.

-

Evaluating (K7)

-

Applying (K 4)

Suggested Readings

1. G C Beri – Business Statistics, 3rd ed, TATA McGrawHill.
2. Chandrasekaran & Umaparvathi-Statistics for Managers, 1st edition, PHI Learning
3. Davis , Pecar – Business Statistics using Excel, Oxford
4. Ken Black – Business Statistics, 5th ed., Wiley India
5. Levin and Rubin – statistics for Management, 7th ed., Pearson
6. Lind, Marchal, Wathen – Staistical techniques in business and economics, 13th ed, McGrawHill
7. Newbold, Carlson, Thorne – Statistics for Business and Economics, 6th ed., Pearson
8. S. C.Gupt

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MARKETING MANAGEMENT

BMB 105

Course Credit: 3

Contact Hours: 40 hours

Course Objectives

1. Assess market opportunities by analyzing customers, competitors, collaborators, context, and the strengths and weaknesses of a company.
2. Understand consumers' requirements and their behaviors.
3. Develop effective marketing strategies to achieve organizational objectives.
4. Communicate and defend your recommendations and critically examine and build upon the recommendations of your classmates both quantitatively and qualitatively.
5. Develop the understanding the current global and digital aspect of marketing.

Unit 1(6 hours)

Introduction: Nature and scope of marketing, Various marketing orientations, Need, Want, Demand, Elements of Marketing mix, customer value and the value delivery process.

Understanding Consumer Behavior: Buying motives, factors influencing buying behavior, buying habits, stages in consumer buying decision process, types of consumer buying decisions.

Unit 2 (8 hours)

Market segmentation, Targeting and Positioning: Meaning, Factors influencing segmentation, Market Aggregation, Basis for segmentation, Segmentation of Consumer. Targeting: Meaning, Basis for identifying target customers, Target Market Strategies. Positioning: Meaning, product differentiation strategies, tasks involved in positioning. Branding: Concept of Branding, Brand Types, Brand equity, Branding Positioning.

Unit 3 (8 hours)

Product Decisions: Concept, P

Page 13 Data Table

S.No	Course Outcome	Bloom's taxonomy
1	CO1. Remember and Comprehend basic marketing concepts.	• Remembering (k1) • Knowledge (K 2)
2	CO2. Understand marketing Insights on application of basic marketing concepts.	• Synthesizing (K6) • Comprehending(K3)
3	CO3. Able to Apply and develop Marketing Strategies and Plans	• Applying (K 4)
4	CO4. Understand and Analyzing Business/ Consumer Markets and ability Identify & evaluate Market Segments and Targeting	• Analyzing (K 5)
5	CO5. Develop skills to understand the current global and digital aspect of marketing.	• Evaluating (K7)

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Course Outcomes:

Upon the successful completion of this course, the student will be able to:

S.No

Course Outcome

Bloom's taxonomy

1

CO1. Remember and Comprehend basic marketing concepts.

- Remembering (k1)
- Knowledge (K 2)

2

CO2. Understand marketing Insights on application of basic marketing concepts.

- Synthesizing (K6)
- Comprehending(K3)

3

CO3. Able to Apply and develop Marketing Strategies and Plans

- Applying (K 4)

4

CO4. Understand and Analyzing Business/ Consumer Markets and ability Identify & evaluate Market Segments and Targeting

- Analyzing (K 5)

5

CO5. Develop skills to understand the current global and digital aspect of marketing.

- Evaluating (K7)

Recommended Text Books :

1. Marketing Management: A South Asian Perspective - Kotler, Keller, Kevin 15/e, Pearson Education, 2016.
2. Marketing Management - Ramaswamy V. S. & Namakumari S, 6/e, Sage Publication India Pvt Ltd., 2018.
3. Marketing Management - Tapan Panda, 5/e, Excel Publication, 2007.
4. Fundamentals of Marketing Management - Etzel M. J, B J Walker & William J. Stanton, 14/e, McGrawHill Education Publishers, 2015.
5. Marketing: Asian Edition Paul Bainies, Chris Fill Kelly Page third edition, Oxford.

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CREATIVITY, INNOVATION AND ENTREPRENEURSHIP

BMB 106

Course Credit: 2

Contact Hours: 20 hours

COURSE OBJECTIVES:

1. The purpose of this course is to expose the student to the basic concepts of entrepreneurship, functions of entrepreneurs and problems faced by them in the real world
2. To provide insights to students in converting an Idea into an opportunity and develop an understanding of various funding sources for a startup
3. Familiarize the students with SME sector activities and the role of government agencies supporting SMEs.
4. To understand the process of a Business Plan

Unit 1 (4 Hours)

Creativity and Innovation: Meaning, the difference between innovation and creativity, Innovation types & Platforms, Business Model Innovation, Service Innovation, Design-led innovation, Improvisation, Large firm Vs. Start-up innovation, Co-creation and open innovation, developing an innovation strategy, Sources of innovation, Innovation Environment, Creative Destruction

Unit 2 (5 Hours)

Entrepreneurship: Meaning, definition and concept, Factors affecting entrepreneurship, characteristics and skills of an entrepreneur, entrepreneur v/s manager. Types of entrepreneurs, functions of an entrepreneur, entrepreneurial decision-process. Social, rural and women Entrepreneurship, Case Studies of Successful Entrepreneurs, Entrepreneurial Failure

Unit 3 (5 Hours)

Entrepreneurial Eco-System in India: Overview of MSME, Role of Central Govt & State G

Page 15 Data Table

	Course Outcome	Expected Levels of Learning as per Bloom's Taxonomy for Assessment of Course Outcome.
CO 1:	Remember and comprehend basic concepts of creativity, innovation and entrepreneurship	• Remembering (K1) • Knowledge (K 2) • Comprehending(K3)
CO2:	Develop knowledge on Entrepreneurial Finance, Assistance and role of Entrepreneurial Development Agencies	• Applying (K 4) • Analyzing (K 5) • Evaluating (K7)
CO3:	Develop understanding of converting an Idea to an opportunity and develop understanding of various funding sources	• Analyzing (K 5) • Evaluating (K7)
CO4:	Gain in depth knowledge of innovation and its various sources	• Remembering (K1) • Knowledge (K 2) • Analyzing (K 5)
CO5:	Develop understanding of various dimensions of innovation along with current trends and general awareness of innovation and startup	• Knowledge (K 2) • Synthesizing (K6) • Evaluating (K7)

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COURSE OUTCOMES

Course Outcome

Expected Levels of Learning as
per Bloom's Taxonomy for
Assessment of Course Outcome.

CO 1:

Remember and comprehend basic
concepts of creativity, innovation
and entrepreneurship

•

Remembering (K1)

•

Knowledge (K 2)

•

Comprehending(K3)

CO2:

Develop knowledge on
Entrepreneurial Finance,
Assistance and role of
Entrepreneurial Development
Agencies

•

Applying (K 4)

-

Analyzing (K 5)

-

Evaluating (K7)

CO3:

Develop understanding of
converting an Idea to an
opportunity and develop
understanding of various funding
sources

-

Analyzing (K 5)

-

Evaluating (K7)

CO4:

Gain in depth knowledge of
innovation and its various sources

-

Remembering (K1)

-

Knowledge (K 2)

-

Analyzing (K 5)

CO5:

Develop understanding of various
dimensions of innovation along with
current trends and general awareness
of innovation
and startup

-

Knowledge (K 2)

-

Synthesizing (K6)

-

Evaluating (K7)

Suggested Readings

1. Roy: Entrepreneurship, OUP
2. Ahmad, Ali and Bhatt, Punita.: Entrepreneurship in Developing and Emerging Economies, SAGE Publishing India
3. Mitra, Jay: The Business of Innovation, 2017, SAGE Publishing
4. Entrepreneurship 10th Ed (Indian Edition) 2016 by Robert Hirsch Michael Peters Dean Shepherd, McGraw Hill
5. Khanka, S.S.; Entrepreneurial Development; S. Chand and Co.
6. Kumar, Arya; Entrepreneurship; Pearson Education.
7. Desai, Vasant; Dynamics of Entrepreneur

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BUSINESS COMMUNICATION

BMB 107

Course Credit: 3

Contact Hours: 40 hours

Course Objectives

1. To understand business communication strategies and principles for effective communication in domestic and international business situations.
2. To understand and appropriately apply modes of expression, i.e., descriptive, expositive, narrative, scientific, and self-expressive, in written, visual, and oral communication.
3. To develop the ability to research and write a documented paper and/or to give an oral presentation.
4. To develop the ability to communicate via electronic mail, Internet, and other technologies for presenting business messages.
5. To understand and apply basic principles of critical thinking, problem solving, and technical proficiency in the development of exposition and argument.

UNIT I : (8 Hours)

Introduction: Role of communication – defining and classifying communication – purpose of communication – process of communication –characteristics of successful communication – importance of communication in management – communication structure in organization – communication in crisis barriers to communication. Case Studies

UNIT II: (8 Hours)

Oral communication: What is oral Communication – principles of successful oral communication –what is conversation control – reflection and empathy: two sides of effective oral communication – effective listening – non – verbal communication. Written communication: Purpose of writing –

Page 17 Data Table

S. No.	Course Outcome	Bloom's Taxonomy
1 2 3 4 5.	CO1. Apply business communication strategies and principles to prepare effective communication for domestic and international business situations. CO2. Analyse ethical, legal, cultural, and global issues affecting business Communication. CO3. Develop an understanding of appropriate organizational formats and channels used in business communications CO4. Gaining an understanding of emerging electronic modes of communication. CO5. Developing effective verbal and non verbal communication skills.	Applying (K4) Analyse (K5) Knowledge (K2) Comprehending(K3) Remembering(K1)/ Applying (K4)

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17

Course Outcomes

Upon successful completion of this course, the student should be able to:

S. No.

Course Outcome

Bloom's Taxonomy

1

2

3

4

5.

CO1. Apply business communication strategies and principles to prepare effective communication for domestic and international business situations.

CO2. Analyse ethical, legal, cultural, and global issues affecting business Communication.

CO3. Develop an understanding of appropriate organizational formats and channels used in business communications

CO4. Gaining an understanding of emerging electronic modes of communication.

CO5. Developing effective verbal and non verbal communication skills.

Applying (K4)

Analyse (K5)

Knowledge (K2)

Comprehending(K3)

Remembering(K1)/

Applying (K4)

Suggested Readings:

1. Bovee&Thill – Business Communication Essentials A Skill – Based Approach to Vital Business English. Pearson.
2. Kulbhushan Kumar & R.S. Salaria, Effective Communication Skills, Khanna Publishing House, Delhi
3. Bisen&Priya – Business Communication (New Age International Publication)
4. Kalkar, Suryavanshi, Sengupta-Business Communication(Orient Blackswan)
5. Varinder Bhatia, Business Communications, Khanna Publishing House
6. Business Communication: Skill, Concepts And Applications – P D Chaturvedi, MukeshChaturvedi Pearson Education.
7. AshaKaul, Business Communication, Prentice Hall of India.

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IT SKILLS-1

BMB 151

Course Credit: 3

Contact Hours: 40

Course Objectives

1. To provide knowledge about the functioning of computers and its uses for managers
2. To provide hands on learning on Internet and its applications
3. To provide hands on learning on Word processing software
4. To provide hands on learning of applications on Spreadsheet software
5. To provide hands on learning on Presentation software

UNIT I (05 hours) Conceptual Framework

Hardware: (a) Input devices - keyboard, printing devices, voice speech devices, scanner, MICR, OMR, Bar code reader, digital camera etc. (b) Output devices - Visual Display UNIT, printers, plotters (c) Storage Devices – Magnetic storage devices, Optical storage devices, Flash Memory.

Software: Types of software with examples; Introduction to languages, compiler, interpreter and Assembler, Operating System Functions, Types and Classification, Elements of GUI based operating system. Network and Internet: Types of computer networks (LAN, WAN and MAN), Netiquettes, Basic services over Internet like WWW, FTP, Telnet, Gopher, URL, Domain names, Web Browsers, Multimedia and its applications: Concepts of Text, Graphics, Animation, Audio, Images, Video. Multimedia Application in Education, Entertainment, Marketing. Names of common multimedia file formats,

UNIT II : Windows and Users Interface (Lab Work)- 7 hours

Windows operating System: Introduction and characteristics, Elements of GUI. Using Mouse, My

Page 19 Data Table

S. No.	Course Outcome	Bloom's Taxonomy
1 2 3 4 5	CO1. Gain in depth knowledge about the functioning of computers and its uses for managers CO2. Learn to use Internet and its applications CO3. Understand and implement Word processing software CO4. Learn applications on Spread sheet softwares CO5. Analyse and learn Presentation software	Knowledge (K2) Applying (K4) Synthesizing (K6) Applying (K4) Knowledge (K2) Analyse (K5)

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UNIT V: Presentation Software (lab Work) – 8 hours

Interface of the Presentation Package: Creating, Opening and Saving Presentations. Professional Look of the Presentation: Working in different Design & Views, Working with Slides. Formatting and Editing: Text, Image and Paragraph formatting, Checking Spelling and Correcting Typing Mistakes, Making Notes Pages and Handouts, Drawing and Working with Objects, Adding Clip Art and other pictures, Designing Slide Shows, Running and Controlling a Slide Show, Printing Presentations.

Course Outcomes

Upon successful completion of this course, the student should be able to:

S. No.

Course Outcome

Bloom's Taxonomy

1

2

3

4

5

CO1. Gain in depth knowledge about the functioning of computers and its uses for managers

CO2. Learn to use Internet and its applications

CO3. Understand and implement Word processing software

CO4. Learn applications on Spread sheet softwares

CO5. Analyse and learn Presentation software

Knowledge (K2)

Applying (K4)

Synthesizing (K6)

Applying (K4)

Knowledge (K2)

Analyse (K5)

Suggested Readings

1. Nasib Singh Gill – Handbook of Computer Fundamentals, Khanna Publishing House, Delhi
2. Shrivastava-Fundamental of Computer& Information Systems (Wiley Dreamtech)
3. Leon A and Leon M - Introduction to Computers (Vikas, 1st Edition).
4. ITL ESL – Introduction to Information Technology (Pearson, 2nd Edition).
6. Introduction to Computers, Norton P. (TATA McGraw Hill)
7. L

Page 20 Data Table

S. No.	Course Outcome	Bloom's Taxonomy
1 2	CO1. Gain in depth knowledge on innovative idea for product or services in form of a project report. CO2. To apply innovative idea, its feasibilities and detail descriptions.	Knowledge (K2) Applying (K4)

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MINI PROJECT -1

BMB 152

(Scanning of ideas for opportunities)

Course Credit -2

Course Objective-

1. To develop an innovative idea for a product or service in form of a project report.
2. To understand the importance and relevance of innovative ideas, its feasibilities and detailed descriptions.

Project/Practical work / Seminar

In the first semester, the students (individuals or teams of 2-3) are required to develop an innovative idea for a product or service and a project report to be prepared on that idea under the guidance of a faculty member. The report will be prepared individually and this report will consist of importance and relevance of the innovative idea, its feasibilities and detailed descriptions. The report will be evaluated by one external examiner appointed by the university. Student has to present his output in a seminar.

S. No.

Course Outcome

Bloom's Taxonomy

1

2

CO1. Gain in depth knowledge on innovative idea for product or services in form of a project report.

CO2. To apply innovative idea, its feasibilities and detail descriptions.

Knowledge (K2)

Applying (K4)

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BVA 151 : SPORTS AND YOGA

Objective of the Course:

- To maintain mental and physical wellness upright and develop ability in the students to cope up with the stress arising in the life.
- To create space in the curriculum to nurture the potential of the students in sports/games/yoga etc.

To introduce a practice oriented introductory course on the subject. More involved / advanced course may come up in subsequent years of study

PART A : SPORTS & GAME

Some form of Athletics would be compulsory for all students, unless restricted due to medical / physical reasons. In addition to this, student has to opt for at least one game out of the remaining mentioned below.

A fair theoretical knowledge and a reasonable amount of field / site practice of the chosen games will be essential.

1. Athletics

Compulsory

2. Volleyball

3. Basketball

4. Handball

5. Football

6. Badminton

7. Kabaddi

8. Kho-kho

9. Table tennis

10. Cricket

PART B: YOGA

a. Introduction of Yoga

Introduction of Yoga, Origin of Yoga, Aims and Objective of Yoga, Patanjali

Yoga darshan, Hath yoga, Gheranda Samhita, Karm yoga, Gyan yoga.

b. Asanas, Pranayam and Meditation Practices

Meaning of Asanas, Objective of Asanas, rules and regulations of Asanas and Pranayams, Types of Yogasana

Yogic postures: Standing Posture, Sitting posture, Supine posture, Prone posture, balancing Postures, Pranayam according to Patanjali and Hath Yoga, Meditaion Mudras.

c. Scie

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General Guidelines

1. Institutes must assign minimum of three periods in the Time Table for the activities of Sports/Yoga.

2. Institutes must provide field/facility and offer a minimum of five choices of the Games/Sports.

3. Institutes are required to provide sports instructor / yoga teacher to mentor the students.

Student must be made familiar with the terminologies, rules/regulations, dimension/ marking of the play field/area and general knowledge of national/ international level facts/figures related to the chosen game.

Assessment:

The Institute must assign coordinator / subject teacher for the subject, for every batch/group of the students, who would be responsible for coordinating the required activities and keep watch on the level of student's participation in the chosen game.

Coordinator/mentor would be responsible for the award of the sessional marks based upon following components.

I.

Level of understanding and general awareness

(20 %)

II.

Involvements in the Practice Sessions

(50 %)

III. Regularity, Sincerity and Discipline

(20 %)

IV.

Participation in University level / District level / State level / National

Level events

(10 %)

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Semester II

Business Environment & Legal Aspect of Business

BMB 201

Course Credit: 3

Contact Hours: 40 hours

Course Objectives:

1. The basic objective of the course is to develop understanding and provide knowledge about business environment to the management students.
2. To promote basic understanding on the concepts of Business Environment and international business environment.
3. To provide basic understanding of law of contract
4. To impart basic understanding of provisions of Companies Act concerning incorporation and regulation of business organizations.
5. To appraise the students on the leading practical application oriented case studies – relevant and updated and analyzing case laws in arriving at conclusions facilitating business decisions.

Unit I - (10Hrs)

Introduction to Micro Environment –

Meaning of Business & Business Environment,, Types of Business Organizations , SWOT analysis , Types of Environment-Internal to the Enterprise (Value System, Management Structure and Nature, Human Resource, Company Image and Brand Value, Physical Assets, Facilities, Research & Development, Intangibles, Competitive Advantage), External to the Enterprise , Micro- Suppliers, Customers, Market Intermediaries; Macro- Demography, Natural, Legal & Political, Technological,) Michael Porter's Five Forces Analysis, Competitive Strategies

Unit II - (6 Hrs)

Macro Factors: Economic, Socio-Cultural, Competitive & International Environment –

Economy, Competi

Page 24 Data Table

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1) Develop understanding and fundamental knowledge about business environment	Remembering (k1) Knowledge (K 2)
2	CO2) Develop understanding on the concepts of Business Environment and international business environment.	K2 Knowledge
3	CO3) Develop basic understanding of law of contract	K2 Knowledge

S. No.	Course Outcome	Bloom's Taxonomy
4	CO4) understanding of provisions of Companies Act concerning incorporation and regulation of business organizations	K2 Knowledge
5	CO5) Able to analyze case laws in arriving at conclusions facilitating business decisions.	K4 Applying K5 Analysing

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UNIT IV (8hrs)

Companies Act Definition, characteristics and kinds of companies, steps in formation of company.

Memorandum of Association, Articles of Association, prospectus. Directors: appointment, power, duties and liabilities, meeting and resolutions: types of meetings. Auditor: appointment, rights and liabilities, modes of winding up of a company.

UNITV (8 hrs)

Consumer Protection Act: Definitions - Aims and objectives, Consumer protection councils, Redressal agencies and penalties for violation.

The Information Technology Act: Definition, Digital Signature, Electronic Governance, Attribution, Acknowledgment and Dispatch of Electronic Records, Sense Electronic Records and Sense Digital Signatures, Regulation of Certifying Authorities, Digital Signature Certificates, Duties of Subscribers, Penalties and Offences.

S. No.

Course Outcome

Bloom's Taxonomy

1

CO1)

Develop

understanding

and

fundamental

knowledge about business environment

Remembering (k1)

Knowledge (K 2)

2

CO2) Develop understanding on the concepts of Business

Environment and international business environment.

K2 Knowledge

3

CO3) Develop basic understanding of law of contract

K2 Knowledge

4

CO4) understanding of provisions of Companies Act concerning incorporation and regulation of business organizations

K2 Knowledge

5

CO5) Able to analyze case laws in arriving at conclusions facilitating business decisions.

K4 Applying

K5 Analysing

Suggested Reading

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HUMAN RESOURCE MANAGEMENT

BMB 202

Course Credit: 3

Contact Hours: 40 hours

Course Objectives: In this course the students will learn the basic concepts and frameworks of Human Resource Management (HRM) and understand the role that HRM has to play in effective business administration. It will provide an insight as to how to use Human Resource as a tool to implement strategies.

UNIT I: (7 Hours)

Essentials of HRM: Functions of HRM, HRM vs.HRD, Strategic HRM: Meaning and Roles in Strategy formulation and implementation, Barriers to strategic HRM, Linking HR strategy with business strategy, Roles of HR Manager, roles of HR in merger and acquisitions, Technology & HR and changing roles of HR due to technology, HRM linkage with TQM & productivity. Case Studies

UNIT II: (8 Hours)

Human Resource Planning and Employee Hiring : Meaning of job Analysis, job design, Human Resource Planning, methods demand forecasting for manpower planning, factors influencing HRP, Employee hiring- methods of Recruitment, Employee selection, process of employee selection, recent trends in recruitment. Case Studies

UNIT III: (8 Hours)

Employee Training & Development: Meaning importance of Training, types and methods and types of training, career planning, promotion, transfer, demotion and separation, Performance Appraisal: Meaning and types of appraisal, Job Evaluation: Meaning and methods of job evaluation. Case Studies

UNIT IV: (9 Hours)

Compensation Management

Page 26 Data Table

S. No.	Course Outcome	Bloom's Taxonomy
1	CO1. Synthesize the role of human resources management as it supports the success of the organization including the effective development of human capital as an agent for organizational change.	K6 Synthesizing
2	CO2. Demonstrate knowledge of laws that impact behaviour in relationships between employers and employees that ultimately impact the goals and strategies of the organization.	K2 Knowledge
3	CO3. Understand the role of employee benefits and compensation as a critical component of employee performance, productivity and organizational effectiveness.	K3 Comprehending
4	CO4. Show evidence of the ability to analyze, manage and problem solve to deal with the challenges and complexities of the practice of collective bargaining.	K5 Analysing
5	CO5. Demonstrate knowledge of practical application of training and employee development as it impacts organizational strategy and competitive advantage.	K2 Knowledge K4 Applying

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COURSE OUTCOME

S. No.

Course Outcome

Bloom's Taxonomy

1

CO1. Synthesize the role of human resources management as it supports the success of the organization including the effective development of human capital as an agent for organizational change.

K6 Synthesizing

2

CO2. Demonstrate knowledge of laws that impact behaviour in relationships between

employers and employees that ultimately impact the goals and strategies of the organization.

K2 Knowledge

3

CO3. Understand the role of employee benefits and compensation as a critical component of employee performance, productivity and organizational effectiveness.

K3 Comprehending

4

CO4. Show evidence of the ability to analyze, manage and problem solve to deal with the challenges and complexities of the practice of collective bargaining.

K5 Analysing

5

CO5. Demonstrate knowledge of practical application of training and employee development as it impacts organizational strategy and competitive advantage.

K2 Knowledge

K4 Applying

Suggested Readings

1. V.S.P.Rao, Human Resource Management (Text and Cases) Himalaya Publications, Thirteenth Edition.
2. Durai Praveen, Human Resource Management Pearson Publication, 2nd Edition.
3. Gary Dessler and Biju Varkkey Human Resource Management, Person Publication, 2013, 14th Edition.
4. Seema Sanghi, Human Resource Management, Vikas Publications, 2014, 5th Edition.
5. K. Aswathappa, Human Resource Management, McGraw Hill Education,

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BUSINESS RESEARCH METHODS

BMB 203

Course Credit: 3

Contact Hours: 40 hours

Course objectives

1. Understand the concept / fundamentals of research and their types.
2. Understand the practical application of various research techniques.
3. Understand the importance of scaling & measurement techniques and sampling techniques
4. Understand the importance of coding, editing, tabulation and analysis in doing research.
5. Understanding and applying the concept of statistical analysis which includes ANOVA technique and technique of report writing.

Unit 1 (8 Sessions)

Research: – Definition, Meaning, Importance types and Qualities of Research; Research applications in functional areas of Business, Emerging trends in Business research.

Research & the Scientific Method: Characteristics of scientific method. Steps in Research Process

Concept of Scientific Enquiry: – Formulation of Research Problem – Management Question – research Question – Investigation Question

Research Proposal – Elements of a Research Proposal, Drafting a Research Proposal, evaluating a research proposal.

Unit 2 (8 Sessions)

Research design: Concept, Features of a good research design, Use of a good research design;

Qualitative and Quantitative research approaches, Comparison – Pros and Cons of both approaches.

Exploratory Research Design: Concept, Types: Qualitative techniques – Projective Techniques, Depth Interview, Experience Survey, Focus Groups, Observation.

Descriptive R

Page 28 Data Table

Course Outcomes	Blooms Taxonomy
CO1. Knowledge of concept / fundamentals for different types of research.	• Knowledge (K 2)
CO2. Applying relevant research techniques.	• Remembering (K1) • Applying (K 4)
CO3. Understanding relevant scaling & measurement techniques and should use appropriate sampling Techniques	• Comprehending (K 3) • Applying (K 4)
CO4.Synthesizing different techniques of coding, editing, tabulation and analysis in doing research.	• Analyzing (K 5) • Synthesizing (K 6)
CO5.Evaluating statistical analysis which includes ANOVA technique and prepare research report.	• Evaluating (K 7)

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Non Probability Sample: Judgment Sampling, Convenience Sampling, Purposive Sampling, Quota Sampling & Snowballing Sampling methods. Determining size of the sample – Practical considerations in sampling and sample size, sample size determination.

Unit 5 (8 Sessions)

Data Analysis: Editing, Coding, Tabular representation of data, frequency tables, Construction of frequency distributions, Graphical Representation of Data: Appropriate Usage of Bar charts, Pie charts, Histogram.

Hypothesis: Qualities of a good Hypothesis –Framing Null Hypothesis & Alternative Hypothesis. Concept of Hypothesis Testing – Logic & Importance. Analysis of Variance: One way and two way Classifications. Mechanism of Report Writing- Report Preparation: Types, Report Structure: preliminary section, main report, interpretation of results, suggestions and recommendations, limitations of the study, Report formulation.

COURSE OUTCOME

Course Outcomes

Blooms Taxonomy

CO1. Knowledge of concept / fundamentals for different types of research.

-

Knowledge (K 2)

CO2. Applying relevant research techniques.

-

Remembering (K1)

-

Applying (K 4)

CO3. Understanding relevant scaling & measurement techniques

and

should

use

appropriate

sampling

Techniques

-

Comprehending (K 3)

-

Applying (K 4)

CO4.Synthesizing different techniques of coding, editing, tabulation and analysis in doing research.

-

Analyzing (K 5)

-

Synthesizing (K6)

CO5.Evaluating

stati

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FINANCIAL MANAGEMENT AND CORPORATE FINANCE

BMB 204

Course Credit: 3

Contact Hours: 40 hours

Course Objectives: This course is intended to introduce the basic theory, concepts and practical applications in corporate finance and to enable students to analyse various corporate decisions. The course objectives are outlined below:

- 1) To understand the fundamentals, various models and agency problems of Corporate Finance.
- 2) To acquire knowledge about various techniques used for analysing various long-term projects.
- 3) To have an understanding about various capital structure techniques and selecting best source of finance.
- 4) To have an understanding of various dividend models and its applicability.
- 5) To acquaint students about corporate valuation in mergers and acquisitions.

UNIT I (8 Hrs)

Introduction to Finance & Corporate Finance: Corporate Finance & its scope, Corporate Governance and Agency Problem, Finance & Corporate Strategy, Time Value of Money, Risk and Return. Types of Financial Markets: Capital Market, Factors affecting Financial Markets, Linkages between Economy & Financial Markets, Integration of Indian Financial Markets with Global Financial Markets.

UNIT II (8 Hrs)

Investment and Financing Decision: Capital Budgeting, Nature of investment decisions, Risk Analysis in investment decisions, Concept of Opportunity Cost, Cost of Debenture, Preference and Equity capital, Composite Cost of Capital, Cash Flows as Profit and components of Cash

Page 30 Data Table

S.No	Course Outcome	Bloom's Taxonomy
1. 2. 3. 4. 5.	CO1 Understand the different basic concepts / Models of Corporate Finance and Governance CO2 Understand the practical application of the time value of money and evaluating long-term investment decisions CO3 Develop analytical skills to select the best source of capital, structure and leverage. CO4 Understand the use and application of different models for a firm's optimum dividend payout. CO5 Understand the recent trends of mergers and acquisitions and its valuation	• Knowledge (K2) • Remembering(K1) • Analyzing (K5) • Evaluating(K7) • Analyzing(K5) • Synthesizing(K6) • Comprehending(K3) • Applying(K4) • Comprehending(K3) • Synthesizing (K6)

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S.No

Course Outcome

Bloom's Taxonomy

1.

2.

3.

4.

5.

CO1 Understand the different basic concepts / Models of Corporate Finance and Governance

CO2 Understand the practical application of the time value of money and evaluating long-term investment decisions

CO3 Develop analytical skills to select the best source of capital, structure and leverage.

CO4 Understand the use and application of different models for a firm's optimum dividend payout.

CO5 Understand the recent trends of mergers and acquisitions and its valuation

- Knowledge (K2)
- Remembering(K1)
- Analyzing (K5)
- Evaluating(K7)
- Analyzing(K5)
- Synthesizing(K6)
- Comprehending(K3)
- Applying(K4)

- Comprehending(K3)
- Synthesizing (K6)

Suggested Readings

- 1) Khan and Jain - Financial Management (Tata McGraw Hill, 7th Ed.)
- 2) Pandey I M - Financial Management (Vikas, 11th Ed.)
- 3) William HakkaBettnerCarcello- Financial and Management Accounting (TMH-16th Ed.)
- 4) Sheebakapil-Fundamental of financial management (Wiley,2015)
- 5) Prasanna Chandra - Fundamentals of Financial Management (TMH, 9th Ed.)
- 6) Bark DemazoThampy- Financial Management (Pearson,2nd Ed.)
- 7) R P Rustagi - Financial Management (Galgotia, 2000, 2nd revised ed.)
- 8) Damodaran, A., Applied Corporate Finance, 3rd Edition, Wiley, 2012
- 9) Ravi.M Kishore – Financial Management (Taxman, 7th Ed)
- 10) Fundamentals to Financial Management, Brigham & Houston, 14/e, Cengage Learning
- 11) Van Horne - Financia

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OPERATIONS MANAGEMENT

BMB 205

Course Credit: 3

Contact Hours: 40 hours

Course Objectives:-

1. To understand the role of Operations in overall Business Strategy of the firm.
2. To understand the application of operations management policies and techniques to the service sector as well as manufacturing firms.
3. To identify and evaluate the key factors and their interdependence of these factors in the design of effective operating systems.
4. To understand the trends and challenges of Operations Management in the current business environment.
5. To familiarize the students with the techniques for effective utilization of operational resources and managing the processes to produce good quality products and services at competitive prices.

UNIT –I (7 sessions) Production Concepts:

Introduction, meaning, nature and scope of production and operations management. Difference between production and operations management. Productivity, factors affecting productivity and productivity measurement. Work study— Method study and work measurement. Production Technology – Types of manufacturing processes. Plant location and types of plant layout.

UNIT –II (8 sessions) Operations Concepts:

Services scenario in India, difference between product and service, characteristics of services, classification of services, product and service design, factors affecting service design, service designing process, service blueprinting, service capacity planning. Dimension

Page 32 Data Table

S.No.	Course Outcomes	Bloom's Taxonomy
CO1.	Understand the role of Operations in overall Business Strategy of the firm - the application of OM policies and techniques to the service sector as well as manufacturing firms.	• Knowledge (K2) • Comprehending (K 3) • Remembering (K1)
CO2.	Understand and apply the concepts of Material Management, Supply Chain Management and TQM perspectives.	• Knowledge (K2) • Remembering (K1) • Applying (K4)
CO3.	Identify and evaluate the key factors and their interdependence of these factors in the design of effective operating systems.	• Comprehending (K3) • Applying (K4)
CO4.	Analyze / understand the trends and challenges of Operations Management in the current business environment.	• Analyzing (K5)
CO5.	Apply techniques for effective utilization of operational resources and managing the processes to produce good quality products and services at competitive prices.	• Synthesizing (K6) • Evaluating (K7)

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UNIT-V (7 sessions) Productivity and Quality:

TQM, Deming's 14 principles, Juran's quality trilogy, PDCA cycle, KAIZEN, quality circles, 7QC

tools and its 7 new management tools, ISO 9000-2000 clauses, six sigma, Total Productive

Maintenance (TPM), 5S. Case Studies

Expected Course Outcomes:

S.No. Course Outcomes

Bloom's Taxonomy

CO1.

Understand the role of Operations in overall

Business Strategy of the firm - the application of

OM policies and techniques to the service sector as

well as manufacturing firms.

•

Knowledge (K2)

•

Comprehending (K 3)

•

Remembering (K1)

CO2.

Understand and apply the concepts of Material Management, Supply Chain Management and TQM perspectives.

-

Knowledge (K2)

-

Remembering (K1)

-

Applying (K4)

CO3.

Identify and evaluate the key factors and their interdependence of these factors in the design of effective operating systems.

-

Comprehending (K3)

-

Applying (K4)

CO4.

Analyze / understand the trends and challenges of Operations Management in the current business environment.

-

Analyzing (K5)

CO5.

Apply techniques for effective utilization of operational resources and managing the processes to produce good quality products and services at competitive prices.

-

Synthesizing (K6)

-

Evaluating (K7)

Suggested Readings:-

1. Aswathappa, K. & Bhat, K.S.-- Production and Operations Management (Himalaya Publishing House, 2nd Edition)
2. Chase, R.B., Shankar, R. & Jacobs

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QUANTITATIVE TECHNIQUES FOR MANAGER

BMB 206

Course Credit: 3

Contact Hours: 40 hours

Course Objectives

1. Understand the importance of the use of OR application in decision Making environment
2. To formulate LPP and Obtain Graphical Solutions & Acquire General idea of the Simplex method.
3. To understand and solve transportation & assignment models.
4. To know optimal sequence model and understand concepts of queuing theory.
5. To identify right time for replacement of equipment and understand project management techniques

Unit I (6 Sessions)-Operations Research & Decision Making Environments

Operations Research:- Uses, Scope and Applications of Operation Research in managerial decision-making .Decision-making environments:- Decision-making under certainty, uncertainty and risk situations; Decision tree approach and its applications.

Unit II (6 Sessions)-Linear Programming Problem

Linear programming: Mathematical formulations of LP Models for product-mix problems; graphical and simplex method of solving LP problems.

Unit III (10 Sessions)- Transportation Problem & Assignment model

Transportation problem: Various methods of finding Initial basic feasible solution-North West Corner Method, Least Cost Method & VAM Method and optimal solution-Stepping Stone & MODI Method, Maximization Transportation Problem

Assignment model: Hungarian Algorithm and its applications, Maximization Assignment Problem.

Unit IV (10 Sessions)-Sequencing & Qu

Page 34 Data Table

CO1	Be able to understand the characteristics of different types of decision-making environments and the appropriate decision making approaches and tools to be used in each type.	Knowledge (K2)/ Remembering (K1)
CO2	To formulate linear programming problem and to find optimal solution by graphical simplex method.	Knowledge (K2)
CO3	Be able to build and solve Transportation Models and Assignment Models also to solve game theory problems by understanding pure and mix strategies.	Applying (K 4)

CO1	Be able to understand the characteristics of different types of decision-making environments and the appropriate decision making approaches and tools to be used in each type.	Knowledge (K2)/ Remembering (K1)
CO4	To assign optimal sequence of difference jobs on different machines and develop understanding of queuing theory concepts.	Applying (K 4)
CO5	To implement replacement of equipments at right time and able to implement project management concepts like CPM, PERT to reduce cost and time.	Synthesizing (K6)/ Evaluating (K7)

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Course Outcomes

CO1

Be able to understand the characteristics of different types of decision-making environments and the appropriate decision making approaches and tools to be used in each type.

Knowledge (K2)/

Remembering (K1)

CO2

To formulate linear programming problem and to find optimal solution by graphical simplex method.

Knowledge (K2)

CO3

Be able to build and solve Transportation Models and Assignment Models also to solve game theory problems by understanding pure and mix strategies.

Applying (K 4)

CO4

To assign optimal sequence of difference jobs on different machines and develop understanding of queuing theory concepts.

Applying (K 4)

CO5

To implement replacement of equipments at right time and able to implement project management concepts like CPM, PERT to reduce cost and time.

Synthesizing (K6)/

Evaluating (K7)

Suggested Readings

1. R. Panneerselvam - Operations Research (PHI, 2nd Edition)
2. Sharma J K - Operations Research (Pearson, 3rd Edition)
3. Apte-Operation Research and Quantitative Techniques (Excel Books)
4. S Kalawathy-Operation Research (Vikas IVth Edition)
5. Natarajan- Operation Research(Pearson)
6. Singh & Kumar—Operation Research(UDH Publisher edition 2013)
7. Taha Hamdy - Operations Research - An Introduction (Prentice-Hall, 9th edition)
8. Vohra - Quantitative Techniques in Management (Tata McGraw-Hill, 2nd)
9. Kothari - Quantitative Techniques (Vikas 1996, 3rd Edition).

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COST AND MANAGEMENT ACCOUNTING

BMB 207

COURSE OBJECTIVES

1. To provide an in depth knowledge of the detailed procedures and documentation involved in cost ascertainment systems.
2. To understand the concepts of Financial Management and its application for managerial decision making.
3. Understand the cost and management accounting techniques for evaluation, analysis and application in managerial decision making.
4. Compare and contrast marginal and absorption costing methods in respect of profit reporting;
5. Apply marginal and absorption costing approaches in job, batch and process environments;
6. Prepare and interpret budgets and standard costs and variance statements;
7. Identify and apply the concepts of Financial Management

Credit 3

Contact Hour: 40

UNIT - I (10 Hours)

Meaning, nature and scope of Management Accounting; Difference between management accounting and financial accounting, Cost concepts: , Cost Unit, Cost Control and Cost Reduction; Components of total Cost, Cost Sheet, Classification of costs, Types and methods of costing, Inventory Management, Labour Cost, Overheads, Activity based costing.

UNIT - II (8 Hours)

Cost-Volume-Profit Analysis: Marginal cost, Contribution per unit and Total contribution. Profit-Volume Ratio, Break-even Point : Margin of safety. Decision Making such as : Key Factor, Pricing, Product Profitability, Dropping a product line, Make or Buy, Export Order, Sell or Proces

Page 36 Data Table

CO1	Understanding and Basic Knowledge of Cost Accounting and		Knowledge (K2)/ Remembering (K1)
	explaining Basic Accounting Procedure ,A	,	
	Balancing of Accounts & Management Accounting.		
CO2	Computation of Fixed, Variable, Semi-Fixed And Semi-Variable Cost Concepts. Analyzes The Relationship Between The Cost-Volume And Profit Contribution Margin, Margin Of Safety, Security Ratio, Profit Margin Concepts. Critically analyze and provide recommendations to improve the business operations using accounting techniques.		Applying (K 4)
CO3	Identify problems associated with relying on financial accounting information for internal decision making		Comprehending (K3) Applying (K4)
CO4	Computing of Budgeting And Operating Budgets. Applying the concepts of Management Accounting For Businesses		Applying (K 4)
CO5	Computation of Standard Variation Analysis Through Standard Costs Understanding and calculation of Standard Cost Concept		Synthesizing (K6)/ Evaluating (K7)

Page 36 Content

36

UNIT –V (10 Hours)

Process costing, concepts of normal loss, abnormal loss, abnormal effectiveness. Preparation of process accounts, normal loss account, abnormal loss account, abnormal gain account. Process costing with opening and closing WIP; equivalent units (using FIFO) and Cost allocation. Joint and by products : Allocation of joint costs based on Physical units method, Relative market value methods (Sales value at split off method and Net realizable value method). Introduction to the concept of Target Costing, Life Cycle Costing, Quality Costing, and Activity based Costing.

Course Outcomes

CO1

Understanding and Basic Knowledge of Cost Accounting and explaining Basic Accounting Procedure ,Apply cost concepts, Balancing of Accounts & Management Accounting.

Knowledge (K2)/

Remembering (K1)

CO2

Computation of Fixed, Variable, Semi-Fixed And Semi-Variable Cost Concepts. Analyzes The Relationship Between The Cost-Volume And Profit Contribution Margin, Margin Of Safety, Security Ratio, Profit Margin Concepts. Critically analyze and provide recommendations to improve the business operations using accounting techniques.

Applying (K 4)

CO3

Identify problems associated with relying on financial accounting information for internal decision making

Comprehending (K3)

Applying (K4)

CO4

Computing of Budgeting And Operating Budgets. Applying the concepts of Management Accounting For Businesses

Applying (K 4)

CO5

Computation

Page 37 Content

37

MANAGEMENT INFORMATION SYSTEMS

BMB 208

Course Credit: 2

Contact Hours: 20 hours

Course Objective

1. To help the students understand the importance of information management in business and management
2. To provide understanding about different types of information systems in business
3. To apply the theory and concepts in practical with help of software
4. To understand various security and ethical issues with Information Systems
5. To provide hands on learning of applications on Spreadsheet and database software

UNIT -1 (6 Hours)

Management Information Systems - Need, Purpose and Objectives, Contemporary Approaches to MIS, Information as a strategic resource, Use of information for competitive advantage, MIS as an instrument for the organizational change. Information Technology – Characteristics and emerging trends, IT Capabilities and their organizational impact, IT enabled services. Transaction Processing

System: Characteristics and its importance

UNIT -II (6 Hours)

Information, Management and Decision Making - Attributes of information and its relevance to Decision Making, Types of information. Models of Decision Making - Classical, Administrative and Herbert Simon's Models. Management Support Systems: Decision Support Systems, Group Decision Support Systems, and Executive Information Systems.

UNIT -III (8 Hours)

Managing Data Resources- The need for data management, Challenges of data management, Data independence, Data redundancy

Page 38 Data Table

CO1	Be able to understand the importance of information management in business and management.	Knowledge(K2) / Remembering (K1)
CO2	To understand and formulate different types of information systems in business	Knowledge (K2)
CO3	Be able to apply the theory and concepts in practical with help of software	Applying (K 4)
CO4	To apply various security and ethical issues with Information Systems	Applying (K 4)
CO5	To synthesize applications on Spread sheet and database software	Synthesizing (K6)/ Evaluating (K7)

Page 38 Content

38

Course Outcomes

CO1

Be able to understand the importance of information management in business and management.

Knowledge(K2)

/ Remembering (K1)

CO2

To understand and formulate different types of information systems in business

Knowledge (K2)

CO3

Be able to apply the theory and concepts in practical with help of software

Applying (K 4)

CO4

To apply various security and ethical issues with

Information Systems

Applying (K 4)

CO5

To synthesize applications on Spread sheet and

database software

Synthesizing (K6)/

Evaluating (K7)

Suggested Readings

1. Management Information System – James 'O' Brian
2. Management Information Systems, Laudon and Laudon, 7th Edition, Pearson Education Asia
3. Management Information Systems, Jawadekar, Tata McGraw Hill
4. Analysis and Design of Information Systems, Rajaraman, Prentice Hall
5. Database Management Systems: A Business-Oriented Approach Using ORACLE, MySQL and MS Access, by Sotirios Zygiari
6. Computer Applications in Business (CBCS) by Dr. Sushil Kumar Sharma & Ms. Mansi Bansal (Taxmann)
7. Excel 2019 All-In-One: Master the new features of Excel 2019 / Office 365, Lokesh Lalwani (BPB)

Page 39 Data Table

S. No.	Course Outcome	Bloom's Taxonomy
1 2 3	CO1. To gain knowledge of pivot table and understand the validating & auditing techniques CO2. Learn to use different charting techniques in MS Excel CO3. Learn to use different formatting techniques in MS Excel	Knowledge (K2) Applying (K4) Synthesizing (K6) Applying (K4) Knowledge (K2)

Page 39 Content

39

IT SKILLS-2

BMB 251

Lab work

20 Hours

Course Objective

1. To develop pivot table and understand the validating & auditing techniques
2. To understand different charting techniques in MS Excel
3. To understand different formatting techniques in MS Excel

Unit I (Lab work on spreadsheet)

Pivot Table: Developing Pivot Table, Analyzing data using goal seek and solver, Scenarios Create named scenarios. Show, edit, delete scenarios, Creating a scenario summary report. Validating and Auditing: Set, edit validation criteria for data entry in a cell range like: whole number, decimal, list, date, time, Trace precedent, dependent cells. Identify cells with missing dependents. Creating applications in Spreadsheet and Macros.

Unit II (Lab work on spreadsheet) 15 Hours

Creating and formatting Charts: Understanding chart types, column chart, bar chart, line chart, pie chart, XY Scatter chart , Area chart, surface chart, bubble chart. Create a combined chart like: column and line, column and area. Change the chart type for a defined data series, Add, delete a data series in a chart, Re-position chart title, legend, data labels. Change scale of value axis: minimum, maximum number to display, major interval. Change display units on value axis without changing data source: hundreds, thousands, millions. Format columns, bars, pie slices, plot area, chart area to display an image.

References

Excel Data Analysis: Modeling and Simulation , Hector Guerrero

Page 40 Data Table

S. No.	Course Outcome	Bloom's Taxonomy
1 2	CO1. To gain knowledge of issues challenge of the industry CO2. Learn to prepare report on the application of emerging technologies in the selected industry	Knowledge (K2) Applying (K4) Synthesizing (K6)

Page 40 Content

40

MINI PROJECT -2

BMB 252

(Business Ideas Validation & Fesibility)

Course Credit: 2

Seminar by students

Objective –

1. To validate the idea which was identified in the last semester.
2. To identify the issues & challenges of the identified industry / market.
3. To prepare a report on the emerging technologies in the selected industry.

In second semester, the students are required to validate the idea which was screened in the previous semester & presented. The validation report shall be a detailed analysis considering the market feasibility, select a Lean Canvas for making your assumptions , test yours assumptions around the identified market, the future scope of the selected product or service, test your value proposition, and prepare a project report. Preference should be given to the application of emerging technologies in the

selected industry. It may consist of Fintech, Blockchain, Financial Services, Data Science, Social Entrepreneurship or any other suitable area of interest. The report will be prepared individually. The report will be evaluated by one external examiner appointed by university.

Feasibility analysis of the idea (market, technical and financial analysis).

COURSE OUTCOME

S. No.

Course Outcome

Bloom's Taxonomy

1

2

CO1. To gain knowledge of issues challenge of the industry

CO2. Learn to prepare report on the application of emerging technologies in the selected industry

Knowledge (K2)

Ap

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Page 1

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY

LUCKNOW

Teaching and Evaluation Scheme

For

Second Year

MBA (Common)

AS PER AICTE MODEL CURRICULUM & NEP 2020

(Effective from the Academic Session: 2025-26)

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Page 2

PREAMBLE

The Evaluation Scheme and Syllabus for Second Year MBA (Common) programme is designed with a view to enhance the skills, knowledge & leadership of management graduates in order to maximize the employment opportunities in various functional areas. The guidelines of Model Curriculum of AICTE are duly considered by incorporating relevant emerging areas in all specializations offered during the program. It has also followed the guidelines of New Education Policy (NEP) to emphasis on skill building through project and practical work specifying the learning outcomes for each subject and used Bloom's hierarchical model as expected indicators of

learning levels. The specified levels of learning outcomes are indicative and could be used suitably for assessment and evaluation.

The management education is dynamic and driven by socio-economic and technological changes as well as innovations. Hence, it is expected that, latest updates from research, industry practices and cases must be discussed extensively during teaching to achieve the desired levels of knowledge and skills with practical outlook among graduating students. The detailed syllabus has introduced case studies and latest updates also.

Visits to Manufacturing Units, malls, ware house & logistic hubs, Ports etc. and some short duration live Projects will be helpful to buddy managers in gaining the feel real corporate culture and working practices. It will help in orienting the student

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Page 3

GUIDELINES FOR SUMMER TRAINING (III SEMESTER) AND
RESEARCH PROJECT REPORT (RPR) (IV SEMESTER)

Summer Training Project Report (STPR)

Objective:

The internship module aims to provide the student with:



A practice-oriented and 'hands-on' working experience in the real world or industry, and



To enhance the student's learning experience.



An opportunity to develop a right work attitude, self-confidence, interpersonal skills and



Ability to work as a team in a real organisational setting.



An opportunity to further develop and enhance operational, customer service and other



Life-long knowledge and skills in a real-world work environment.



Pre-employment training opportunities and an opportunity for the company or



Organisation to assess the performance of the student and to offer the student an employment



Opportunity after his/her graduation, if it deems fit.

Outcomes:

At the end of the course, a student will be able to:

CO – 1: Understand the application of knowledge and skill sets acquired from the course and workplace in the assigned job function/s.

CO – 2: Apply real life challenges in the workplace by analyzing work environment and conditions, and selecting appropriate skill sets acquired from the course.

CO – 3: Create critical thinking and problem-solving skills by analyzing underlying issue/s.

CO – 4: Develop the ability to harness resources by analyzing challenges and considering opportunities.

CO – 5: Understa

Page 44 Content

Page 4

exploring feasible solutions.

3.

During the course of training, the organization (where the student is undergoing training) will assign a problem/project to the student.

4.

The student, after the completion of training will submit a report to the College/Institute which will form part of the third semester examination. However, the report must be submitted by the end of September 30.

5.

The report (based on training and the problem/project studied) prepared by the student will be known as Summer Training Project Report. The report should reflect in depth study of a micro problem, ordinarily assigned by the organization where the student undergoes training. Relevant tables and bibliography should support it. One comprehensive chapter must be included about the organization where the student has undergone training. This should deal with brief history of the organization, its structure, performance products/services and problem faced. The average size of report ordinarily will be of minimum 100 pages in standard font size (12) and double spacing. Two neatly typed (one sided only) and soft bound copies of the report will be submitted to the College/Institute.

The report will be typed on A-4 size paper.

6.

The report will have two certificates, one from the Faculty guide and another from reporting officer of the organization where the student has undergone training. These certificates should be attached in the beginning of the report.

7.

The Su

Page 45 Data Table

Evaluation Criteria	Marks
Understanding of Objectives with topic	20
Understanding of Relevance of topic	20
Interpretation & Analysis	10
Presentation	10
Query handling	10

Page 45 Content

Page 5

Project Report Evaluation:

Evaluation Criteria

Marks

Understanding of Objectives with topic

20

Understanding of Relevance of topic

20

Interpretation & Analysis

10

Presentation

10

Query handling

10

Research Project Report (RPR)

In fourth semester, the candidates will have to submit a Research Project Report on a problem/topic (from the specialization areas) to be assigned by the MBA department under the supervision of a core faculty member of the department. The Research Project Report will carry 300 marks. The evaluation of the project report will be done by two examiners (external & internal). The evaluation will consist of (1) Evaluation of Project Report (2) Presentation and Viva Voce.

The evaluation of Project Report will comprise of 100 marks and would be evaluated by the internal guide. The evaluation of Viva Voce of Project would comprise of 200 marks and would be evaluated by an external examiner appointed by University and one internal examiner appointed by institute. The average of the marks awarded by the 2 examiners will be taken into account for the results. In case the difference in the marks given by the examiners is 30 or more, the project report will be referred to a third examiner. In such cases the average of two closer awards (given by three examiners) will be taken into account for the results. The report will contain the objectives and scope of the study. Research Methodology, use and importance of the stu

Page 46 Data Table

SN	CODE	SUBJECT	PERIODS			INTERNAL EVALUATION SCHEME	
			L	T	P	CT	TA
1	BMB 301	Strategic Management	3	0	0	20	10
2		Elective- 1 Specialization Group 1	3	0	0	20	10
3		Elective -2 Specialization Group 1	3	0	0	20	10
4		Elective -3 Specialization Group 1	3	0	0	20	10
5		Elective -1 Specialization Group 2	3	0	0	20	10
6		Elective -2 Specialization Group 2	3	0	0	20	10
7		Elective -3 Specialization Group 2	3	0	0	20	10
8	BMB 308	Summer Training Project Report & Viva Voce	0	4*	0	0	30
		TOTAL					

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Page 6

MBA II Year Teaching and Evaluation Scheme

W.E.F. Academic Session 2025-26

(In Accordance with AICTE Model Curriculum & New Education Policy)

III SEMESTER

SN

CODE

SUBJECT

PERIODS

INTERNAL

EVALUATION

SCHEME

END

SEMESTER TOTAL CREDIT

L

T

P CT TA PS Total

TE

PE

1

BMB

301

Strategic Management

3

0

0

20

10

0

30

70

0

100

3

2

Elective- 1

Specialization Group 1

3

0

0

20

10

0

30

70

0

100

3

3

Elective -2

Specialization Group 1

3

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0

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10

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30

70

0

100

3

4

Elective -3

Specialization Group 1

3

0

0

20

10

0

30

70

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100

3

5

Elective -1

Specialization Group 2

3

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20

10

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30

70

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3

6

Elective -2

Specialization Group 2

3

0

0

20

10

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30

70

0

100

3

7

Elective -3

Specialization Group 2

3

0

0

20

10

0

30

70

0

100

3

8

BMB

308

Summer Training

Project Report

& Viva Voce

0

4*

0
0
30
0
30
0
70
100
4
TOTAL
800
25

■ Students will be guided & assessed by Faculty Guide for internship report.

Page 47 Data Table

SN	CODE	SUBJECT	PERIOD S			INTERNAL EVALUATION SCHEME	
			L	T	P	CT	TA
1	BVE 401	Universal Human Values and Professional Ethics	3	0	0	20	10
2		Elective-4 Specialization Group-1	3	0	0	20	10
3		Elective 5 Specialization Group-1	3	0	0	20	10
4		Elective -4 Specialization Group-2	3	0	0	20	10
5		Elective -5 Specialization Group-2	3	0	0	20	10
6	BMB 406	Research Project Report & Viva Voce	0	6*	0	0	100
		TOTAL					

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Page 7

MBA II Year Teaching and Evaluation Scheme

W.E.F. Academic Session 2025-26

(In Accordance with AICTE Model Curriculum & New Education Policy)

Semester IV

SN

CODE

SUBJECT

PERIOD

S

INTERNAL

EVALUATION

SCHEME

END

SEMESTER TOTAL CREDIT

L

T

P CT TA PS Total

TE

PE

1

BVE 401

Universal

Human

Values and

Professional

Ethics

3

0

0

20

10

0

30

70

0

100

3

2

Elective-4

Specialization

Group-1

3

0

0

20

10

0

30

70

0

100

3

3

Elective 5

Specialization

Group-1

3

0

0

20

10

0

30

70

0

100

3

4

Elective -4

Specialization

Group-2

3

0

0

20

10

0

30

70

0

100

3

5

Elective -5

Specialization

Group-2

3

0

0

20

10

0

30

70

0

100

3

6

BMB 406

Research

Project

Report

& Viva Voce

0

6*

0

0

100

0

100

0

200

300

10

TOTAL

800

25

*Students will be guided and assessed by Faculty Guide for Research project

Page 48 Data Table

SN	CODE	SUBJECT
1	BMB MK 01	CONSUMER BEHAVIOUR AND NEURO MARKETING
2	BMB MK 02	MARKETING & WEB ANALYTICS
3	BMB MK 03	SALES AND DISTRIBUTION MANAGEMENT

Page 48 Data Table

SN	CODE	SUBJECT
1	BMB HR 01	TALENT ACQUISITION MANAGEMENT
2	BMB HR 02	EMPLOYEE RELATIONS AND LABOUR LAWS
3	BMB HR 03	PERFORMANCE AND REWARD MANAGEMENT

Page 48 Data Table

SN	CODE	SUBJECT
1	BMB FM 01	INVESTMENT AND PORTFOLIO MANAGEMENT
2	BMB FM 02	FINANCIAL PLANNING AND TAX MANAGEMENT
3	BMB FM 03	FINANCIAL AND CREDIT RISK ANALYSIS

Page 48 Data Table

SN	CODE	SUBJECT
1	BMB OM 01	SUPPLY CHAIN AND LOGISTICS MANAGEMENT
2	BMB OM 02	BUSINESS PROCESS REENGINEERING
3	BMB OM 03	QUALITY MANAGEMENT

Page 48 Data Table

SN	CODE	SUBJECT
1	BMB IB 01	INTERNATIONAL BUSINESS MANAGEMENT
2	BMB IB 02	EXPORT IMPORT DOCUMENTATION
3	BMB IB 03	TRADE AND GEO-POLITICS

Page 48 Data Table

SN	CODE	SUBJECT
1	BMB IT 01	SOFTWARE ENGINEERING AND MANAGEMENT
2	BMB IT 02	EMERGING TECHNOLOGIES FOR BUSINESS
3	BMB IT 03	DATABASE MANAGEMENT SYSTEM

Page 48 Data Table

SN	CODE	SUBJECT
1	BMB CM 01	PRINCIPLES AND PRACTICES OF CO-OPERATION
2	BMB CM 02	CO-OPERATIVE LEGISLATION
3	BMB CM 03	CREDIT CO-OPERATIVES

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Page 8

ELECTIVES IN III SEM

Specialization Group: MARKETING (MK)

SN

CODE

SUBJECT

1

BMB MK 01 CONSUMER BEHAVIOUR AND NEURO MARKETING

2

BMB MK 02 MARKETING & WEB ANALYTICS

3

BMB MK 03 SALES AND DISTRIBUTION MANAGEMENT

Specialization Group: HUMAN RESOURCE MANAGEMENT (HR)

SN

CODE

SUBJECT

1

BMB HR 01

TALENT ACQUISITION MANAGEMENT

2

BMB HR 02

EMPLOYEE RELATIONS AND LABOUR LAWS

3

BMB HR 03

PERFORMANCE AND REWARD MANAGEMENT

Specialization Group: FINANCIAL MANAGEMENT (FM)

SN

CODE

SUBJECT

1

BMB FM 01

INVESTMENT AND PORTFOLIO MANAGEMENT

2

BMB FM 02

FINANCIAL PLANNING AND TAX MANAGEMENT

3

BMB FM 03

FINANCIAL AND CREDIT RISK ANALYSIS

Specialization Group: OPERATION MANAGEMENT (OM)

SN

CODE

SUBJECT

1

BMB OM 01

SUPPLY CHAIN AND LOGISTICS MANAGEMENT

2

BMB OM 02

BUSINESS PROCESS REENGINEERING

3

BMB OM 03

QUALITY MANAGEMENT

Specialization Group: INTERNATIONAL BUSINESS (IB)

SN

CODE

SUBJECT

1

BMB IB 01

INTERNATIONAL BUSINESS MANAGEMENT

2

BMB IB 02

EXPORT IMPORT DOCUMENTATION

3

BMB IB 03

TRADE AND GEO-POLITICS

Specialization Group: INFORMATION TECHNOLOGY (IT)

SN

CODE

SUBJECT

1

BMB IT 01

SOFTWARE ENGINEERING AND MANAGEMENT

2

BMB IT 02

EMERGING TECHNOLOGIES FOR BUSINESS

3

BMB IT 03

DATABASE MANAGEMENT SYSTEM

Specialization Group: COOPERATIVE MANAGEMENT (CM)

SN

CODE

SUBJECT

1

BMB CM 01

PRINCIPLES AND PRACTICES OF CO-OPERATION

2

BMB CM 02

CO-OPERATIVE LEGISLATION

3

BMB CM 03

CREDIT C

Page 49 Data Table

SN	CODE	SUBJECT
4	BMB MK 04	SERVICES & RETAIL MARKETING
5	BMB MK 05	B 2 B MARKETING

Page 49 Data Table

SN	CODE	SUBJECT
4	BMB HR 04	HR ANALYTICS
5	BMB HR 05	ORGANIZATIONAL CHANGE AND DEVELOPMENT

Page 49 Data Table

SN	CODE	SUBJECT
4	BMB FM 04	BEHAVIOURAL FINANCE
5	BMB FM 05	STRATEGIC FINANCIAL MANAGEMENT

Page 49 Data Table

SN	CODE	SUBJECT
4	BMB OM 04	SERVICE OPERATIONS MANAGEMENT
5	BMB OM 05	PROJECT AND SOURCING MANAGEMENT

Page 49 Data Table

SN	CODE	SUBJECT
4	BMB IB 04	MANAGING GLOBAL SUPPLY CHAINS
5	BMB IB 05	INTERNATIONAL FINANCE

Page 49 Data Table

SN	CODE	SUBJECT
4	BMB IT 04	E-BUSINESS
5	BMB IT 05	BUSINESS DATA WAREHOUSING AND DATA MINING

Page 49 Data Table

SN	CODE	SUBJECT
4	BMB CM 04	NON-CREDIT CO-OPERATIVES
5	BMB CM 05	CO-OPERATIVE ACCOUNTING & AUDIT

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ELECTIVES IN IV SEM

Specialization Group: MARKETING (MK)

SN

CODE

SUBJECT

4

BMB MK 04 SERVICES & RETAIL MARKETING

5

BMB MK 05 B 2 B MARKETING

Specialization Group: HUMAN RESOURCE MANAGEMENT (HR)

SN

CODE

SUBJECT

4

BMB HR 04 HR ANALYTICS

5

BMB HR 05 ORGANIZATIONAL CHANGE AND DEVELOPMENT

Specialization Group: FINANACIAL MANAGEMENT (FM)

SN

CODE

SUBJECT

4

BMB FM 04

BEHAVIOURAL FINANCE

5

BMB FM 05

STRATEGIC FINANCIAL MANAGEMENT

Specialized Group : OPERATION MANAGEMENT (OM)

SN

CODE

SUBJECT

4

BMB OM 04

SERVICE OPERATIONS MANAGEMENT

5

BMB OM 05

PROJECT AND SOURCING MANAGEMENT

Specialized Group : INTERNATIONAL BUSINESS (IB)

SN

CODE

SUBJECT

4

BMB IB 04

MANAGING GLOBAL SUPPLY CHAINS

5

BMB IB 05

INTERNATIONAL FINANCE

Specialized Group : INFORMATION TECHNOLOGY (IT)

SN

CODE

SUBJECT

4

BMB IT 04

E-BUSINESS

5

BMB IT 05

BUSINESS DATA WAREHOUSING AND DATA MINING

Specialization Group : COOPERATIVE MANAGEMENT (CM)

SN

CODE

SUBJECT

4

BMB CM 04

NON-CREDIT CO-OPERATIVES

5

BMB CM 05

CO-OPERATIVE ACCOUNTING & AUDIT

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Page 10

III SEMESTER

STRATEGIC MANAGEMENT

BMB 301

Course Credit: 3

Total Hours: 40

Course Objectives: The objective of this course is to:

- Understand the fundamental concepts, models, and processes of strategic management.
- Analyze the external and internal environments of organizations using relevant frameworks and tools.
- Formulate effective business, corporate, and functional strategies based on situational analysis.

- Evaluate and choose appropriate strategies using strategic analysis techniques.
- Implement strategies effectively through structured programs and organizational alignment.
- Evaluate strategic performance and apply controls to ensure strategic success and sustainability.

Unit 1 (5 Hours)

Introduction: meaning nature, scope, and importance of strategy; Model of strategic management, Strategic Decision-Making Process.

Corporate Governance: Composition of the board, Role and Responsibilities of the board of directors, Trends in corporate governance, Corporate Social Responsibility. Case Studies and Latest Updates.

Unit 2 (8 hours)

Environmental Scanning: Understanding the Macro Environment: PESTEL Analysis,

Industrial Organization (IO) & the Structure Conduct Performance (SCP) approach, Porter's

Five Forces Model, Understanding the Micro Environment: Resource Based View (RBV)

Analysis, VRIO Framework, Using resources to

Page 51 Data Table

	Course Outcome	Bloom's Level
CO1	Define and explain the fundamental concepts, models, and processes of strategic management	K2: Understanding
CO2	Analyze external and internal environments using various frameworks to assess organizational positioning.	K4: Analyzing
CO3	Apply situational analysis techniques to formulate competitive, corporate, and functional strategies in diverse business contexts.	K3: Applying;
CO4	Evaluate strategic options using analytical tools for informed strategy choice.	K5: Evaluating
CO5	Design and develop strategic implementation plans for effective strategy execution.	K6: Creating
CO6	Assess and control organizational performance to ensure strategic success and sustainability.	K4: Analyzing

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Page 11

Unit 4 (9 hours)

Strategy Choice and Analysis: Scenario Analysis Process, Tools & Techniques of strategic

Analysis: BCG Matrix, Ansoff Grid, GE Nine Cell Planning Grid, McKinsey's 7'S framework.

Strategy implementation: Developing Programs, Budget and Procedures, Stages of

Corporate Development, Organizational Life cycle, Organizational Structures: Matrix,

Network & Modular/Cellular; Reengineering and Strategy implementation, Leadership and

corporate culture, Case Studies and Latest Updates.

Unit 5 (5 hours)

Strategy Evaluation & Control: Evaluation & Control process, Measuring performance:

types of controls, activity based costing, enterprise risk management, primary measures of corporate performance, balance scorecard approach to measure key Performance, responsibility centers, Benchmarking, Problems in measuring Performance & Guidelines for proper control. Strategic Audit of a Corporation. Case Studies and Latest Updates.

Course Outcomes

Course Outcome

Bloom's Level

CO1

Define and explain the fundamental concepts, models, and processes of strategic management

K2: Understanding

CO2

Analyze external and internal environments using various frameworks to assess organizational positioning. K4: Analyzing

CO3

Apply situational analysis techniques to formulate competitive, corporate, and functional strategies in diverse business contexts.

K3: Applying;

CO4

Evaluate strategic options using analytical tools for informed str

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Suggested Readings

1.

Wheelen, L. Thomas and Hunger, David J.; Concepts in Strategic Management and Business Policy, Pearson Education,

2.

Stewart Clegg, Chris Carter, Martin Kornberger & Jochen Schweitzer : Strategy - Theory and Practice.(SAGE Publishing India)

3.

Kazmi, Azhar; Business Policy and Strategic Management; McGraw-Hill Education.

4.

David, Fred; Strategic Management: Concepts and Cases; PHI Learning.

5.

Thomson, Arthur A. and Strickland, A. J.; Strategic Management: Concept and Cases;
McGraw Hill Education,

6.

Jauch, L.F., and Glueck, W.F.; Business Policy and Strategic Management; McGraw-
Hill Education.

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III SEMESTER ELECTIVES

MARKETING ELECTIVES

CONSUMER BEHAVIOUR AND NEURO MARKETING

BMB MK 01

Course Credit: 3

Total Hours: 40

Course Objectives: The objective of this course is to:

- Understand the fundamental concepts, theories, and models of consumer behaviour and their relevance in marketing decision-making.
- Analyze the psychological, personal, social, and cultural factors influencing consumer decision-making.
- Examine consumer behaviour across digital platforms and modern consumption contexts.
- Understand the principles, techniques, and applications of neuromarketing in analyzing consumer behaviour.
- Evaluate the ethical and legal considerations related to neuromarketing and consumer influence.
- Apply consumer insights and neuromarketing tools to develop evidence-based marketing strategies.

Unit 1 (5 Hours)

Introduction: Consumer Behavior & the Marketing Mix, Consumer Goals, Consumer

Decision Making Process: Need Recognition, Search for information, Pre purchase evaluation of alternatives, Purchase, Consumption, Post consumption Evaluation & Divestment, Interruption in buying process & their effects, Customer involvement, Consumer Journey through the World of Technology.

Unit 2 (8 hours)

Psychological factors: Perception, learning, motivation, attitude formation and change, personality, and lifestyle. Personal factors: Demographics, life cycle, occupation, and income.
Social fac

Page 54 Data Table

	Course Outcome	Bloom's Level
CO1	Define and explain key concepts, theories, and models of consumer behaviour and neuromarketing.	K1: Remembering, K2: Understanding
CO2	Analyze the impact of psychological, social, personal, and cultural factors on consumer buying behaviour.	K4: Analyzing
CO3	Apply frameworks of consumer behaviour to develop marketing strategies suitable for traditional and digital markets.	K3: Applying
CO4	Evaluate neuromarketing techniques and their application in strategic marketing decisions, considering ethical and regulatory concerns.	K5: Evaluating
CO5	Design and develop integrated marketing strategies using consumer behaviour insights and neuromarketing data for real-world business scenarios.	K6: Creating
CO6	Critically assess emerging trends in digital consumer behaviour and recommend appropriate marketing interventions.	K5: Evaluating

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Introduction to Neuromarketing: Meaning and scope of neuromarketing. Principles of neuroscience relevant to marketing: Brain functions, emotions, and decision-making.

Neuromarketing techniques: Eye-tracking, fMRI, EEG, biometrics, facial coding. Applications of neuromarketing in advertising, branding, packaging, and pricing strategies. Benefits and limitations of neuromarketing. Ethical issues, criticisms, and regulatory frameworks related to neuromarketing.

Case Studies.

Unit 5 (7 hours)

Consumer Behaviour and Neuromarketing Strategies: Translating consumer insights into marketing strategy. Designing customer experiences using neuromarketing findings. Sensory marketing and subconscious consumer triggers. Role of neuromarketing in retail, FMCG, luxury brands, and service sectors. Developing integrated consumer behaviour and neuromarketing strategies for real-world challenges. Case Study.

Course Outcomes

Course Outcome

Bloom's Level

CO1

Define and explain key concepts, theories, and models of consumer behaviour and neuromarketing.

K1: Remembering,

K2: Understanding

CO2

Analyze the impact of psychological, social, personal, and cultural factors on consumer buying behaviour.

K4: Analyzing

CO3

Apply frameworks of consumer behaviour to develop marketing strategies suitable for traditional and digital markets.

K3: Applying

CO4

Evaluate neuromarketing techniques and their application in strategic marketing decisions, considering ethical

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MARKETING AND WEB ANALYTICS

BMB MK 02

Course Credit: 3

Total Hours: 40

Course Objectives: The objective of this course is to:

- To introduce the concept, scope, and importance of Marketing Analytics in data-driven decision-making.
- To equip students with pricing, sales forecasting, and customer analysis techniques using quantitative models.
- To enable understanding and application of various retailing, advertising, and web analytics methods.
- To familiarize students with tools and frameworks used in social media and digital analytics platforms.
- To develop competence in interpreting qualitative insights from digital interactions through modern web tools and methods.

Unit 1: (4 Hours)

Introduction to Marketing Analytics: Meaning, Characteristics, Advantages & Disadvantages of Marketing Analytics, Market Data Sources: Primary & Secondary Market Sizing: Stakeholders, Applications, Top-down & Bottom-up Approaches, PESTLE Analysis & Porter's Five Forces Model for market evaluation

Unit 2: (8 Hours)

Pricing Analytics: Pricing Policy & Objectives, Demand Estimation: Price Elasticity, Linear &

Power Demand Curves, Optimizing Pricing and Subjective Demand Curves, Pricing Multiple Products ,Price Bundling: Pure vs. Mixed Bundling Nonlinear Pricing & Profit Maximization Strategies, Revenue Management: Markdown Pricing & Uncertainty Handling, Price Skimming and Sales Promotions

Unit 3: (7 Hours)

Sales Forecasting & Customer Anal

Page 57 Data Table

	Course Outcome	Bloom's Level
CO1	Understand the meaning, characteristics, and applications of marketing analytics and market evaluation tools such as PESTLE and Porter's Five Forces.	K2 (Understand)
CO2	Apply pricing analytics techniques including bundling, nonlinear pricing, and revenue management to solve pricing problems.	K3 (Apply),
CO3	Use regression models and forecasting techniques to estimate sales and customer lifetime value.	K4 (Analyze),
CO4	Analyze & Evaluate customer behavior through segmentation, clustering, and retail metrics like RFM and basket analysis.	K5 (Evaluate)
CO5	Evaluate the effectiveness of online advertising and interpret web analytics metrics using tools such as Google Analytics.	K5 (Evaluate)
CO6	Develop insights using social media analytics, A/B testing, and qualitative tools like heuristic evaluations and surveys.	K6 (Create)

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Instagram, Twitter, LinkedIn, YouTube Organic vs. Paid Traffic, Benchmarking

Web Analytics Tools: A/B Testing, Online Surveys, Web Crawling & Indexing, NLP Techniques for Micro-text Analysis Google Website Optimizer: Working & Implementation.

Qualitative Techniques: Heuristic Evaluation, Site Visits, Surveys (Post-visit & Online)Web

Analytics 2.0 vs. 1.0,Competitive Intelligence, Website Traffic Trends & Overlap

Suggested Readings

1. Marketing Analytics: Data-Driven Techniques with Microsoft Excel by Wayne L Winston, Wiley India Pvt. Ltd.

2. Marketing Analytics: Strategic Models and Metrics by Stephan Sorger, Create Space Publishing

3. Marketing Engineering and Analytics by Gary Lilen, Arvind Rangaswamy,and Arnaud De Bruyn, Decision Pro, Inc.

4. Marketing Metrics by Dugar Anurag, SAGE Publishing India\

5. Avinash Kaushik, Web Analytics 2.0: The Art of Online Accountability and Science of Customer Centricity,
6. Clifton B., Advanced Web Metrics with Google Analytics, Wiley Publishing, Inc. 2nd ed.
7. Kaushik A., Web Analytics 2.0, The Art of Online Accountability and Science of Customer Centricity, Wiley Publishing, Inc. 1st ed.
8. Sterne J., Web Metrics: Proven methods for measuring web site success, John Wiley and Sons
9. Annmarie Hanlon, Digital Marketing, SAGE Publishing India

Course Outcomes

Course Outcome

Bloom's Level

CO1

Understand the meaning, characteristics, and applications of marketing analytics and market evaluation t

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SALES & DISTRIBUTION MANAGEMENT

BMB MK 03

Course Credit: 3

Total Hours: 40

Course Objectives: The objective of this course is to:

- Provide students with a conceptual understanding of the role of selling in marketing, personal selling, types of sales personnel, and the process of effective selling.
- Develop managerial competencies in sales force recruitment, training, motivation, compensation, forecasting, territory management, productivity assessment, and performance appraisal.
- Enable students to plan and control sales operations using forecasting methods, budgeting, resource allocation, sales planning processes, CRM tools, and ethical practices.
- Equip students with knowledge and analytical skills for designing, managing, and evaluating distribution channels, channel partners, and retail-wholesale networks.
- Familiarize students with modern logistics, supply chain processes, IT-enabled distribution, and emerging trends such as e-commerce, green logistics, and reverse logistics.
- Cultivate skills to analyze real-world sales and distribution problems and formulate innovative, technology-driven strategies for effective sales and distribution management.

Unit1: (4 Hours)

Introduction to Sales: Role of selling in marketing, Personal selling, Types of sales personnel, Characteristics of a successful salesman, Process of effective selling.

UNIT 2: (8 Hours)

Sales Force Management: Types of sales organizations and their struc

Page 59 Data Table

	Course Outcome	Bloom's Level
CO1	Define and explain the fundamentals of selling, personal selling, and characteristics of sales personnel in the marketing function.	K1: Remembering,
CO2	Apply sales force management concepts in recruitment, training, motivation, forecasting, and productivity management within various sales organizations.	K3: Applying
CO3	Analyze sales planning and control processes including forecasting, budgeting, sales strategy formulation, ethical practices, and the role of technology in sales management.	K4: Analyzing
CO4	Evaluate distribution channel structures, partner selection processes, retail-wholesale strategies, and conflict management for effective channel operations.	K5: Evaluating
CO5	Develop strategies for managing logistics, modern distribution systems, and supply chain networks incorporating emerging trends and technology-driven innovations.	K6: Creating
CO6	Critically assess the use of omnichannel distribution, e-commerce models, and IT tools such as ERP and SCM systems for enhancing distribution efficiency and customer value.	K5: Evaluating

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strategies, Mobile retailing, Social commerce, AI/ML in retail, Ethical and legal issues in retail,

Sustainability in retailing, Future trends

Suggested Readings

1. Still, R.R., Cundiff, E.W. and Govani, N.A.P.; Sales Management; Pearson Education
2. Venugopal, Pingali, Sales and Distribution Management, SAGE Publishing
3. Berman, Evans, Retail Management Strategic approach, Pearson
4. Chaudhary Prashant , Selling and Negotiation, SAGE Publishing
5. Tapan Panda: Sales and Distribution Management, OUP.
6. Havaladar, K.K., and Cavale, V.M.; Sales and Distribution Management; McGraw- Hill Education.

Course Outcomes

Course Outcome

Bloom's Level

CO1 Define and explain the fundamentals of selling, personal selling, and characteristics of sales personnel in the marketing function.

K1:

Remembering,

CO2

Apply sales force management concepts in recruitment, training, motivation, forecasting, and productivity management within various sales organizations.

K3: Applying

CO3

Analyze

sales

planning

and

control

processes

including

forecasting, budgeting, sales strategy formulation, ethical practices, and the role of technology in sales management.

K4: Analyzing

CO4

Evaluate

distribution

channel

structures,

partner

selection

processes, retail-wholesale strategies, and conflict management for effective channel operations.

K5: Evaluating

CO5

Develop strategies for managing logistics, modern distribution systems, and supply ch

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Human Resource Management Electives

Talent Acquisition Management

BMB HR 01

Credit: 3

Total Hours: 40

Course Objectives

1. To familiarize students with the principles and practices of Talent Management.
2. To develop an understanding of talent acquisition, development, retention, and compensation strategies.
3. To provide insight into global talent practices,
4. To prepare students to manage talents in organizational settings.
5. To explore the strategic role of HR and leadership in talent optimization.

Unit 1: Introduction to Talent Management

(6 Hr)

Concept and evolution of Talent Management; Talent Value Chain: Acquisition, Retention, & Development; Talent Management Models (Zinger Model, Six Principles of TM); Talent Practices: India, USA, Europe, other Asian countries; Strategic relevance of talent for competitive advantage; Cases and latest updates.

Unit 2: Talent Acquisition & Retention Strategies

(10 Hr)

Talent Acquisition lifecycle: Job analysis, sourcing, assessment; Use of Psychometric tools in recruitment; Best practices in recruitment of various MNCs; Employer Branding and Employee Value Proposition (EVP); Employee Engagement & Retention Strategies; Cases and EVP design exercise.

Unit 3: Talent Development for Global Leadership

(8Hr)

Talent development methods: Coaching, Mentoring, Training; Global Leadership Competencies (GLOBE study, case analysis); Integrated

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COs	Outcome	Bloom's Level
CO1	Remember and understand talent management practices at national and global levels	Remember (K1)
CO2	Apply talent management principles to real-world organizational problems	Apply (K3)
CO3	Analyze the interplay between engagement, motivation, and retention	Analyze (K4)
CO4	Evaluate leadership and HRM roles in managing high-potential talent	Evaluate (K5)
CO5	Design and create comprehensive talent management strategies using analytics	Create (K6)

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Suggested Readings

Textbooks:

1. Berger, L. A., & Berger, D. R. The Talent Management Handbook. Association for Talent Development, USA.
2. Joshi & Vohra, Talent Management, Cengage India Private Limited
3. Gerard Assey, Case Studies in Human Resources & Talent Management
4. Goel Dewakar, Performance Appraisal and Compensation Management, PHI

Reference Books:

1. Silzer, R., & Dowell, Strategy-Driven Talent Management: A Leadership Imperative, Jossey-Bass.
2. Goleman, D. Coaching with Compassion, Harvard Business Review.
3. Joel Alemibola Elegbe, Talent Management in the Developing World: Adopting a Global Perspective, Routledge

Course Outcomes

COs

Outcome

Bloom's Level

CO1 Remember and understand talent management practices at national and global levels

Remember

(K1)

CO2 Apply talent management principles to real-world organizational problems

Apply (K3)

CO3 Analyze the interplay between engagement, motivation, and retention

Analyze (K4)

CO4 Evaluate leadership and HRM roles in managing high-potential talent

Evaluate (K5)

CO5 Design and create comprehensive talent management strategies using analytics

Create (K6)

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Employee Relations, Employee Relations Management Tools, Approaches		to Understand IR, the
Trends of Industrial Relations in India, Factors Leading the Present State of Industrial Relations,		
Impact of Globalization on the Liberalized Economy	Emerging challenges of IR in India, Linking	

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EMPLOYEE RELATIONS AND LABOR LAWS

BMB HR 02

Course Credit: 3

Total Hours: 40

Course Objectives:

1. To Provide conceptual framework of Industrial Relation
2. To make students aware with the Indian Labor legislation
3. To make students aware with the basic requirements and mandate of labor legislations
4. To help the students to understand the existing framework of Industrial Relation and Labor legislation.

Unit 1: (9 Hours)

Employee Relations Management (ERM) & Industrial Relation: Introduction and Importance of Employee Relations, Employee Relations Management Tools, Approaches to Understand IR, the Trends of Industrial Relations in India, Factors Leading the Present State of Industrial Relations, Impact of Globalization on the Liberalized Economy Emerging challenges of IR in India, Linking Industrial Relations with economic growth of a country, Trade Unionism: Development of trade unionism, functions, type and structure of trade union, Why Employees Join Trade Unions , Trade Unions in the Eyes of the Management, Politics and Trade Unions, Outside Leadership of Trade Unions problems & suggestive remedial measures of trade unions, The Trade Unions Act 1926 & Amendment Bill, 2019 : Objective, Recognition and registration, Industrial Democracy & Participative Management. Case Studies

Unit2: (8 Hours)

Collective Bargaining: Significance, types & procedure of Collective bargaining Discipline: The Industrial E

Page 63 Data Table

Course Outcomes	Learning Levels as per Bloom's Taxonomy for Evaluation and Assessment
CO 1: Knowledge of Industrial Relation framework	K1(Remember)
CO 2: Competency to understand the importance of Employee Relation within the perspective of Industrial Relation	K2(Understand)
CO 3: Application of relevant Laws of HR management	K3(Apply)
CO 4: Competency to interpreted and implement the Labor Laws within organization	K5 (Evaluate)
CO 5: Competency mapping to use Collective Bargaining and Grievance redressal Mechanism	K6(Create)

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COURSE OUTCOME

Course Outcomes

Learning

Levels

as

per

Bloom's

Taxonomy for Evaluation and Assessment

CO

1:

Knowledge

of

Industrial

Relation

framework

K1(Remember)

CO 2: Competency to understand the importance

of Employee Relation within the perspective of

Industrial Relation

K2(Understand)

CO 3: Application of relevant Laws of HR

management

K3(Apply)

CO 4: Competency to interpreted and implement

the Labor Laws within organization

K5 (Evaluate)

CO 5: Competency mapping to use Collective

Bargaining and Grievance redressal Mechanism

K6(Create)

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Performance and Reward Management

BMB HR 03

Course Credit: 3 Total Hours: 40

Course Objective:

1. To create an understanding of the key concepts of performance management and contemporary methods for administering compensation and rewards in practices.
2. To articulate the benefits of using a performance development plan and the consequences of not having one in place.
3. To distinguish the elements of an effective, integrated performance development system.
4. To devise “SMART” annual performance objectives (e.g., objectives that are specific, measurable, attainable, relevant and track able).
5. To familiarize the students with the concept of competency mapping and understanding its role in career development.
6. To familiarize students with various aspects of compensation system in India and make them understand various issues linked with the process of fixing salary dearness allowance, bonus, incentive scheme and benefits.

Unit1: (7 hours)

Introduction to Performance Management System :Meaning, Uses and purpose of Performance Management, Performance Management vs Performance Appraisal, Performance management and its challenges in current scenario, Performance management as a System and Process, Establishing Performance Criterion of developing an Effective Appraisal System, Criteria (KRA, KSA VS KPI).

Case Studies

Unit2: (9 hours)

Managin

Page 65 Data Table

CO 1: Knowledge of Performance Management and Performance Appraisal	K2(Understand)
CO 2: Competency to use the Performance Management	K3(Apply)
CO 3: Knowledge about the Compensation and Reward Systems	K2(Understand)
CO 4: Developing & implementing the effective reward systems in the organization	K6(Create)
CO 5: Ability to explain the relevance of competency mapping and understanding its linkage with career development	K5(Evaluate)

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Unit 5: (8 Hours)

Compensation: Method of pay and Allowances, Pay structure: Basic Pay, DA, HRA, Gross Pay, Take home pay etc. Incentive schemes; Methods of payment: Time and piece rate. Fringe benefits & other allowances: Overtime, City compensatory, Travelling etc. Regulatory compliance: Introductions, Wage and Pay commissions, Overview of minimum wages Act- 1948 and Equal Remuneration Act-1976. Profit Sharing options; Case Studies.

Course Outcomes & Bloom's Taxonomy

CO

1:

Knowledge

of

Performance

Management and Performance Appraisal

K2(Understand)

CO 2: Competency to use the Performance

Management

K3(Apply)

CO 3: Knowledge about the Compensation

and Reward Systems

K2(Understand)

CO 4: Developing & implementing the

effective reward systems in the organization

K6(Create)

CO 5: Ability to explain the relevance of

competency mapping and understanding its

linkage with career development

K5(Evaluate)

Suggested Reading:

1. T V Rao (2007). Performance Management and Appraisal Systems: HR Tools for Global Competitiveness (Response Books)
2. Michael, Armstrong (1999). Performance Management. Kogan Page.
3. Shrinivas R Kandula (2006). Performance Management : Strategies , Intervention & Drivers. Pearson
4. Chadha, P. (2003). Performance Management: It's About Performing Not Just Appraising. McMillan India Ltd.
5. B D Singh (2012). Compensation and Reward Management, Excel Book
6. Robert Bacal (2007). Perform

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Financial Management (FM) Electives

Investment and Portfolio Management

BMB FM 01

Credits 3

Total Hours 40

Course Objective: This course will emphasize an understanding of the economic forces that influence the pricing of financial assets with an objective

1. To understand the fundamental concepts of investment theory and their application in portfolio selection and management.
2. To apply appropriate financial models and formulae to solve business problems related to active portfolio management.
3. To analyze securities using various tools and techniques to assess risk-return trade-offs and investment performance.
4. To evaluate different investment avenues and portfolio strategies to enhance returns and manage risks effectively.
5. To develop the ability to construct and manage efficient portfolios by integrating theoretical knowledge with practical insights.

Unit I Investments (10 hrs)

Overview of Capital Market: Market of securities, Stock Exchange and New Issue Markets - their nature, structure, functioning and limitations; Trading of securities: equity and debentures/ bonds.

Securities trading - Types of orders, margin trading, clearing and settlement procedures.

Regularity systems for equity markets, Type of investors, Aim & Approaches of Securit

Page 67 Data Table

S. No.	Course Outcome	Bloom's Taxonomy
1.	CO 1: Understand about various investment avenues.	■ K2(Understand)
2.	CO 2: Use the value of assets and manage investment portfolio.	■ K3(Apply)
3.	CO 3 : Use & Evaluate of various Models of Investment and its application	■ K5(Evaluate)
4.	CO 4: Understand and create various investment strategies on the basis of various market conditions.	■ K6(Create)
5.	CO 5: Measure riskiness of a stock or a portfolio position.	■ K3 (Apply)

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Unit V Active Portfolio Management (8Hrs)

Portfolio Management and Performance Evaluation: Performance Evaluation of existing portfolio, Sharpe, Treynor and Jensen measures; Finding alternatives and revision of portfolio;

Portfolio Management and Mutual Fund Industry

Course Outcomes

S. No.

Course Outcome

Bloom's Taxonomy

1.

CO 1: Understand

about

various

investment avenues.

■ K2(Understand)

2.

CO 2: Use the value of assets and manage

investment portfolio.

■ K3(Apply)

3.

CO 3 : Use & Evaluate of various Models

of Investment and its application

■ K5(Evaluate)

4.

CO 4: Understand and create various

investment strategies on the basis of

various market conditions.

■ K6(Create)

5.

CO 5: Measure riskiness of a stock or a

portfolio position.

■ K3 (Apply)

Suggested Readings:

Text Books :

1) Ranganatham - Security Analysis and Portfolio Management (Pearson Education, 2nd Ed.)

2) Chandra P - Investment Analysis and Portfolio Management (Tata McGraw Hill, 3rd Ed)

3) Bhatt- Security Analysis and Portfolio Management (Wiley ,1st Ed)

4) Pandian P - Security Analysis and Portfolio Management (Vikas, 1st Ed.)

5) Bodie, Kane, Marcus & Mohanti - Investment and Indian Perspective (TMH, 10th Ed)

Reference Books

1. William F. Sharpe, Gordon J.Alexander and Jeffery V.Bailey: Investments,

(Prentice Hall, 6th Ed).

2. Donald E. Fischer and Ronald J. Jorda

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Tax Planning & Management

BMB FM 02

Credit 3

Total Hours 40

COURSE OBJECTIVES:

1. To understand the principles, structure, and key issues related to different types of taxes in the Indian economy.
2. To apply tax policies and provisions in evaluating personal and corporate tax obligations.
3. To analyze the role of taxation in promoting economic growth and industrial development.
4. To evaluate the impact of direct and indirect taxes, including GST, on corporate decision-making and financial planning.
5. To develop the ability to design tax-efficient strategies that align with legal provisions and contribute to sound fiscal management.

Unit 1 : Fundamental Concepts

(6Hours)

Introduction: Definition, Canons of Taxation Person, Assesse, Income, Previous Year, Assessment Year, Income Tax Important Dates and Forms. Residential Status & Tax Incidence: Individual Income Exempted from Tax.

Unit 2 : Heads of Income and provisions

(10Hours)

Heads of Income – Salaries, Income from House Property, Profits & Gains from Business or Profession, Capital Gains, Income from Other sources., Clubbing of incomes, Calculation of Taxable Income, Tax Calculation including Surcharge and Marginal relief, Deduction, Rebate, Relief, Set Off & Carry Forward of Losses – Principles, Meaning, inter-sources & inter-head

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S. No.	Course Outcome	Bloom's Taxonomy
1.	CO 1: Understand about various Tax provisions and Tax planning	K2(Understand)
2.	CO 2: Use of tax planning concerning various business and managerial and strategic activities	K3(Apply)
3.	CO 3: Evaluate the various Tax saving options & Rates	K5(Evaluate)
4.	CO 4: Measure Corporate Tax and Taxation in case of business restructuring	K3(Apply)

S. No.	Course Outcome	Bloom's Taxonomy
5.	CO 5: Understand how GST can be calculated & managed.	K3 (Apply)

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Course Outcomes

S. No.

Course Outcome

Bloom's

Taxonomy

1.

CO 1: Understand about various Tax provisions and

Tax planning

K2(Understand)

2.

CO 2: Use of tax planning concerning various business

and managerial and strategic activities

K3(Apply)

3.

CO 3: Evaluate the various Tax saving options &

Rates

K5(Evaluate)

4.

CO 4: Measure Corporate Tax and Taxation in case of

business restructuring

K3(Apply)

5.

CO 5: Understand how GST

can be calculated & managed.

K3 (Apply)

SUGGESTED READINGS

Text Books :

1. Dr. Vinod K. Singhania & Dr. Monica Singhania Students Guide to Income Tax

(Taxmann Publication, Latest Edition according to assessment year

2. Dr.B.K. Agarwal & Dr. Rajeev Agarwal Tax Planning and Management

(Nirupam Publication, Latest Edition according to assessment year)

3. Paolo M. Panteghini Corporate Taxation in a Dynamic World (Springer, Latest

Edition)

4. Girish Ahuja & Ravi Gupta Direct Tax Laws & Practice (Bharat Law House, Latest Edition)

5. Datey V.S. – Indirect Taxes – Law & Practice (Taxman ,Latest Edition) 6.E.

A. Srinivas Corporate Tax Planning(Tata McGraw Hill, Latest Edition)

Reference Books & Journals :

1. Dr.Vinod K. Singhania & Dr. Kapil Singhania Students Guide to Income Tax (Taxmann Publication ,Latest Edition)

2. Parthasarathy Corporate Governance: Principles, Mechanisms & Practice (Wiley, Latest Edition)

3. H. P. Ranina Corporate Taxation (Orient Law House, Lat

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Financial Credit and Risk Analysis

BMB FM 03

Course Credit: 3

Total Hours 40

Course Objectives:

1. To understand the fundamentals of risk management, its context, and the prevailing perceptions and stereotypes associated with it.
2. To apply risk management principles as an enabler for effective decision-making and business performance.
3. To analyze the role of risk culture in integrating risk management into organizational processes.
4. To evaluate how effective risk management contributes to sustainable competitive advantage.
5. To develop a proactive risk culture that positions risk management as a strategic enabler of business success.

Course Outcome: On completion of this course, students will be able to:

1. Understand the fundamental principles of credit assessment, creditworthiness evaluation, and risk analysis in financial decision-making.
2. Apply quantitative and qualitative techniques to conduct credit appraisal, financial statement analysis, and borrower risk profiling.
3. Analyze credit risk models and interpret key indicators for assessing default probabilities and portfolio risk.
4. Evaluate credit policies, risk mitigation strategies, and portfolio management practices to enhance financial stability.

5. Develop effective credit strategies and sound risk management frameworks to minimize exposure and ensure sustainable financial performance.

UNIT I : Introduction

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CO 1: Understand about various types of financial credit.	K2(Understand)
CO 2: Understand the credit risk and its rating.	K1(Remember)
CO 3 : Use of credit commitments and its application	K3(Apply)
CO 4: Evaluate the risk management and corporate governance.	K5 (Evaluate)
CO 5: Develop the metrics of Measuring riskiness of a stock or a portfolio position.	K5 (create)

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Various types of Bank Guarantees: Performance Guarantee, Financial Guarantees, Deferred Payment Guarantees, Types of Performance and Financial Guarantees, Assessment of Bank Guarantees Limit, Period of Claim under Guarantee.

UNIT IV : Operational Risk Overview (08 hours)

Risk & Uncertainty, Financial Sector, Risk Types, Operational Risk Management- Recruitment & Training, Work flow Design, Work Flow Documentation, Delegation of Authority, Independent Internal Audit, Independent Compliance Function, Independent Risk Management Function, System Audit, Corporate Governance, Whistle Blower Policy, Risk Management Culture.

UNIT V : Credit Analysis & Rating (08 hours)

Importance of credit analysis, Stages of credit analysis profitability analysis and pricing of loans, Credit risk analysis (Debt ratios and risk of leverage), Analysis of working capital, liquidity , operating and cash cycle risk .

Credit Rating: Measurement of Risk, Objective of Rating, Internal & External Rating, Model Credit Rating, Methodology of Rating, Internal & External Comparison, Model Rating Formats.

Course Outcomes: After successful completion of this course students will be able to

CO 1: Understand about various types of financial credit.

K2(Understand)

CO 2: Understand the credit risk and its rating.

K1(Remember)

CO 3 : Use

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OPERATIONS MANAGEMENT ELECTIVES

SUPPLY CHAIN & LOGISTICS MANAGEMENT

BMB OM 01

Course Credit: 3

Contact Hours: 40

Course Objectives: The objective of this course is to:

- Develop a comprehensive understanding of fundamental concepts, processes, and strategies involved in logistics and supply chain management.
- Analyze the role of logistics management in achieving competitive advantage through efficient transportation, warehousing, and distribution systems.
- Evaluate supply chain performance using appropriate measurement tools and frameworks like the SCOR model.
- Examine global supply chain strategies, their design considerations, and the challenges of managing international logistics networks.
- Understand and assess the role of technology, CRM integration, sustainability, and green practices in enhancing supply chain effectiveness.
- Apply contemporary supply chain strategies and logistics practices to real-world scenarios for designing and optimizing supply chains.

Unit 1 (8 Hours)

Supply Chain Concepts: Objectives of a Supply Chain, Stages of Supply chain, Value Chain Process, Cycle view of Supply Chain Process, Key issues in SCM, logistics & Supply Chain Drivers and obstacles, Supply chain strategies, strategic fit, Best practices in SCM, Obstacles of streamlined SCM.

Unit 2 (8 Hours)

Logistics :Evolution, Objectives, Components and Functions of Logistics Management, Distribution related Issues and Challenges; Gain

Page 73 Data Table

	Course Outcome	Bloom's Level
CO1	Define and explain the key concepts, stages, strategies, and processes of supply chain management and logistics.	K2: Understanding
CO2	Apply the principles of logistics and supply chain strategies to analyze distribution challenges, transportation functions, and warehousing decisions.	K3: Applying
CO3	Analyze supply chain performance using performance dimensions, SCOR Model, and identify the bullwhip effect and its reduction strategies.	K4: Analyzing
CO4	Evaluate global supply chain strategies, outsourcing decisions (3PL, 4PL), and reverse logistics in the context of network design and operational challenges.	K5: Evaluating

	Course Outcome	Bloom's Level
CO5	Design and develop integrated supply chain models using CRM linkages, IT infrastructure, green supply chain principles, and sustainable logistics solutions for business growth.	K6: Creating
CO6	Critically assess best practices and obstacles in supply chain management for continuous process improvement and strategic fit in dynamic business environments.	K5: Evaluating

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sustainability.

Course Outcomes

Course Outcome

Bloom's Level

CO1 Define and explain the key concepts, stages, strategies, and processes of supply chain management and logistics.

K2: Understanding

CO2

Apply the principles of logistics and supply chain strategies to analyze distribution challenges, transportation functions, and warehousing decisions.

K3: Applying

CO3

Analyze supply chain performance using performance dimensions, SCOR Model, and identify the bullwhip effect and its reduction strategies.

K4: Analyzing

CO4

Evaluate global supply chain strategies, outsourcing decisions (3PL, 4PL), and reverse logistics in the context of network design and operational challenges.

K5: Evaluating

CO5

Design and develop integrated supply chain models using CRM linkages, IT infrastructure, green supply chain principles, and sustainable logistics solutions for business growth.

K6: Creating

CO6

Critically assess best practices and obstacles in supply chain management for continuous process improvement and strategic fit in dynamic business environments.

K5: Evaluating

Suggested Readings:

1. Chopra, Sunil, Meindl, Peter and Kalra, D. V.; Supply Chain Management: Strategy, Planning and Operation; Pearson Education
2. Altekhar, Rahul V.; Supply Chain Management: Concepts and Cases; PHI Learning

Reference Books

3. Ballou, Ronald H.; Supply Chain Management; Pearson Education
4. Sahay, B.S.; Supply Chain Management; Macmillan
5. Ballou

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BUSINESS PROCESS RE-ENGINEERING

BMB OM 02

Course Credit: 3

Contact Hours: 40

Course Objectives: The objective of this course is to:

- Explain the fundamental concepts, scope, and relevance of Business Process Re-engineering (BPR) in modern organizations.
- Analyze business processes using appropriate modeling techniques and process mapping tools.
- Identify and evaluate inefficiencies and bottlenecks in existing processes to determine reengineering opportunities.
- Examine the methodologies, strategies, and tools required to implement BPR successfully.
- Evaluate organizational challenges, change management strategies, and risk mitigation approaches involved in BPR.
- Develop innovative, sustainable, and technology-driven re-engineered processes for organizational competitiveness.

Unit 1: (8 Hours)

Introduction to Business Process Re-engineering: Concept, definition, and evolution of BPR, Objectives, significance, and scope of BPR in operations management. Principles and philosophy of BPR. Traditional vs. Re-engineered Processes. Role of BPR in enhancing organizational competitiveness. Critical success factors and myths related to BPR.

Unit 2: (8 Hours)

Business Process Mapping and Modeling: Tools and Techniques. Process documentation and

process benchmarking. Identifying core and support processes. Analyzing process performance and identifying bottlenecks. Process innovation vs. process improvement. Evaluating BPR readiness

Page 75 Data Table

	Course Outcome	Bloom's Level
CO1	Define and explain the fundamental concepts, objectives, principles, and scope of Business Process Re-engineering in operations management.	K2: Understanding
CO2	Apply process mapping techniques and process modeling tools to analyze existing business processes and identify reengineering opportunities.	K3: Applying
CO3	Analyze organizational processes to detect inefficiencies, bottlenecks, and areas requiring reengineering, considering operational and strategic factors.	K4: Analyzing
CO4	Evaluate BPR methodologies, frameworks, and IT-enabled tools for successful process transformation in diverse industries.	K5: Evaluating
CO5	Design and develop innovative, technology-driven, and sustainable reengineered processes aligning with organizational strategies and competitiveness.	K6: Creating
CO6	Critically assess the role of change management and risk mitigation strategies in successful BPR implementation, using real-world organizational examples.	K5: Evaluating

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Case Study.

Suggested Readings

1. Hammer, M., & Champy, J. (2009). Reengineering the Corporation: Manifesto for Business Revolution, Harper Business.
2. Davenport, T. H. (1993). Process Innovation: Reengineering Work through Information Technology, Harvard Business School Press.
3. Harrington, H. J. (1991). Business Process Improvement: The Breakthrough Strategy for Total Quality, Productivity, and Competitiveness, McGraw-Hill.
4. Sethi, P. (2010). Business Process Reengineering: Text and Cases, Excel Books.
5. Rosing, M. von, White, S., & Cummins, F. (2015). Business Process Management: Practical Guidelines to Successful Implementations, Morgan Kaufmann.
6. Saxena, P. (2013). Process Reengineering and Workflow Automation, Tata McGraw-Hill.
7. Sharma, R. B. (2015). Business Process Reengineering and Change Management, Global India Publications.

Course Outcomes

Course Outcome

Bloom's Level

CO1

Define and explain the fundamental concepts, objectives, principles, and scope of Business Process Re-engineering in operations management.

K2: Understanding

CO2

Apply process mapping techniques and process modeling tools to analyze existing business processes and identify reengineering opportunities.

K3: Applying

CO3

Analyze organizational processes to detect inefficiencies, bottlenecks, and areas requiring reengineering, considering operational and strategic factors.

K4: Analyzing

CO4

Evaluate BPR methodologies, frameworks, and IT-enabl

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QUALITY MANAGEMENT

BMB OM 03

Course Credit: 3

Contact Hours: 40

Course Objectives: The objective of this course is to:

1. To provide a comprehensive understanding of the fundamental concepts, principles, and evolution of quality management.
2. To familiarise learners with quality philosophies, frameworks, and globally recognised quality management practices.
3. To develop the ability to apply quality improvement tools, techniques, and statistical methods for process optimisation.
4. To enhance analytical skills for evaluating total quality management approaches, Six Sigma, and benchmarking strategies.
5. To enable learners to assess and implement national and international quality standards, audits, and certifications for organisational excellence.

Unit 1(8Hours)

Quality Concepts: Evolution of Quality Management, Concepts of Quality, Quality Control v/s Quality assurance , Dimensions of Quality, Principles of Quality, Deming's, Juran's & Crosby's Quality Philosophy, Quality Cost, Quality Leadership, Role of Top Management.

Unit 2 (8 Hours)

Quality Management System (QMS) & Process Quality Improvement : Basics of QMS, 7 QC tools, Regression Control Charts, Process Capability and Analysis, Measurement system Analysis, Design and Analysis of Experiment (DOE), Acceptance sampling plan, Different Cost associated with Quality like Assurance cost, Failure cost , prevention cost, rectification cost, appraisal cost, Process failure

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Sl	CO Code	CO Statement	Bloom's K Level
1	CO1	Define the fundamental concepts, scope, and significance of business in a war economy.	K1 – Remember
2	CO2	Explain the impact of wars on global and national business environments, trade policies, and supply chains.	K2 – Understand
3	CO3	Apply economic theories to assess how wars influence production, employment, and corporate strategies.	K3 – Apply
4	CO4	Analyse business opportunities, risks, and sectoral shifts arising during and after wartime economies.	K4 – Analyse
5	CO5	Evaluate ethical, social, and long-term economic implications of war- driven business strategies.	K5 – Evaluate

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2. Lt. Gen. H. Lal, "Total Quality Management", Eastern Limited
3. Greg Bounds, "Beyond Total Quality Management", McGraw Hill
4. Menon, H.G, "TQM in New Product manufacturing", McGraw Hill
5. D. C. Montgomery, Introduction to Statistical Quality Control, John Wiley & Sons,
6. J Evans and W Lindsay, The Management and Control of Quality, Thomson
7. Besterfield, D H et al., Total Quality Management, 3rd Edition, Pearson Education,
8. D. C. Montgomery and G C Runger, Applied Statistics and Probability for Engineers, John Wiley & Sons

Course Outcome:

SI

CO

Code

CO Statement

Bloom's K

Level

1

CO1 Define the fundamental concepts, scope, and significance of business in a war economy.

K1 –

Remember

2

CO2 Explain the impact of wars on global and national business environments, trade policies, and supply chains.

K2 –

Understand

3

CO3 Apply economic theories to assess how wars influence production, employment, and corporate strategies.

K3 – Apply

4

CO4 Analyse business opportunities, risks, and sectoral shifts arising during and after wartime economies.

K4 – Analyse

5

CO5 Evaluate ethical, social, and long-term economic implications of war-driven business strategies.

K5 – Evaluate

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INTERNATIONAL BUSINESS (IB) ELECTIVES

INTERNATIONAL BUSINESS MANAGEMENT

BMB IB 01

Course Credits 3

Contact Hours 40

Course objectives

1. To introduce the basic concepts, importance, and theories of international business and trade.

2. To explain various trade policies, tools, and India's role in global trade.
3. To help students understand global business environments and political systems.
4. To teach the fundamentals of international marketing, including market research and strategy.
5. To give knowledge of global business strategies, institutions, and international operations

Unit 1: Introduction to International Business and Trade Theories (8 Hours)

Nature, scope, and importance of international business, introduction to EPRG & LPG frameworks, major modes of market entry: exporting, joint ventures, FDI, licensing, wholly owned subsidiaries, theoretical foundations: classical and modern trade theories, Leontief paradox, PLC theory, national competitive advantage, and factor mobility theory.

Unit 2: Trade Policy & Commercial Instruments (6 Hours)

Instruments of commercial policy: tariffs, quotas, subsidies, non-tariff barriers, economic multiplier effects (with numerical examples), trade protectionism, India's foreign trade policy, Make in India initiatives, LPG policy framework.

Unit 3: Business Environment & Political Economy (10 Hours)

International business environment analysis: PESTEL coverin

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CO	Course Outcome (CO)	Bloom's K Levels
CO1	Apply international trade theories to analyze global markets.	Understand(K1) and Apply(K3)
CO2	Evaluate the impact of trade policies on international business.	Evaluate(K5)
CO3	Analyze global business environments including political and cultural factors.	Analyze(K4)
CO4	Evaluate international marketing strategies for diverse markets.	Evaluate(K5)
CO5	Formulate global business strategies incorporating trade institutions and operations.	Create(K6)

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Geocentric); Environmental factors affecting international marketing – Political, Economic, Social, Cultural, Technological, Legal; Hofstede's cultural dimensions and their impact on marketing decisions; International market research and segmentation; International product and pricing decisions. Latest updates and cases.

Unit 5: International Strategy, Institutions and Operations (8 Hours)

Global business strategy: standardization vs localization, international expansion strategies including M&A, alliances, licensing, franchising, international marketing, supply chain and logistics management, use of Incoterms, role of trade promotion bodies in India (EXIM Bank,

ECGC), global institutions and trade agreements: WTO, GATT, IMF, World Bank, TRIPS, TRIMS, GATS, regional economic blocs: EU, ASEAN, NAFTA, SAARC, emergence and strategies of multinational firms, Indian export promotion schemes and SEZ policies.

Outcomes

CO

Course Outcome (CO)

Bloom's K Levels

CO1 Apply international trade theories to analyze global markets.

Understand(K1) and

Apply(K3)

CO2 Evaluate the impact of trade policies on international business.

Evaluate(K5)

CO3 Analyze global business environments including political and cultural factors.

Analyze(K4)

CO4 Evaluate international marketing strategies for diverse markets.

Evaluate(K5)

CO5 Formulate global business strategies incorporating trade institutions and operations.

Create(K6)

Suggested Readings

1. Srin

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EXPORT-IMPORT DOCUMENTATIONS

BMB IB 02

Course Credits: 3

Contact Hours: 40

Course Objectives:

1. To explain the structure and procedures involved in Indian export-import operations;
2. To equip students with practical knowledge of documentation as per Indian EXIM policy;
3. To introduce Indian digital trade platforms such as DGFT, ICEGATE, and e-Sanchit;
4. To familiarize students with customs, shipping, and payment procedures in India;
5. To develop skills to independently prepare and verify export-import documentation.

Unit 1: Indian EXIM Framework and Registration

(8Hr)

Latest Foreign Trade Policy (FTP) overview; objectives and key provisions; role of Directorate

General of Foreign Trade (DGFT), Export-Import Bank of India (EXIM Bank), Export Promotion Councils (EPCs), Special Economic Zones (SEZs); types of exports and imports; registration processes including Import Export Code (IEC), Registration Cum Membership Certificate (RCMC) and Authorized Dealer Code (AD Code); overview of Start-up and Micro, Small and Medium Enterprises (MSME) support in India's FTP.

Unit 2: Commercial and Regulatory Documentation

(8Hr)

Commercial documents: proforma invoice; commercial invoice; packing list; certificate of origin from Directorate General of Foreign Trade (DGFT) and chamber of commerce; inspection and quality certificates (Food Safety and St

Page 81 Data Table

COs	Course Outcome	Knowledge Level (K- Level)
CO1	Explain the legal and regulatory framework of export-import in India	K2 – Understanding
CO2	Identify and describe commercial, shipping, banking, and regulatory documents used in India	K1 – Remembering
CO3	Apply documentation procedures using Indian portals such as DGFT and ICEGATE	K3 – Applying
CO4	Analyze documentation requirements under various Indian trade schemes and customs processes	K4 – Analyzing
CO5	Design a complete export-import document set for a sample Indian trade transaction	K6 – Creating

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documents: bill of lading; airway bill; multimodal transport documents; marine insurance in India.

Unit 4: Banking, Payment & Foreign Exchange Documents

(9Hr)

Payment modes in Indian trade: advance payment; documents against payment (DP); documents against acceptance (DA); letter of credit (LC) process under Uniform Customs and Practice for Documentary Credits (UCPDC) norms; Reserve Bank of India (RBI) and Foreign Exchange Management Act (FEMA) guidelines on foreign exchange (forex) management; Authorized Dealer (AD) bank's role; banking documents: electronic Bank Realization Certificate (e-BRC); Foreign Inward Remittance Certificate (FIRC); Goods Receipt (GR) form; Shipping Declaration Form (SDF); steps for foreign currency realization and repatriation.

Unit 5: Customs Procedures and Digital Trade Platforms

(8Hr)

Overview of Indian Customs Act; customs clearance process for exports and imports; role of

Customs House Agent (CHA); Indian Customs Electronic Gateway (ICEGATE) registration; uploading documents online (shipping bill, electronic Certificate of Origin (e-CoO), electronic Bank Realization Certificate (e-BRC)); Indian Single Window System; trade analytics and compliance monitoring tools.

Course Outcomes (COs)

COs

Course Outcome

Knowledge Level (K-

Level)

CO1 Explain the legal and regulatory framework of export-import in India K2 – Understanding

CO2 I

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GEO POLITICS AND TRADE

BMB IB 03

Course Credits 3

Contact Hours 40

Course Objectives

1. To introduce the basics of geopolitics and how it shapes global trade.
2. To explain how conflicts and political issues disrupt international trade.
3. To understand the role of energy and resources in trade decisions.
4. To explore how global alliances and trade blocs influence trade flows.
5. To examine new risks in global trade and future geopolitical challenges.

Unit 1: Introduction to Geopolitics and the Global Trade Order (8 Hours)

Meaning and scope of geopolitics, relationship between geography, power, and economics, historical evolution of geopolitical trade (Silk Road to WTO), role of nation-states in trade influence, geostrategy and economic nationalism in trade decisions.

Unit

2:

Geopolitical

Conflicts

and

Trade

Disruptions

(8

Hours)

Strategic use of trade restrictions: sanctions, embargoes, and tariffs, conflicts and wars impacting

trade flows (e.g., Russia-Ukraine, Israel-Palestine), disruption of global supply chains due to political instability, control over critical raw materials (oil, gas, rare earths), case studies: Strait of Hormuz, Suez Canal, South China Sea. Latest updates.

Unit

3:

Resource

Geopolitics

and

Energy

Security

(8

Hours)

Global energy politics: pipelines, ports, and control zones, political economy of OPEC+ and fuel pricing, climate politics: carbon tariffs, green

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CO	Course Outcome (CO)	Bloom's K Levels
CO1	Explain the relationship between geography, power, and international trade.	Remember (K1) and Understand(K2)
CO2	Analyze the impact of conflicts and political disruptions on global supply chains.	Analyze(K4)
CO3	Evaluate the geopolitical influence of energy, water, and agricultural resources on trade.	Evaluate(K5)
CO4	Assess the role of power blocs and alliances in shaping regional and global trade flows.	Evaluate(K5)
CO5	Recommend strategies for navigating future trade risks in a fragmented world order.	Create(K6)

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satellite-based trade surveillance, maritime control, future outlook: multipolar world, global fragmentation, trade resilience strategies. Latest updates.

Course Outcomes

CO

Course Outcome (CO)

Bloom's K Levels

CO1 Explain the relationship between geography, power, and international trade.

Remember (K1) and

Understand(K2)

CO2 Analyze the impact of conflicts and political disruptions on global supply chains.

Analyze(K4)

CO3 Evaluate the geopolitical influence of energy, water, and agricultural resources on trade.

Evaluate(K5)

CO4 Assess the role of power blocs and alliances in shaping regional and global trade flows.

Evaluate(K5)

CO5 Recommend strategies for navigating future trade risks in a fragmented world order.

Create(K6)

Suggested Readings

1. Mavroidis, Petros C. (2024). Industrial Policy, National Security, and the Perilous Plight of the WTO. Oxford University Press
 2. Baru, Sanjaya. (2016). India and the World: Essays on Geoeconomics and Foreign Policy. Academic Foundation.
 3. Dadush, Uri. (2024). Geopolitics, Trade Blocks, and the Fragmentation of World Commerce. Lexington Books.
 4. Scholten, Daniel (Ed.). (2023). Handbook on the Geopolitics of the Energy Transition. Edward Elgar Publishing.
- Sottilotta, Cecilia Emma, Campisi, Julian, Leitner, Johannes, & Meissner, Hannes (2025). The Routledge Handbook of Political Risk (1st ed.). Routledge

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INFORMATION TECHNOLOGY (IT) ELECTIVES

SOFTWARE ENGINEERING AND MANAGEMENT

BMB IT 01

Credits: 3

Contact Hours 40 L

Course Objectives



To understand the fundamentals of information system development and system design processes.



To apply structured analysis and documentation techniques such as DFDs, ERDs, and data dictionaries.



To develop practical skills in application prototyping, project planning, cost estimation, and resource analysis.



To evaluate design methodologies in data management and e-commerce system architectures.



To identify security and auditing requirements in software systems and suggest appropriate control mechanisms.



To implement best practices for software quality assurance, system maintenance, and business continuity.

Unit - I: 6hrs

Introduction to Information System Development: Overview of System Analysis and Design; Business System Concepts; Categories of Information Systems; Strategies for System Development; Implementation and Evaluation of Systems; Managing the Application Development Portfolio

Unit - II: 8 hrs

Analysis Techniques & Tools: Study of Existing Systems - Information Requirement Analysis, System Documentation; Structured Analysis - Data Flow Tools: Data Flow Diagrams (DFD), Data Dictionary Concepts; Application Prototype Development - Steps, Uses, Tools, and Strategies; Computer-Aided System Tools (CASE); Project Planning Fundamentals

Page 85 Data Table

S. No	Course Outcome	Bloom's Taxonomy
CO1	Understand the lifecycle stages of system development and distinguish between types of information systems.	K2 (Understanding)
CO2	Apply structured analysis techniques to design business applications and document systems using appropriate tools.	K3 (Applying)
CO3	Analyze cost estimation models, risk management strategies, and project planning techniques in software development.	K5 (Evaluating)
CO4	Evaluate database design principles and e-commerce architectures in the context of online business systems.	K5 (Evaluating)
CO5	Develop and justify software control, auditing, and continuity plans to ensure quality and system integrity.	K6 (Creating)

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Business Applications; Design of Online Catalogues; File Organization and Design Techniques;
Database Concepts and Design

Unit - IV: 8 hrs

Systems Analysis and Design in E-Commerce: E-Commerce Models: B2B, B2C, and C2C;
Advantages and Disadvantages of E-Commerce Systems; E-Commerce System Architectures;
Security Considerations in E-Commerce

Unit - V: 10 hrs

Business System Development and Implementation: System Testing and Quality Assurance;
Documentation for Systems; Implementation and Development Processes; Hardware and Software
Selection Criteria; System Maintenance and Support
Security Control and Auditing: Security and Auditing of Information Systems; Objectives and
Techniques of Information System Controls; Auditing Information Systems; Disaster Recovery and
Business Process Continuity Planning

Course Outcomes (COs)

S.

No

Course Outcome

Bloom's

Taxonomy

CO1 Understand the lifecycle stages of system development and distinguish
between types of information systems.

K2

(Understanding)

CO2 Apply structured analysis techniques to design business applications and
document systems using appropriate tools.

K3 (Applying)

CO3 Analyze cost estimation models, risk management strategies, and project
planning techniques in software development.

K5 (Evaluating)

CO4 Evaluate database design principles and e-commerce architectures in the
context of online business systems.

K5 (Evaluating)

CO5 Develop and justify software control, au

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EMERGING TECHNOLOGIES FOR BUSINESS

BMB IT 02

Credits: 3

Contact Hours 40 L

Course Objectives



To understand the evolution of emerging technologies and their relevance to business strategy in the era of Industry 4.0.



To examine the architecture and application of data science, big data, and cloud technologies in business contexts.



To analyze the integration of AI, IoT, and computer vision in domains such as manufacturing, retail, and healthcare.



To explore the disruptive impact of blockchain, 3D printing, and advanced computing technologies.



To assess immersive technologies like AR, VR, MR, and their roles in customer experience and business transformation.



To evaluate ethical, cybersecurity, and sustainability challenges in deploying emerging technologies.

Unit - I: 8 Hours

Foundations of Digital & Emerging Technologies: Evolution of technology and the Fourth Industrial Revolution (Industry 4.0); Introduction to digital transformation in business; Overview of enabling technologies and business drivers; Human-machine interaction; future trends in business technology adoption; Introduction to programmable networks, devices, and automation

Unit - II: 8 Hours

Data Science, Big Data & Cloud Computing: Understanding data, information, data value chain, and types; The data pipeline: collection, cleaning, storage, analysis, curation, and visualization; Big Data: key concepts and business use cases;

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S. No	Course Outcome	Bloom's Taxonomy
CO1	Understand the strategic importance of emerging technologies in modern business transformation.	K2 (Understanding)
CO2	Apply data science, cloud, and big data concepts in real-world analytics scenarios.	K3 (Applying)

S. No	Course Outcome	Bloom's Taxonomy
CO3	Analyze how AI, IoT, and computer vision can be leveraged for intelligent automation and services.	K4 (Analyzing)
CO4	Evaluate the application and implications of blockchain, 3D printing, and other disruptive technologies.	K5 (Evaluating)
CO5	Develop strategic insights for integrating AR/VR and address ethical, security, and sustainability concerns.	K6 (Creating)

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AR, VR, MR & Virtual Try-On in Business: Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR): differences, technologies, and development; Immersive experiences in business: marketing, training, product design, simulation; Virtual Try-On: concepts, enabling technologies (AR/AI), applications in retail, fashion, beauty, home décor; Data-driven personalization and customer engagement via immersive technologies; Challenges and opportunities of implementing immersive solutions

Unit - V: 6 Hours

Ethics, Security, Sustainability & Leadership in Emerging Tech: Data privacy, security, and compliance (GDPR, other regulations); Cybersecurity issues in cloud, AI, IoT, and blockchain adoption; Ethics and bias in AI, big data, and immersive technologies; Sustainable technology adoption and digital responsibility; Leadership and future workplace skills: managing innovation, change management, digital strategy

Course Outcomes (COs)

S.

No

Course Outcome

Bloom's

Taxonomy

CO1 Understand the strategic importance of emerging technologies in modern business transformation.

K2

(Understanding)

CO2 Apply data science, cloud, and big data concepts in real-world analytics scenarios.

K3 (Applying)

CO3 Analyze how AI, IoT, and computer vision can be leveraged for intelligent automation and services.

K4 (Analyzing)

CO4 Evaluate the application and implications of blockchain, 3D printing, and other disruptive technologies.

K5 (Evaluating)

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DATABASE MANAGEMENT SYSTEM

BMB IT 03

Credits: 3

Contact Hours 40 L

Course Objectives



To understand the core principles and architecture of database systems.



To develop ER models and relational schema for real-world applications.



To formulate complex SQL queries and apply normalization techniques.



To analyze query optimization and transaction management strategies.



To examine database security, recovery, and concurrency mechanisms.



To explore emerging concepts in distributed databases and data warehousing.

Unit - I: 6 hrs

Introduction to Database Systems: Basic concepts: Data, information, fields, records, files, databases; Evolution and need for database systems in business; Data models: Hierarchical, network, relational, object-oriented; File-oriented systems vs. Database systems; Three-layered architecture of DBMS; Applications and advantages/disadvantages of DBMS

Unit - II: 8 hrs

Data Modelling & Database Design: Entity-Relationship (ER) Model - Entities, attributes, relationships, constraints, ER diagrams, extended features (generalization, specialization, aggregation); Relational database model - Structure, keys (primary, foreign), integrity constraints, Normalization (1NF, 2NF, 3NF, BCNF); Relational schema and table design; Role and responsibilities of the Database Administrator (DBA)

Unit - III: 8 hrs

Relational Query Languages: Structured Query Language (SQL) -Data Definition

Page 89 Data Table

S. No	Course Outcome	Bloom's Taxonomy
CO1	Understand fundamental database concepts and the need for DBMS in modern organizations.	K2 (Understanding)
CO2	Apply data modeling techniques to design relational databases and use SQL for data manipulation.	K3 (Applying)
CO3	Analyze database normalization and indexing methods for data integrity and performance.	K4 (Analyzing)
CO4	Evaluate transaction management, backup, and recovery mechanisms.	K5 (Evaluating)
CO5	Design and implement secure, scalable databases and understand trends in warehousing and data mining.	K6 (Creating)

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Course Outcomes (COs)

S.

No

Course Outcome

Bloom's

Taxonomy

CO1 Understand fundamental database concepts and the need for DBMS in modern organizations.

K2

(Understanding)

CO2 Apply data modeling techniques to design relational databases and use SQL for data manipulation.

K3 (Applying)

CO3 Analyze database normalization and indexing methods for data integrity and performance.

K4 (Analyzing)

CO4 Evaluate transaction management, backup, and recovery mechanisms.

K5 (Evaluating)

CO5 Design and implement secure, scalable databases and understand trends in warehousing and data mining.

K6 (Creating)

Suggested Readings:

■ Chakrabarti- Advance Database Management System (Wiley Dreamtech)

- Beynon- Davies P- Database Systems (Palgrave, 2003)
- Karthikeyan- Understanding Database Management System (Acme Learning)
- Hoffer- Modern Database Management (Pearson Education, 6th edition)
- Alexis and Leo- Database Management System (Vikas, 2003.)
- Majumdar and Bhattacharya- Database Management System (Tata Mc Graw Hill, 1996).
- Navathe- e-Fundamentals of Database Systems (Pearson Education, 3rd Ed.)

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CO OPERATIVE MANAGEMENT ELECTIVES

PRINCIPLES AND PRACTICES OF CO-OPERATION

BMB CM 01

Course Credit: 3 Contact Hours: 40

Course Objectives

1. To impart knowledge about different Co-operative thoughts
2. To introduce to the evolution of Co-operative movement in India
3. To acquire knowledge about the various Committees on Co-operatives
4. To impart knowledge in Co-operatives and other economic enterprises
5. To familiarize the students on the recent developments in Co-operatives

UNIT I (9 Hrs)

History of Co-operative Thought: Pre-Rochdale Thought; Robert Owen, Dr. William King, and Charles Fourier; Rochdale Model; Post-Rochdale Cooperative Thought; Dr. Warbasse, Charles Gide, Raiffeisen and Schulz; Capitalism, Socialism and Co- operation as Economic Systems; ICA - Cooperative Identity Statement 1995; Definition, Principles and Values of Cooperation.

UNIT II (8 Hrs)

Co-operative Management: Objectives and Functions of Co-operative Management; Democratic Structure in Cooperatives; Membership, General Body, Board, and CEO; Bye Laws and their Amendment; Special features of Co-operative Leadership; Cooperative elections; Need for professional management in Co-operatives.

UNIT III (8 Hrs)

Co-operative Administration: Role of State and Union Government in Co -operative Administration; Role, powers and functions of the Registrar; Co-operative Department set up

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	S.No			Course Outcomes			Bloom's Technology
1			CO1. The students will understand the history and principles co-operation.			Knowledge (K2)	
2			CO2. To Understand the different school of co-operative thoughts.			Remembering (K1)	
3			CO3. The students gain knowledge about the co-operation and other form of economic organization.			Comprehending (K3)	
4			CO4. To know about the origin & development of foreign co-operatives.			Knowledge (K2)	
5			CO5. To make students understand about the co-operative movement in India.			Knowledge (K2)	

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Co-operation in Foreign Countries: Co-operative Credit Movement in Germany; Consumer Co-operatives in U.K and Sweden; Dairy Co-operatives in Denmark.

Course Outcome

After successful completion of this course students will be able to,

S.No

Course Outcomes

Bloom's Technology

1

CO1. The students will understand the history and principles co-operation.

Knowledge (K2)

2

CO2. To Understand the different school of co-

operative thoughts.

Remembering (K1)

3

CO3. The students gain knowledge about the co-operation and other form of economic organization.

Comprehending (K3)

4

CO4. To know about the origin & development of foreign co-operatives.

Knowledge (K2)

5

CO5. To make students understand about the co-operative movement in India.

Knowledge (K2)

Suggested Readings

- Dr. B.S. Mathur, Co-operation in India - Sahitya Bhawan, New Delhi, 2015.
- John Winfred and V.Kulandaiswamy, Co-operative Thought, Rainbow Publications, 1987.
- Dr.O.R.Krishnaswami and Dr. V. Kulandaiswamy, Co-operation Concept and Theory, Arudra Academy, 2000.
- R.D. Bedi, Theory, History and Principles of Co-operation, R. Lall Book Depot, Meerut, 2014.
- Dr.V. Kulandaiswamy, Co-operative Dairying in India, Rainbow Publications, 1986.

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CO-OPERATIVE LEGISLATION

BMB CM 02

Course Credit: 3 Contact Hours: 40

Course Objectives

1. To understand the history of Co-operative legislation in India
2. To familiarize with essential provisions of Uttar Pradesh Cooperative Societies Act.
3. To learn the State aid, duties and privileges given to registered societies.
4. To impart the regulatory provisions relating to Co-operatives audit, inquiry and inspection.
5. To know the functions and recruitment process of Uttar Pradesh Sahkari Seva Mandal.

UNIT I (9 Hrs)

Legal Framework for Cooperatives: History of Co-operative Legislation in India; Co-operative Credit Societies Act 1904; Co-operative Societies Act 1912; Model Co-operative Societies Bill 1957; Model Co-operative Societies Bill 1991; Self-reliant Co-operative Societies Act 1999; Multi-state Co-

operative Societies Act 2002; 97th Constitution Amendment 2011.

UNIT II (9 Hrs)

The Uttar Pradesh Cooperative Societies Act, 1965 and Rules, 1968: Provision relating to Registration; Amendment of Bye Laws; Division and Amalgamation; Qualifications; Obtaining Membership; Rights and Liabilities of Members; Management; Qualification and Disqualification of Board.

UNIT III (6 Hrs)

Registration of Cooperative Societies: Duties and Privileges of Registered Societies; Properties and funds of Registered Societies; Net Profit Distribution; Audit; Inquiry; Inspecti

Page 93 Data Table

	S.No			Course Outcomes			Bloom's Technology
1			CO1.Gain knowledge on History of Co-operative Legislation and its importance.			Remembering (K1)	
2			CO2. Apply the registration procedure and management of co-operatives.			Comprehending (K3)	
3			Cos. Knowing State aid, duties and privileges of Registered Societies.			Knowledge (K2)	
4			CO4. Describe and Practice of audit report, inquiry and investigation procedures of co-operatives.			Knowledge (K2)	
5			CO5. Acquaintance on Cooperative Recruitment and Selection Procedure.			Comprehending (K3)	

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Course Outcome

After successful completion of this course students will be able to,

S.No

Course Outcomes

Bloom's Technology

1

CO1. Gain knowledge on History of Co-operative Legislation and its importance.

Remembering (K1)

2

CO2. Apply the registration procedure and management of co-operatives.

Comprehending (K3)

3

Cos. Knowing State aid, duties and privileges of Registered Societies.

Knowledge (K2)

4

CO4. Describe and Practice of audit report, inquiry and investigation procedures of co-operatives.

Knowledge (K2)

5

CO5. Acquaintance on Cooperative Recruitment and Selection Procedure.

Comprehending (K3)

Suggested Readings

1. Vidwans M.D, Cooperative Law in India. Sahitya Bhavan Publishers, New Delhi, 1956.
2. Goel. B.B. Cooperative Legislation Trends and Dimensions, Deep and Deep Publication New Delhi. (2013).
3. The Uttar Pradesh Co-operative Societies Act. 1965 and rules 1968.
4. The Uttar Pradesh Co-operative Societies (Amendment) Act, 2020.
5. The Uttar Pradesh Co-operative Societies Employees Service Regulations, 1975.
6. Weeraman P.E. The Effect of Cooperatives Law on the Autonomy of Cooperatives in South East Asia, New Delhi, ICA. 1989.
7. Weeraman P.E. A Model Cooperatives Societies Law with the Authors Commentary,
8. New Delhi, ICA, 1994.

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CREDIT CO-OPERATIVES

BMB CM 03

Course Credit: 3 Contact Hours: 40

Course Objectives

1. To understand the origin and development of Co-operative credit.
2. To gain knowledge about various commissions and committees on Co-operative Credit.
3. To know about Co-operative credit structure (ST, MT and LT)
4. To discuss the role, importance and challenges of credit Co-operatives in India

UNIT I (9 Hrs)

Agricultural Co-operative Credit: Classification of Cooperatives; Agriculture and Non Agriculture Cooperatives; Cooperative Credit; Cooperative Banking Structure: S.T, M.T & L.T; Principles of Good Credit System; Advantages of Co- operative Credit; Constitution and Functioning of PACS; Crop Loans; Re- organization of PACS; S.T. Loan Policy and Procedures; Linking of Credit with Marketing; M.T. Credit - Purpose and Security; NFS Lending; and Micro Credit.

UNIT II (9 Hrs)

Co-operative Development: Meaning; Economic Planning and Cooperatives; Co-operatives in Five Year Plans; Recommendations of various Cooperative Committees: Mehta Committee; Mirdha Committee; AIRCSC; AIRCRC; CRAFTCARD; ACRC; and Task Force on ST and LT Co-operative Credit Structure (Vaidyanathan Committee), and Narasimham Committee (Prudential Norms).

UNIT III (8 Hrs)

DCCB & SCB: Significance of DCCBs and SCBs in ST Cooperative Credit; Constitution and Working; Mobilization of Deposits; Lending Oper

Page 95 Data Table

	S.No			Course Outcomes			Bloom's Technology
1			CO1. Identify co-operative credit methods and structure of credit.			Remembering (K1)	
2			CO2.Describe the co-operative development and Recommendation of various committees.			Knowledge (K2)	
3			CO3. To gain knowledge about functions of CCB and state co-operative bank.			Knowledge (K2)	

	S.No			Course Outcomes			Bloom's Technology
4			CO4. To know the significance of long term structure.			Remembering (K1)	
5			CO5. Describe the constitutions and functions of non-Agricultural credit co-operatives.			Comprehending (K3)	

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Course Outcome

After successful completion of this course students will be able to,

S.No

Course Outcomes

Bloom's Technology

1

CO1. Identify co-operative credit methods and structure of credit.

Remembering (K1)

2

CO2. Describe the co-operative development and Recommendation of various committees.

Knowledge (K2)

3

CO3. To gain knowledge about functions of CCB and state co-operative bank.

Knowledge (K2)

4

CO4. To know the significance of long term structure.

Remembering (K1)

5

CO5. Describe the constitutions and functions of non-Agricultural credit co-operatives.

Comprehending (K3)

Suggested Readings

1. Dr. B.S. Mathur. Co-operation in India - Sahitya Bhawan, 2015.
2. R.D.Bedi, Theory. History and Principles of Co-operation, R. Lall Book Depot, Meerut, 2000.
3. B.L Mathur, Rural Development and Co-operation, RBSA Publishers. 2006.
4. C.R.Reddy, Rural Banking in India. Rainbow Publications, 1987.
5. Nakkiran and John Winfred. A. Co-operative Banking In India, Rainbow Publications, 1988.

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IV SEMESTER

UNIVERSAL HUMAN VALUES AND PROFESSIONAL ETHICS

BVE 401

Credit: 3

Total Hours: 40

Course Objectives

1. To help students distinguish between values and skills, and understand the need, basic guidelines, content and process of value education.
2. To help students initiate a process of dialog within themselves to know what they 'really want to be' in their life and profession
3. To help students understand the meaning of happiness and prosperity for a human being.
4. To facilitate the students to understand harmony at all the levels of human living, and live accordingly.
5. To facilitate the students in applying the understanding of harmony in existence in their profession and lead an ethical life

Course Outcomes

1. Understand the significance of value inputs in a classroom, distinguish between values and skills, understand the need, basic guidelines, content and process of value education, explore the meaning of happiness and prosperity and do a correct appraisal of the current scenario in the society
2. Distinguish between the Self and the Body, understand the meaning of Harmony in the Self the Co-existence of Self and Body.
3. Understand the value of harmonious relationship based on trust, respect and other naturally acceptable feelings in human-human relationships and explore their role in ensuring a harmonious society
4. Understand the harmony in nature and existence, and wor

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UNIT-1: Course Introduction - Need, Basic Guidelines, Content and Process for Value Education

(8 Hours)

Understanding the need, basic guidelines, content and process for Value Education, Self-Exploration—what is it? - its content and process; 'Natural Acceptance' and Experiential Validation— as the mechanism for self-exploration, Continuous Happiness and Prosperity- A look at basic Human Aspirations, Right understanding, Relationship and Physical Facilities- the basic requirements for fulfillment of aspirations of every human being with their correct priority, Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario, Method to fulfill the above human aspirations: understanding and living in harmony at various levels.

UNIT-2: Understanding Harmony in the Human Being - Harmony in Myself (8Hours)

Understanding human being as a co-existence of the sentient 'I' and the material 'Body', Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha, Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer), Understanding the characteristics and activities of 'I' and harmony in 'I', Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail, Programs to ensure Sanyam and Swasthya.

UNIT-3: Understanding Harmony in the Family and Society- Harmony in Human-Hu

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Suggested Readings

1.

R R Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.

2.

Ivan Illich, 1974, Energy & Equity, The Trinity Press, Worcester, and Harper Collins, USA

3.

E.F. Schumacher, 1973, Small is Beautiful: a study of economics as if people mattered, Blond & Briggs, Britain.

4.

Sussan George, 1976, How the Other Half Dies, Penguin Press. Reprinted 1986, 1991

5.

Donella H. Meadows, Dennis L. Meadows, Jorgen Randers, William W. Behrens III,

1972, Limits to Growth – Club of Rome's report, Universe Books.

6.

A Nagraj, 1998, Jeevan Vidya Ek Parichay, Divya Path Sansthan, Amarkantak.

7.

P L Dhar, RR Gaur, 1990, Science and Humanism, Commonwealth Publishers.

8.

A N Tripathy, 2003, Human Values, New Age International Publishers.

9.

Subhas Palekar, 2000, How to practice Natural Farming, Pracheen (Vaidik)

KrishiTantraShodh, Amravati.

10. E G Seebauer & Robert L. Berry, 2000, Fundamentals of Ethics for Scientists & Engineers, Oxford University Press

11. M Govindrajran, S Natrajan & V.S. Senthil Kumar, Engineering Ethics (including Human Values), Eastern Economy Edition, Prentice Hall of India Ltd.

12. B P Banerjee, 2005, Foundations of Ethics and Management, Excel Books

13. B L Bajpai, 2004, Indian Ethos and Modern Management, New Royal Book Co., Lucknow. Reprinted 2008

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IV SEMESTER ELECTIVES

MARKETING GROUP (MK)

SERVICE & RETAIL MARKETING

BMB MK 04

Credit: 3

Total Hours: 40

Course Objectives: The objective of this course is to:

- Understand the fundamental concepts, characteristics, and strategic frameworks of services marketing.
- Analyze consumer behavior in services and apply principles for service design and delivery.
- Evaluate pricing, promotion, and service quality aspects to ensure customer satisfaction and loyalty.
- Gain insights into the nature of industrial marketing and different types of industrial products.
- Apply knowledge of organizational buying behavior, models, and influencing factors in industrial markets.

Unit 1 (6 Hours)

Introduction to Services Marketing: Introduction: Definition, Characteristics and Classification of Services, Difference between Product and Services Marketing, Paradigms in Services Marketing,

Present Marketing Environment, Services Marketing Mix: Understanding the 7 P's, Strategies for Services Marketing: Segmentation, Targeting & Positioning, Differentiation.

Unit 2 (6 Hours)

Understanding Consumer Behavior and Service Design Understanding Consumer Behavior: Services vis-à-vis goods, Consumer Behavior in Services, Customer Expectations and Perceptions of Services – Evaluation of services. Service Development Design & Standards: New Service Development Process – Basic service to potential service, Customer Defined Service Standards, Demand

Page 100 Data Table

	Course Outcomes	Bloom's Level
CO1	Understand the fundamental concepts of services marketing in the evolving marketing environment.	K2: Understand
CO2	Analyze consumer behavior in services, identify customer expectations and perceptions, and apply service design and development strategies.	K4: Analyze
CO3	Evaluate service delivery systems, pricing, promotion mix, and service performance using quality frameworks.	K5: Evaluate
CO4	Create various Retail Formats & classifications.	K6 : Create
CO5	Examine organizational buying processes and analyze factors affecting Retailing Business and its evaluation .	K5: Evaluate

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Unit 5 (10)

Merchandise Management, Pricing, and Supply Chain: Merchandise planning process, Assortment planning and category management, Retail buying process; vendor selection, Inventory management and stock turnover Pricing objectives and strategies in retail, Supply chain management in retail, Role of technology: barcoding, RFID, POS

Suggested Readings:

1. Services marketing, Zeithaml Valerie and Mary jo Bitner, Gremler & Pandit, Tata McGraw Hill. Services Marketing, Lovelock, Christopher, Prentice Hall
2. Services Marketing: Concepts, Planning and Implementation, Adrian Payne
3. Chetan Bajaj, Rajnish Tuli & Nidhi V. Srivastava; Retail Management; Oxford University Press
4. Swapna Pradhan; Retail Management: Text and Cases; McGraw Hill
5. S.C. Bhatia; Retailing Management; Atlantic Publishers & Distributors
6. U.C. Mathur; Retail Marketing; Excel Books

Course Outcomes

Course Outcomes

Bloom's Level

CO1 Understand the fundamental concepts of services marketing in the evolving marketing environment.

K2: Understand

CO2

Analyze consumer behavior in services, identify customer expectations and perceptions, and apply service design and development strategies.

K4: Analyze

CO3 Evaluate service delivery systems, pricing, promotion mix, and service performance using quality frameworks.

K5: Evaluate

CO4 Create various Retail Formats & classifications.

K6 : Create

CO5 Examine organizational buying processes and analyze fa

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B2B MARKETING

BMB MK 05

Course Credit: 3

Total Hours: 40

Course Objectives:

1. Introduce the fundamental concepts and characteristics of business-to-business (B2B) marketing.
2. Explain the nature of organizational buying behavior and the decision-making processes involved in B2B markets.
3. Provide an understanding of B2B marketing strategies, including market segmentation, targeting, and positioning.
4. Familiarize students with various B2B communication tools and distribution channel strategies.
5. Equip learners with strategic tools and frameworks to analyze and manage marketing performance in business markets.

UNIT I: Introduction to B2B Marketing (8 Hours)

Concept of Business Marketing and Business Market Customers; Market Structure and Business

Environment; Characteristics of Business Marketing; Strategic Role of Marketing in Business Context; Types of Commercial Enterprises; Commercial and Institutional Customers. Case studies.

UNIT II: Organizational Buying and Buyer Behaviour (8 Hours)

Organizational Buyers' Decision Process: Stepwise Model, Process Flow Model; Characteristics of Business Markets; Government as a Customer; Commercial Enterprises within Business Markets; Commercial and Institutional Customers' Roles and Buying Behavior. Case studies.

UNIT III: B2B Marketing Strategy (8 Hours)

Strategy Making and Strategy Management in B2B; Industrial Product Strategy; Managing Products and Services for Business Ma

Page 102 Data Table

CO No.	Course Outcome	Knowledge Level (Bloom's Taxonomy)
CO1	Explain the foundational concepts of B2B marketing and the nature of business markets.	K2 – Understand
CO2	Analyze the behavior of organizational buyers and decision- making frameworks.	K4 – Analyze
CO3	Apply strategic marketing tools to develop B2B product and channel strategies.	K3 – Apply
CO4	Evaluate segmentation, targeting, and positioning strategies for business markets.	K5 – Evaluate
CO5	Design integrated B2B marketing communication plans and distribution channels.	K6 – Create

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Course Outcomes (COs)

CO

No.

Course Outcome

Knowledge Level (Bloom's

Taxonomy)

CO1

Explain the foundational concepts of B2B marketing and the nature of business markets.

K2 – Understand

CO2

Analyze the behavior of organizational buyers and decision-making frameworks.

K4 – Analyze

CO3

Apply strategic marketing tools to develop B2B product and channel strategies.

K3 – Apply

CO4

Evaluate segmentation, targeting, and positioning strategies for business markets.

K5 – Evaluate

CO5

Design integrated B2B marketing communication plans and distribution channels.

K6 – Create

Suggested Readings

1. Michael D. Hutt, Dheeraj Sharma, Thomas W. Speh, B2B Marketing: A South Asian Perspective, Cengage, 2014.
2. Sharad Sarin, Business Marketing: Concepts and Cases, McGraw Hill, 2013.
3. James C. Anderson, Das Narayandas, James A. Narus, D.V.R. Seshadri, Business Market Management (B2B), Pearson, 2010.
4. Robert Vitale, Waldemar Pfoertsch, Joseph Giglierano, Business to Business Marketing, Pearson, 2011.
5. Krishna K. Havaladar, Business Marketing: Text and Cases, McGraw Hill, 2014.
6. Armstrong, Gary and Philip Kotler, Principles of Marketing, Prentice Hall, 2006.

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HUMAN RESOURCE MANAGEMENT (HR) GROUP

HR ANALYTICS

BMB HR 04

Credit: 3

Total Hours: 40

COURSE OBJECTIVES:

1. This course introduces the student to the theory, concepts, and business application of HR analytics, and the ability to track, store, retrieve, analyze and interpret HR data to support decision making.
2. The student will use applicable benchmarks/metrics to conduct research and statistical analyses related to Human Resource Planning and Recruitment and Selection.
3. Employ appropriate software to record, maintain, retrieve and analyze Performance and training effectiveness.
4. Apply quantitative and qualitative analysis to understand and design compensation system.

5. Demonstrate how to connect HR results to business results.

UNIT 1

8 Hours

Introduction to HR Analytics: Evolution of HR Analytics, HR information systems and data sources, Evolution of HR Analytics; HR Metrics and HR Analytics; Intuition versus analytical thinking; HRMS/HRIS and data sources; Analytics frameworks like LAMP, HR Scorecard & Workforce Scorecard.

UNIT 2

8 Hours

Human Resource Planning and forecasting: Quantitative and Qualitative Dimensions of HR Planning, Methods and Techniques of HR Demand Forecasting, Data Base for Manpower Forecasting.

Recruitment and Selection Analytics: Evaluating Reliability and validity of selection models, Finding ou

Page 104 Data Table

Course Outcomes	Learning Levels as per Bloom's Taxonomy for Evaluation and Assessment
CO 1: Apply HR Analytical techniques in the areas of HRP, recruitment and selection, Compensation and Benefits and Training etc.	Applying (K 3)
CO2: Demonstrate HR function in adding value in business terms.	Analyzing (K 4)
CO3: Utilise soft factors in a people management context and convert them into measurable variables.	Applying (K 3)
CO4: Design a Metrics and Analysis index for recruitment, performance and or a training and development context	K6 Create
CO5: Predict the issues using the available HR data and formulate the best strategies.	K 5 Evaluating

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Course Outcomes

Learning Levels as per Bloom's Taxonomy
for Evaluation and Assessment

CO 1: Apply HR Analytical techniques in the
areas of HRP, recruitment and selection,
Compensation and Benefits and Training etc.

Applying (K 3)

CO2: Demonstrate HR function in adding
value in business terms.

Analyzing (K 4)

CO3: Utilise soft factors in a people management context and convert them into measurable variables.

Applying (K 3)

CO4: Design a Metrics and Analysis index for recruitment, performance and or a training and development context

K6 Create

CO5: Predict the issues using the available HR data and formulate the best strategies.

K 5 Evaluating

Suggested Readings

1. Bhattacharya Kumar Dipak, HR Analytics Understanding Theories and Applications, SAGE Publishing
2. Banerjee Pratyush, Pandey Jatin and Gupta Manish (2019), Practical Applications of HR Analytics, SAGE Publishing
3. Sesil. J, Applying advanced analytics to HR management decisions: Methods for recruitment, managing performance and improving knowledge management. Prentice Hall.
4. Barnett K, Berk J, Human Capital Analytics. Word Association Publication. Fitz■Enz J,
5. The HR Analytics: Predicting the Economic Value of your Company's Human Capital Investments, AMACOM.

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ORGANIZATIONAL DEVELOPMENT AND CHANGE MANAGEMENT

BMB HR 05

Credit: 3

Total Hours :40

Course Objectives

1. To evaluate organizational dynamics, integrating historical evolution, current trends, and power management strategies in diverse contexts.
2. To apply theories of planned change, diagnose organizational readiness, and implement effective change management strategies across levels.
3. To create comprehensive stress management and safety programs, considering cultural variations, grievance procedures, and psychological contract implications

Unit 1 : ORGANIZATIONAL DEVELOPMENT - 8 hrs

Defining Organizational Development (OD) Concepts, Nature and Scope of O.D. Characteristics of OD, Evolution and Process of Organizational Development, Dynamics of planned change,

triggers for change strategies for implementing organizational change, Strategies of change, Inter-Disciplinary Nature of OD, Designing Interventions The OD Practitioner, Client- Consultant relationship, Ethics in OD, Recent trends in OD.

Unit 2 Action Research and OD 8hrs

Managing OD Process: Diagnosis, Nature of OD intervention; Creating Parallel Learning Structures, O.D. Interventions: Team Interventions – Inter group interventions – Personal, Interpersonal and group process interventions – Com

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Course Outcomes

On completion of the course the student will be able to:

- CO1 : Students will be able to remember and understand the basic concepts of Organizational development & Change (K1)
- CO2: Employing theories and models of planned change, students will evaluate organizational readiness, identify sources of resistance, and craft efficient change management strategies across various organizational tiers (K2)
- CO 3: Analysing stressors, preventive measures, grievance procedures, and occupational safety protocols (K3)
- CO 4: Assess the dynamics of organizational development (OD) by integrating its characteristics, historical evolution, and current trends to develop strategies for effectively managing power dynamics within organizational contexts. (K4)
- CO 5: students will create comprehensive stress management and employee safety programs, considering cultural variations and the implications of psychological contracts. (K5)

References

1. Vohra, N., Robbins, S. P., & Judge, T. A. (2022). Organizational Behavior, 18th Edition
2. Organizational Dynamics and Intervention: Tools for Changing the Workplace. New Delhi: Prentice-Hall of India Private Limited.
3. Anderson, L.D. (2013). Organizational Development. 2nd Edition. Sage.
4. Bowditch, J.L. and Buono, A.F. (1994). A premier on organizational behavior. New York. Wiley.
5. Bruke, W.W. (1992). Organizational Development. Boston, Little Brow

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FINANACIAL MANAGEMENT (FM) GROUP
BEHAVIOURAL FINANCE

BMB FM 04

Credit: 3

Total Hours 40

Objectives:

1. To understand the foundations of rational finance and behavioural finance, and the challenges behavioural finance poses to traditional financial theories.
2. To apply knowledge of cognitive and emotional biases in making informed and rational investment decisions
3. To analyze the impact of overconfidence, herd behaviour, and other social factors on financial markets division and investor decisions.
4. To evaluate the principles of value investing through the lens of behavioural biases and investor psychology.
5. To develop informed investment strategies by integrating behavioural insights into financial decision-making and market outcomes.

Unit I

8 Hours

Introduction to Behavioral Finance: Foundations of behavioral Finance, behavioral Finance Vs Conventional Finance

Rational Markets Hypothesis and Challenge of Behaviouralist : Intellectual Underpinnings,

The rise of Rational Market Hypothesis, Impact on wall street and the choices, The Challenge of Behaviouralist, Synthesis and Future Horizons. Impact on Capital Market

Foundations of Rational Finance: Introduction, Neoclassical, Economics, Rational Preferences,

Utility maximization, Relevant information; Expected Utility Theory, Modern Portfolio Theory,

Capital Asset Pricing Model, E

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S. No	Course Outcome	Bloom's Taxonomy
CO1	Understand the foundations of rational finance and behavioural finance.	K2 (Understanding)
CO2	Apply knowledge of cognitive and emotional biases in making informed and rational investment decisions.	K3 (Applying)
CO3	Analyze the impact of overconfidence, herd behaviour, and other social factors on financial markets division and investor decisions	K4 (Analyzing),
CO4	Evaluate the principles of value investing through the lens of behavioural biases and investor psychology.	K5 (Evaluating)
CO5	Develop informed investment strategies by integrating behavioural insights into financial decision-making and market outcomes	K6 (Creating)

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theory, psychographic models, basic ingredients of sound investment philosophy, guidelines for overcoming psychological biases.

UNIT V

Hrs 8

Market outcomes: Size effect and seasonality, Momentum and reversal, post- earnings announcement drift, value premium, premium puzzle, excessive volatility, Bubbles, Behavioral Asset Pricing Model.

Value Investing: Central tenets of value investing, Evidence and prospects of value investing, Strategies of well-known value investors, academic research on value investing.

S.

No

Course Outcome

Bloom's

Taxonomy

CO1 Understand the foundations of rational finance and behavioural finance.

K2

(Understanding)

CO2 Apply knowledge of cognitive and emotional biases in making informed and rational investment decisions.

K3 (Applying)

CO3 Analyze the impact of overconfidence, herd behaviour, and other social factors on financial markets division and investor decisions

K4 (Analyzing),

CO4 Evaluate the principles of value investing through the lens of behavioural biases and investor psychology.

K5 (Evaluating)

CO5 Develop informed investment strategies by integrating behavioural insights into financial decision-making and market outcomes

K6 (Creating)

Reference Books

1. Chandra Prasanna (2016): Behavioral Finance, Tata Mc Graw Hill Education(India) Private Limited, Chennai.
2. Acket Lucy F and DeavesRichard(2011): Understanding Behavior

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STRATEGIC FINANCIAL MANAGEMENT

BMB FM 05

Credits: 3

Hours 40 L

Objectives:

1.

To understand the basic knowledge of financial management and its functions, frameworks, and tools of strategic financial management

2.

To apply financial strategies for investment, financing, and dividend decisions in alignment with organizational goals.

3.

To analyze capital structure, cost of capital, risk-return trade-offs, and their impact on shareholder value to evaluate alternative investment proposals.

4.

To evaluate the knowledge of factors influencing dividend, capital structure, working capital of a firm.

5.

To develop innovative financial strategies for value creation, long-term sustainability, and competitive advantage.

Unit I 8 Hours

Strategic Financial Management: Objectives and Functions. Valuation of Securities- Approaches to corporate valuation, valuation of equities including cash flow, dividend earning approach, valuation of debts and preference shares.

Unit II 8 Hours

Capital Structure: factors affecting capital structure, Capital structure theories. Operating Leverage, Financial Leverage and Combined Leverage, EBIT and EPS analysis, ROI and ROE analysis.

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S. No	Course Outcome	Bloom's Taxonomy
CO1	Understand the basic knowledge of financial management and its functions.	K2 (Understanding)
CO2	Apply financial strategies for investment, financing, and dividend decisions in alignment with organizational goals.	K3 (Applying)

S. No	Course Outcome	Bloom's Taxonomy
CO3	Analyze capital structure, cost of capital, risk-return trade-offs, and their impact on shareholder value to evaluate alternative investment proposals..	K4 (Analyzing), K5
CO4	Evaluate the knowledge of factors influencing dividend, capital structure, working capital of a firm	K5 (Evaluating)
CO5	Develop innovative financial strategies for value creation, long-term sustainability, and competitive advantage .	K6 (Creating)

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S.

No

Course Outcome

Bloom's

Taxonomy

CO1 Understand the basic knowledge of financial management and its functions.

K2

(Understanding)

CO2 Apply financial strategies for investment, financing, and dividend decisions in alignment with organizational goals.

K3 (Applying)

CO3

Analyze capital structure, cost of capital, risk-return trade-offs, and their impact on shareholder value to evaluate alternative investment proposals..

K4 (Analyzing),

K5

CO4 Evaluate the knowledge of factors influencing dividend, capital structure, working capital of a firm

K5 (Evaluating)

CO5

Develop innovative financial strategies for value creation, long-term sustainability, and competitive advantage

.

K6 (Creating)

Reference Books:

1. Rajiv Srivastava and Anil Misra- Financial Management-OXFORDUniversity Press.
2. Prasannachandra -Financial Management- Tata Mc GrawHill
3. I M Pandey -Financial Management-Vikas Publishing House Pvt. Lt. New Delhi
4. Lawrence D. Schall &Charles W. Haley -Introduction to Financial Management-McGraw Hill.

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OPERATION MANAGEMENT (OM) GROUP

SERVICE OPERATIONS MANAGEMENT

BMB OM 04

Course Credit: 3

Contact Hours: 40

Course Objectives: The objective of this course is to:

- Understand the fundamental concepts, nature, and scope of service operations management across diverse service sectors.
- Analyze the design, development, and delivery of service processes using various operational frameworks and models.
- Evaluate service quality management techniques, capacity planning, and demand-supply alignment strategies.

.

Unit 1: (8 Hours)

Introduction to Service Operations: Nature and characteristics of services, Classification of service industries, Evolution of service operations management, Service Economy and Growth of Services. Service encounter and service delivery system, Service process matrix and service positioning strategies. Role of service operations in organizational competitiveness.

Unit 2: (8 Hours)

Service process design: flow diagrams, blueprinting, process analysis. Service capacity planning and management. Managing demand and capacity mismatches. Service Process Management: Queue management and waiting line models, Service facility location decisions, Facility layout and service environment design. Employee scheduling and service staffing strategies.

Unit 3: (8 Hours)

Service Quality and Productivity: Concept of service quality and SERVQUAL model, Gap model of service quality, Techniques for measuring service quality

Page 113 Data Table

	Course Outcome	Bloom's Level
CO1	Define and explain the fundamental concepts, nature, and scope of service operations and their role in competitive advantage.	K2: Understanding
CO2	Apply service process design and management techniques including capacity planning, queue management, and facility location strategies to real-world service organizations.	K3: Applying
CO3	Analyze service quality using models like SERVQUAL and Gap Model, and develop strategies to manage productivity and customer satisfaction.	K4: Analyzing
CO4	Evaluate technology applications, CRM systems, and self- service technologies for enhancing service operations and customer experiences.	K5: Evaluating
CO5	Design and develop service operations strategies focusing on innovation, sustainability, and performance excellence across various service sectors.	K6: Creating
CO6	Critically assess service recovery strategies, demand-supply alignment, and the challenges of digital service delivery models in a competitive business environment.	K5: Evaluating

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Suggested Readings

1. Fitzsimmons, J. A., Fitzsimmons, M. J., & Bordoloi, S. (2014). Service Management: Operations, Strategy, Information Technology, McGraw-Hill.
2. Chandrasekaran, K. (2016). Essentials of Services Marketing, Tata McGraw-Hill.
3. Johnston, R., & Clark, G. (2008). Service Operations Management: Improving Service Delivery, Pearson.
4. Apte, G. (2011). Services Marketing, Oxford University Press India.
5. Jha, S. M. (2015). Services Marketing, Himalaya Publishing House.
6. Lovelock, C., & Wirtz, J. (2021). Services Marketing: People, Technology, Strategy, Pearson.
7. Zeithaml, V. A., Bitner, M. J., Gremler, D. D. (2020). Services Marketing: Integrating Customer Focus Across the Firm, McGraw-Hill.

Course Outcomes

Course Outcome

Bloom's Level

CO1 Define and explain the fundamental concepts, nature, and scope of service operations and their role in competitive advantage.

K2: Understanding

CO2 Apply service process design and management techniques

including capacity planning, queue management, and facility location strategies to real-world service organizations.

K3: Applying

CO3 Analyze service quality using models like SERVQUAL and Gap Model, and develop strategies to manage productivity and customer satisfaction.

K4: Analyzing

CO4 Evaluate technology applications, CRM systems, and self-service technologies for enhancing service operations and customer experiences.

K5: Evaluating

CO5 Design and develop service op

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PROJECT AND SOURCING MANAGEMENT

BMB OM 05

Course Credit: 3

Contact Hours: 40

Course Objectives: The objective of this course is to:

- Understand the fundamental concepts, processes, and strategies involved in sourcing management and project management.
- Analyze supplier evaluation methods, vendor rating techniques, and supplier development strategies in sourcing decisions.
- Examine price determination mechanisms and negotiation strategies in sourcing and procurement.
- Understand project characteristics, project life cycle stages, budgeting processes, and various cost structures in projects.
- Apply project scheduling techniques, for effective project planning, monitoring, and control.
- Evaluate real-world sourcing and project management practices to recommend strategic improvements and innovations.

Unit 1 (8 Hours)

Sourcing Management: Introduction to Sourcing, Sourcing v/s Procurement, Purchasing: Purchasing Cycle, 8 R's of Purchasing, Role of a Purchasing Manager, Risks associated with purchasing process and its mitigation, Concept and applications of Make or Buy

Decision:. Case Studies

Unit 2 (8 Hours)

Evaluating Suppliers' Efficiency: Vendor Rating, Selection and Development: Need

for Measuring Supplier Performance, Categorization of Suppliers, Suppliers Evaluation Methods/ Vendor Rating Methods, Supplier Selection Process, Vendor Rating process, Factors Affecting the Selection of Optimal Suppliers or Vendor Rating,

Page 115 Data Table

	Course Outcome	Bloom's Level
CO1	Define and explain fundamental concepts related to sourcing management, purchasing cycles, vendor management, and project management frameworks.	K1: Remembering, K2: Understanding
CO2	Apply vendor rating methods, supplier selection criteria, and pricing strategies in real-world sourcing scenarios.	K3: Applying
CO3	Analyze supplier efficiency, negotiation processes, and risks associated with sourcing and purchasing operations.	K4: Analyzing
CO4	Evaluate project budgeting methods, cost estimation techniques, financing options, and project resource allocation strategies.	K5: Evaluating
CO5	Design and develop project schedules, using Work Breakdown Structure (WBS), network analysis, PERT, CPM, and Earned Value Analysis for tracking and controlling projects.	K6: Creating
CO6	Critically assess project termination processes, project performance metrics, and the role of negotiation and supplier management in overall project success.	K5: Evaluating

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Unit 5 (8 Hours)

Project Scheduling, Network Analysis & Control : Steps in Project Scheduling and Network design, Gantt Chart, Work Breakdown Structure (WBS) , Identifying and application of the Nodes and Activities, Activity on Arrow (AoA) and Activities on Node (AoN) methods, Application of PERT and CPM, Planning- Monitoring and Control Cycle, Tracking through Gantt chart. Earned Value Analysis (EVA): Planned Value (PV), Earned Value (EV), Cost Variance (CV), Schedule Variance (SV), Cost performance Index (CPI), Schedule performance Index (SPI). Project Termination: Types of Terminations, Project Termination Process. Case Studies

Suggested Readings

1. Dobler, D. W., jr, L. L., & Burt, D. N., Purchasing and Materials Management. New Delhi: Tata McGraw-Hill Publishing Company Limited
2. Gopalakrishnan P., Purchasing and Materials Management, Tata McGraw-Hill Publishing Company Ltd, New Delhi.
3. David N. Burt, Sheila Petcavage, Richard Pinkerton: Proactive Purchasing in the

Supply Chain: The Key to World-Class Procurement, McGraw Hill

Course Outcomes

Course Outcome

Bloom's Level

CO1

Define and explain fundamental concepts related to sourcing management, purchasing cycles, vendor management, and project management frameworks.

K1: Remembering,

K2: Understanding

CO2 Apply vendor rating methods, supplier selection criteria, and pricing strategies in real-world sourcing scenarios.

K3: Applying

CO3 Analyze supplier efficiency, negotia

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INTERNATIONAL MANAGEMENT (IB) GROUP

MANAGING GLOBAL SUPPLY CHAINS

BMB IB 04

Course Credits: 3

Contact Hours 40

Course Objectives

1. To gain a comprehensive understanding of the principles and frameworks underpinning global supply chains and their strategic importance in international business.
2. To critically analyze the complexities and challenges in designing, operating, and managing supply chains across diverse geographical and cultural settings.
3. To develop practical skills to design, optimize, and implement global supply chain networks leveraging Indian and global business scenarios.
4. To explore risk identification, mitigation strategies, and sustainable practices in supply chain operations.
5. To examine the role of emerging technologies and innovations shaping the future of supply chains and supply chain management decision-making.

Unit 1: Introduction to Global Supply Chains (8Hrs)

Definition, scope, and evolution of global supply chains; Key components: sourcing, manufacturing, logistics, distribution, and customers; Importance and strategic role of global supply chains in competitive advantage; Supply chain flows: product, information, financial, and reverse flows; Globalization trends impacting supply chains in India and worldwide; Case study analysis of

Indian companies with global supply chains.

Unit 2: Supply Chain Design and Network Configuratio

Page 117 Data Table

CO No.	Course Outcome Description	K-Level (Bloom's Taxonomy)
CO1	Recall and explain key concepts, components, and significance of global supply chains	Remembering (K1)
CO2	Analyze and design effective supply chain networks tailored to Indian and global contexts	Analyzing (K4)
CO3	Apply operational tools and techniques for managing inventory, logistics, and partner coordination	Applying (K3)
CO4	Evaluate risks and develop sustainability strategies for efficient global supply chains	Evaluating (K5)
CO5	Assess the impact of emerging technologies and innovations on supply chain management and performance	Evaluating (K5)

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Unit 5: Technology and Innovation in Global Supply Chains (8Hrs)

Digital transformation in supply chains; Emerging technologies: AI, blockchain, robotics, and their applications in supply chains; Role of Industry 4.0 and latest trends in supply chain innovation; Challenges and opportunities of technology adoption in Indian firms; Case study on digital supply chains in Indian retail and manufacturing sectors

Course Outcomes (COs)

CO

No.

Course Outcome Description

K-Level (Bloom's

Taxonomy)

CO1

Recall and explain key concepts, components, and significance of global supply chains

Remembering (K1)

CO2

Analyze and design effective supply chain networks tailored to Indian and global contexts

Analyzing (K4)

CO3

Apply operational tools and techniques for managing inventory,

logistics, and partner coordination

Applying (K3)

CO4

Evaluate risks and develop sustainability strategies for efficient
global supply chains

Evaluating (K5)

CO5

Assess the impact of emerging technologies and innovations on
supply chain management and performance

Evaluating (K5)

Suggested Readings

1. Mahajan, M. I. (2015). Export-import procedures and documentation (26th ed.). Snow White Publications Pvt Ltd.
2. Balagopal, T. A. S. (2016). Export management (22nd ed.). Himalaya Publishing House.
3. Cherunilam, F. (2007). International trade and export management (4th rev. ed.). PHI Learning.
4. Purushothaman, S., & Saikumari,

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INTERNATIONAL FINANCE

BMB IB 05

Course Credits 3

Contact Hours 40

Course Objectives

1. To provide a comprehensive understanding of the international financial environment, markets, and instruments.
2. To analyze the dynamics of foreign exchange markets, exchange rate determination, and currency risks.
3. To develop skills in managing international financial operations, including multinational financing and investment decisions.
4. To evaluate the impact of global economic policies, international monetary systems, and balance of payments on business decisions.
5. To explore techniques for managing foreign exchange risk and decision-making in multinational corporate finance.

Unit I: Introduction to International Finance (6 Hrs)

Definitions and distinctions among International Trade, International Business, and International Finance; Theories of International Trade; International Trade Financing in India; Balance of

Payments of India (Using data from RBI website); International Monetary Systems: Gold Standard, Gold Exchange Standard, Bretton Woods System, Current Monetary System, European Monetary Union.

Unit II: International Financial Institutions

(6 Hrs)

Overview of International Financial Institutions: Roles of International Monetary Fund (IMF), World Bank, Asian Development Bank (ADB), BRICS Development Bank, European Bank for Reconstruction and Development (EBRD).

Unit III: Foreign Exchange Management (10 Hrs)

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Page 119 Data Table

Cos	Course Outcomes	K-Level (Bloom's Taxonomy)
CO1	Understand and explain fundamental concepts of international finance, trade, and monetary systems	Remembering (K1)
CO2	Analyze foreign exchange markets, exchange rate determination, and forecasting models	Analyzing (K4)
CO3	Apply foreign exchange risk management techniques, including hedging and derivatives	Applying (K3)
CO4	Evaluate multinational financial management decisions such as international investment and project appraisal	Evaluating (K5)
CO5	Assess the impact of political risk, emerging technologies, and international regulations on global financial operations	Evaluating (K5)

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Unit V: International Investment and Challenges (8 Hrs)

International Portfolio Investment: Foreign Investment Analysis, International Bond Investing, Direct Investment Strategies, Optimal International Asset Allocation; International Foreign exchange risk: types of exposure including transaction exposure, translation exposure, and economic exposure; impact of currency volatility on multinational corporations and cross-border cash flows; political and sovereign risk: effects of political instability, abrupt policy changes, and expropriation; regulatory and compliance challenges.

Course Outcomes

Cos

Course Outcomes

K-Level (Bloom's

Taxonomy)

CO1 Understand and explain fundamental concepts of international

finance, trade, and monetary systems

Remembering (K1)

CO2 Analyze foreign exchange markets, exchange rate determination, and forecasting models

Analyzing (K4)

CO3 Apply foreign exchange risk management techniques, including hedging and derivatives

Applying (K3)

CO4 Evaluate multinational financial management decisions such as international investment and project appraisal

Evaluating (K5)

CO5 Assess the impact of political risk, emerging technologies, and international regulations on global financial operations

Evaluating (K5)

Suggested Readings:

1. Eun, Cheol S., & Resnick, Bruce G. International Financial Management. McGraw Hill Education (Latest Edition)
2. Hill, W. L. Charles & Jain, A.K. (2008). International Business (6th

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INFORMATION TECHNOLOGY (IT) GROUP

E-BUSINESS

BMB IT 04

Credits: 3

Contact Hours 40 L

Course Objectives



To understand the foundations of the digital economy and distinguish between various e-business and e-commerce models.



To explore tools and strategies for launching and managing an online business.



To analyze mobile commerce, IT-enabled services, and pervasive computing in business.



To examine the structure, implementation, and impact of e-governance models.



To evaluate ethical, regulatory, and legal issues governing digital business and service

delivery.



To assess emerging technological trends shaping the future of e-business and governance.

Unit - I: 8 hrs

The Digital Economy and E-Commerce Foundations: Understanding the Internet Economy -

Evolution and characteristics of the digital economy, Current trends and future outlook;

Introduction to E-Commerce - Definitions and scope, Historical emergence and milestones; E-

Commerce Business Models - B2B, B2C, C2C, C2B, B2A, C2A models, Hybrid and emerging models, Case studies of major e-commerce ventures

Unit - II: 8 hrs

Setting Up and Managing an E-Business: Launching an E-Business - Steps in starting an online

business, Tools and platforms for e-business; Building and Operating an E-Commerce Website -

Web development technologies, User interface and experience design; Customer Acquisition and

Retention - Digital marketing and net-based promoti

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S. No	Course Outcome	Bloom's Taxonomy
CO1	Understand digital economy concepts and apply e-business models in diverse business contexts.	K2 (Understanding)
CO2	Analyze strategies for setting up and managing e-commerce platforms.	K4 (Analyzing)
CO3	Evaluate applications of ITES, mobile commerce, and legal frameworks in e-business.	K5 (Evaluating)
CO4	Interpret and assess e-governance strategies and their impact on public service delivery.	K4 (Analyzing)
CO5	Formulate insights into future trends in e-business using AI, blockchain, and ethical frameworks.	K6 (Creating)

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Unit - V: 12 hrs

E-Governance Models and Applications: Categories and Models of E-Governance - Technology

policy,

infrastructure,

training,

and

consulting

funds,

Models:

Digital

Governance,

Broadcasting/Wider Dissemination, Critical Flow, Interactive Service, Govt-to-Citizen-to-Govt;

Major E-Governance Service Areas: Public grievance redressal (telephone, ration card, land records, police records), Rural and urban digital service delivery

Contemporary Issues and Future Trends: Trends in E-Business and E-Governance - Artificial intelligence in e-business, Big data and analytics, Blockchain and smart contracts; Sustainability and Ethical Considerations -Data privacy, green IT, social responsibility

Course Outcomes (COs)

S.

No

Course Outcome

Bloom's

Taxonomy

CO1 Understand digital economy concepts and apply e-business models in diverse business contexts.

K2

(Understanding)

CO2 Analyze strategies for setting up and managing e-commerce platforms.

K4 (Analyzing)

CO3 Evaluate applications of ITES, mobile commerce, and legal frameworks in e-business.

K5 (Evaluating)

CO4 Interpret and assess e-governance strategies and their impact on public service delivery.

K4 (Analyzing)

CO5 Formulate insights into future trends in e-business using AI, blockchain, and ethical frameworks.

K6 (Creating)

Suggested Readings:

■ Joseph, E- Commerce: A Managerial Perspective, Prentice Hall

■ Stain, L.D., Web Security: Step-by Step Reference

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BUSINESS DATA WAREHOUSE AND DATA MINING

BMB IT 05

Credits: 3

Contact Hours 40 L

Course Objectives



To understand the role and architecture of data warehouses in business intelligence.



To develop multidimensional data models and explore OLAP operations for business analytics.



To analyze preprocessing steps for mining and implement feature extraction methods.



To apply various data mining techniques for classification, clustering, and association rule mining.



To examine advanced mining areas including text, web, spatial, and multimedia mining.



To evaluate ethical, legal, and implementation issues associated with data mining.

Unit - I: 6 hrs

Introduction to Data Warehousing and Data Mining: Fundamentals and Definitions: Data warehousing and its business value, Introduction and evolution of data mining, goals of data mining, Myths about data mining, The Data Mining process; Business Relevance - Data warehousing vs. OLTP systems, Roles of data warehousing and mining in business decisions; Basic Architecture and Components - Overall architecture of data warehouse systems, Layers: staging, integration, access, Enterprise Data Warehouse (EDW); Knowledge Discovery in Databases (KDD)

- Knowledge Extraction through Data Mining, Steps in KDD process and business applications

Unit - II: 7 hrs

Data Warehouse Modeling and Implementation: Multidimensional Data Modeling - Star, snowflake, and fact constellation

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S. No	Course Outcome	Bloom's Taxonomy
CO1	Understand data warehousing components and their role in business intelligence.	K2 (Understanding)
CO2	Apply OLAP and multidimensional modeling techniques for strategic insights.	K3 (Applying)
CO3	Analyze data preprocessing methods and their impact on mining accuracy.	K4 (Analyzing)
CO4	Evaluate and compare various mining algorithms and their business applications.	K5 (Evaluating)
CO5	Develop solutions using mining techniques while addressing ethical and deployment considerations.	K6 (Creating)

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high-dimensional data, outlier detection, Applications in segmentation, targeting, fraud detection

Unit - V: 10 hrs

Advanced Mining Topics and Applications: Web, Text, and Multimedia Mining - Concepts and business applications; Spatial and Temporal Data Mining - Techniques and relevant uses; Business Intelligence and Case Studies -CRM, financial analytics, marketing, social media, retail, insurance; Trends in Data Mining -Big Data, cloud data warehousing, real-time analytics, AI-driven mining Data Mining Implementation and Ethics: Evaluation and Validation - Accuracy, overfitting, underfitting, cross-validation; Business Integration - Aligning mining outcomes with business strategy, User adoption and deployment; Privacy, Security, and Ethical Issues - Data privacy challenges, security in warehousing/mining, Regulations and best practices, Ethical implications in data analysis and usage

Course Outcomes (COs)

S.

No

Course Outcome

Bloom's

Taxonomy

CO1 Understand data warehousing components and their role in business intelligence.

K2

(Understanding)

CO2 Apply OLAP and multidimensional modeling techniques for strategic insights.

K3 (Applying)

CO3 Analyze data preprocessing methods and their impact on mining accuracy.

K4 (Analyzing)

CO4 Evaluate and compare various mining algorithms and their business applications.

K5 (Evaluating)

CO5 Develop solutions using mining techniques while addressing ethical and deployment considerations.

K6

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COOPERATIVE MANAGEMENT (CM) GROUP

NON-CREDIT CO-OPERATIVES

BMB CM 04

Course Credit: 3 Contact Hours: 40

Course Objectives

1. To gather the functions of Milk Producers Co-operatives.
2. To acquire the various types of Marketing Co-operatives.
3. To understand the functions of Consumer Co-operatives.
4. To know about the Industrial and Processing Co-operatives.
5. To acquire knowledge about various types of Non-Credit Co-operatives.

UNIT I (8 Hrs)

Marketing Co-operatives: Origin and Development of Marketing Co-operatives in India; Primary and Apex Co-operative Marketing Societies in India; their Constitution and Functioning; Various forms of Government assistance; NAFED; and Regulated Markets.

UNIT II (8 Hrs)

Consumer Co-operatives: Need and Importance; Origin and Development; Structure; Working of Primary, District, Apex Consumer Co-operatives; NCCF; Supermarkets; Recent development and problems in consumer Co-operatives.

UNIT III (8 Hrs)

Dairy Co-operatives: Place of Dairy Co-operatives in Indian Economy; Structure of Dairy Co-operatives; NDDDB; AMUL Pattern; Working and Functions of National Cooperative Dairy Federation; State Cooperative Milk Producers Federation; District Cooperative Milk Producers Union; and Primary Cooperative Milk Producers Societies; Operation Flood Schemes; Recent developments and problems in dairy Co-operatives.

UNIT IV (7H)

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	S.No			Course Outcomes			Bloom's Technology
1			CO1. Imbibe the methods of marketing by Primary Agricultural Producers Co-operative Marketing Society Ltd			Remembering (K1)	

	S.No			Course Outcomes			Bloom's Technology
2			CO2. Gather knowledge regarding the method of processing of milk by Dairy Co-operatives.			Knowledge (K2)	
3			CO3. Know the impact of consumer Co-operatives in regulating the prices of consumer goods			Knowledge (K2)	
4			CO4. Understand the scope of employment opportunities created by industrial Co-operatives in rural India			Remembering (K1)	
5			CO5. Know the role of miscellaneous and service Co-operatives in the upliftment of community as a whole.			Comprehending (K3)	

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Colleges; Fisheries Co-operatives; Forest Produce Co-operatives (LAMPS) and New Generation Cooperatives.

Course Outcome

After successful completion of this course students will be able to,

S.No

Course Outcomes

Bloom's Technology

1

CO1. Imbibe the methods of marketing by Primary

Agricultural Producers Co-operative Marketing Society Ltd

Remembering (K1)

2

CO2. Gather knowledge regarding the method of processing

of milk by Dairy Co-operatives.

Knowledge (K2)

3

CO3. Know the impact of consumer Co-operatives in regulating the prices of consumer goods

Knowledge (K2)

4

CO4. Understand the scope of employment opportunities created by industrial Co-operatives in rural India

Remembering (K1)

5

CO5. Know the role of miscellaneous and service Co-operatives in the upliftment of community as a whole.

Comprehending (K3)

Suggested Readings

1. B.L. Mathur, Rural Development and Co –operation, RBSA Publishers, 2006.
2. B.S. Mathur, Co-operation in India. Sahitya Bhawan, 2000.
3. R.D.Bedi, Theory, History and Principles of Co-operation, R. Call Book Depot, 2000.
4. S. Shanmuga Sundaram, Weavers Co-operatives. Rainbow, Publications, 1987.
5. T.N. Hajela, Principles, Problems and Practice of Co-operation. Ane's Student Edition, 2000.

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CO-OPERATIVE ACCOUNTING & AUDIT

BMB CM 05

Course Credit: 3 Contact Hours: 40

Course Objectives

1. To understand the concept of general and Co-operative audit.
2. To classify the verification and valuation of various assets and liabilities.
3. To know the role, duties and responsibilities of Co-operative auditors.
4. To enable for auditing system in different types of Co-operatives.
5. To understand concept of errors and frauds.

UNIT I (8 Hrs)

Accounting of PACS: Accounting records to be maintained by PACS; Basic accounting principles and policies; Steps for preparation of financial statements; Accounting entries for credit activities; Accounting entries for non-credit activities; Accounting entries for common activities; Preparation of final accounts; Understanding the financial statements of PACS; Computerization of PACS;

Accounting under CBS environment.

UNIT II (8 Hrs)

Accounting of ZSBs: Accounting rules and procedures for the Zila Sahakari Bank Ltd; Head Office

Accounting; Rules and procedures for the Zila Sahakari Bank Ltd; DCCB Branches; Reconciliation

procedures; Accounting of LDBs; Accounting rules and procedures for the Uttar Pradesh Bhumi

Vikas Bank Ltd; Head Office Accounting; Rules and procedures for the Uttar Pradesh Bhumi Vikas

Bank Ltd; Branches; and reconciliation procedures in LDBs.

UNIT III (8 Hrs)

Cooperative Audit: Definition; Objective

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	S.No			Course Outcomes			Bloom's Technology
1			CO1. Gain knowledge on general and Co-op audit.			Remembering (K1)	
2			CO2. Acquired skills to apply in verification and valuation of assets and liabilities.			Comprehending (K3)	
3			CO3. Learn the duties and responsibilities of general and Co-operative auditor.			Knowledge (K2)	
4			CO4. Expand knowledge on conducting audit programme in cooperatives.			Knowledge (K2)	
5			CO5. Identify the embezzlement of Errors and Frauds.			Comprehending (K3)	

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UNIT V (8 Hrs)

Audit Programme for Selected Societies: Various stages of Audit; Audit procedures for

Cooperative Credit Institutions; Marketing Societies; Consumer Stores; Housing Societies; Milk

Producers Societies; Industrial Cooperatives; Classification on the Reserve Bank of India Standard;

Preparation of Final Audit Memorandum and its Enclosures.

Course Outcome

After successful completion of this course students will be able to,

S.No

Course Outcomes

Bloom's Technology

1

CO1. Gain knowledge on general and Co-op audit.

Remembering (K1)

2

CO2. Acquired skills to apply in verification and valuation of assets and liabilities.

Comprehending (K3)

3

CO3. Learn the duties and responsibilities of general and Co-operative auditor.

Knowledge (K2)

4

CO4. Expand knowledge on conducting audit programme in cooperatives.

Knowledge (K2)

5

CO5. Identify the embezzlement of Errors and Frauds.

Comprehending (K3)

Suggested Readings

1. Manual on Common Accounting System (CAS) for Primary Agricultural Credit Societies (PACS), NABARD.
2. R.G. Saxena. Principles and Practice of Auditing, Himalaya Publishing House, New Delhi, 2011.
3. D.R. Kapoor, Hand Book of Co-operative Audit, Anmol Publications Pvt Ltd. 1998.
4. B.N. Tandon, S. Sudhhranan, S. Sundhora Babu, A Hand Book of Practical Auditing, S. Chand and Company Ltd, 2006.
5. B.N. Tandon, A Hand Book of Practical Auditing, S. Chand and Company Ltd, 2007.

DOCUMENT CHUNK 29: Docimp (PDF) [PDF Document] | URL: <https://www.vsicskanpur.org/attention/bca/docimp.pdf>

Docimp (Official Document)

****Source Document URL:**** <https://www.vsicskanpur.org/attention/bca/docimp.pdf>

****Total Document Pages:**** 2

DOCUMENT CHUNK 30: Bca Bba Fee (PDF) [PDF Document] | URL:
<https://www.vsicskanpur.org/docs/BCA-BBA-Fee.pdf>

Bca Bba Fee (Official Document)

****Source Document URL:**** <https://www.vsicskanpur.org/docs/BCA-BBA-Fee.pdf>

****Total Document Pages:**** 1

DOCUMENT CHUNK 31: VSICS [HTML Page] | URL: <https://www.vsicskanpur.org/committee.html>

Mr. Nitin Mishra

Computer Application

Mr. Sanjay Kumar Tiwari

Computer Application

Mrs. Shweta Agnihotri

Business Management

Mr. Praateek Arora

Business Management

Literal Committee

The Literal committee inspires and nurtures mindset of individual's which makes them to shine throughout their academic career. Literal committee provides a platform for the students to enhance their intellectual and independent thinking skills which will imbibe a sense of confidence among themselves. The committee aims to inculcate among students a flair for their language and to shape their enticement. Literal committee emphasize to guide individuals in developing critical and analytical literary skills

Organizing Committee

The heart of any Institute is its library. Our library is well stocked with a variety of books that has more than 2000 titles in over 11,000 multiple copies. It is regularly updated with the latest titles from India as well as overseas.

Social Committee

The social committee aims to provide a healthy and collegial environment by contributing to the society. We do this by empowering people to manage their own lives; promoting people's independence; supporting people's social inclusion and participation in society; and helping people to keep safe and well.

Sports Committee

Only a healthy body can develop and maintain a healthy mind. Hence the Institute Promotes fitness and health as a regular activity. The Institute boasts of a compound field for Football, Cricket, Hockey, Boxing & Athletics in addition to a Badminton and Handball courts. The Institute provides facilities for the conduction of the above mentioned games. The Institute also has an indoor games room with Chess, Carrom and Table Tennis facilities.

Editorial Committee

We are looking forward to excavate the hidden talents of the college to showcase their outstanding charisma in the form of new creations.

Media & Publication Committee

The Media & Publications Committee is responsible for capturing and archiving photographic memories of events. We are responsible for photographing various events including games, meetings and social gatherings. The goal is preserve these images for all the VSICS family members to enjoy.

Placement & Alumni Committee

Placement Cell interacts with the reputed organizations all over the country for arranging campus interviews for the final year students. We make efforts to organize technical seminars, workshops and corporate expectation sessions. Industry personnel are invited periodically to enrich the knowledge of our student community with the latest technological innovations and industry practices. We produce graduates and Postgraduates who are well equipped to handle the working norms of the industry and commerce in the public and private sectors and helping every student to define his/her career interest through individual counseling through experts.

Cultural Committee

The cultural committee in our college Dr.VSICS has been introduced to encourage student's interest, participation and responsibilities in the ingenious fields through a medium of creative art and literacy curriculum and scope. It's to help students in knowing their hidden talents and to push their boundaries to create a new version of themselves.

DOCUMENT CHUNK 32: Bba New Syllabus (PDF) [PDF Document] | URL:
<https://www.vsicskanpur.org/docs/BBA-new-syllabus.pdf>

Bba New Syllabus (Official Document)

Source Document URL: <https://www.vsicskanpur.org/docs/BBA-new-syllabus.pdf>

Total Document Pages: 44

Page 1 Data Table

Yea r	Sem.	Subject	Part	Paper Code	Paper Name	Credit
1	I	Course/ paper-1	A	F010101T	Business Economics	3
			B		Basic Accounting	3
	I	Course/ paper-2	A	F010102T	Business Statistics	3
			B		Principles of Management	3
	I	Course/ paper-3	A	F010103T	Business Ethics and Governance	3
			B		Computer Applications	3

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
1	II	Course/ paper-4	A	F010201T	Organisational Behavior	3
			B		Business Finance	3
	II	Course/ paper-5	A	F010202T	Human Resource Development	3
			B		Marketing Theory and Practices	3
	II	Course/ paper-6	A	F010203T	Business Mathematics	3
			B		Advertising Management	3
2	III	Course/ paper-7	A	F010301T	Management & Cost Accounting	3
			B		Business Law	3
	III	Course/ paper-8	A	F010302T	Production Management	3
			B		Business Policy	3
	III	Course/ paper-9	A	F010303T	Business Communication	3
			B		Business Environment	3
2	IV	Course/ paper-10	A	F010401T	Supply Chain Management	3
			B		Research Methodology	3
	IV	Course/ paper-11	A	F010402T	Specialised Accounting	3
			B		Consumer Behaviour	3
	IV	Course/ paper-12	A	F010403T	Investment Analysis & Portfolio Management	3
			B		Company Law	3
3	V	Course/ paper-13	A	F010501T	Income Tax	3
			B		Marketing Communication	3
	V	Course/ paper-14	A	F010502T	Entrepreneurship and small business management	3
			B		Sales management	3
	V	Course/ paper-15	A	F010503T	Industrial Relations & Labour Laws	3
			B		Company Accounts	3
3	VI	Course/ paper-16	A	F010601T	Project Management	3

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
			B		Goods & Service Tax	3
	VI	Course/ paper-17	A	F010602T	Auditing	3
			B		International Trade	3
	VI	Course/ paper-18	A	F010603T	Strategic Management	3
			B		Training and Development	3

Page 1 Content

C.S.J.M.U, KANPUR

COURSE: BBA (Syllabus as amended Under National Education Policy 2020)

Year

r

Sem.

Subject

Part

Paper Code

Paper Name

Credit

1

I

Course/

paper-1

A

F010101T

Business Economics

3

B

Basic Accounting

3

I

Course/

paper-2

A

F010102T

Business Statistics

3

B

Principles of Management

3

I

Course/

paper-3

A

F010103T

Business Ethics and Governance

3

B

Computer Applications

3

1

II

Course/

paper-4

A

F010201T

Organisational Behavior

3

B

Business Finance

3

II

Course/

paper-5

A

F010202T

Human Resource Development

3

B

Marketing Theory and Practices

3

II

Course/

paper-6

A

F010203T

Business Mathematics

3

B

Advertising Management

3

2

III

Course/

paper-7

A

F010301T

Management & Cost Accounting

3

B

Business Law

3

III

Course/

paper-8

A

F010302T

Production Management

3

B

Business Policy

3

III

Course/

paper-9

A

F010303T

Business Communication

3

B

Business Environment

3

2

IV

Course/

paper-10

A

F010401T

Supply Chain Management

3

B

Research Methodology

3

IV

Course/

paper-11

A

F010402T

Specialised Accounting

3

B

Consumer Behaviour

3

IV

Course/

paper-12

A

F010403T

Investment Analysis & Portfolio Management

3

B

Company Law

3

3

V

Course/

paper-13

A

F010501T

Income Tax

3

B

Marketing Communication

3

V

Course/

paper-14

A

F010502T

Entrepreneurship and small business management

3

B

Sales management

3

V

Course/

paper-15

A

F010503T

Industria

Page 2 Data Table

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
1	I	Course/ paper-1	A	F010101T	Business Economics	3
			B		Basic Accounting	3
	I	Course/ paper-2	A	F010102T	Business Statistics	3
			B		Principles of Management	3
	I	Course/ paper-3	A	F010103T	Business Ethics and Governance	3
			B		Computer Applications	3

Page 2 Content

BBA First Year (First Semester)

Year

Sem.

Subject

Part

Paper Code

Paper

Name

Credit

1

I

Course/

paper-1

A

F010101T

Business Economics

3

B

Basic Accounting

3

I

Course/

paper-2

A

F010102T

Business Statistics

3

B

Principles of Management

3

I

Course/

paper-3

A

F010103T

Business Ethics and Governance

3

B

Computer Applications

3

Page 3 Data Table

Programme/Class: Degree		Year:First		Semester: First	
Course/ paper-1 (A)					
Course Code: F010101T		Course Title: Business Economics			
Course outcomes: The aim of the course is to build knowledge and understanding business economics among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about business economics. The outcome of the course will be as follows – To provide knowledge about business economics. To provide knowledge about Demand Analysis. To Determine Production and cost analysis. To Make aware with pricing and profit management.					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 2-0-0					
Unit	Topics				No. of Lectures Total=30
I	Introduction to Business Economics: Nature and Scope of Business Economics, its relationship with other subjects. Fundamental Economic Tools-Opportunity cost concept, Incremental concept, Principle of time perspective, discounting principle and Equi-marginal principle.				6

Programme/Class: Degree		Year:First		Semester: First	
II	Demand Analysis: Concept of Demand &types of demand; Law of demand, Demand determinants. Price, Income & Substitution effects, Elasticity of demand: meaning, types, measurement and significance in managerial decisions, Revenue concepts, Concept of demand forecasting and methods of demand forecasting.				8
III	Production and Cost Analysis: Meaning, Production function, Law of variable proportion and laws of return to scale, Various cost concepts and classification, Cost output relationship in short run & long run, Short run and long run Cost curves, Economics and diseconomies of scale.				7
IV	Pricing: Nature of market, Types of markets and their characteristics, Pricing under different market structures–Perfect, Monopoly, Oligopoly and Monopolistic competition, Price discrimination under monopoly competition. Profit Management & Inflation: Profit, Functions of profit, Profit maximization, Break even analysis. Elementary idea of Inflation and National income.				9

Programme/Class: Degree		Year:First		Semester: First	
Suggested Readings: 1. Varsney & Maheshwari, Managerial Economics 2. Mote Paul & Gupta, Managerial Economics: Concepts & cases 3. D.N.Dwivedi, Managerial Economics 4. D.C.Huge, Managerial Economics 5. Peterson & Lewis, Managerial Economics					
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.					
Suggested equivalent online courses:					
Further Suggestions:					

Page 3 Content

Programme/Class: Degree

Year:First

Semester: First

Course/ paper-1 (A)

Course Code: F010101T

Course Title: Business Economics

Course outcomes:

The aim of the course is to build knowledge and understanding business economics among the student.

The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about business economics. The outcome of the course will be as follows –

To provide knowledge about business economics. To provide knowledge about Demand Analysis. To Determine Production and cost analysis.

To Make aware with pricing and profit management.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of

Lectures

Total=30

I

Introduction to Business Economics: Nature and Scope of Business Economics, its relationship with other subjects. Fundamental Economic Tools-Opportunity cost concept, Incremental concept, Principle of time perspective, discounting principle and Equi-marginal principle.

6

II

Demand Analysis: Concept of Demand & types of demand; Law of demand, Demand determinants. Price, Income & Substitution effects, Elasticity of demand: meaning, types, measurement and significance in managerial decisions, Revenue concepts, Concept of demand forecasting and methods of demand forecasting.

8

III

Production and Cost Analysis: Meaning, Production function, Law of variable proportion and laws of re

Page 4 Data Table

Programme/Class: Degree		Year:First		Semester: First	
Course/ paper-1 (B)					
Course Code: F010101T		Course Title: Basic Accounting			

Programme/Class: Degree		Year:First		Semester: First	
Course outcomes: The aim of the course is to build knowledge and understanding principles of accounting among the students. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Accounting. The outcome of the course will be as follows – To Introduce about Accounting Principles and other aspects of accounting. To provide knowledge about rectification of errors. To make able about valuation of stocks. To make aware with share and Debenture.					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 2-0-0					
Unit	Topics				No. of Lectures Total=30
I	Introduction: Meaning and process of accounting, Basic terminology of accounting, Difference between accounting & book keeping. Importance & limitations of accounting, Various users of accounting information, Accounting Principles: Conventions & Concepts.				6

Programme/Class: Degree		Year:First		Semester: First	
II	Accounting equation, Dual aspect of accounting, Types of accounts, Rules of debit & credit, Preparation of Journal and Cash book including banking transactions, Ledger and Trial balance, Subsidiary books of accounts. Rectification of errors, Preparation of bank reconciliation statement, Bills of exchange and promissory notes.				10
III	Valuation of stocks, Accounting treatment of depreciation, Reserves and provisions, Preparation of final accounts along with adjustment entries.				8
IV	Issue of shares and debentures, Issue of bonus shares and right issue, Redemption of preference shares and debentures.				6
Suggested Readings: 1. Agarwal B.D., Advanced Accounting 2. Chawla & Jain, Financial Accounting 3. Chakrawarti K.S., Advanced Accounts. 4. Gupta R.L. & Radhaswamy, Fundamentals of Accounting 5. Jain & Narang, Advanced Accounts					
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.					

Programme/Class: Degree		Year:First		Semester: First	
Suggested equivalent online courses:					
Further Suggestions:					

Page 4 Content

Programme/Class: Degree

Year:First

Semester: First

Course/ paper-1 (B)

Course Code: F010101T

Course Title: Basic Accounting

Course outcomes:

The aim of the course is to build knowledge and understanding principles of accounting among the students. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Accounting. The outcome of the course will be as follows –

To Introduce about Accounting Principles and other aspects of accounting. To provide knowledge about rectification of errors. To make able about valuation of stocks. To make aware with share and Debenture.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of

Lectures

Total=30

I

Introduction: Meaning and process of accounting, Basic terminology of accounting,

Difference between accounting & book keeping. Importance & limitations of accounting, Various users of accounting information, Accounting Principles:

Conventions & Concepts.

6

II

Accounting equation, Dual aspect of accounting, Types of accounts, Rules of debit & credit, Preparation of Journal and Cash book including banking transactions, Ledger and Trial balance, Subsidiary books of accounts.

Rectification of errors, Preparation of bank reconciliation statement, Bills of exchange and promissory notes.

10

III

Valuation of stocks, Accounting treatment of depreciation, Reserves and pr

Page 5 Data Table

Programme/Class: Degree		Year:First		Semester: First	
Course/ paper-2 (A)					
Course Code: F010102T		Course Title: Business Statistics			
Course outcomes: The aim of the course is to build knowledge and understanding of Business Statistics among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Business Statistics. The outcome of the course will be as follows – To provide knowledge about basic concepts of Statistics. To provide knowledge measurement of central tendency. To give an overview of correlation and regression analysis. To make able to know the sampling					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 2-0-0					
Unit	Topics				No. of Lectures Total=30

Programme/Class: Degree		Year:First		Semester: First	
I	Introduction: Concept, features, significance & limitations of statistics, Types of data, Classification & Tabulation, Frequency distribution & graphical representation.				6
II	Measures of Central Tendency (Mean, Median, Mode), Measures of Variation (Range, Quartile Deviation, Mean Deviation and Standard Deviation), Significance & properties of a good measure of variation, Measures of Skewness & Kurtosis.				10
III	Correlation and Regression: Meaning and Types of Correlation, simple correlation, Scatter diagram method, Karl Pearson's Coefficient of correlation, Significance of correlation, Regression concept, Regression concept, Regression equations and Regression coefficient.				8

Programme/Class: Degree		Year:First		Semester: First	
IV	Probability: Concept, Events, Addition Law, Conditional Probability, Multiplication Law & Introduction to Bayes' theorem [Simple numerical]. Introduction to Probability Distribution: Binomial, Poisson and Normal. Sampling: Method of sampling, Sampling and non-sampling errors, Introduction to Test of hypothesis, Type-I and Type-II Errors, Large sample tests. Introduction to MS Excel and its use in Business statistics.				6
Suggested Readings: 1. Agarwal B.D., Advanced Accounting 2. Chawla & Jain, Financial Accounting 3. Chakrawarti K.S., Advanced Accounts. 4. Gupta R.L. & Radhaswamy, Fundamentals of Accounting 5. Jain & Narang, Advanced Accounts					
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.					
Suggested equivalent online courses:					
Further Suggestions:					

Page 5 Content

Programme/Class: Degree

Year:First

Semester: First

Course/ paper-2 (A)

Course Code: F010102T

Course Title: Business Statistics

Course outcomes:

The aim of the course is to build knowledge and understanding of Business Statistics among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Business Statistics. The outcome of the course will be as follows –

To provide knowledge about basic concepts of Statistics. To provide knowledge measurement of central tendency. To give an overview of correlation and regression analysis. To make able to know the sampling and probability.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of

Lectures

Total=30

I

Introduction: Concept, features, significance & limitations of statistics, Types of data, Classification & Tabulation, Frequency distribution & graphical representation.

6

II

Measures of Central Tendency (Mean, Median, Mode), Measures of Variation (Range, Quartile Deviation, Mean Deviation and Standard Deviation), Significance & properties of a good measure of variation, Measures of Skewness & Kurtosis.

10

III

Correlation and Regression: Meaning and Types of Correlation, simple correlation, Scatter diagram method, Karl Pearson's Coefficient of correlation, Significance of correlation, Regression concept, Regression concept, Regression equ

Page 6 Data Table

Programme/Cla s: Degree		Year:First			Semester: First	
Course/ paper-2 (B)						
Course Code: F010102T			Course Title: Principles of Management			
Course outcomes: The aim of the course is to build knowledge and understanding about principles of management among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about management. The outcome of the course will be as follows – To provide knowledge about management and its principles. To provide knowledge about Managerial functions. To make aware with management thinkers and their contributions.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree		Year:First			Semester: First	
I	Introduction: Concepts, objectives, nature, scope and significance of management, Contribution of Taylor, Weber and Fayol in management, Management Vs. administration..					6
II	Planning: Concept, objectives, nature, importance and limitations of planning, planning process Concept of Decision Making and its Importance, forms, techniques and process.					8
III	Organizing: Concept, objectives, nature of organizing, Types of Organization, Delegation of authority, Authority and responsibility, Centralization and Decentralization, Span of Control.					6
IV	Directing: Concept, principles & aspects of directing, Concept and types of Coordination, Concept of leadership, Supervision, Motivation and Communication. Controlling: Concept, Principles, Process and Techniques of Controlling, Relationship between planning and controlling					10

Programme/Class s: Degree		Year:First			Semester: First	
Suggested Readings: 1. Pagare Dinkar, Principles of Management 2. Prasad L.M., Principles and Practice of Management 3. Satya Narayan and Raw VSP, Principles and Practice of Management 4. Srivastava and Chunawalla, Management Principles and Practice						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 6 Content

Programme/Class: Degree

Year:First

Semester: First

Course/ paper-2 (B)

Course Code: F010102T

Course Title: Principles of Management

Course outcomes:

The aim of the course is to build knowledge and understanding about principles of management among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about management. The outcome of the course will be as follows – To provide knowledge about management and its principles.

To provide knowledge about Managerial functions. To make aware with management thinkers and their contributions.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Introduction: Concepts, objectives, nature, scope and significance of management,

Contribution

of

Taylor,

Weber

and

Fayol

in

management, Management Vs. administration..

6

II

Planning: Concept, objectives, nature, importance and limitations of planning, planning process Concept of Decision Making and its Importance, forms, techniques and process.

8

III

Organizing: Concept, objectives, nature of organizing, Types of Organization, Delegation of authority, Authority and responsibility,

Centralization and Decentralization, Span of Control.

6

IV

Directing: Concept, principles & aspects of directing, Concept and types of

Coordination, Concept of leadership, Supervision, Motivation a

Page 7 Data Table

Programme/Cla s: Degree		Year:First			Semester: First	
Course/ paper-3 (A)						
Course Code: F010103T			Course Title: Business Ethics and Governance			
Course outcomes: The aim of the course is to build knowledge and understanding Business Ethics among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Business Ethics. The outcome of the course will be as follows – To develop understanding of business ethics and values. To provide relationship between ethics and corporate excellence. To give an overview about Gandhian philosophy and social responsibility.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree		Year:First			Semester: First	
I	Introduction: Concept and nature of ethics; ethics, values and behavior; development of ethics, relevance of ethics and values in business, Arguments against business ethics.					6
II	Work life in Indian Philosophy: Indian ethos for work life, Indian values for the work place, Work-life balance, Ethos of Vedanta in management, Hierarchism as an organizational value.					8
III	Relationship between Ethics & Corporate Excellence, Corporate Mission Statement, Code of Ethics, Organizational Culture, TQM. Gandhian Philosophy of Wealth Management, Philosophy of Trusteeship, Gandhi's Seven Greatest Social Sins, Concept of knowledge management and wisdom management.					8
IV	Corporate Social Responsibility-So cial Responsibility of business with respect to different stakeholders, Arguments for and against Social responsibility of business, Social Audit.					8

Programme/Class: Degree		Year:First			Semester: First	
Suggested Readings: 1. Kaur Tripat, Values & Ethics in Management, Galgotia Publishers. 2. Chakraborty S.K., Human values for Managers 3. McCarthy, F.J., Basic Marketing 4. Chakraborty S.K., Ethics in Management: A Vedantic Perspective, Oxford University Press.						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 7 Content

Programme/Class: Degree

Year:First

Semester: First

Course/ paper-3 (A)

Course Code: F010103T

Course Title: Business Ethics and Governance

Course outcomes:

The aim of the course is to build knowledge and understanding Business Ethics among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Business Ethics. The outcome of the course will be as follows –

To develop understanding of business ethics and values.

To provide relationship between ethics and corporate excellence.

To give an overview about Gandhian philosophy and social responsibility.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Introduction: Concept and nature of ethics; ethics, values and behavior; development of ethics, relevance of ethics and values in business, Arguments against business ethics.

6

II

Work life in Indian Philosophy: Indian ethos for work life, Indian values for the work place, Work-life balance, Ethos of Vedanta in management, Hierarchism as an organizational value.

8

III

Relationship between Ethics & Corporate Excellence, Corporate Mission Statement, Code of Ethics, Organizational Culture, TQM. Gandhian Philosophy of Wealth Management, Philosophy of Trusteeship, Gandhi's Seven Greatest Social Sins, Concept of knowledge management and wisdom management.

8

IV

Corporate

Page 8 Data Table

Programme/Cla s: Degree			Year:First		Semester: First	
Course/ paper-3 (B)						
Course Code: F010103T		Course Title: Computer Applications				
Course outcomes: The aim of the course is to build knowledge, understanding Computer Applications among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Computer Applications. The outcome of the course will be as follows – To provide knowledge about computer and its application. To provide knowledge about components and working on computer. To give an overview about software system and Data base management.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year:First		Semester: First	
I	Computer: An Introduction, Computers in Business. Elements of Computer system, Indian computing Environment, Management of data processing systems in Business organizations, Programmes development cycle, flow charting, Input Output analysis Programming Concept, Software Development process.					8
II	Components of a computer system, Generation of computer and computer languages, personal computers in Business, PC-software Packages, An Introduction to Disk. Operating system and windows, GUI, Other system software's.					7
III	Text Processing, software, Introduction to spreadsheet software, creation of spreadsheet application, Range, formulas, function data base functions in spreadsheet, Graphics on spreadsheet, modes of data processing, Report generation, Presentation graphics, Creating a presentation.					7

Programme/Cla s: Degree			Year:First		Semester: First	
IV	Computer software system, software development process, files design & Report design, Data files types, Master & Transaction file. Data Hierarchy & data file structure, Use of files in Programming. Relevance of Data base management system, data base manager, data communication, networking, LAN & WAN, Real Time Sharing, On line & off line processing.					8
Suggested Readings: 1. P. K. Sinha & P. Sinha, Computer Fundamentals, BPB Publication 2. V. Rajaraman, Computer Fundamentals, PHI 3. Tannenbaum, Computer Applications and Networks 4. 'O' Brien, Management Information Systems						

Programme/Class s: Degree			Year:First		Semester: First	
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 8 Content

Programme/Class: Degree

Year:First

Semester: First

Course/ paper-3 (B)

Course Code: F010103T

Course Title: Computer Applications

Course outcomes:

The aim of the course is to build knowledge, understanding Computer Applications among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Computer Applications. The outcome of the course will be as follows –

To provide knowledge about computer and its application.

To provide knowledge about components and working on computer.

To give an overview about software system and Data base management.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of

Lectures

Total=30

I

Computer: An Introduction, Computers in Business. Elements of Computer system, Indian computing Environment, Management of data processing systems in Business organizations, Programmes development cycle, flow charting, Input Output analysis Programming Concept, Software Development process.

8

II

Components of a computer system, Generation of computer and computer languages, personal computers in Business, PC-software Packages, An Introduction to Disk. Operating system and windows, GUI, Other system software's.

7

III

Text Processing, software, Introduction to spreadsheet software, creation of spreadsheet application, Range, formulas, function data base functions

Page 9 Data Table

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
1	II	Course/ paper-4	A	F010201T	Organizational Behavior	3
			B		Business Finance	3
	II	Course/ paper-5	A	F010202T	Human Resource Development	3
			B		Marketing Theory and Practices	3
	II	Course/ paper-6	A	F010203T	Business Mathematics	3
			B		Advertising Management	3

Page 9 Data Table

Programme/Class: Degree		Year:First		Semester: Second	
Course/ paper-4 (A)					

Programme/Class: Degree		Year:First		Semester: Second	
Course Code: F010201T		Course Title: Organizational Behavior			
Course outcomes: The aim of the course is to build knowledge and understanding of Organizational Behavior among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Organizational Behavior. The outcome of the course will be as follows – To provide knowledge about Organizational Behavior. To provide knowledge about individual and group behavior. To give overview about change in organization and QWL.					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 2-0-0					
Unit	Topics				No. of Lectures Total=30
I	Introduction: Nature and scope of OB, Challenges and opportunities for OB, Organization Goals, Models of OB, Impact of Global and Cultural diversity on OB.				7

Programme/Class: Degree		Year:First		Semester: Second	
II	Individual Behavior: concept, Personality, Perception and its role in individual decision making, Learning, Motivation, Hierarchy of needs theory, Theory X and Y, Motivation-Hygiene theory, Vroom's expectancy theory.				8
III	Behavior Dynamics: Interpersonal behavior, Communication, Transaction Analysis, The Johari Window, Leadership, Its Theories and prevailing leadership styles in Indian Organizations. Group Behavior: Definition and classification of Groups, Types of Group Structures, Group decision making, Teams Vs Groups, Contemporary issues in managing teams, Inter-group problems in organizational group dynamics, Management of conflict.				8
IV	Management of Change: Change and Organizational development, Resistance to change, Approaches to managing organizational change, Organizational effectiveness, Organizational culture, Power and Politics in Organization, Quality of work life, Recent advances in OB.				7

Programme/Class: Degree		Year:First		Semester: Second	
Suggested Readings: 1 Bennis, W.G., Organization Development 2. Breech Islwar, Organizational-The Framework of Management 3. Dayal, Keith, organizational Development 4. Sharma, R.A., organizational Theory and Behavior 5. Prasad, L.M., organizational Behavior					
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.					
Suggested equivalent online courses:					
Further Suggestions:					

Page 9 Content

BBA First Year (Second Semester)

Year

Sem.

Subject

Part

Paper Code

Paper Name

Credit

1

II

Course/

paper-4

A

F010201T

Organizational Behavior

3

B

Business Finance

3

II

Course/

paper-5

A

F010202T

Human Resource Development

3

B

Marketing Theory and Practices

3

II

Course/

paper-6

A

F010203T

Business Mathematics

3

B

Advertising Management

3

Programme/Class: Degree

Year:First

Semester: Second

Course/ paper-4 (A)

Course Code: F010201T

Course Title: Organizational Behavior

Course outcomes:

The aim of the course is to build knowledge and understanding of Organizational Behavior among the

student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Organizational Behavior. The outcome of the course will be as follows –

To provide knowledge about Organizational Behavior.

To provide knowledge about individual and group behavior.

To given overview about change in organization and QWL.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Introduction: Nature and scope of OB, Challenges and opportunities

for OB, Organization Goals, Models of OB, Impact of Global and Cultural diversity on OB.

7

II

Individual Behavior: concept, Personality, Perception and its role in individual decision making,

Learning, Motivation, Hierarchy of needs theory, Theor

Page 10 Data Table

Programme/Cla s: Degree			Year:First		Semester: Second	
Course/ paper-4 (B)						
Course Code: F010201T		Course Title: Business Finance				

Programme/Cla s: Degree			Year:First		Semester: Second	
Course outcomes: The aim of the course is to build knowledge and understanding of Business Finance among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Business Finance. The outcome of the course will be as follows – To provide knowledge about business finance and investment decisions. To provide knowledge about financing and dividend decision. To give an overview about working capital.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year:First		Semester: Second	
I	Introduction to Finance: Finance functions, objectives of financial management- Profitability vs. Shareholder wealth maximization. Time Value of Money – Compound interest and terminal values, Present values, relationship between present value and Discount rate, present value and future value of an annuity, amortizing a loan Valuation of bonds. Calculating return from stocks investment, perpetual dividend growth model					10
II	Nature of Capital Budgeting decisions, Payback, NPV, IRR and ARR methods and their practical applications. Financing Decision: Making capital structure decisions- concept of leverage, operating, financial and combined leverage, EBIT-EPS Analysis, cash flow ability to service debt, Pecking order of financing					7

Programme/Cla s: Degree			Year:First		Semester: Second	
III	Procedural and regulatory aspect of paying dividend Concepts of Relevance and Irrelevance of dividend Walter's Model, Gordon's model. Passive and residual Dividend policy Factors affecting dividend policy, alternative forms of dividend-bonus, stock split , share-buy back					7
IV	Concepts of working capital, operating cycle concept, estimating working capital requirements, Approaches to the financing of current Assets, sources of short-term financing, cost of alternative sources of financing working capital Contents of an Annual Report and Financial Ratio analysis.					6
Suggested Readings: 1. Maheshwari S.N., Financial Management 2. Khan and Jain, Financial Management 3. Singh H.K., Business Finance						

Programme/Class s: Degree			Year:First		Semester: Second	
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						

Page 10 Content

Programme/Class: Degree

Year:First

Semester: Second

Course/ paper-4 (B)

Course Code: F010201T

Course Title: Business Finance

Course outcomes:

The aim of the course is to build knowledge and understanding of Business Finance among the student.

The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about

Business Finance. The outcome of the course will be as follows –

To provide knowledge about business finance and investment decisions.

To provide knowledge about financing and dividend decision.

To give an overview about working capital.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Introduction to Finance: Finance functions, objectives of financial management- Profitability vs. Shareholder wealth maximization.

Time Value of Money – Compound interest and terminal values, Present values, relationship between present value and Discount rate, present value and future value of an annuity, amortizing a loan

Valuation of bonds. Calculating return from stocks investment, perpetual dividend growth model

10

II

Nature of Capital Budgeting decisions, Payback, NPV, IRR and ARR methods and their practical applications.

Financing Decision: Making capital structure decisions- concept of leverage, operating, financial and combined leverage, EBIT-EPS Analysis, cash flow ability to service de

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Further Suggestions:

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Page 12 Data Table

Programme/Class: Degree		Year:First		Semester: Second	
Course/ paper-5 (A)					
Course Code: F010202T		Course Title: Human Resource Development			

Programme/Class: Degree		Year:First		Semester: Second	
Course outcomes: The aim of the course is to build knowledge and understanding of Human Resource Development among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Human Resource Development. The outcome of the course will be as follows – To provide knowledge about HRD concepts and other aspects. To provide knowledge about potential appraisal. To give an overview about Job Enrichment and Quality circles. To make aware with hum					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 2-0-0					
Unit	Topics				No. of Lectures Total=30
I	HRD: Concept, importance, benefits and its distinction from HRM, focus of HRD System, Structure of HRD System, Role of HRD manpower. Management Development: Concept, need, management development methods.				7

Programme/Class: Degree		Year:First		Semester: Second	
II	Potential Appraisal: Concept, need, objectives, methods and Obstacles. Training: Meaning, role, assessing needs for training, organizing training programmes, training methods, evaluation of Training.				7
III	Job Enrichment: Concept, Principles, steps for job enrichment, hurdles in job enrichment, making job enrichment effective, job and work redesign. Quality Circles: Concept, structure, training in quality circle, problem solving techniques, role of management, trade union and workers, quality circles in India.				10
IV	HRA: Introduction, scope, limitations, methods. Management of careers. Stress Management: Definition, potential, sources of stress, consequences of stress, managing stress.				6
Suggested Readings: 1. Dipak Kumar Bhattacharya, Human Resource Management 2. Arun Monappa, Managing Human Resource 3. P.Subba Rao, Essential of HRM and Industrial Relations 4. C.B. Memoria, Personnel Management					

Programme/Class: Degree		Year:First		Semester: Second	
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.					
Suggested equivalent online courses:					
Further Suggestions:					

Page 12 Content

Programme/Class: Degree

Year:First

Semester: Second

Course/ paper-5 (A)

Course Code: F010202T

Course Title: Human Resource Development

Course outcomes:

The aim of the course is to build knowledge and understanding of Human Resource Development among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Human Resource Development. The outcome of the course will be as follows –

To provide knowledge about HRD concepts and other aspects.

To provide knowledge about potential appraisal.

To give an overview about Job Enrichment and Quality circles.

To make aware with human resource accounting.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

HRD: Concept, importance, benefits and its distinction from HRM, focus of HRD System, Structure of HRD System, Role of HRD manpower. Management Development: Concept, need, management development methods.

7

II

Potential Appraisal: Concept, need, objectives, methods and Obstacles. Training: Meaning, role, assessing needs for training, organizing training programmes, training methods, evaluation of Training.

7

III

Job Enrichment: Concept, Principles, steps for job enrichment, hurdles in job enrichment, making job enrichment effective, job and work redesign. Quality Circles: Concept, structure, training in quality circle, prob

Page 13 Data Table

Programme/Class: Degree		Year: First		Semester: Second	
Course/ paper-5 (B)					
Course Code: F010202T		Course Title: Marketing Theory and Practices			

Programme/Class: Degree		Year: First		Semester: Second	
Course outcomes: The aim of the course is to build knowledge and understanding of Marketing management among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Marketing Theory and Practices. The outcome of the course will be as follows – To provide knowledge about Marketing Theory and Practices. To provide knowledge about market segmentation and marketing mix. To give an overview about marketing research.					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 2-0-0					
Unit	Topics				No. of Lectures Total=30
I	Introduction to Marketing: Definition, nature, scope & importance of Marketing Management, Core concepts of marketing. Orientation of marketers towards the market place: production concept, product concept, selling concept, modern marketing concept & societal concept.				7

Programme/Class: Degree		Year: First		Semester: Second	
II	Consumer Behaviour: Concept, Importance and factors influencing consumer behaviour. Marketing Information System: Meaning, Importance and Scope. Marketing Research: Importance, Scope & Process.				7
III	Market segmentation: Concept, basis & importance of segmentation in marketing; Market Targeting: Concept, Types & Importance; Positioning: Concept, Importance, Brand positioning & Repositioning.				7
IV	Marketing Mix: Product – Product Mix, New Product development, types of product, Product life cycle, Branding and packaging. Distribution – Concept, importance & different types of distribution channels. Pricing – Setting the price, adapting the price & price discrimination. Promotion – Importance and nature of promotion mix tools.				9
Suggested Readings: 1. Philip Kotler, Marketing Mgt. (PHI) 2. Etzet, Walker, Stanton, Marketing 3. Rajan Saxena, Marketing Management					

Programme/Class: Degree		Year: First		Semester: Second	
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.					
Suggested equivalent online courses:					
Further Suggestions:					

Page 13 Content

Programme/Class: Degree

Year: First

Semester: Second

Course/ paper-5 (B)

Course Code: F010202T

Course Title: Marketing Theory and Practices

Course outcomes:

The aim of the course is to build knowledge and understanding of Marketing management among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Marketing Theory and Practices. The outcome of the course will be as follows –

To provide knowledge about Marketing Theory and Practices.

To provide knowledge about market segmentation and marketing mix.

To give an overview about marketing research.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Introduction to Marketing: Definition, nature, scope & importance of Marketing Management, Core concepts of marketing. Orientation of marketers towards the market place: production concept, product concept, selling concept, modern marketing concept & societal concept.

7

II

Consumer Behaviour: Concept, Importance and factors influencing consumer behaviour. Marketing Information System: Meaning, Importance and Scope. Marketing Research: Importance, Scope & Process.

7

III

Market segmentation: Concept, basis & importance of segmentation in marketing; Market Targeting: Concept, Types & Importance; Positioning: Concept, Importance, Brand positioning & Repositioning.

Page 14 Data Table

Programme/Cla s: Degree			Year:First		Semester: Second	
Course/ paper-6 (A)						
Course Code: F010203T		Course Title: Business Mathematics				

Programme/Cla s: Degree			Year:First		Semester: Second	
Course outcomes: The aim of the course is to build knowledge and understanding of Business Mathematics among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Business Mathematics. The outcome of the course will be as follows – To provide knowledge about Mathematics and its use in business. To make able about mathematical calculations. To learn about the use of set theory and calculus in business.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year:First		Semester: Second	
I	Matrix: Introduction, Square Matrix, Row Matrix, Column Matrix Diagona Matrix, Identity Matrix, Addition, Subtraction & Multiplication of Matrix, Use of Matrix in Business, Mathematical Induction. Inverse of Matrix, Rank of Matrix, Solution to a system of equation by the ad-joint matrix methods & Guassian Elimination Method.					19
II	Percentage, Ratio and Proportion, Average, Mathematical Series- Arithmetic, Geometric & Harmonic, Simple Interest & Compound Interest					8
III	Set theory: Notation of Sets, Singleton Set, Finite Set, Infinite Set, Equal Set Null Set, Subset, Proper Subset, Universal Set, Union of Sets, Inter- section of Sets, Use of set theory in business, Permutation & Combination.					7

Programme/Cla s: Degree			Year:First		Semester: Second	
IV	Concept of Differentiation and Integration, Maxima and Minima in Differentiation, Application of Differentiation & Integration in Business (No proof of theorems, etc.)					6
Suggested Readings: 1. Mehta & Madnani, Mathematics for Economics 2. Mongia, Mathematics for Economics 3. Zamiruddin, Business Mathematics 4. Raghavachari, Mathematics for Management						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 14 Content

Programme/Class: Degree

Year:First

Semester: Second

Course/ paper-6 (A)

Course Code: F010203T

Course Title:

Business Mathematics

Course outcomes:

The aim of the course is to build knowledge and understanding of Business Mathematics among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Business Mathematics. The outcome of the course will be as follows –

To provide knowledge about Mathematics and its use in business.

To make able about mathematical calculations.

To learn about the use of set theory and calculus in business.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Matrix: Introduction, Square Matrix, Row Matrix, Column Matrix Diagonal

Matrix, Identity Matrix, Addition, Subtraction & Multiplication of

Matrix, Use of Matrix in Business, Mathematical Induction.

Inverse of Matrix, Rank of Matrix, Solution to a system of equation
by the ad-joint matrix methods & Guassian Elimination Method.

9

II

Percentage, Ratio and Proportion, Average, Mathematical Series-
Arithmetic, Geometric & Harmonic, Simple Interest & Compound

Interest

8

III

Set theory: Notation of Sets, Singleton Set, Finite Set, Infinite Set, Equal Set Null Set, Subset, Proper Subset, Universal Set, Union of Sets, Intersection of Sets, Use of set theory in business, Permutation & C

Page 15 Data Table

Programme/Cla s: Degree			Year: First		Semester: Second	
Course/ paper-6 (B)						
Course Code: F010203T		Course Title: Advertising Management				
Course outcomes: The aim of the course is to build knowledge and understanding of advertisement among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about advertising Management. The outcome of the course will be as follows – To provide knowledge about advertisement and its use in business. To make able about advertisement concept and its management. To learn about the use of advertisement in business.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year: First		Semester: Second	
I	Advertising: Introduction, Scope, importance in business : Role of advertising in social and economic development of India: Ethics and truths in Indian Advertising.					9
II	Integrated Communication Mix (IMC)-meaning, importance; Communication meaning, importance, process, communication mix- components, role in marketing, Branding-meanin g, importance in advertising.					8
III	Promotional objectives – importance determination of promotional objectives, setting objective DAGMAR; Advertising Budget importance, establishing the budget- approaches allocation of budget.					7

Programme/Cla s: Degree			Year: First		Semester: Second	
IV	Advertising Copy-meaning components types of advertising copy, importance of creativity in advertising; Media planning-i mportance, strategies, media mix. Advertising research – importance, testing advertising effectiveness market testing for ads; International Advertising- importance, international Vs local advertising.					6
Suggested Readings: ■ Advertising and Promotion George E. Beich & Michael A. Belch. T.M.H. ■ Advertising Management, Concept and Cases Manendra Mohan,TMH ■ Advertising Management Rajeev Batra, PHI						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						

Programme/Class s: Degree			Year: First		Semester: Second	
Further Suggestions:						

Page 15 Content

Programme/Class: Degree

Year: First

Semester: Second

Course/ paper-6 (B)

Course Code: F010203T

Course Title:

Advertising Management

Course outcomes:

The aim of the course is to build knowledge and understanding of advertisement among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about advertising Management. The outcome of the course will be as follows –

To provide knowledge about advertisement and its use in business.

To make able about advertisement concept and its management.

To learn about the use of advertisement in business.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Advertising: Introduction, Scope, importance in business : Role of advertising in social and economic development of India: Ethics and truths in Indian Advertising.

9

II

Integrated Communication Mix (IMC)-meaning, importance;

Communication meaning, importance, process, communication mix-components, role in marketing, Branding-meaning, importance in advertising.

8

III

Promotional objectives – importance determination of promotional objectives, setting objective DAGMAR; Advertising Budget importance, establishing the budget- approaches allocation of budget.

7

IV

Advertising Copy-meaning components types of advertising copy, importance of creativity in advertising; Media pl

Page 16 Data Table

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
2	III	Course/ paper-7	A	F010301T	Management & Cost Accounting	3
			B		Business Law	3
	III	Course/ paper-8	A	F010302T	Production Management	3
			B		Business Policy	3
	III	Course/ paper-9	A	F010303T	Business Communication	3
			B		Business Environment	3

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BBA Second Year (Third Semester)

Year

Sem.

Subject

Part

Paper Code

Paper Name

Credit

2

III

Course/

paper-7

A

F010301T

Management & Cost Accounting

3

B

Business Law

3

III

Course/

paper-8

A

F010302T

Production Management

3

B

Business Policy

3

III

Course/

paper-9

A

F010303T

Business Communication

3

B

Business Environment

3

Page 17 Data Table

Programme/Cla ss: Degree			Year: Second		Semester: Third	
Course/ paper-7 (A)						
Course Code: F010301T		Course Title: Management & Cost Accounting				

Programme/Cla s: Degree			Year: Second		Semester: Third	
Course outcomes: The objective of this paper is to give the basic knowledge about the Management and cost accounting						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 3-3-0						
Unit	Topics					No. of Lectures Total=30
I	Management Accounting- purpose, nature and focus on decision making. Income statement and Balance Sheet of a manufacturer, Classification of costs by behavior, traceability , controllability, relevance and function. Cost concepts for service companies. Manufacturing Management principles- TQM, JIT, Theory of constraints and continuous improvement.					8

Programme/Cla s: Degree			Year: Second		Semester: Third	
II	Manufacturing and Job order Cost Accounting: Flow of manufacturing activities. Cost Accounting system using perpetual inventory system Job Order Cost accounting , Job order cost sheet , Adjusting over applied and under applied overhead, multiple overhead application rates Process cost accounting: Organisation of process operations , Equivalent Units of production, Process cost summary,					7
III	Cost- Volume – Profit Analysis: Identifying cost behavior- fixed, variable, mixed costs, step-wise Costs and their determination. Break-Even Analysis, Sensitivity analysis, computing multiproduct B.E.P					8

Programme/Cla s: Degree			Year: Second		Semester: Third	
IV	Budgeting process imperatives for analysis, focus , evaluation, employee motivation and communication. Master budget components –operating budgets , capital expenditure budget and financial budgets , flexible budgets and standard costs, computing labour, material and overhead variances Managerial decisions in addition or deletion of produc t./department , make or buy, sell or process, selecting sales mix using relevant costs.					7
Suggested Readings: 1. Maheshwari S.N., Advanced Problem and Solutions in Cost Accounting 2. Khan & Jain, Management Accounting 3. Gupta, S.P., Management Accounting						
Suggested Continuous Evaluation Methods:						
Suggested equivalent online courses:						

Programme/Class s: Degree			Year: Second		Semester: Third	
Further Suggestions:						
.....						
.....						
.....						
.....						

Page 17 Content

Programme/Class: Degree

Year: Second

Semester: Third

Course/ paper-7 (A)

Course Code: F010301T

Course Title: Management & Cost Accounting

Course outcomes:

The objective of this paper is to give the basic knowledge about the Management and cost accounting

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-3-0

Unit

Topics

No. of

Lectures

Total=30

I

Management Accounting- purpose, nature and focus on decision making. Income statement and Balance Sheet of a manufacturer, Classification of costs by behavior, traceability , controllability, relevance and function. Cost concepts for service companies.

Manufacturing Management principles- TQM, JIT, Theory of constraints and continuous improvement.

8

II

Manufacturing and Job order Cost Accounting: Flow of manufacturing activities.

Cost Accounting system using perpetual inventory system

Job Order Cost accounting , Job order cost sheet , Adjusting over applied and

under applied overhead, multiple overhead application rates

Process cost accounting: Organisation of process operations , Equivalent Units of production, Process cost summary,

7

III

Cost- Volume – Profit Analysis: Identifying cost behavior- fixed, variable, mixed costs, step-wise Costs and their determination. Break-Even Analysis, Sensitivity analysis, computing multiproduct B.E.P

8

IV

Budgeting process imperatives for analysis, focus , evaluation, employee

Page 18 Data Table

Programme/Cla s: Degree			Year: Second		Semester: Third	
Course/ paper-7 (B)						
Course Code: F010301T		Course Title: Business Law				
Course outcomes: The objective of this paper is to give the basic knowledge about the rules and regulation of execution of Business.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 3-2-1						
Unit	Topics					No. of Lectures Total=30
I	The Indian Contract Act 1872: Scope of the Act, Essential of A Valid Contract, Agreement, Performance of Contracts, Breach of Contract & Remedies, Quasi-Contracts					8

Programme/Cla s: Degree			Year: Second		Semester: Third	
II	The Sale of Good Act, 1930: Formation of Contract, Conditions & Warranties, Rights of an Unpaid Seller, Performance of the Contract of Sale					7
III	The Negotiable Instruments Act, 1881: Nature and Types of negotiable instruments, Negotiation and Assignment, Holder-in-Due Course, Dishonor and Discharge of Negotiable Instrument; Arbitration					8
IV	The Companies Act, 1956: Nature and Type of Companies, Formation of Companies, Memorandum and Articles of Association, Prospectus, Share capital, Membership, Meetings and Winding-Up					7
Suggested Readings: 1. Avatar Singh, Company Law 2. Khergamwalla, JS, The Negotiable Instrument Act 3. Ramaya A, A Guide to Companies Act 4. Tuteja SK, Business Law for Managers						
Suggested Continuous Evaluation Methods:						

Programme/Class s: Degree			Year: Second		Semester: Third	
Suggested equivalent online courses:						
Further Suggestions:						

Page 18 Content

Programme/Class: Degree

Year: Second

Semester: Third

Course/ paper-7 (B)

Course Code: F010301T

Course Title: Business Law

Course outcomes:

The objective of this paper is to give the basic knowledge about the rules and regulation of execution of Business.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-2-1

Unit

Topics

No. of

Lectures

Total=30

I

The Indian Contract Act 1872: Scope of the Act, Essential of A Valid Contract, Agreement, Performance of Contracts, Breach of Contract & Remedies, Quasi-Contracts

8

II

The Sale of Good Act, 1930: Formation of Contract, Conditions &

Warranties, Rights of an Unpaid Seller, Performance of the Contract of Sale

7

III

The Negotiable Instruments Act, 1881: Nature and Types of negotiable instruments, Negotiation and Assignment, Holder-in-Due Course, Dishonor and Discharge of Negotiable Instrument; Arbitration

8

IV

The Companies Act, 1956: Nature and Type of Companies, Formation of Companies, Memorandum and Articles of Association, Prospectus, Share capital, Membership, Meetings and Winding-Up

7

Suggested Readings:

1. Avatar Singh, Company Law
2. Khargamwalla, JS, The Negotiable Instrument Act
3. Ramaya A, A Guide to Companies Act
4. Tuteja SK, Business Law for Managers

Suggested Continuous Evaluation Methods:

Page 19 Data Table

Programme/Cla s: Degree			Year: Second		Semester: Third	
Course/ paper-8 (A)						
Course Code: F010302T		Course Title: Production Management				
Course outcomes: The objective of this paper is to give the basic knowledge about the Production Management in industry.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 3-2-1						

Programme/Cla s: Degree			Year: Second		Semester: Third	
Unit	Topics					No. of Lectures Total=30
I	Introduction to Production Management; History of Production Management; Definitions of Production Management; Production Process; Production: The Heart of an Organization; Objectives of Production Management; Scope of Production Management; Importance of Technology in Production					8
II	Concept of Forecasting; Purpose of Sales Forecasting, Basic Elements of Forecasting, Importance of Forecasting, Objectives of Forecasting, Classification of Forecasting ; Qualitative and Quantitative Techniques of Forecasting					7
III	Product Selection; Definitions of Product Design and Development; Need for Product Design and Development, Origin of the Product Idea and Selection from Various Alternatives, Choosing among Alternative Products, Modifying the Existing Products, Sources of Product					8

Programme/Class s: Degree			Year: Second		Semester: Third	
IV	Nature of Production Planning and Control (PPC): Types of Plans, Elements of Production Planning, Strategy of Production Planning, Aggregate Planning; Main Functions of Production Planning and Control (PPC)					7
Suggested Readings: 1. Production Management by Telsang Martand S Chand Publication						
Suggested Continuous Evaluation Methods:						
Suggested equivalent online courses:						
Further Suggestions:						

Page 19 Content

Programme/Class: Degree

Year: Second

Semester: Third

Course/ paper-8 (A)

Course Code: F010302T

Course Title: Production Management

Course outcomes:

The objective of this paper is to give the basic knowledge about the Production Management in industry.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-2-1

Unit

Topics

No. of

Lectures

Total=30

I

Introduction to Production Management: History of Production Management; Definitions of Production Management; Production Process; Production: The Heart of an Organization; Objectives of Production Management; Scope of Production Management; Importance of Technology in Production

8

II

Concept of Forecasting: Purpose of Sales Forecasting, Basic Elements of Forecasting, Importance of Forecasting, Objectives of Forecasting, Classification of Forecasting ; Qualitative and Quantitative Techniques of Forecasting

7

III

Product Selection; Definitions of Product Design and Development: Need for Product Design and Development, Origin of the Product Idea and Selection from Various Alternatives, Choosing among Alternative Products, Modifying the Existing Products, Sources of Product

8

IV

Nature of Production Planning and Control (PPC): Types of Plans, Elements

of Production Planning, Strategy of Production Planning, Aggregate Planning; Main Functions of Production Planning and Control (PPC)

7

Suggested Readings:

1.Product

Page 20 Data Table

Programme/Class: Degree		Year: Second		Semester: Third	
Course/ paper-8 (B)					
Course Code: F010302T		Course Title: Business Policy			
Course outcomes: 1. To understand the nature of top management functions and their imperatives. 2. To deduce strategy 3. To perform SWOT analysis 4. Recommend strategy 5. Perform analysis of corporate business portfolio and recommend changes.					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 3-2-1					
Unit	Topics				No. of Lectures Total=30

Programme/Class: Degree		Year: Second		Semester: Third	
I	Business Policy as a field of study- objectives in knowledge, skills and attitudes. Top management functions, roles and responsibilities. The concept of strategy- meaning and importance. Deducing strategy and articulating a summary statement of strategy. Strategic Intent: Vision, mission, business definition. Strategic and financial objectives. Components of strategic formulation process.				8
II	Analysis of External environment – PESTLE Industry analysis using Porter's model, Analysis of competitive environment, ETOP Analysis of Internal capability: Functional, PLC and Value Chain approach to Internal analysis, Mckinsey's 7S framework, Resource audit SWOT analysis Relevance of social responsibility and power politics in strategy formulation				7

Programme/Class: Degree		Year: Second		Semester: Third	
III	Business Strategies and Corporate strategies. External Vs. Internal growth options, their conditions and prerequisites. Five Generic Competitive strategies business strategies Growth strategies- concentration, concentric, horizontal, vertical, Retrenchment- turnaround, divestment and liquidation strategies Modes of entering foreign markets, Multi-domestic and global strategy				8
IV	Strategic Choice: Concept of Portfolio balance, Display matrices – BCG, Directional policy, GE etc. Factors contributing to business Strength as well as industry attractiveness.				7
Suggested Readings: 1. Gluek & Jauch, Corporate Strategy 2. Hatton & Hatton, Strategic Management 3. Christian, Anderson, Bower Business Policy 4. McCarthy, IninChiello, Curran Business Policy & Strategy 5. Azhar Kazmi, Business Policy					
Suggested Continuous Evaluation Methods:					
Suggested equivalent online courses:					

Page 20 Content

Programme/Class: Degree

Year: Second

Semester: Third

Course/ paper-8 (B)

Course Code: F010302T

Course Title: Business Policy

Course outcomes:

1. To understand the nature of top management functions and their imperatives.
2. To deduce strategy
3. To perform SWOT analysis
4. Recommend strategy
5. Perform analysis of corporate business portfolio and recommend changes.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-2-1

Unit

Topics

No. of

Lectures

Total=30

I

Business Policy as a field of study- objectives in knowledge, skills and attitudes. Top management functions, roles and responsibilities.

The concept of strategy- meaning and importance. Deducing strategy and articulating a summary statement of strategy.

Strategic Intent: Vision, mission, business definition. Strategic and financial objectives.

Components of strategic formulation process.

8

II

Analysis of External environment – PESTLE

Industry analysis using Porter's model, Analysis of competitive environment, ETOP

Analysis of Internal capability: Functional, PLC and Value Chain approach

to Internal analysis, Mckinsey's 7S framework, Resource audit

SWOT analysis

Relevance of social responsibility and power politics in strategy formulation

7

III

Business Strategies and Corporate strategies.

External Vs. Internal growth options, their conditions and prerequisites.

Five Generic Competitive strategies bu

Page 21 Content

Further Suggestions:

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Page 22 Data Table

Programme/Cla s: Degree			Year: Second		Semester: Third	
Course/ paper-9 (A)						
Course Code: F010303T		Course Title: Business Communication				
Course outcomes: The objective of this paper is to give the basic knowledge about the Business Communication						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 3-3-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year: Second		Semester: Third	
I	Introduction: Meaning and objective of Business communication, Forms of Communication, Communication model and process, Principles of Effective Communication					8
II	Corporate Communication: Formal and Informal Communication, Networks, Grapevine, Barriers in Communication, Groups discussion, Mock Interviews, Seminars, Individual and Group Presentations					7
III	Essential of effective Business letters, Writing Important Business letters including correspondence with Bank and Insurance companies; Oral & Non- verbal communication: Principles of Oral Presentation, Factors affecting Presentation, effective Presentation skills, conducting Surveys; Body Language, Para Language, Effective Listening, Interviewing skill, Writing Resume, Letter and Application;					8

Programme/Class s: Degree			Year: Second		Semester: Third	
IV	Modern forms of communication, International communication, Cultural sensitiveness and cultural context, Writing and presenting in international situations					7
Suggested Readings: 1. Bapat & Davar, A Text book of Business Correspondence 2. Bhende D.S., Business Communication 3. David Berio, The Process of Communication 4. Gowd & Dixit, Advance Commercial Correspondence 5. Gurky J.M., A Reader in Human Communication						
Suggested Continuous Evaluation Methods:						
Suggested equivalent online courses:						
Further Suggestions:						

Page 22 Content

Programme/Class: Degree

Year: Second

Semester: Third

Course/ paper-9 (A)

Course Code: F010303T

Course Title: Business Communication

Course outcomes:

The objective of this paper is to give the basic knowledge about the Business Communication

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-3-0

Unit

Topics

No. of

Lectures

Total=30

I

Introduction: Meaning and objective of Business communication, Forms of Communication, Communication model and process, Principles of Effective Communication

8

II

Corporate Communication: Formal and Informal Communication, Networks, Grapevine, Barriers in Communication, Groups discussion, Mock Interviews, Seminars, Individual and Group Presentations

7

III

Essential of effective Business letters, Writing Important Business letters including correspondence with Bank and Insurance companies; Oral & Non-verbal communication: Principles of Oral Presentation, Factors affecting Presentation, effective Presentation skills, conducting Surveys; Body Language, Para Language, Effective Listening, Interviewing skill, Writing Resume, Letter and Application;

8

IV

Modern forms of communication, International communication, Cultural sensitiveness and cultural context, Writing and presenting in international situations

7

Suggested Readings:

1. Bapat & Davar, A Text book of Business Correspondence
2. Bhende D.S., Business Communication
3. Davi

Page 23 Data Table

Programme/Cla s: Degree			Year: Second		Semester: Third	
Course/ paper-9 (B)						
Course Code: F010303T		Course Title: Business Environment				
Course outcomes: The objective of this paper is to give the basic knowledge about the business environment in industry						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 3-2-1						
Unit	Topics					No. of Lectures Total=30
I	Introduction: Concept, Significance and Components of Business environment, Factor affecting Business Environment, Micro and Macro environment.					8
II	Economic Systems: Capitalism, Socialism, Communism, Mixed Economy- Public Sector & Private Sector					7

Programme/Cla s: Degree			Year: Second		Semester: Third	
III	Industrial Policy-B rief historical perspective; New industrial policy of India, Socio-economic implications of Liberalization, Privatization and Globalization					8
IV	Role of Government in Regulation and Development of Business; Monetary and Fiscal Policy; EXIM Policy, FEMA; Overview of International Business Environment, Trends in World Trade: WTO-Objectives and role in international trade.					7
Suggested Readings: 1. Francis Cherunilum, Business Environment 2. K. Aswathapa, Business Environment						
Suggested Continuous Evaluation Methods:						
Suggested equivalent online courses:						
Further Suggestions:						

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Programme/Class: Degree

Year: Second

Semester: Third

Course/ paper-9 (B)

Course Code: F010303T

Course Title: Business Environment

Course outcomes:

The objective of this paper is to give the basic knowledge about the business environment in industry

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-2-1

Unit

Topics

No. of

Lectures

Total=30

I

Introduction:

Concept,

Significance

and

Components

of

Business

environment, Factor affecting Business Environment, Micro and Macro environment.

8

II

Economic Systems: Capitalism, Socialism, Communism, Mixed Economy-

Public Sector & Private Sector

7

III

Industrial Policy-Brief historical perspective; New industrial policy of India,

Socio-economic implications of Liberalization, Privatization and

Globalization

8

IV

Role of Government in Regulation and Development of Business; Monetary and Fiscal Policy; EXIM Policy, FEMA; Overview of International Business Environment, Trends in World Trade: WTO-Objectives and role in international trade.

7

Suggested Readings:

1. Francis Cherunilum, Business Environment
2. K. Aswathapa, Business Environment

Suggested Continuous Evaluation Methods:

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Suggested equivalent online courses:

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Page 24 Data Table

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
2	IV	Course/ paper-10	A	F010401T	Supply Chain Management	3
			B		Research Methodology	3
	IV	Course/ paper-11	A	F010402T	Specialized Accounting	3
			B		Consumer Behaviour	3
	IV	Course/ paper-12	A	F010403T	Investment Analysis & Portfolio Management	3
			B		Company Law	3

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BBA Second Year (Fourth Semester)

Year Sem.

Subject

Part

Paper Code

Paper Name

Credit

2

IV

Course/

paper-10

A

F010401T

Supply Chain Management

3

B

Research Methodology

3

IV

Course/

paper-11

A

F010402T

Specialized Accounting

3

B

Consumer Behaviour

3

IV

Course/

paper-12

A

F010403T

Investment Analysis & Portfolio Management

3

B

Company Law

3

Page 25 Data Table

Programme/Class: Degree		Year: Second		Semester: Fourth	
Course/ paper- 10 (A)					
Course Code: F010401T		Course Title: Supply Chain Management			

Programme/Class: Degree		Year: Second		Semester: Fourth	
Course outcomes: The objective of this paper is to give the basic knowledge about the Supply Chain Management for goods and services					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 3-2-1					
Unit	Topics				No. of Lectures Total=30
I	Introduction, Definition of Supply Chain Management, Evolution of the Concept of Supply Chain Management, Key Drivers of Supply Chain Management, Typology of Supply Chains, Cycle View of Supply Chain, Problems in SCM and Suggested Solutions				8
II	Introduction, Three Components of SCM, Demand Management, Demand Forecasting; Introduction, Supply Management, Evolution of ERP, Concept of ERP in SCM, Quick Response and Accurate Response System in SCM, Use of Other Planning Strategies				7
III	Introduction, Understanding the Benchmarking Concept, Benchmarking Process, Benchmarking Procedure				8

Programme/Class: Degree		Year: Second		Semester: Fourth	
IV	Introduction, New Developments in Supply Chain Management, Outsourcing Supply Chain Operations, Co-Maker ship, The Role of E- Commerce in Supply Chain Management, Green Supply Chain Management, Distribution Resource Planning, World Class Supply Chain Management				7
Suggested Readings: 1. Supply Chain Management by Michel H Hungo 2. Supply Chain Management by Sunil Chopra					
Suggested Continuous Evaluation Methods:					
Suggested equivalent online courses:					
Further Suggestions:					

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Programme/Class: Degree

Year: Second

Semester: Fourth

Course/ paper- 10 (A)

Course Code: F010401T

Course Title: Supply Chain Management

Course outcomes:

The objective of this paper is to give the basic knowledge about the Supply Chain Management for goods and services

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-2-1

Unit

Topics

No. of

Lectures

Total=30

I

Introduction, Definition of Supply Chain Management, Evolution of the Concept of Supply Chain Management, Key Drivers of Supply Chain Management, Typology of Supply Chains, Cycle View of Supply Chain, Problems in SCM and Suggested Solutions

8

II

Introduction, Three Components of SCM, Demand Management, Demand Forecasting; Introduction, Supply Management, Evolution

of

ERP,

Concept of ERP in SCM, Quick Response and Accurate Response System in SCM, Use of Other Planning Strategies

7

III

Introduction, Understanding the Benchmarking Concept, Benchmarking Process, Benchmarking Procedure

8

IV

Introduction, New Developments in Supply Chain Management, Outsourcing Supply Chain Operations, Co-Maker ship, The Role of E- Commerce in Supply Chain Management, Green Supply Chain Management, Distribution Resource Planning, World Class Supply Chain Management

7

Suggested Readings:

1. Supply Chain Management by Michel H Hungo
 2. Supply Chain Management by Sunil Chopra
- Suggested Continuous Evaluation Methods:

Page 26 Data Table

Programme/Cla s: Degree		Year: Second			Semester: Fourth	
Course/ paper- 10 (B)						
Course Code: F010401T			Course Title: Research Methodology			
Course outcomes: The objective of this paper is to give the basic knowledge about the Research Methodology						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 3-3-0						
Unit	Topics					No. of Lectures Total=30
I	Introduction: Meaning of Research, Objectives of Research, Types of Research Research Process, Research Problem formulation; Research Design: Features of a good research design; Different Research Designs; Measurement in Research; Data types; Sources of Error.					, 8

Programme/Cla s: Degree		Year: Second			Semester: Fourth	
II	Sampling Design: Census & Sample Surveys; Steps in Sampling Design; Types of Sample designs-Probabilit y & Non Probability sampling.					7
III	Processing & Analysis of Data: Processing operations; problems in processing; types of analysis, Hypothesis Testing: Chi-square test, Z-test, t-test, F-test.					8
IV	Presentation: Diagrams; graphs; charts. Report writing; Layout of Research report; Types of Reports; Mechanism of writing a Research report; Precautions for writing report. Use of MS Office in Report writing and making Presentations					7
Suggested Readings: 1. C.R. Kothari, Research Methodology 2. Banerjee S. and Roy Ramendu, Fundamentals of Research Methodology						
Suggested Continuous Evaluation Methods:						
Suggested equivalent online courses:						

Programme/Class: s: Degree		Year: Second			Semester: Fourth	
Further Suggestions:						

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Programme/Class: Degree

Year: Second

Semester: Fourth

Course/ paper- 10 (B)

Course Code: F010401T

Course Title: Research Methodology

Course outcomes:

The objective of this paper is to give the basic knowledge about the Research Methodology

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-3-0

Unit

Topics

No. of

Lectures

Total=30

I

Introduction: Meaning of Research, Objectives of Research, Types of Research, Research Process, Research Problem formulation; Research Design: Features of a good research design; Different Research Designs; Measurement in Research; Data types; Sources of Error.

8

II

Sampling Design: Census & Sample Surveys; Steps in Sampling Design; Types of Sample designs-Probability & Non Probability sampling.

7

III

Processing & Analysis of Data: Processing operations; problems in processing;

types of analysis, Hypothesis Testing: Chi-square test, Z-test, t-test, F-test.

8

IV

Presentation: Diagrams; graphs; charts. Report writing; Layout of Research

report; Types of Reports; Mechanism of writing a Research report;

Precautions for writing report. Use of MS Office in Report writing and

making Presentations

7

Suggested Readings:

1. C.R. Kothari, Research Methodology

2. Banerjee S. and Roy Ramendu, Fundamentals of Research Methodology

Suggested Continuous Evaluation Methods:

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Page 27 Data Table

Programme/Class: Degree		Year: Second		Semester: Fourth	
Course/ paper- 11 (A)					
Course Code: F010402T		Course Title: Specialized Accounting			
Course outcomes: The objective of this paper is to give the basic knowledge about the specialized Accounting					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 3-3-0					
Unit	Topics				No. of Lectures Total=30
I	Accounting of Non-trading Institutions, Joint Venture and Consignment				8
II	Accounts of Banking companies and General Insurance companies				7

Programme/Class: Degree		Year: Second		Semester: Fourth	
III	Department account and Branch account.Accounts related to Hire Purchase and Installment payment transactions, Royalty Accounts				8
IV	Partnership Accounts: Final Account, Reconstitution of Partnership firms: admission, retirement and death of a partner, Dissolution of Partnership (Excluding insolvency of Partner)				7
Suggested Readings: 1. Agarwal, B.D., Advanced Accounting 2. Chawla & Jain, Financial Accounting 3. Chakrawarti, K.S., Advanced Accounts 4. Shukla, M.B., Financial Analysis and Business Forecasting 5. Jain & Naranag, Advanced Accounts					
Suggested Continuous Evaluation Methods:					
Suggested equivalent online courses:					
Further Suggestions:					

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Programme/Class: Degree

Year: Second

Semester: Fourth

Course/ paper- 11 (A)

Course Code: F010402T

Course Title: Specialized Accounting

Course outcomes:

The objective of this paper is to give the basic knowledge about the specialized Accounting

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-3-0

Unit

Topics

No. of

Lectures

Total=30

I

Accounting of Non-trading Institutions, Joint Venture and Consignment

8

II

Accounts of Banking companies and General Insurance companies

7

III

Department account and Branch account.Accounts related to Hire Purchase and Installment payment transactions, Royalty Accounts

8

IV

Partnership Accounts: Final Account, Reconstitution of Partnership firms: admission, retirement and death of a partner, Dissolution of Partnership (Excluding insolvency of Partner)

7

Suggested Readings:

1. Agarwal, B.D., Advanced Accounting
2. Chawla & Jain, Financial Accounting
3. Chakrawarti, K.S., Advanced Accounts
4. Shukla, M.B., Financial Analysis and Business Forecasting
5. Jain & Naranag, Advanced Accounts

Suggested Continuous Evaluation Methods:

.....

Suggested equivalent online courses:

.....

Further Suggestions:

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Page 28 Data Table

Programme/Class: Degree		Year: Second		Semester: Fourth	
Course/ paper-11 (B)					
Course Code: F010402T		Course Title: Consumer Behaviour			
Course outcomes: The objective of this paper is to give the basic knowledge about the consumer behaviour					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 3-3-0					
Unit	Topics				No. of Lectures Total=30
I	CB, Consumer research process.CB models: Economic model, Psycho-analytic model, Sociological model, Howard & Seth model, Nicosia model, Engel-Kollat-Blackwell model.				8
II	Individual determinants: Perceptual process, consumer learning process Introduction: Concept, importance and scope of CB, need for studying, consumer attitude formation, attitude measurement, meaning and nature of personality, self concept.				7

Programme/Class: Degree		Year: Second		Semester: Fourth	
III	Influences & Consumer Decision making: Family, reference group, personal, social and cultural influence on CB, Consumer Decision making process, Consumer Communication process, consumer satisfaction.				8
IV	Industrial Buying Behaviour: Participants, characteristics of industrial markets, factors influencing industrial markets, stages of industrial buying process, Customer and marketing of services.				7
Suggested Readings: 1. Suja. R. Nair, Consumer Behaviour in Indian Perspective 2. Schiffman & Kanuk, Consumer Behaviour 3. Louden & Bitta, Consumer Behaviour 4. Bennet & Kasarji, Consumer Behaviour					
Suggested Continuous Evaluation Methods:					
Suggested equivalent online courses:					
Further Suggestions:					

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Programme/Class: Degree

Year: Second

Semester: Fourth

Course/ paper-11 (B)

Course Code: F010402T

Course Title: Consumer Behaviour

Course outcomes:

The objective of this paper is to give the basic knowledge about the consumer behaviour

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-3-0

Unit

Topics

No. of

Lectures

Total=30

I

CB, Consumer research process. CB models: Economic model, Psycho-analytic model, Sociological model, Howard & Seth model, Nicosia model, Engel-Kollat-Blackwell model.

8

II

Individual determinants: Perceptual process, consumer learning process

Introduction: Concept, importance and scope of CB, need for studying, consumer attitude formation, attitude measurement, meaning and nature of personality, self concept.

7

III

Influences & Consumer Decision making: Family, reference group, personal, social and cultural influence on CB, Consumer Decision making process, Consumer Communication process, consumer satisfaction.

8

IV

Industrial Buying Behaviour: Participants, characteristics of industrial markets, factors influencing industrial markets, stages of industrial buying process, Customer and marketing of services.

7

Suggested Readings:

1. Suja. R. Nair, Consumer Behaviour in Indian Perspective
2. Schiffman & Kanuk, Consumer Behaviour
3. Louden & Bitta, Consumer Behaviour
4. Bennet & Kasarji, Consumer Behaviour

Suggested Continuous Evaluation

Page 29 Data Table

Programme/Classes: Degree			Year: Second		Semester: Fourth	
Course/ paper-12 (A)						
Course Code: F010403T		Course Title: Investment Analysis & Portfolio Management				
Course outcomes: The objective of this paper is to give the basic knowledge about the investment analysis and portfolio management subject 3(A)						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-2-1						
Unit	Topics					No. of Lectures Total=30
I	Investments: Nature, scope, objective and Process of investments analysis, concept of return and risk analysis, measurement of return and risk: Systematic and Unsystematic Risk.					8

Programme/Cla s: Degree			Year: Second		Semester: Fourth	
II	Investment Alternatives: Investment instrument of Capital Market and Money Market, Valuation of Fixed and Variable securities Non Security forms of Investment, Government Securities, Mutual Fund, Real Estate and Gold.					7
III	Fundamental Analysis: Economic analysis industry analysis and company analysis Technical Analysis: Trends, indicators, indices and moving average applied in technical analysis. Efficient Market Hypothesis: weak, semi-strong and strong market and its testing techniques					8
IV	Portfolio Management: Meaning, importance and objectives of portfolio and portfolio management, Risk and Return- Definition types and importance. Portfolio Analysis: Risk Measurement; Estimating rate of return and standard deviation of portfolio returns; Effects of Combining securities;					7

Programme/Class s: Degree			Year: Second		Semester: Fourth	
Suggested Readings: 1. Security analysis and Portfolio Management by Punithavathy Pandian						
Suggested Continuous Evaluation Methods:						
Suggested equivalent online courses:						
Further Suggestions:						

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Programme/Class: Degree

Year: Second

Semester: Fourth

Course/ paper-12 (A)

Course Code: F010403T

Course Title: Investment Analysis & Portfolio Management

Course outcomes:

The objective of this paper is to give the basic knowledge about the investment analysis and portfolio management subject 3(A)

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-2-1

Unit

Topics

No. of

Lectures

Total=30

I

Investments: Nature, scope, objective and Process of investments analysis, concept of return and risk analysis, measurement of return and risk: Systematic and Unsystematic Risk.

8

II

Investment Alternatives: Investment instrument of Capital Market and Money Market, Valuation of Fixed and Variable securities Non Security forms of Investment, Government Securities, Mutual Fund, Real Estate and Gold.

7

III

Fundamental Analysis: Economic analysis industry analysis and company analysis Technical Analysis: Trends, indicators, indices and moving average applied in technical analysis. Efficient Market Hypothesis: weak, semi-strong and strong market and its testing techniques

8

IV

Portfolio Management: Meaning, importance and objectives of portfolio and portfolio management, Risk and Return- Definition types and importance.

Portfolio Analysis: Risk Measurement; Estimating rate of return and standard deviation of portfolio returns; Effects of Combining securities;

7

Suggested Readi

Page 30 Data Table

Programme/Cla s: Degree			Year: Second		Semester: Fourth	
Course/ paper- 12 (B)						
Course Code: F010403T		Course Title: Company Law				
Course outcomes: The objective of this paper is to give the basic knowledge about the Company Law						
Credits: 3				Compulsory		

Programme/Cla s: Degree			Year: Second		Semester: Fourth	
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 3-2-1						
Unit	Topics					No. of Lectures Total=30
I	Introduction: Definition and Kinds of Company, Promotion and Incorporation of Companies; Memorandum of Association, Articles of Association, Prospectus.					8
II	Shares, Share Capital, Members, Transfer and Transmission of shares, Directors- Managing Director, Whole Time Director					7
III	Capital Management; Borrowing powers, mortgages and charges, debentures, Company Meetings-kinds, quorum, voting, resolutions, minutes					8
IV	Majority Powers and minority rights, Prevention of oppression and mismanagement, winding up of companies, its Kinds and Conduct					7

Programme/Class s: Degree			Year: Second		Semester: Fourth	
Suggested Readings: 1. Grower L.C.B., Principles of Modern Company Law 2. Ramaiya A., Guide to the Companies Act 3. Singh, Avtar, Company Law 4. Kuchhal, S.C., Modern Indian Company Law 5. Kapoor, N.D., Company Law						
Suggested Continuous Evaluation Methods:						
Suggested equivalent online courses:						
Further Suggestions:						

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Programme/Class: Degree

Year: Second

Semester: Fourth

Course/ paper- 12 (B)

Course Code: F010403T

Course Title: Company Law

Course outcomes:

The objective of this paper is to give the basic knowledge about the Company Law

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 3-2-1

Unit

Topics

No. of

Lectures

Total=30

I

Introduction: Definition and Kinds of Company, Promotion and Incorporation of Companies; Memorandum of Association, Articles of Association, Prospectus.

8

II

Shares, Share Capital, Members, Transfer and Transmission of shares, Directors- Managing Director, Whole Time Director

7

III

Capital Management; Borrowing powers, mortgages and charges, debentures, Company Meetings-kinds, quorum, voting, resolutions, minutes

8

IV

Majority Powers and minority rights, Prevention of oppression and mismanagement, winding up of companies, its Kinds and Conduct

7

Suggested Readings:

1. Grower L.C.B., Principles of Modern Company Law
2. Ramaiya A., Guide to the Companies Act
3. Singh, Avtar, Company Law
4. Kuchhal, S.C., Modern Indian Company Law
5. Kapoor, N.D., Company Law

Suggested Continuous Evaluation Methods:

.....

Suggested equivalent online courses:

.....

Page 31 Data Table

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
3	V	Course/ paper-13	A	F010501T	Income Tax	3
			B		Marketing Communication	3

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
	V	Course/ paper-14	A	F010502T	Entrepreneurship and small business management	3
			B		Sales management	3
	V	Course/ paper-15	A	F010503T	Industrial Relations & Labour Laws	3
			B		Company Accounts	3

Page 31 Data Table

Programme/Classes: Degree			Year: Third		Semester: Fifth	
Course/ paper-13 (A)						
Course Code: F010501T		Course Title: Income Tax				
Course outcomes: The aim of the course is to build knowledge, understanding about income tax among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Income Tax. The outcome of the course will be as follows – To provide knowledge about Income Tax Act. To provide knowledge about gross income and taxable income. To give an overview about different deductions and exemptions.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total = 30
I	Indian Income Tax Act, 1961: Basic Concepts - Income, Agriculture Income, Casual Income, Assessment Year, Previous Year, Gross Total Income, Total Income, Person, Tax Evasion, Tax Avoidance.					8
II	Basis of Charge: Scope of Total Income, Residence and Tax Liability, Income which does not form part of Total Income.					6
III	Heads of Income: Income from Salaries, Income from House Property. Profit and Gains of Business or Profession, Capital Gains, Income from other sources.					10
IV	Aggregation of Income, Set off and Carry forward of losses, deductions from gross total Income, Computation of total Income and Tax liability.					6

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Suggested Readings: 1. Mehrotra, H.C., Income Tax Law and Account 2. Prasad, Bhagwati, Income Tax Law and Practice 3. Chandra Mahesh and Shukla D.C., Income Tax Law and Practice 4. Agarwal, B.K., Income Tax 5. Jain, R.K., Income Tax						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestio ns:.....						

Page 31 Content

BBA Third Year (Fifth Semester)

Yea

r

Sem.

Subject

Part

Paper Code

Paper Name

Credit

3

V

Course/

paper-13

A

F010501T

Income Tax

3

B

Marketing Communication

3

V

Course/

paper-14

A

F010502T

Entrepreneurship and small business management

3

B

Sales management

3

V

Course/

paper-15

A

F010503T

Industrial Relations & Labour Laws

3

B

Company Accounts

3

Programme/Class: Degree

Year: Third

Semester: Fifth

Course/ paper-13 (A)

Course Code: F010501T

Course Title: Income Tax

Course outcomes:

The aim of the course is to build knowledge, understanding about income tax among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about Income Tax. The outcome of the course will be as follows –

To provide knowledge about Income Tax Act.

To provide knowledge about gross income and taxable income.

To give an overview about different deductions and exemptions.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of

Lectures

Total = 30

I

Indian Income Tax Act, 1961: Basic Concepts - Income, Agriculture

Income, Casual Income, Assessment Year, Previous Year, Gross Total Income, Total Income, Person, Tax Evasion, Tax Avoidance.

8

II

Basis of Charge: Scope of Total Income, Residence and Tax Liability, Income which does not form part of Total Income.

6

III

Heads of Inco

Page 32 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Course/ paper-13 (B)						
Course Code: F010501T		Course Title: Marketing Communication				

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Course outcomes: The aim of the course is to build knowledge, understanding and skills in marketing communication among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about IMC and advertising and their role in over all promotion strategies of the firm. The outcome of the course will be as follows – ■ Apply an IMC approach in the development of an overall advertising and promotional plan. ■ Enhance creativity, critical thinking an						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
I	Marketing Communication: Meaning and its objectives, Integrated Marketing Communication (IMC): concepts and process, IMC promotion Mix, Advertising - Meaning, objectives its role and functions, Classification of advertising, economic, social and ethical issues in advertising, DAGMAR approach, STP strategies in advertising, Advertising Agencies,					7
II	Process in Advertising: Consumer and mental process in buying, AIDA model, Hierarchy of effects model, Information processing model, Advertising Budget – Top down and Build up approach, methods of advertising – Affordable method, arbitrary allocation method, percentage of sales method, competitive parity method, Objective and Task method.					7

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
III	Advertising Creativity: Meaning of creativity, Creative strategy, Creative tactics, Advertising Appeals, USP theory of creativity, Copywriting: Meaning and Definition of Copywriting, The Copywriter, Copywriting for Print, Copywriting guidelines, Radio Copywriting, TV Copywriting, Writing for the Web, Tips for writing good web content					8
IV	Media Planning and Strategy: Media Types and their characteristics; Setting Media objectives; Steps involved in media planning,					8
	evaluation of media, media scheduling strategy, Evaluation of advertising effectiveness – need and purpose of evaluation, pre-testing and post testing techniques, Advertising research, decision areas in international advertising.					

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Suggested Readings: 1. George E Belch & Michael A Belch: Advertising and promotion- An integrated Marketing Communication P erspective-McGra w Hill Education 2. Chunawala & Sethia : Foundations of Advertising Theory & Practice; Himalaya Publishing House 3. Copley Paul: Marketing Communications Management Concepts & theories, Cases and Practices; Butterworth Heinemann Publication. 4. Aaker, David A. et al., Advertising Management, PHI,						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 32 Content

Programme/Class: Degree

Year: Third

Semester: Fifth

Course/ paper-13 (B)

Course Code: F010501T

Course Title: Marketing Communication

Course outcomes:

The aim of the course is to build knowledge, understanding and skills in marketing communication among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about IMC and advertising and their role in over all promotion strategies of the firm. The outcome of the course will be as follows –



Apply an IMC approach in the development of an overall advertising and promotional plan.



Enhance creativity, critical thinking and analytical ability through developing an integrated marketing communication campaign

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of

Lectures

Total=30

I

Marketing Communication: Meaning and its objectives, Integrated Marketing Communication (IMC): concepts and process, IMC promotion Mix, Advertising - Meaning, objectives its role and functions, Classification of advertising, economic, social and ethical issues in advertising, DAGMAR approach, STP strategies in advertising, Advertising Agencies,

7

II

Process in Advertising: Consumer and mental process in buying, AIDA model, Hierarchy of effects model, Information processing model, Advertising Budget – Top down and Build

up approach, methods of advertising – Afford

Page 33 Data Table

Programme/Class: Degree		Year: Third		Semester: Fifth	
Course/ paper-14 (A)					
Course Code: F010502T		Course Title: Entrepreneurship and small business management			
Course outcomes: The aim of the course is to develop concept of entrepreneur and entrepreneurship among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about entrepreneurship and small businesses. The outcome of the course will be as follows – To provide knowledge about entrepreneurial concept To provide knowledge about entrepreneurship development, EDPs and support system. To give an overview about project and project report prepa					
Credits: 3			Compulsory		
Max. Marks: 25+75			Min. Passing Marks:		
Total No. of Lectures -Tutorials-Practical (in hours per week): L-T-P: 2-0-0					
Unit	Topics				No. of Lectures Total=30

Programme/Class: Degree		Year: Third		Semester: Fifth	
I	Entrepreneurship: Concept, Role & Importance in Indian Economy, Theories of Entrepreneurship, Entrepreneurs – Evolution of concept Types of entrepreneurs, traits of entrepreneur, entrepreneurs Vs managers, Entrepreneurs, problems faced by entrepreneurs, Women Entrepreneurs, Rural Entrepreneurs				8
II	Entrepreneurial Development and Institutional Support System: Entrepreneurship development, Concept and Significance, Entrepreneurial Development Programmes (EDP), problems of EDP, Institutional support to entrepreneurs, Arrangement of finance and support from financial institutions				8
III	Business Idea: Environmental analysis, Search for business idea, Identification of projects, Selection of project, Project formulation, Project report, project appraisal.				8
IV	Small Business: Definitions, MSMED Act 2006, Strategic Planning and its steps for small business, Incentives and subsidies available to small business, forms of ownership, Registration as SSI				6

Programme/Class: Degree		Year: Third		Semester: Fifth	
Suggested Readings: 1. Entrepreneurship 10th Ed (Indian Edition) 2016 by Robert Hisrich Michael Peters Dean Shepherd, McGraw Hill 2. Khanka, S.S.; Entrepreneurial Development; S. Chand and Co. 3. Kumar, Arya; Entrepreneurship; Pearson Education. 4. Desai, Vasant; Dynamics of Entrepreneurial Development and Management; Himalaya Publishing 5. Blundel, R. and Lockett, N.; Exploring Entrepreneurship Practices and Perspectives; Oxford Publications.					
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.					
Suggested equivalent online courses:					
Further Suggestions:					

Page 33 Content

Programme/Class: Degree

Year: Third

Semester: Fifth

Course/ paper-14 (A)

Course Code: F010502T

Course Title: Entrepreneurship and small business management

Course outcomes:

The aim of the course is to develop concept of entrepreneur and entrepreneurship among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about entrepreneurship and small businesses. The outcome of the course will be as follows –

To provide knowledge about entrepreneurial concept

To provide knowledge about entrepreneurship development, EDPs and support system.

To give an overview about project and project report preparation

To give an overview about the small business.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Entrepreneurship: Concept, Role & Importance in Indian Economy,
Theories of Entrepreneurship, Entrepreneurs – Evolution of concept
Types of entrepreneurs, traits of entrepreneur, entrepreneurs Vs
managers, Entrepreneurs, problems faced by entrepreneurs, Women
Entrepreneurs, Rural Entrepreneurs

8

II

Entrepreneurial Development and Institutional Support System:
Entrepreneurship development, Concept and Significance,
Entrepreneurial Development Programmes (EDP), problems of EDP,
Institutional support to entrepreneurs, Arrangement of finance and
support from financial institutions

Page 34 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Course/ paper-14 (B)						

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Course Code: F010502T		Course Title: Sales management				
Course outcomes: The aim of the course is to build knowledge, understanding and skills in sales management among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about sales management. The outcome of the course will be as follows – To provide knowledge about sales personnel and salesmanship. To provide knowledge about personal selling and focus light on the different perspectives of managing sales force. To give an overview about i						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
I	Introduction to Sales Management: Concept, Evolution of sales function, Objectives of sales management positions, Functions of Sales manager and their relation with other executives.					8
II	Salesmanship: Theories of personal selling, Types of Sales executives, Qualities of sales executives, Personal selling process, Showroom & exhibition,					8
III	Sales Organization and Relationship: Purpose of sales organization, Types of sales organization structures, Sales department external relations, Distributive network relations. Sales Force Management: Recruitment and Selection, Sales Training, Sales Compensation.					8
IV	Distribution Network Management: Types of Marketing Channels, Factors affecting the choice of channel, Types of middleman and their characteristics, Concept of physical distribution system.					6

Programme/Class s: Degree			Year: Third		Semester: Fifth	
Suggested Readings: 1. Cundiff, Still, Govoni, Sales Management 2. Pradhan, Jakate, Mali, Salesmanship & Publicity 3. S.A. Chunawalla, Sales Management						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 34 Content

Programme/Class: Degree

Year: Third

Semester: Fifth

Course/ paper-14 (B)

Course Code: F010502T

Course Title: Sales management

Course outcomes:

The aim of the course is to build knowledge, understanding and skills in sales management among the student.

The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about sales management. The outcome of the course will be as follows –

To provide knowledge about sales personnel and salesmanship.

To provide knowledge about personal selling and focus light on the different perspectives of managing sales force.

To give an overview about importance of sales force in organization.

To give an overview about concept of distribution channels.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Introduction to Sales Management: Concept, Evolution of sales function, Objectives of sales management positions, Functions of Sales manager and their relation with other executives.

8

II

Salesmanship: Theories of personal selling, Types of Sales executives, Qualities of sales executives, Personal selling process, Showroom & exhibition,

8

III

Sales Organization and Relationship: Purpose of sales organization, Types of sales organization structures, Sales department external relations, Distributive network relations.

Sales Force Management: Recruitment and

Page 35 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Course/ paper-15 (A)						
Course Code: F010503T		Course Title: Industrial Relations & Labour Laws				

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Course outcomes: This course will help students identify and develop an overview of industrial relations. It also help in acquiring knowledge and understanding of Industrial Labour and General Laws. ■ Knowledge of Industrial Relation framework ■ Competency to understand the importance of Employee Relation within the perspective of Industrial Relation ■ Knowledge about relevant Laws of HR management ■ Competency to interpreted and implement the Labour Laws within organization ■ Competency to use						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30
I	Industrial Relations: Role - Importance - Trade Unions - Industrial disputes and their Resolutions.					6

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
II	Participative Management: Structure - Scope - Collective Bargaining - Works Committee - Joint Management Councils - Pre-Requisite for successful participation - Role of Government in Collective Bargaining.					8
III	Industrial unrest: Employee dissatisfaction - Grievances - Disciplinary Action - Domestic Enquiry - Strikes - lockout - Prevention of Strikes - Lockouts. Discipline: Positive, negative discipline, disciplinary procedure, Absenteeism, Turnover, Dismissal and Discharge.					8
IV	Factories Act: Meaning, Definition – Welfare – Safety – Health Measures. Workmen's Compensation Act and International Labor Organization - Role and Function, General provisions of Bonus Act and Gratuity Act					8

Programme/Class s: Degree			Year: Third		Semester: Fifth	
Suggested Readings: 5. Sreenivasan M.R - Industrial Relations & Labor legislations. 6. Aswathappa K - Human Resource and Personnel Management. 7. Subba Rao P - Human Resource Management and Industrial Relations. 8. Monoppa - Industrial Relations. 5. S.C. Srivastava, Industrial Relation of Labour Laws.						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 35 Content

Programme/Class: Degree

Year: Third

Semester: Fifth

Course/ paper-15 (A)

Course Code: F010503T

Course Title: Industrial Relations & Labour Laws

Course outcomes:

This course will help students identify and develop an overview of industrial relations. It also help in acquiring knowledge and understanding of Industrial Labour and General Laws.



Knowledge of Industrial Relation framework



Competency to understand the importance of Employee Relation within the perspective of Industrial Relation



Knowledge about relevant Laws of HR management



Competency to interpreted and implement the Labour Laws within organization



Competency to use Collective Bargaining and Grievance redressal Mechanism

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of

Lectures

Total=30

I

Industrial Relations: Role - Importance - Trade Unions - Industrial disputes and their Resolutions.

6

II

Participative Management: Structure - Scope - Collective Bargaining - Works Committee - Joint Management Councils - Pre-Requisite for successful participation - Role of Government in Collective Bargaining.

8

III

Industrial unrest: Employee dissatisfaction - Grievances - Disciplinary Action - Domestic Enquiry - Strikes - lockout - Prevention of Strikes - Lockouts. Discipline: Positive, negative discipline, disciplinary procedure,

Absenteeism, Turnover, Dismissal
and Discha

Page 36 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
Course/ paper-15 (B)						
Course Code: F010503T		Course Title: Company Accounts				
Course outcomes: The aim of the course is to build knowledge, understanding and skills in the area of company accounts among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about accounting practices relevant to companies. The outcome of the course will be as follows- ■ To understand joint stock companies and knowledge about shares and debentures. ■ To have understanding about final accounts and accounting practices related to amalg						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year: Third		Semester: Fifth	
I	Joint Stock Companies: Its types and share capital, Issue, Forfeiture and Re-issue of shares, Redemption of preference shares, Issue and Redemption of Debenture.					7
II	Final Accounts: Including Computation of managerial Remuneration and disposal of profit.					7
III	Accounting for Amalgamation of companies as per Accounting Standard 14, Accounting for Internal reconstruction, Liquidation of Company.					8
IV	Consolidated Balance Sheet of Holding Companies with one Subsidiary only, Statement of Affairs and Deficiency/Surplus, Receivers Receipt and Payment A/c.					8
Suggested Readings: 1. Gupta R.L. Radhaswamy M, Company Accounts 2. Maheshwari, S.N., Corporate Accounting 3. Monga J.R., Ahuja, Girish, and Sehgal Ashok, Financial Accounting 4. Shukla, M.C., Grewal T.s. and Gupta, S.C. Advanced Accounts						

Programme/Class s: Degree			Year: Third		Semester: Fifth	
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 36 Content

Programme/Class: Degree

Year: Third

Semester: Fifth

Course/ paper-15 (B)

Course Code: F010503T

Course Title: Company Accounts

Course outcomes:

The aim of the course is to build knowledge, understanding and skills in the area of company accounts among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about accounting practices relevant to companies. The outcome of the course will be as follows-



To understand joint stock companies and knowledge about shares and debentures.



To have understanding about final accounts and accounting practices related to amalgamation.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Joint Stock Companies: Its types and share capital, Issue, Forfeiture and Re-issue of shares, Redemption of preference shares, Issue and Redemption of Debenture.

7

II

Final Accounts: Including Computation of managerial Remuneration and disposal of profit.

7

III

Accounting for Amalgamation of companies as per Accounting Standard 14, Accounting for Internal reconstruction, Liquidation of Company.

8

IV

Consolidated Balance Sheet of Holding Companies with one Subsidiary only, Statement of Affairs and Deficiency/Surplus, Receivers Receipt and Payment A/c.

8

Suggested Readings:

1. Gupta R.L. Radhaswamy M, Company Accounts
2. Maheshwari, S.N.,

Page 37 Data Table

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
3	VI	Course/ paper-16	A	F010601T	Project Management	3
			B		Goods & Service Tax	3
	VI	Course/ paper-17	A	F010602T	Auditing	3
			B		International Trade	3
	VI	Course/ paper-18	A	F010603T	Strategic Management	3

Year	Sem.	Subject	Part	Paper Code	Paper Name	Credit
			B		Training and Development	3

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BBA Third Year (Sixth Semester)

Year

Sem.

Subject

Part

Paper Code

Paper Name

Credit

t

3

VI

Course/

paper-16

A

F010601T

Project Management

3

B

Goods & Service Tax

3

VI

Course/

paper-17

A

F010602T

Auditing

3

B

International Trade

3

VI

Course/

paper-18

A

F010603T

Strategic Management

3

B

Training and Development

3

Page 38 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Course/ paper-16 (A)						
Course Code: F010601T		Course Title: Project Management				
Course outcomes: ■ Students will be able to understand the characteristics of Project and Project Management Knowledge ■ The students will understand the managerial process along with tools & techniques used in Project management Knowledge ■ Students will understand the scheduling and monitoring process in Project. They will be able to apply PERT and CPM method for project scheduling ■ Students will understand the perspectives in which optimum decisions are to be taken in case of risks with plan						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30
I	Definitions & Characteristics of Project, Types of Projects, Project Life Cycle, Project Management Process: Introduction, Tools & Techniques of Project Management. Project Team and Scope of Project Management, Project Organization.					6
II	Project Identification & Selection: Identification, Generation of ideas, Approaches to Project Screening and Selection, Project Rating Index. Market & Demand Analysis Techniques: Survey & Trend Projection Methods. Project Risk Management					8

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
III	Project Costing: Fundamental components of Project Cost, Types of Costs: Direct, Indirect, Recurring, Non-Recurring, Fixed, Variable, Normal, Expedite costs. Project Financing and Budgeting: Sources of Finance, Social Cost Benefit Analysis (SCBA) of Project, Project Scheduling and Network Analysis: Steps in Project Scheduling and Network design, Introduction to CPM and PERT,					8
IV	Monitoring and Control: Planning- Monitoring and Control Cycle. Project Management Information System. Milestone Analysis and Tracking Gantt chart. Earned Value Analysis (EVA): Planned Value(PV), Earned Value (EV), Cost Variance (CV), Schedule Variance (SV), Cost performance Index (CPI), Schedule performance Index (SPI). Project Termination: Types of Terminations, Project Termination Process					8

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Suggested Readings: 1. Project Management- A Managerial Approach: Jack R. Meredith Broyhill Samuel J. Mantel, Jr (JohnWiley & Sons) 2. Project Management : Mr. Sanjiv Marwah- (Wiley Dreamtech) 3. Project- Preparation, Appraisal, Budgeting and Implementation: Chandra Prasanna - (TMH) 4. Project Management Core Text Book : M R Gopalan (Wiley)						
5. Quantitative Techniques in Management : N D Vohra (TMH) 6. Entrepreneurship and Small Business Management : M B Shukla						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						

Programme/Class s: Degree			Year: Third		Semester: Sixth	
Further Suggestions:						
.....						
.....						
.....						
.....						

Page 38 Content

Programme/Class: Degree

Year: Third

Semester: Sixth

Course/ paper-16 (A)

Course Code: F010601T

Course Title: Project Management

Course outcomes:



Students will be able to understand the characteristics of Project and Project Management Knowledge



The students will understand the managerial process along with tools & techniques used in Project management Knowledge



Students will understand the scheduling and monitoring process in Project. They will be able to apply PERT and CPM method for project scheduling



Students will understand the perspectives in which optimum decisions are to be taken in case of risks with planned activities in project

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of

Lectures

Total=30

I

Definitions & Characteristics of Project, Types of Projects, Project Life Cycle, Project

Management Process: Introduction, Tools & Techniques of Project Management.

Project Team and Scope of Project Management, Project Organization.

6

II

Project Identification & Selection: Identification, Generation of ideas, Approaches to

Project Screening and Selection, Project Rating Index. Market & Demand Analysis

Techniques: Survey & Trend Projection

Methods. Project Risk Management

8

III

Project Costing: Fundamental components of Project Cost, Types of Costs: Direct,

Indirect, Recurring, Non-Recurring, Fixed, Variable, Normal, Expedit

Page 39 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Course/ paper-16 (B)						
Course Code: F010601T		Course Title: Goods & Service Tax				

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Course outcomes: The aim of the course is to build knowledge and understanding about GST among the student. The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about GST. The outcome of the course will be as follows – To provide knowledge about indirect taxes before GST. To provide knowledge about registration and documentation process under GST. To give an overview about tax exemptions. To give an overview about filing of GSTR.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
I	Introduction : Constitutional framework of Indirect Taxes before GST(Taxation Powers of Union & State Government); Concept of VAT: Meaning, Variants and Methods; Major Defects in the structure of Indirect Taxes prior to GST; Overview of GST ; Structure of GST (SGST, CGST, UTGST & IGST); GST Council.					7
II	Supply of Goods and Services - Definition of supply; Place of Supply: Intra-State and Inter-State supply; Composite and Mixed supply; Import and Export; Supplies of goods and services liable to be reverse charged; Time of supply; Nil rated supplies, Zero rated supplies, Exempted supplies & Non- GST supplies.					7
III	Registration and Documentation: (A) Registration-P ersons liable to registration; Compulsory registration; Procedure of Registration; Exemption from Registration; Composition Scheme.					8

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
	(B) Documentation- Tax Invoice; Bill of Supply; Receipt Voucher; Payment Voucher; Refund Voucher; Debit Note; Credit Note. Returns: GSTR 1 and GSTR 2, Monthly / Quarterly Return, Annual Return; Time and procedure of filing of Returns.					
IV	Input Tax Credit: Introduction, Concept of Input Service Distributor, Legal Formalities for an ISD, Distribution of Credit, Claiming Input Tax Credit for inputs goods, Claiming Input Tax Credit for Capital Goods Payment of Tax- (a) Through Input Tax Credit (b) By cash / bank after generation of online Challan. E-Way Bill: Introduction, Preparation of E-Way Bill, Important Points for Transporter. GST Portal: Introduction, GST Eco-system, GST Suvidha Provider (GSP), Uploading Invoices					8

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Suggested Readings: 1. Anandaday Mishra, GST Law & Procedure, Taxman. 2. Goods and Service Tax Acts. 3. Relevant Goods and Services Tax Rules. Nitya Tax Associates Basics of GST Taxman 4. Publication on GST by the Institute of Chartered Accountants of India (www.icaai.org) 5. Publication on GST by the Central Board of Excise and Customs (www.cbec.org). 6. Nitya Tax Associates Basics of GST Taxman						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 39 Content

Programme/Class: Degree

Year: Third

Semester: Sixth

Course/ paper-16 (B)

Course Code: F010601T

Course Title: Goods & Service Tax

Course outcomes:

The aim of the course is to build knowledge and understanding about GST among the student.

The course seeks to give detailed knowledge about the subject matter by instilling them basic ideas about GST. The outcome of the course will be as follows –

To provide knowledge about indirect taxes before GST.

To provide knowledge about registration and documentation process under GST.

To give an overview about tax exemptions.

To give an overview about filing of GSTR.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of

Lectures

Total=30

I

Introduction : Constitutional framework of Indirect Taxes before GST(Taxation Powers of Union & State Government); Concept of VAT: Meaning, Variants and Methods; Major Defects in the structure of Indirect Taxes prior to GST; Overview of GST ; Structure of GST (SGST, CGST, UTGST & IGST); GST Council.

7

II

Supply of Goods and Services - Definition of supply; Place of Supply: Intra-State and Inter-State supply; Composite and Mixed supply; Import and Export; Supplies of goods and services liable to be reverse charged; Time of supply; Nil rated supplies, Zero rated supplies, Exempted supplies & Non-GST supplies.

7

III

Registration and Documentation:

(A) Registration-Persons

Page 40 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Course/ paper-17 (A)						
Course Code: F010602T		Course Title: Auditing				
Course outcomes: The aim of the course is to build knowledge and understanding about Auditing among the student. The outcome of the course will be as follows – To provide knowledge about Auditing and its different types. To provide knowledge about audit procedure and audit of limited companies. Students will get an overview about special audit recent trends in auditing.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
I	Introduction: Meaning and objectives of Auditing, Types of Audit, Internal Audit, Audit Programme, Audit Notebook, Routine Checking and Test Checking					6
II	Internal Check System: Internal Control, Audit Procedure: Vouching, Verification of Assets and Liabilities.					7
III	Audit of Limited Companies: Company Auditor - Appointment, Powers, Duties and Liabilities. Auditor's Report and Audit Certificate.					7
IV	Special Audit: Audit of Banking Companies, Audit of Insurance Companies, Audits of Educational Institutions, Audit of Cooperative Societies, Efficiency Audit, Social Audit etc. Recent trends in Auditing: Nature and Significance of Cost Audit, Tax Audit, Management Audit					10
Suggested Readings: 1. Basu B.K., An insight with Auditing 2. Gupta Kamal, Contemporary Auditing						

Programme/Class s: Degree			Year: Third		Semester: Sixth	
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 40 Content

Programme/Class: Degree

Year: Third

Semester: Sixth

Course/ paper-17 (A)

Course Code: F010602T

Course Title: Auditing

Course outcomes:

The aim of the course is to build knowledge and understanding about Auditing among the student. The outcome of the course will be as follows –

To provide knowledge about Auditing and its different types.

To provide knowledge about audit procedure and audit of limited companies.

Students will get an overview about special audit recent trends in auditing.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Introduction: Meaning and objectives of Auditing, Types of Audit, Internal Audit, Audit Programme, Audit Notebook, Routine Checking and Test Checking

6

II

Internal Check System: Internal Control, Audit Procedure: Vouching, Verification of Assets and Liabilities.

7

III

Audit of Limited Companies: Company Auditor - Appointment, Powers, Duties and Liabilities. Auditor's Report and Audit Certificate.

7

IV

Special Audit: Audit of Banking Companies, Audit of Insurance Companies, Audits of Educational Institutions, Audit of Cooperative Societies, Efficiency Audit, Social Audit etc.

Recent trends in Auditing: Nature and Significance of Cost Audit, Tax Audit, Management Audit

10

Suggested Readings:

1. Basu B.K., An insight with Auditing
2. Gupta Kamal, Contemporary Auditing

Suggested Con

Page 41 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Course/ paper-17 (B)						
Course Code: F010602T		Course Title: International Trade				

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Course outcomes: The aim of the course is to build knowledge and understanding about International Trade among the student. The outcome of the course will be as follows – ■ To provide knowledge about different methods of international trade. ■ To provide knowledge about international economic institutions. ■ Students will get an overview about India foreign trade and India's trade policy.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30
I	Introduction: Basics of international trade, international trade theories, drivers of international trade, restraining forces, recent trends in world trade.					6
II	Foreign trade: Foreign trade & economic growth, balance of trade, balance of payments, free trade, forms and restrictions.					7

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
III	International economic institutions: IMF, World Bank, WTO (in brief), Regional economic groupings - NAFTA, EU, ASEAN, SAARC.					7
IV	India's foreign trade: Recent trends in India's foreign trade, institutional infrastructure for export promotion in India, projects & consultancy exports. Trade Policy: India's Trade policy, export assistance, marketing plan for exports.					10
Suggested Readings: 3. Varshney & Bhattacharya, International Marketing						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						

Programme/Class s: Degree			Year: Third		Semester: Sixth	
Further Suggestions:						

Page 41 Content

Programme/Class: Degree

Year: Third

Semester: Sixth

Course/ paper-17 (B)

Course Code: F010602T

Course Title: International Trade

Course outcomes:

The aim of the course is to build knowledge and understanding about International Trade among the student. The outcome of the course will be as follows –



To provide knowledge about different methods of international trade.



To provide knowledge about international economic institutions.



Students will get an overview about India foreign trade and India's trade policy.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Introduction: Basics of international trade, international trade theories, drivers of international trade, restraining forces, recent trends in world trade.

6

II

Foreign trade: Foreign trade & economic growth, balance of trade, balance of payments, free trade, forms and restrictions.

7

III

International economic institutions: IMF, World Bank, WTO (in brief), Regional economic groupings - NAFTA, EU, ASEAN, SAARC.

7

IV

India's foreign trade: Recent trends in India's foreign trade, institutional infrastructure for export promotion in India, projects & consultancy exports. Trade Policy: India's Trade policy, export assistance, marketing plan for exports.

10

Suggested Readings:

3. Varshney & Bhattacharya, International Marketing

Suggested Continuous Ev

Page 42 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Course/ paper-18 (A)						
Course Code: F010603T		Course Title: Strategic Management				

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Course outcomes: The aim of the course is to build knowledge and understanding about Strategic Management among the student. The outcome of the course will be as follows – ■ To develop analytical and administrative skills required for implementation of strategy. ■ To understand the Strategic management process and variables to address in change process ■ To relate the considerations involved in designing organizational structure ■ To assess organizational culture and apply best practices to deve						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
I	Interrelationship of formulation and implementation of strategy. Strategic management Process. Key tasks in implementing strategy, Role of implementers at different levels. Principal managerial components of strategy execution process- Building capable organization, staffing the organization, building core competencies and competitive capabilities, Matching organization structure to strategy-Value chain activities to perform /outsource, Centralization Vs decentralization in decision making, Prov					6
II	Managing internal operations: marshalling resources, instituting policies and procedures, adopting best practices for continuous improvement, TQM and six sigma quality programmes, installing information and operating systems, Tying rewards and incentives to execution					8

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
III	Corporate Culture and leadership: perpetuating the culture, aspects of healthy, unhealthy culture and adaptive culture, changing a problem culture – symbolic and substantial culture changing actions. Grounding the culture in core values and ethics					8
IV	Strategic Evaluation and Control- Types of organizational controls – feed forward, real time and feedback, meaning of Strategic and operational controls and their types, steps in strategic control process, Evaluation techniques for strategic control, Evaluation techniques for operational control, Balance Score card					8

Programme/Class s: Degree			Year: Third		Semester: Sixth	
Suggested Readings: 7. Lawrence, R. Jauch and William F. Glueck; Strategic Management and Business Policy, - McGraw – Hill 8. Wheelen & Hunger, Concepts in Strategic Management and Business Policy, 12th edition, Pearson Education. 9. Kazmi, Azhar, (2008), Strategic Management and Business Policy, 3rd Edition, McGraw Hill Education. 10. R. Srinivasan, Strategic Management the Indian context, Prentice Hall of India 11. L. M. Prasad – Strategic Management – Sultan Chand						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, Group Discussions. This will instill in student a sense of decision making and practical learning.						

Page 42 Content

Programme/Class: Degree

Year: Third

Semester: Sixth

Course/ paper-18 (A)

Course Code: F010603T

Course Title: Strategic Management

Course outcomes:

The aim of the course is to build knowledge and understanding about Strategic Management among the student. The outcome of the course will be as follows –



To develop analytical and administrative skills required for implementation of strategy.



To understand the Strategic management process and variables to address in change process



To relate the considerations involved in designing organizational structure



To assess organizational culture and apply best practices to develop strategy supportive culture



To appreciate the importance of values and ethics in organizational culture.



To exercise strategic leadership in effecting change.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Interrelationship of formulation and implementation of strategy.

Strategic management Process. Key tasks in implementing strategy,

Role of implementers at different levels.

Principal managerial components of strategy execution process-

Building capable organization, staffing the organization, building core competencies and competitive capabilities, Matching organization

structure to strategy-Value chain activities to perform /outsource,

Centralization Vs decentralization in decision mak

Page 43 Content

Suggested equivalent online courses:

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Further Suggestions:

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Page 44 Data Table

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Course/ paper-18 (B)						
Course Code: F010603T		Course Title: Training and Development				
Course outcomes: ■ The field of Training and Development and its role in opti mizing performanc e. ■ Applying theoretical concepts and models to training design.						
■ Designing training interventions using a variety of methodologies. ■ Evaluating the effectiveness of training & development interventions. ■ Assessing whether training & development is a viable career option.						
Credits: 3				Compulsory		
Max. Marks: 25+75				Min. Passing Marks:		
Total No. of Lectu res-Tutorials-Prac tical (in hours per week): L-T-P: 2-0-0						
Unit	Topics					No. of Lectures Total=30

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
I	Introduction: Concepts and Rationale of Training and Development; Difference between Training, Development & Education, overview of training and development systems; organizing training department; training and development policies; Requisites of Effective Training.					7
II	Training Needs Assessment (TNA): Meaning of TNA, Purpose and Methods of TNA, the Need Assessment Process – Organizational Analysis, Person Analysis, Task Analysis, Output of TNA. Learning Theories.					7

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
III	Designing, Conducting & Evaluation of Training Program: Areas of training, Types of training, System's Approach to Training, Training Methods, Designing a training program, contents & scheduling, study material, selecting a trainer, deciding method of training, Types of Teaching Aids in Training, Training Evaluation & Methods of Training Evaluation, Training Effectiveness Models - Kirkpatrick Model of Training Effectiveness, CIRO Model.					8
IV	Executive Development: Importance of Executive Development, Steps in the organization of a management Development Program/ Executive Development Program, Methods/ Techniques of Management Development Program, Special Issues in Training & Development – Legal Issues, Cross Cultural Preparation, Managing Workforce Diversity, Sensitivity Training, Succession Planning.					8

Programme/Cla s: Degree			Year: Third		Semester: Sixth	
Suggested Readings: 12. Noe, Raymond A., and Amitabh Deo Kodwani, Employee Training and Development, Tata McGraw Hill, 5th Edition, 2012. 13. Rao VSP, Human Resource Management, Excel Books Publication, 3rd Edition. 2013. 14. Rolf, P., and Udai Pareek, Training for Development, Sage Publications Pvt. Ltd. 15. Jack J. Phillips, Hand book of Training Evaluation and Measurement Methods, Routledge. 16. Dayal, Ishwar, Management Training in Organisations, Prentice Hal						
Suggested Continuous Evaluation Methods: In addition to the theoretical inputs the course will be delivered through Assignments, Presentation, and Group Discussions. This will instill in student a sense of decision making and practical learning.						
Suggested equivalent online courses:						
Further Suggestions:						

Page 44 Content

Programme/Class: Degree

Year: Third

Semester: Sixth

Course/ paper-18 (B)

Course Code: F010603T

Course Title: Training and Development

Course outcomes:



The field of Training and Development and its role in optimizing performance.



Applying theoretical concepts and models to training design.



Designing training interventions using a variety of methodologies.



Evaluating the effectiveness of training & development interventions.



Assessing whether training & development is a viable career option.

Credits: 3

Compulsory

Max. Marks: 25+75

Min. Passing Marks:

Total No. of Lectures-Tutorials-Practical (in hours per week): L-T-P: 2-0-0

Unit

Topics

No. of Lectures

Total=30

I

Introduction: Concepts and Rationale of Training and Development; Difference between Training, Development & Education, overview of training and development systems; organizing training department; training and development policies; Requisites of Effective Training.

7

II

Training Needs Assessment (TNA): Meaning of TNA, Purpose and Methods of TNA, the Need Assessment Process – Organizational Analysis, Person Analysis, Task Analysis, Output of TNA. Learning

Theories.

7

III

Designing, Conducting & Evaluation of Training Program: Areas of training, Types of training, System's Approach to Training, Training Methods, Designing a training program, contents & scheduling, study material, selecting a trainer, deciding method of training, Types of Teaching Aids in Training, Trai

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Bca Syllabus 2021 New (Official Document)

Source Document URL: <https://www.vsicskanpur.org/docs/BCA-Syllabus-2021-new.pdf>

Total Document Pages: 45

Page 1 Content

C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Semester – wise breakup of course

Semester-Ist

Course Code

Course Name

L

T

P

C

BCA-1001

Computer Fundamental & Problem solving

Techniques

3

0

0

3

BCA-1002

C Programming

3

0

0

3

BCA-1003

Principle of Management

4

0

0

4

BCA-1004

Business Communication

3

1

0

4

BCA-1005

Mathematics –I

4

0

0

4

BCA-1001P

Computer Laboratory and Practical Work of Office

Automation

0

0

3

2

BCA-1002P

Computer Laboratory and Practical Work of

C Programming

0

0

3

2

22

Semester-IIInd

Course Code

Course Name

L

T

P

C

BCA-2001

Object Oriented Programming Using C++

3

0

0

3

BCA-2002

Internet Technology and Web Design

3

1

0

4

BCA-2003

Organization Behavior

4

0

0

4

BCA-2004

Financial Accounting & Management

3

1

0

4

BCA-2005

Mathematics II

4

0

0

4

BCA-2001P

Computer Laboratory and Practical Work of C++

Programming

0

0

6

3

22

=

Page 2 Content

C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Semester – wise breakup of course

Semester-IIIrd

Course Code

Course Name

L

T

P

C

BCA-3001

Python Programming

3

0

0

3

BCA-3002

Data Structure Using C & C++

3

0

0

3

BCA-3003

Operating System

3

1

0

4

BCA-3004

Digital Electronics & Computer Organization

3

1

0

4

BCA-3005

Elements of Statistics

3

1

0

4

BCA-3001P

Computer Laboratory and Practical Work of Python

0

0

3

2

BCA-3002P

Computer Laboratory and Practical Work of DS

0

0

3

2

22

Semester-IVth

Course Code

Course Name

L

T

P

C

BCA-4001

Computer Graphics & Animation

3

1

0

4

BCA-4002

Database Management System

3

0

0

3

BCA-4003

Software Engineering

3

1

0

4

BCA-4004

Optimization Techniques

3

1

0

4

BCA-4005

Mathematics-III

4

0

0

4

BCA-4001P

Computer Graphics & DBMS Laboratory

0

0

6

3

22

Page 3 Content

C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Semester – wise breakup of course

Semester-Vth

Course Code

Course Name

L

T

P

C

BCA-5001

Knowledge Management

3

1

0

4

BCA-5002

Java Programming and Dynamic Webpage

Design

3

0

0

3

BCA-5003

Computer Network

3

1

0

4

BCA-5004

Numerical Methods

3

1

0

4

BCA-5005

Minor Project

0

1

2

2

BCA-5006

Viva-Voice on Summer Training

0

0

2

1

BCA-5002P

Computer Laboratory and Practical Work of

Java Programming & Dynamic Webpage

Design

0

0

6

3

21

Semester-VIth

Course Code

Course Name

L

T

P

C

BCA-6001

Information & Cyber Security

4

0

0

4

BCA-6002

Internet Of Things

3

1

0

4

BCA-6003

E-Commerce

4

0

0

4

BCA-6004

Data Science and Machine Learning

3

1

0

4

BCA-6005

Major Project

0

3

6

5

BCA-6006

Presentation/Seminar based on Major Project

1

22

Page 4 Content

C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Semester – wise breakup of course

Semester-Ist

Course Code

Course Name

External

Internal

Total

L

T

P

C

BCA-1001

Computer Fundamental &

Problem solving

techniques

75

25

100

3

0

0

3

BCA-1002

C Programming

75

25

100

3

0

0

3

BCA-1003

Principle of Management

75

25

100

4

0

0

4

BCA-1004

Business Communication

75

25

100

3

1

0

4

BCA-1005

Mathematics -I

75

25

100

4

0

0

4

BCA-1001P

Computer Laboratory and

Practical Work of Office

Automation

-

-

50

0

0

3

2

BCA-1002P

Computer Laboratory and

Practical Work of C

Programming

-

-

50

0

0

3

2

600

22

Semester-IIInd

Course Code

Course Name

External

Internal

Total

L

T

P

C

BCA-2001

Object Oriented

Programming Using C++

75

25

100

3

0

0

3

BCA-2002

Internet Technology and

Web Design

75

25

100

3

1

0

4

BCA-2003

Organization Behavior

75

25

100

4

0

0

4

BCA-2004

Financial Accounting &

Management

75

25

100

3

1

0

4

BCA-2005

Mathematics II

75

25

100

4

0

0

4

BCA-2001P

Computer Laboratory

and Practical Work of

C++ Programming

-

-

100

0

0

6

3

600

22

Semester-IIIrd

Page 5 Content

C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Semester – wise breakup of course

Course Code

Course Name

External

Internal

Total

L

T

P

C

BCA-3001

Python Programming

75

25

100

3

0

0

3

BCA-3002

Data Structure Using C

& C++

75

25

100

3

0

0

3

BCA-3003

Operating System

75

25

100

3

1

0

4

BCA-3004

Digital Electronics &

Computer Organization

75

25

100

3

1

0

4

BCA-3005

Elements of Statistics

75

25

100

3

1

0

4

BCA-3001P

Computer Laboratory and

Practical Work of Python

-

-

50

0

0

3

2

BCA-3002P

Computer Laboratory and

Practical Work of DS

-

-

50

0

0

3

2

600

22

Semester-IVth

Course Code

Course Name

External

Internal

Total

L

T

P

C

BCA-4001

Computer Graphics &

Animation

75

25

100

3

1

0

4

BCA-4002

Database Management

System

75

25

100

3

0

0

3

BCA-4003

Software Engineering

75

25

100

3

1

0

4

BCA-4004

Optimization Techniques

75

25

100

3

1

0

4

BCA-4005

Mathematics-III

75

25

100

4

0

0

4

BCA-4001P

Computer Graphics &

DBMS Laboratory

100

0

0

6

3

-

-

600

22

Page 6 Content

C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Semester – wise breakup of course

Semester-Vth

Course Code

Course Name

External

Internal

Total

L

T

P

C

BCA-5001

Knowledge

Management

75

25

100

3

1

0

4

BCA-5002

Java Programming and

Dynamic Webpage Design

75

25

100

3

0

0

3

BCA-5003

Computer Network

75

25

100

3

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BCA-5004

Numerical Methods

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BCA-5005

Minor Project

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BCA-5006

Viva-Voice on Summer

Training

BCA-5002P

Computer Laboratory and

Practical Work of Java

Programming & Dynamic

Webpage Design

600

21

Semester-VIth

Course Code

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BCA-6006

Presentation/Seminar

based on Major Project

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22

BCA-6001

Information and Cyber

Security

BCA-6002

Internet Of Things

75

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BCA-6003

E-Commerce

75

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BCA-6004

Data Science and Machine

Learning

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BCA-6005

Major Project

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Page 7 Content

C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-1001

Computer Fundamental & Problem solving techniques

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UNIT-I

Introduction to Computers

Introduction, Characteristics of Computers, Block diagram of computer. Types of computers and features, Mini Computers, Micro Computers, Mainframe Computers, Super Computers. Types of Programming Languages (Machine Languages, Assembly Languages, High Level Languages). Data Organization, Drives, Files, Directories, Number Systems Introduction to Binary, Octal, Hexadecimal system Conversion, Binary Arithmetic Simple Addition, Subtraction, Multiplication

UNIT-II

Memory Organization

Types of Memory (Primary And Secondary) RAM, ROM, PROM, EPROM. Secondary Storage Devices (FD, CD, HD, Pen drive) I/O Devices (Scanners, Plotters, LCD, Plasma Display). Cache, Virtual memory, RAID.

UNIT-III

Operating System and Services in O.S.

History, Files and Directories, DOS (Internal and External Commands), Batch Files, Types of Operating System, File Management System. Introduction to Linux – Features of Linux , Components of Linux

UNIT-IV

Problem solving techniques

Understanding the problem, Analyzing the problem, Developing the solution, Algorithm and Flowcharts

- Definition, Characteristics, Expressing Algorithms, Analysis of Algorithms, Advantages and disadvantages, Examples Flowchart: Definition, Define symbols of flowchart, Limitations of Using Flowc

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-1002

C Programming

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UNIT-I

Fundamentals of C programming and Control Structures: History, Structure of a C program, C Conventions, Character Set, Identifiers, Keywords, Simple Data types, Modifiers, Variables, Constants, Operators, Operator precedence. Input and Output operation: Single character input and output, formatted input and output. Control Structures, Conditional statement and switch statement. Goto statement. Looping statement, break and continue, nested for statement.

UNIT-II

Arrays and Functions: Introduction (One and multi-dimensional), Declaration of arrays, Initialization of arrays, processing with arrays. String manipulation, declaration of string arrays, string operations. Functions: Introduction, advantages of functions, Function definition, function call, Actual and formal arguments, local and global variables, function prototypes, types of functions, recursive functions, arrays and functions.

UNIT-III

Searching and Sorting: selection sort, bubble sort, insertion sort, quick sort, merge sort Searching: linear and binary search methods, comparison of sorting and searching methods.

UNIT-IV

Structures and Pointers: Introduction to structures, Advantages of structures, accessing elements of a structure, nested structures, array of structures, functions and structures, Pointers: Introduction, pointer variable, pointer

Page 9 Content

C.S.J.M. University, Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-1003

Principle of Management

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UNIT-I

Nature of Management: Meaning, Definition, nature purpose, importance & Functions, Management as Art, Science & Profession- Management as social System Concepts of management-Administration- Organization, Management Skills, Levels of Management.

UNIT-II

Evolution of Management Thought: Contribution of F.W.Taylor, Henri Fayol, Elton Mayo, Chester Bernard & Peter Drucker to the management thought. Business Ethics, Social Responsibility of business.

UNIT-III

Functions of Management: Part-I Planning – Meaning- Need & Importance, types, Process of Planning, Barriers to Effective Planning, levels – advantages & limitations. Forecasting- Need & Techniques Decision making-Types - Process of rational decision making & techniques of decision making Organizing – Elements of organizing & processes: Types of organizations, Delegation of authority – Need, difficulties Delegation – Decentralization Staffing – Meaning & Importance Direction – Nature – Principles.

UNIT-IV

Functions of Management: Part-II Motivation – Importance – theories, Leadership – Meaning –styles, qualities & function of leader, Controlling - Need, Nature, importance, Process & Techniques, Total Quality Management Coordination – Need – Importance.

UNIT – V

Management of Change: Meaning, Features of change, Force for Change, Models for

Page 10 Content

C.S.J.M.University,Kanpur.

Bachelors of Computer Application

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BCA-1004

Business Communication

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UNIT-I

Means of Communication: Meaning and Definition – Process – Functions – Objectives – Importance – Essentials of good communication – Communication barriers, 7C's of Communication.

UNIT-II

Types of Communication: Oral Communication: Meaning, nature and scope – Principle of effective oral communication – Techniques of effective speech – Media of oral communication (Face-to-face conversation – Teleconferences – Press Conference –Video Conferencing– Demonstration – Radio Recording –Meetings –Grapevine – Group Discussion – Mobile Phone Conversation– Oral report). The art of listening – Principles of good listening.

UNIT-III

Written Communication: Purpose of writing, Clarity in Writing, Principles of Effective writing, Writing an e-mail, SMS.

UNIT-IV

Business Letters & Reports: Need and functions of business letters – Planning & layout of business letter – Kinds of business letters – Essentials of effective correspondence, Purpose, Kind and Objective of Reports, Writing Reports.

UNIT-V

Drafting of business letters : Enquiries and replies – Placing and fulfilling orders – Complaints and follow-up Sales letters – Circular letters Application for employment and resume

Referential Books :

1. Business Communication – K.K.Sinha – Galgotia Publishing Company, New Delhi.

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-1005

Mathematics –I

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UNIT-I

Matrices and Determinants: Matrix, Types of matrices, Addition, subtraction, scalar multiplication of a matrix, product of two matrices, Determinants of a square matrix, Co-factor of element of a square matrix, Adjoint, Inverse of a Square Matrix, Cayley Hamilton theorem (statement only) and problems.

UNIT-II

Limits and Continuity: Limit at a Point, Properties of Limit, Computation of Limits of Various Types of Functions, Indeterminate Forms, L' Hospitals Rule, Continuity at a Point, Continuity Over an Interval.

UNIT-III

Differentiation: Derivatives of Sum, Differences, Product & Quotients, Chain Rule, Derivatives of Composite Functions, Logarithmic Differentiation, Rolle's Theorem, Mean Value Theorem), Maxima & Minima. Taylor's and Maclaurin's Theorem

UNIT-IV

Integration: Fundamental Theorem of Calculus (without proof.), Indefinite Integrals, Methods of Integration Substitution, By Parts, Partial Fractions.

UNIT-V

Vector Algebra: Definition of a vector in 2 and 3 Dimensions; Double and Triple Scalar and Vector Product.

Referential Books :

1. B.S. Grewal, "Elementary Engineering Mathematics", 34th Ed., 1998.
2. "Advanced Engineering Mathematics", S. Chand & Company, 9th Revised Edition, 2001.
3. Shanti Narayan, "Integral Calculus", S. Chand & Company, 1999.
4. Shanti Narayan, "Differentia

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

L T P C

BCA-1001P

Laboratory Computer Fundamental & Problem
solving techniques

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

L T P C

BCA-1002P

Laboratory and Practical Work of C Programming

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

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BCA-2001

Object Oriented Programming Using C++

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UNIT-I

Introduction Introducing Object – Oriented Approach, Relating to other paradigms {Functional, Data decomposition}. Basic terms and ideas Abstraction, Encapsulation, Inheritance, Polymorphism, Review of C, Difference between C and C++ - cin, cout, new, delete, operators.

UNIT-II

Classes and Objects Encapsulation, information hiding, abstract data types, Object & classes, attributes, methods, C++ class declaration, State identity and behaviour of an object, Constructors and destructors, instantiation of objects, Default parameter value, object types, C++ garbage collection, dynamic memory allocation, abstract classes.

UNIT-III

Inheritance and Polymorphism Inheritance, Class hierarchy, derivation – public, private & protected, Aggregation, composition vs classification hierarchies, Polymorphism, Categorization of polymorphism techniques, Method polymorphism, Operator overloading.

UNIT-IV

Generic function Template function, function name overloading, Overriding inheritance methods,
Run time polymorphism, Multiple Inheritance.

UNIT-V

Files and Exception Handling Streams and files, Exception handling

Referential Books:

1. A.R.Venugopal, Rajkumar, T. Ravishanker "Mastering C++", TMH, 1997.
2. S.B.Lippman & J.Lajoie, " C++ Primer", 3rd Edition, Addison Wesley, 200

Page 15 Content

C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-2002

Internet Technology and Web Design

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UNIT-I

Introduction to Internet: Internet, Growth of Internet, Owners of the Internet, Anatomy of Internet,
ARPANET and Internet history of the World Wide Web, basic Internet Terminology, Net etiquette. Internet
Applications – Commerce on the Internet, Governance on the Internet, Impact of Internet on Society –
Crime on/through the Internet.

UNIT-II

Internet Connectivity & Network: Connectivity types: level one, level two and level three connectivity,
modem, dedicated connections through the telephone system, ISDN, Protocol options – Shell, SLIP, PPP,
Service options – E-mail, WWW, News Firewall etc. Network definition, Common terminologies: LAN,
WAN, Node, Host, Workstation, bandwidth, Interoperability, Network administrator, network security.

UNIT-III

Internet Security Management Concepts: Overview of Internet Security, Firewalls, Internet Security,
Management Concepts and Information Privacy.

UNIT-IV

Introduction to Java: The JDK Directory Structure, Java History; Java Features; Structure of Java
Program; Compiling and Interpreting Applications; Java Tokens; Java Character set; Keywords and

Identifiers, Primitive Data types Declarations, Non-Primitive data types; Operators and Expressions;
Implicit and Explicit Type Conversions: The Cast Operator; Control Statements: If- else – if

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code

BCA-2003

Course Name

Organization Behavior

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UNIT-I

Fundamentals of Organizational Behaviour : Nature, Scope, Definition, Fundamental Concepts of Organizational Behaviour; Models of Organizational Behaviour; Emerging aspects of Organizational Behaviour: Meaning Cultural Diversity

UNIT-II

Perception, Attitude, Values and Motivation: Concept, Nature, Process, Importance, Management, Behavioural aspect of Perception. Effects of employee attitudes; Job Satisfaction; Nature and Importance of Motivation; Achievement Motive; Theories of Work Motivation: Maslow's Need Hierarchy Theory, McGregor's Theory 'X' and Theory 'Y'

UNIT-III

Personality : Definition of Personality, Determinants of Personality; Theories of Personality- Trait and Type Theories, The Big Five Trait Theory, Myres-Briggs Indicator; Locus of Control, Type A and Type B Theory of Personality

UNIT-IV

Work Stress : Meaning and definition of Stress, Symptoms of Stress; Sources of Stress: Individual Level, Group Level, Organizational Level; Stressors, Extra Organizational Stressors; Effect of Stress – Burnouts; Stress Management – Individual Strategies, Organizational Strategies

UNIT-V

Group Behaviour and Leadership : Nature of Group, Types of Groups; Nature and Characteristics of team; Team Building, Effective Teamwork; Nature of Leadership, Leadership Styles; Traits of Effect

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-2004

Financial Accounting &Management

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UNIT-I

Overview - Meaning and Nature of Financial Accounting, Scope of Financial Accounting, Financial Accounting & Management Accounting, Accounting concepts & convention, Accounting standards in India.

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UNIT-II

Basics of accounting – Capital & Revenue items, Application of Computer in Accounting Double Entry System, Introduction to Journal, Ledger and Procedure for Recording and Posting, Introduction to Trial Balance, Preparation of Final Account, Profit & Loss Account and related concepts, Balance Sheet and related concept. Ratio analysis.

UNIT-III

Definition nature and Objective of Financial Management, Long Term Sources of Finance, Introductory idea about capitalization, Capital Structure, Concept of Cost of Capital, introduction, importance, explicit & implicit cost, Measurement of cost of capital, cost of debt.

UNIT-IV

Concept & Components of working Capital. Factors Influencing the Composition of working Capital, Objectives of working Capital Management – Liquidity Vs. Profitability and working capital policies. Theory of working capital: Nature and concepts

UNIT-V

Cash Management, Inventory Management and Receivables Management.

Referential Books:

1. Maheshwari & Maheshwari, "An Introduction to Accountancy", 8th Edition, Vikas Publishing House, 2003

2. Gupta R.L., Gupta V.K.

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-2005

Mathematics II

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UNIT-I

SETS

Sets, Subsets, Equal Sets Universal Sets, Finite and Infinite Sets, Operation on Sets, Union, Intersection and Complements of Sets, Cartesian Product, Cardinality of Set, Simple Applications.

UNIT-II

RELATIONS AND FUNCTIONS

Properties of Relations, Equivalence Relation, Partial Order Relation Function: Domain and Range, Onto, Into and One to One Functions, Composite and Inverse Functions.

UNIT-III

PARTIAL ORDER RELATIONS AND LATTICES

Partial Order Sets, Representation of POSETS using Hasse diagram, Chains, Maximal and Minimal Point, Glb, lub, Lattices & Algebraic Systems, Principle of Duality, Basic Properties, Sublattices, Distributed & Complemented Lattices.

UNIT-IV

FUNCTIONS OF SEVERAL VARIABLES

Partial Differentiation, Chain Rule, Extrema of Functions of 2 Variables, Euler's Theorem.

UNIT-V

MULTIPLE INTEGRATION

Double Integral in Cartesian and Polar Coordinates to find Area, Change of Order of Integration, Triple Integral to Find Volume of Simple Shapes in Cartesian Coordinates.

Referential Books:

1. Kolman, Busby and Ross, "Discrete Mathematical Structure", PHI,1996.
2. S.K. Sarkar, "Discrete Maths"; S. Chand & Co.,2000
3. "Discrete Mathematics", Schaum's Outlines

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

CourseCode

CourseName

L T P C

BCA-2001P

Computer Laboratory and Practical Work of

C++ Programming

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Practical will be based on Paper Object Oriented Programming Using C++: Covers

UNIT-III, UNIT-IV, UNIT-V of Syllabus

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

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BCA-3001

Python Programming

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UNIT-I

Python: Features of Python, Environmental setup, Installation and tools required for running, Basic Types

Variable types and operators : Assigning values to variables Multiple Assignments Standard Data Types Set

Map Single line comments using Multi-line comments using triple quote, Data Type Conversion Operators,

Types of Operator, Conditional statement, Looping statements with else-Pass-Break continue.

UNIT-II

Number and List: Accessing values in List-Delete, update List element-Basic List operations-Indexing,

Slicing and Matrices Built in methods and Functions for List-Accessing values in Tuple-Delete, List element-Basic Tuple operations Indexing, Slicing and Matrices Built in methods and Functions for Tuple.

UNIT-III

Dictionary and Function: Accessing values in Dictionary-Updating Dictionary-Deleting Dictionary – elements-Properties of Dictionary keys-Built in Dictionary Functions and Methods Defining Function- Calling function- Pass by reference vs value Function Arguments-Required arguments-Key word arguments-Default arguments-Variable-length arguments Recursion.

UNIT-IV

Modules and Packages: The Time Module and its functions-Calendar modules and its functions-Other modules and Functions Sum and Difference f time and date Import From import statement From import statement Executing modules, Local

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

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BCA-3002

Data Structure Using C &C++

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UNIT-I

Introduction to Data Structure and its Characteristics

Array

Representation of single and multidimensional arrays; Sprase arrays – lower and upper triangular matrices and Tridiagonal matrices with Vector Representation also.

UNIT-II

Stacks and Queues

Introduction and primitive operations on stack; Stack application; Infix, postfix, prefix expressions;

Evaluation of postfix expression; Conversion between prefix, infix and postfix, introduction and primitive operation on queues, D- queues and priority queues.

UNIT-III

Lists

Introduction to linked lists; Sequential and linked lists, operations such as traversal, insertion, deletion searching, Two way lists and Use of headers

UNIT-IV

Trees

Introduction and terminology; Traversal of binary trees; Recursive algorithms for tree operations such as traversal, insertion, deletion; Binary Search Tree

UNIT-V

Graphs: Graph terminology, Representation of graphs, path matrix, BFS (breadth first search), DFS (depth first search), topological sorting, Warshall's algorithm (shortest path algorithm.)

Referential Books:

1. E.Horowitz and S.Sahani, " Fundamentals of Data structures", Galgotia Book source Pvt. Ltd., 2003
2. R.S.Salaria, " Data Structures & Algorithms" , Khanna Book Publishing Co. (P)Ltd.,2002
3. Y.Langsam et. Al., " Data Structures

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

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BCA-3003

Operating System

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UNIT-I

Introduction, What is an operating system, Simple Batch Systems, Multi-programmed Batch systems, Time- Sharing Systems, Personal – Computer Systems, Parallel systems, Distributed systems, Real- Time Systems.

Memory Management: Background, Logical versus physical Address space, swapping, Contiguous allocation, Paging, Segmentation

Virtual Memory: Demand Paging, Page Replacement, Page- replacement Algorithms, Performance of Demand Paging, Allocation of Frames, Thrashing, Other Considerations

UNIT-II

Processes: Process Concept, Process Scheduling, Operation on Processes

CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Multiple – Processor Scheduling.

UNIT-III

Deadlocks: System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock

UNIT-IV

Device Management: Techniques for Device Management, Dedicated Devices, Shared Devices, Virtual Devices; Input or Output Devices, Storage Devices, Buffering.

UNIT-V

Information Management: Introduction, A Simple File system, General Model of a File System, Symbolic File System, Basic File System, Access Control Verification, Logical File System, Physical File system File – System Interface; File Concept, Access Methods, Directory Structure, Pr

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

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BCA-3004 Digital Electronics & Computer Organization

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UNIT-I

Number System & Boolean Algebra Number System: Binary, Octal, Decimal, Hexadecimal; Conversion of Number System; Binary Arithmetic & Complement, Binary Codes: Weighted & Non Weighted, Gray Code, Excess-3 Code. Boolean Function; Boolean Postulates; De-Morgan's Theorem; Boolean Expressions: Sum of Product, Product of Sum, Minimization of Boolean Expressions using K-Map; Logic Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR; Implementations of Logic Functions using Gates; NAND- NOR Implementations; Multilevel gate Implementations.

UNIT-II

Combinational Circuits Adders & Subtractors: Half Adder, Full Adder, Binary Adder, Half

Subtractor, Full Subtractor, Adder Subtractor; Magnitude Comparator: Two Bit Magnitude Comparator, Three Bit Magnitude Comparator; Multiplexer & De-Multiplexer: 4*1 Multiplexer, 8*1 Multiplexer; Decoder & Encoder; Parity Checker & Generator; Code Converter.

UNIT-III

Sequential Circuit: Introduction to Flip Flops: SR, JK, T, D, Master Slave Flip Flops; Conversion of Flip Flops; Characteristic Table & Equation; Edge Triggering & Level Triggering; Excitation Table; State Diagram; State Table; State Reduction; Design of Sequential Circuits.

UNIT-IV

Registers Introduction of Registers; Classification of Registers; Register with Parallel Load; Shift Register

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

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BCA-3005

Elements of Statistics

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UNIT-I

Population, Sample and Data Condensation

Definition and scope of statistics, concept of population and sample with Illustration, Raw data, attributes and variables, classification, frequency distribution, Cumulative frequency distribution.

UNIT-II

Measures of Central Tendency

Concept of central Tendency, requirements of a good measures of central tendency, Arithmetic mean, Median, Mode, Harmonic Mean, Geometric mean for grouped and ungrouped data.

UNIT-III

Measures of Dispersion:

Concept of dispersion, Absolute and relative measure of dispersion, range variance, Standard deviation, Coefficient of variation.

UNIT-IV

Permutations and Combinations

Permutations of 'n' dissimilar objects taken 'r' at a time (with or without repetitions). $nPr = n!/(n-r)!$ (without proof). Combinations of 'r' objects taken from 'n' objects. $nCr = n!/(r!(n-r)!)$ (without proof) .

Simple examples, Applications.

UNIT-V

Sample space, Events and Probability

Experiments and random experiments, Ideas of deterministic and non-deterministic experiments; Definition of sample space, discrete sample space, events; Types of events, Union and intersections of two or more events, mutually exclusive events, Complementary event, Exhaustive event; Simple examples.

Classical definition of probability, Addition theorem of probab

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

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L T PC

BCA-3001P

Computer Laboratory and Practical Work of Python

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Practical will be based on Paper Python Programming: Covers UNIT-II, UNIT-III, UNIT-IV, UNIT-V of Syllabus

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-3002P

Computer Laboratory and Practical Work of DS

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Practical will be based on Paper Data Structure: Covers UNIT-III, UNIT-IV, UNIT-V of Syllabus

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-4001

Computer Graphics and Animation

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UNIT-I

Introduction: Interactive Computer Graphics, Advantages of Interactive Graphics, Representative Uses of Computer Graphics, Conceptual Framework for Interactive Graphics, Classification of Application Development of Hardware and software for computer Graphics.

UNIT-II

Scan Conversion: Scan Converting Lines, Scan Converting Circles, Scan Converting Ellipses.

Clipping: point clipping, Cohen-Sutherland line clipping Algorithm, Midpoint Subdivision Algorithm, polygon clipping (Sutherland-Hodgeman)

UNIT-III

Geometrical Transformation: 2D Transformation (translation, rotation, scaling, reflection and shearing), Homogeneous Coordinates and Matrix Representation of 2D Transformations, Successive and composite 2D Transformations, the Window-to-Viewport Transformations, Introduction to 3D Transformations Matrix.

UNIT-IV

Curves & Surfaces: Polygon Surfaces and polygon meshes, Quadratic and super quadrics surfaces, Spline curve and representation.

UNIT-V

Computer Animation: introduction, Application of animation, Morphing, Keyframe system, Motion specifications in Animation, Types of animation, Sequencing of Animation Design and Fundamental principles of animation.

Referential Books:

1. Foley, Van Dam, Feiner, Hughes, Computer Graphics Principles& practice,2000.
2. D.J. Gibbs & D.C. Ts

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-4002

Database Management System

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UNIT-I

Introduction: Characteristics of database approach, data models, DBMS architecture and data independence.

UNIT-II

E-R Modeling: Entity types, Entity set, attribute and key, relationships, relation types, roles and structural constraints, weak entities, enhanced E-R and object modeling, Sub classes; Super classes, inheritance, specialization and generalization.

UNIT-III

Data Normalization: Functional Dependencies, Normal form up to 5th normal form, Data base design using EER to relational language.

UNIT-IV

Relational Data Model: Relational model concepts, relational constraints, relational algebra

SQL: SQL queries, programming using SQL.

UNIT-V

Concurrency Control: Transaction processing, locking techniques and associated, database recovery, security and authorization. Recovery Techniques, Database Security

Referential Books:

1. Abraham Silberschatz, Henry Korth, S.Sudarshan, "Database Systems Concepts", 4th Edition, McGraw Hill,1997.
2. Jim Melton, Alan Simon, "Understanding the new SQL: A complete Guide", Morgan Kaufmann Publishers,1993.
3. A.K.Majumdar, P. Bhattacharya, "Database Management Systems", TMH,1996.
4. Bipin Desai, "An Introduction to database systems", Galgotia Publications,1991.

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-4003

Software Engineering

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UNIT-I

Software Engineering: Definition and paradigms, A generic view of software engineering.

UNIT-II

Requirements Analysis: Statement of system scope, isolation of top level processes and entities and their allocation to physical elements, refinement and review.

UNIT-III

Designing Software Solutions: Refining the software Specification; Application of fundamental design concept for data, architectural and procedural designs using software blue print methodology and object oriented design paradigm; Creating design document.

UNIT-IV

Software Implementation: Relationship between design and implementation, Implementation issues and programming support environment, Coding the procedural design, Good coding style.

UNIT-V

Software Maintenance: Maintenance as part of software evaluation, reasons for maintenance, types of maintenance (Perceptive, adoptive, corrective), designing for maintainability, techniques for maintenance.

UNIT-VI

Comprehensive examples using available software platforms/case tools, Configuration Management.

Referential Books:

1. K.K.Aggarwal & Yogesh Singh "Software engineering", 2nd Ed., New Age International 2005.
2. I.Sommerville, "Software Engineering", Addison Wesley, 2002.
3. James Peter, W. Pedrycz, "Software Engineering: An Engineering Approach" John Wiley & Sons.

Page 30 Data Table

Game theory		
Introduction, Two-person zero-sum game, pure strategies (Minmax and Maxmin principles),Mixed		
strategies, The rules principles of Dominance, Algebraic method to solve games without saddle		
point, Graphical method to solve the games.		

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-4004

Optimization Techniques

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UNIT-I

Linear programming

Central Problem of linear Programming various definitions included Statements of basic theorem and also their properties, simplex methods, primal and dual simplex method, transport problem.

Assignment problem and its solution. Graphical Method Formulation, Linear Programming Problem.

UNIT-II

Game theory

Introduction, Two-person zero-sum game, pure strategies (Minmax and Maxmin principles),Mixed strategies, The rules principles of Dominance, Algebraic method to solve games without saddle point, Graphical method to solve the games.

UNIT-III

Replacement Theory

Replacement of item that deteriorates replacement of items that fail. Group replacement and individual replacement.

UNIT-IV

PERT and CPM

Project management origin and use of PERT, origin and use of CPM, Applications of PERT and CPM, Project Network, Diagram representation, Critical path calculation by network analysis and critical path method (CPM).

UNIT-V

Job Sequencing

Introduction, solution of sequencing problem Johnson s algorithm for n jobs through 2 machines

Referential Books:

1. Gillet B.E. "Introduction to OperationResearch"
2. Taha,H.A. "Operation Research - anintroduction"
3. Kanti Swarup "OperationResearch"
4. S.D.Sharma "OperationResearch"
5. Hira & Gupta "OperationResearch"

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-4005

Mathematics-III

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UNIT-I

COMPLEX VARIABLES: Complex Number System, Algebra of Complex Numbers, Polar Form, Powers and Roots, Functions of Complex Variables, Elementary Functions.

UNIT-II

VECTOR CALCULUS: Differentiation of Vectors, Scalar and Vector Fields, Gradient, Directional Derivatives, Divergence and Curl and their Physical Meaning.

UNIT-III

FOURIER SERIES: Periodic Functions, Fourier series, Fourier Series of Even and Odd Functions, Half Range Series.

UNIT-IV

ORDINARY DIFFERENTIAL EQUATIONS OF FIRST ORDER: Variable- Separable

Method, Homogeneous Differential Equations, Exact Differential Equations, Linear Differential Equations, Bernoulli's Differential Equations, Differential Equations of First Order and First Degree by Integrating Factor.

UNIT-V

ORDINARY DIFFERENTIAL EQUATIONS OF SECOND ORDER:

Homogenous Differential Equations with Constant Coefficients, Cases of Complex Roots and Repeated Roots, Differential Operator, Solutions by Methods of Direct Formulae for Particular Integrals, Operator Method for Finding Particular Integrals, (Direct Formulae).

Referential Books:

1. A.B. Mathur and V.P. Jaggi, "Advanced Engineering Mathematics", Khanna Publishers, 1999.
2. H.K. Dass, "Advanced Engineering Mathematics", S. Chand & Co., 9th Revised Ed.

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-4001P

Computer Laboratory and Practical Work of Computer

Graphics and DBMS

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Practical will be based on Paper Computer Graphics and DBMS Lab

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-5001

Knowledge Management

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UNIT-I

Business Intelligence and Business Decisions: Modeling Decision Process; Decision support systems; Group decision support and Groupware Technologies.

UNIT-II

Executive Information and support Systems: Business Expert System and AI, OLTO & OLAP; Data Warehousing; Data Marts, Data Warehouse architecture; Tools for data warehousing.

UNIT-III

Multi- Dimensional analysis: Data mining and knowledge discovery; Data mining and Techniques; Data mining of Advance Databases.

UNIT-IV

Knowledge Management Systems: Concept and Structure KM systems, techniques of knowledge management appreciation & limitation.

Referential Books:

1. Decision support system, EIS, 2000
2. W.H.Inmon, "Building Data Warehousing", Willey,1998.
5. Han, Jiawei, Kamber, Michelinal, " Data Mining Concepts & Techniques", Harcourt India, 2001

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-5002

Java Programming and Dynamic Webpage Design

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UNIT-I

Java Programming: Data types, control structured, arrays, strings, and vector, classes (inheritance, package, exception handling) multithreaded programming.

UNIT-II

Java applets, AWT controls (Button, Labels, Combo box, list and other Listeners, menu bar) layout manager, string handling (only main functions)

UNIT-III

JDBC: JDBC Fundamentals, Establishing Connectivity and Working with Connection Interface, Working with Statements, Creating and Executing SQL Statements, Working with ResultSet Objects.

UNIT-IV

Java Servlets: Introduction, HTTP Servlet Basics, The Servlet Lifecycle, Retrieving Information, Sending HTML Information, Session Tracking.

UNIT-V

Java Server Pages: Introducing Java Server Pages, JSP Overview, Setting Up the JSP Environment, Generating Dynamic Content, Using Custom Tag Libraries and the JSP Standard Tag Library, Processing Input and Output.

Referential Books:

1. Patrick Naughton and Herbertz Schildt, "Java-2 The Complete Reference" 199,TMH.
2. Shelley Powers, "Dynamic Web Publishing" 2nd Ed. Techmedia,1998.
3. Ivor Horton, "Beginning Java-2" SPDPublication
4. Jason Hunter, "Java Servlet Programming"O'Reilly
5. Shelley Powers, "Dynamic Web Publishing" 2nd Ed. Techmedia,1998
6. Hans Bergsten, "Java Server Pages", 3rd Ed.O'reilly

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-5003

Computer Network

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UNIT-I

Basic Concepts: Components of data communication, distributed processing, standards and

organizations. Line configuration, topology, Transmission mode, and categories of networks.

OSI and TCP/IP Models: Layers and their functions, comparison of models.

Digital Transmission: Interfaces and Modems: DTE-DCE Interface, Modems, Cable modems.

UNIT-II

Transmission Media: Guided and unguided, Attenuation, distortion, noise, throughput, propagation speed and time, wavelength, Shannon capacity, comparison of media.

UNIT-III

Telephony: Multiplexing, error detection and correction: Many to one, One to many, WDM, TDM, FDM, Circuit switching, packet switching and message switching.

Data link control protocols: Line discipline, flow control, error control, synchronous and asynchronous protocols, character and bit oriented protocols, Link access procedures.

Point to point controls: Transmission states, PPP layers, LCP, Authentication, NCP.

ISDN: Services, Historical outline, subscriber's access, ISDN Layers and broadcast ISDN.

UNIT-IV

Devices: Repeaters, bridges, gateways, routers, The Network Layer; Design issues, Internetworking, Network-Layer in the internet.

UNIT-V

Transport and upper layers in OSI Model: Transport layer functions, connection management, functions of session layers, presentation layer and a

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-5004

Numerical Methods

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UNIT-I

Roots of Equations: Bisections Method, False Position Method, Newton's Raphson Method, Rate of convergence of Newton's method.

UNIT-II

Interpolation and Extrapolation : Finite Differences, The operator E, Newton's Forward and Backward Differences, Newton's dividend differences formulae, Lagrange's Interpolation formula for unequal Intervals.

UNIT-III

Numerical Differentiation Numerical Integration : Introduction, direct methods, maxima and minima of a tabulated function, General Quadratic formula.

UNIT-IV

Solution of Linear Equation: Gauss's Elimination method and Gauss's Siedel iterative method.

UNIT-V

Solution of Differential Equations: Euler's method, Picard's method, Fourth-order Runga – Kutta method.

Referential Books:

1. Scarbourogh, "Numerical Analysis".

1. Gupta & Bose S.C. "Introduction to Numerical Analysis, "Academic Press, Kolkata, 3.

S.S.Shashtri, " Numerical Analysis", PHI

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Bachelors of Computer Application

Course Code Course Name

L T P C

BCA-5005

Minor Project

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Evaluation will be based on Summer Training held after fourth semester and will be Conducted by the college committee only.

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

L T P C

BCA-5006

Viva-Voice on Summer Training

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The viva will be conducted based on summer training of four weeks after the end of fourth Semester and will be Conducted by the college committee only.

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

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BCA-5002P

Computer Laboratory and Practical Work of Java

Programming and Dynamic Webpage Design

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Practical will be based on Paper Java Programming & Dynamic Webpage Design : on Whole

Syllabus

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-6001

Information & Cyber Security

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UNIT-I

Concept of Cyberspace: Netizens Technology, Law and Society Object, Scope of the Information

Technology Act, 2000, Electronic Records and Electronic Commerce., Intrusion Detection System,

Intrusion Prevention System, Public Key Infrastructure.

UNIT-II

Internet Security: Computer Security and Threats, Hacking, Cracking, sneaking, Viruses, Trojan Horses,

malicious code, Worms and Logic Bombs. Network attack and Defence Most Common Attacks, Scripts

Kiddies and Packaged Defense.

UNIT-III

Wireless Network Security : Wireless Network Components, Security issues in Wireless Networks,

Securing a Wireless Network, Mobile Security, The Smartphone Pentest Framework

UNIT-IV

Cyber Laws and Standards: ISO 27001, Cyber Law (Information Technology Act, 2000), International Standards maintained for Cyber Security, Security Audit ,Investigation by Investing Agency, Cyber Security Solutions.

UNIT-V

Security Management: Disaster Recovery, Digital Signature, Ethical Hacking, Penetration Testing, Computer Forensics

Referential Books:

1. GautamKumawat, Ethical Hacking & Cyber Security Course : A Complete Package, Udemmy Course, 2017
2. Georgia Weidman , Penetration testing A Hands-On In t r o d u c t i o n to Hacking, no starch press, 2014
3. Charles P. Pfleeger Shari Lawrence Pfleeger Jonathan Marg

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C.S.J.M.University,Kanpur.

Bachelors of Computer Application

Course Code

BCA-6002

Course Name

Internet Of Things

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UNIT-I

Internet of Things (IoT): Vision, Definition, Conceptual Framework, Architectural view, technology behind IoT, Sources of the IoT, M2M Communication, IoT Examples.

UNIT-II

M2M vs IoT An Architectural Overview: Building architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. Reference Architecture and Reference Model of IoT.

UNIT-III

Hardware for IoT: Sensors, Digital sensors, actuators, radio frequency identification (RFID) technology, wireless sensor networks, participatory sensing technology. Embedded Platforms for IoT:

Embedded computing basics, Overview of IOT supported Hardware platforms.

UNIT-IV

Network & Communication aspects in IoT: Wireless Medium access issues, MAC protocol survey, Survey routing protocols, Sensor deployment & Node discovery.

UNIT-V

Domain specific applications of IoT: Home automation, Industry applications, Surveillance applications, Other IoT application.

Referential Books:

1. Arshdeep Bahga, Vijay Madisetti "Internet of Things (A hands on approach)" 1ST edition, VPI publications, 2014
2. Jeeva Jose, Internet of Things, Khanna Publishing House
3. Michael Miller "The Internet of Things" by Pearson
4. Raj Kamal "INTERNET OF THINGS", McGraw-Hill, 1ST Edition, 2016

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C.S.J.M.University, Kanpur.

Bachelors of Computer Application

Course Code Course Name

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BCA-6003

E-Commerce

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UNIT-I

Introduction to E-Commerce: The Scope of Electronic Commerce, Definition of Electronic Commerce, Electronic E-commerce and the Trade Cycle, Electronic Markets, Electronic Data Interchange, Internet Commerce, E-Commerce in Perspective.

UNIT-II

Business-to-Business Electronic Commerce: Characteristics of B2B EC, Models of B2B EC, Procurement Management Using the Buyer's Internal Marketplace, Just in Time Delivery, Other B2B Models, Auctions and Services from Traditional to Internet Based EDI, Intergration with Back-end Information System, The Role of Software Agents for B2B EC, Electronic marketing in B2B, Solutions of B2B EC, Managerial Issues, Electronic Data Interchange (EDI), EDI: The Nuts

and Bolts, EDI & Business.

UNIT-III

Internet and Extranet : Automotive Network Exchange, The Largest Extranet, Architecture of the Internet, Intranet and Extranet, Intranet software, Applications of Intranets, Intranet Application Case Studies, Considerations in Intranet Deployment, The Extranets, The structures of Extranets, Extranet products & services, Applications of Extranets, Business Models of Extranet Applications, Managerial Issues.

Electronic Payment Systems : Is SET a failure, Electronic Payments & Protocols, Security Schemes in Electronic payment systems, Electronic Credit card system on the Internet, Electr

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Bachelors of Computer Application

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BCA-6004

Data Science and Machine Learning

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UNIT-I

Introduction to Data Science: Evolution of Data Science, Data Science Roles, Stages in a Data Science Project, Applications of Data Science in various fields, Data Security Issues.

UNIT-II

Data Collection and Data Pre-Processing: Data Collection Strategies, Data Pre-Processing Overview , Data Cleaning, Data Integration and Transformation, Data Reduction.

UNIT-III

Exploratory Data Analytics: Descriptive Statistics - Mean, Standard Deviation, Skewness and Kurtosis – Box Plots – Pivot Table – Correlation Statistics – ANOVA.

UNIT-IV

Introduction: Idea of Machines learning from data, Classification of problem – Regression and Classification, Supervised and Unsupervised learning.

UNIT-V

Neural Networks: History,Artificial and biological neural networks,Artificial intelligence and neural

networks,Biological neurons,Models of single neurons,Different neural network models.

Referential Books:

1. Cathy O'Neil and Rachel Schutt , "Doing Data Science", O'Reilly, 2015.
2. David Dietrich, Barry Heller, Beibei Yang, "Data Science and Big data Analytics", EMC 2013
3. Machine Learning, Tom M. Mitchell
4. Introduction to Machine learning, Nils J.Nilsson

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BCA-6005

Major Project

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Evaluation will be based on held after fourth semester and will be

Conducted by the college committee only.

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Bachelors of Computer Application

Course Code Course Name

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BCA-6006

Presentation/Seminar based on Major Project

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Presentation/Seminar based on Major Project will be evaluated by external examiner only.

DOCUMENT CHUNK 34: VSICS [HTML Page] | URL: <https://www.vsicsskanpur.org/course-single.html>

Our Events

Srijan- An Innovation to Creativity

Ever since the advent of human beings, they have been ornamented with an attribute called creativity. To adorn it more, DR. VIRENDRA SWARUP INSTITUTE OF COMPUTER STUDIES created SRIJAN - THE ANNUAL CULTURAL FESTIVAL. It's a compendium of courage, valour, talent, dance, drama, singing, innovation to new ideas, are been portrayed by our students. The extravagant show, been nurtured by sheer talent and enthusiasm live up to the expectations of the spectators. Organized in the winters, where the stage is all set to have a larger than life experience. Students, staff, faculties & the management whole heartedly give their best to make it a mega success.

Winter Carnival - Fun Fiesta

Among the various avenues & moments of learning, the students of Dr. VSICS organized Winter Carnival- Fun Fiesta. Fun Fiesta Envision, Enter prize & Enthrall, one day event to provide our students with opportunity to give space to their ideas & explore their professional aptitude of creating & managing with exuberance & frolic. The events was a practical exposure to in- class studies marking the day by plethora of activities, amusement stalls, lucky draw & many other besides.

Energy - The Athletic Meet

VSEF organizes two-day athletic meet - Energy providing students with the opportunity to relax & rejuvenate exploring an untouched realm in their lives - sports & athletics. The two day event was marked by plethora of track & field events participated by eight teams of VSEF with immense vigour & fervour.
